



# **General Biology (Biol1012)**

## **Lecture Slides**

### **Chapter 2: Biomolecules**



# Chapter 2: Biological Molecules

- At its most fundamental level, **life is made up of matter**
- **Matter** is any substance that **occupies space and has mass**.
- **Elements** are unique forms of matter with specific chemical and physical properties that cannot be broken down into smaller substances by ordinary chemical reactions.
- There are **118** elements, but only **92** occur naturally.

## How many elements are represented in life?

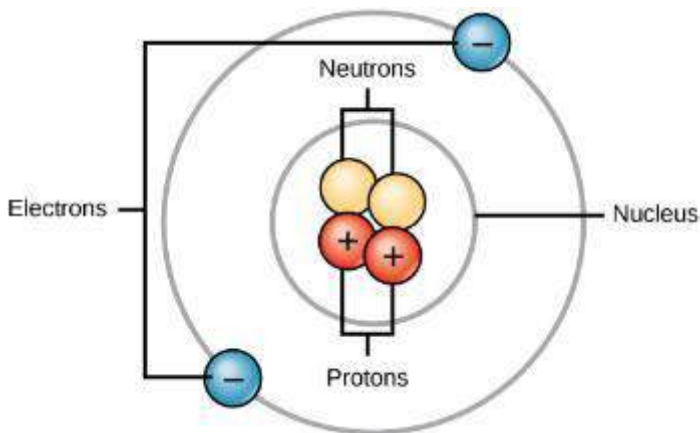
- Of the 92 naturally occurring elements, only fewer than 30 are found in living cells
- The smallest component of an element that retains all of the chemical properties of that element is called an **Atom**
- All atoms contain the subatomic particles:
  - ✓ protons
  - ✓ electrons, and
  - ✓ neutrons (H is the exception, lacking neutrons)



- **A proton:**
  - ✓ is a positively charged particle (+1)
  - ✓ located in the nucleus (core) of the atom
  - ✓ has a mass of 1 dalton
- **An electron:**
  - ✓ is a negatively charged particle (-1)
  - ✓ travels in the space around the nucleus
  - ✓ has a negligible mass
- **A neutron:**
  - ✓ does not have a charge (0)
  - ✓ located in the nucleus (core) of the atom
  - ✓ has a mass of 1 dalton



- *Atoms are made up of protons and neutrons located within the nucleus, and electrons surrounding the nucleus.*
- *Atoms of each element contain a characteristic number of protons and electrons. The number of protons determines an element's atomic number and is used to distinguish one element from another.*



The differences in chemical reactivity between the elements are based on the number and spatial distribution of an atom's electrons.

Atoms that chemically react and bond to each other form **molecules**. Molecules are simply two or more atoms chemically bonded together. Logically, when two atoms chemically bond to form a molecule, their electrons, which form the outermost region of each atom, come together first as the atoms form a **chemical bond**.





- **Chemical bonds** are interactions between atoms, ions or molecules that enable the formation of molecules (a force that holds them together).

## Major types of bonds or interactions:

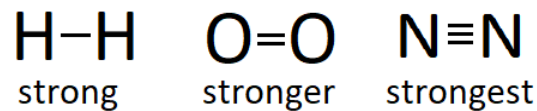
1. **Ionic bond**: formed from the electrostatic attraction between oppositely charged ions in a chemical compound.
  - ✓ Because the number of electrons does not equal the number of protons, each ion has a net charge.
  - ✓ **Cations** are positive ions that are formed by losing electrons.
  - ✓ Negative ions are formed by gaining electrons and are called **anions**.

*Movement of electrons from one element to another is referred to as electron transfer*



## 2. Covalent bond - form when an electron is shared between two atoms

- Covalent bonds are the **strongest** and most common form of chemical bond in living organisms
- ✓ When one pair of electrons are shared, it forms a single bond
- ✓ When two pairs of electrons are shared, it forms a double bond
- ✓ When three pairs of electrons are shared, it forms a triple bond

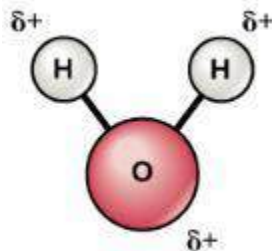




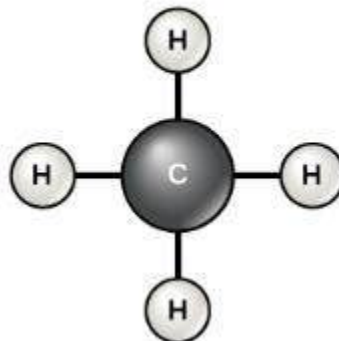
❑ **There are two types of covalent bonds: nonpolar covalent bonds and polar covalent bonds**

- **Nonpolar covalent bonds** are formed when the electrons are shared equally
- **Polar covalent bonds** are formed when the electrons are not shared equally, they spend more time closer to one nucleus than to the other nucleus. This gives the atom where the electron spends more time a small negative charge, and the atom missing the electron more a small positive charge

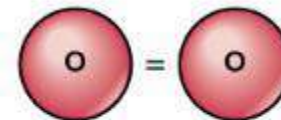
Polar covalent bond



Nonpolar covalent bond



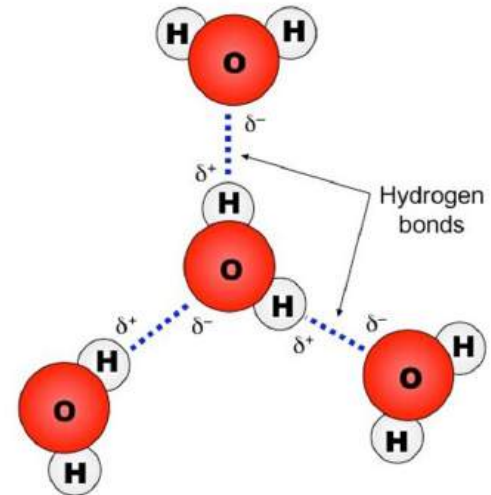
Nonpolar covalent bond





## Hydrogen Bond:

- When polar covalent bonds containing hydrogen form, the hydrogen in that bond has a slightly positive charge because hydrogen's electron is pulled more strongly toward the other element and away from the hydrogen.
- Because the hydrogen is slightly positive, it will be attracted to neighboring negative charges. When this happens, a weak interaction occurs between the  $\delta^+$  of the hydrogen from one molecule and the  $\delta^-$  charge on the more electronegative atoms of another molecule, usually oxygen or nitrogen, or within the same molecule. This interaction is called a **hydrogen bond**





# Important Molecules of life

## Organic molecules:-

- carbohydrates
- lipids
- Proteins
- nucleic acids (DNA, RNA)
- Vitamins

## Inorganic molecules:

- water
- oxygen
- carbon dioxide
- mineral salts



# Macromolecules

Large molecules necessary for life are built from smaller organic molecules and are called biological macromolecules

There are four major classes of biological macromolecules:

1. Carbohydrates
2. Lipids,
3. Proteins, and
4. Nucleic acids (DNA and RNA).



# Carbohydrates



- Also called saccharides
- **most abundant organic** molecules in our biosphere
- composed of carbon, hydrogen and oxygen
- General formula  **$C_n(H_2O)_n$** , where n is the number of carbon atoms in the molecule

## Functions

- Primary source of energy (glucose)
- Energy storage (glycogen, starch)
- Structural component of
  - ✓ *cell membrane (Glycoproteins and Glycolipids),*
  - ✓ *Cell wall of plants (cellulose).*
  - ✓ *cell wall of bacteria (Peptidoglycan),*
  - ✓ *cell wall of fungi (Chitin),*
  - ✓ *exoskeleton of many insects are made up of carbohydrates*



## Classification of Carbohydrates

Based on the number of sugar units carbohydrates can be classified as:

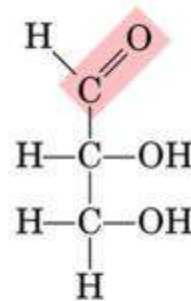
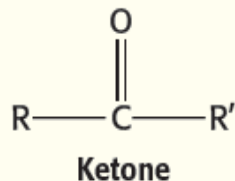
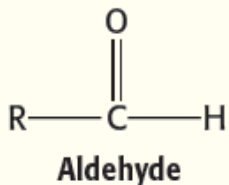
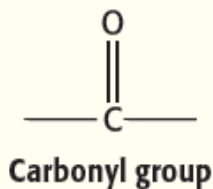
- **Monosaccharides** – composed of one sugar unit
- **Oligosaccharides**: Composed of 2 to 20 sugar units and includes
  - ✓ Disaccharides (composed of two sugar subunits),  
Trisaccharides... etc
- **Polysaccharides** – Composed of more than 20 sugar units



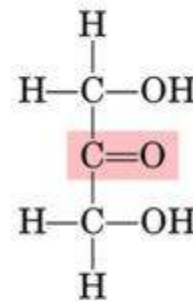


# Monosaccharides

- (mono– = “one”; sacchar– = “sweet”) are simple sugars the most common of which is glucose
- Represented by the general formula **(CH<sub>2</sub>O)<sub>n</sub>** where n is 3 -7
- Most monosaccharide names end with the suffix –ose
- consists of a carbon chain with a number of hydroxyl (OH) groups and either an aldehyde group or a ketone functional group



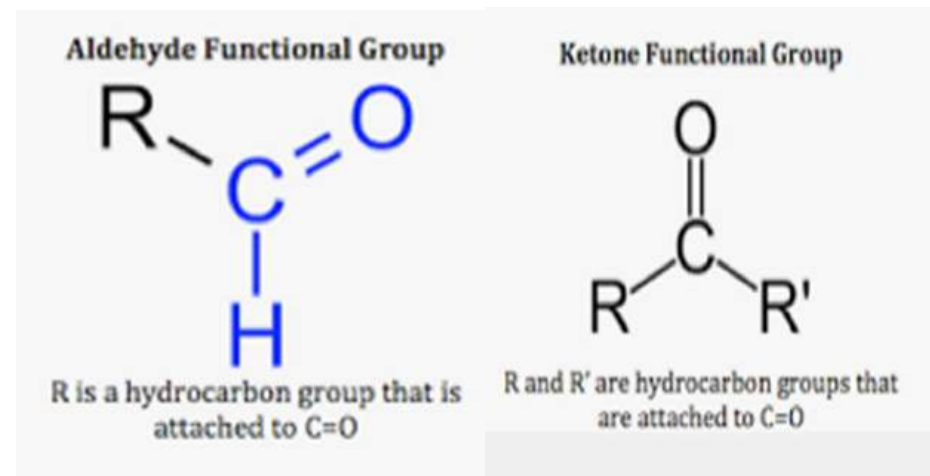
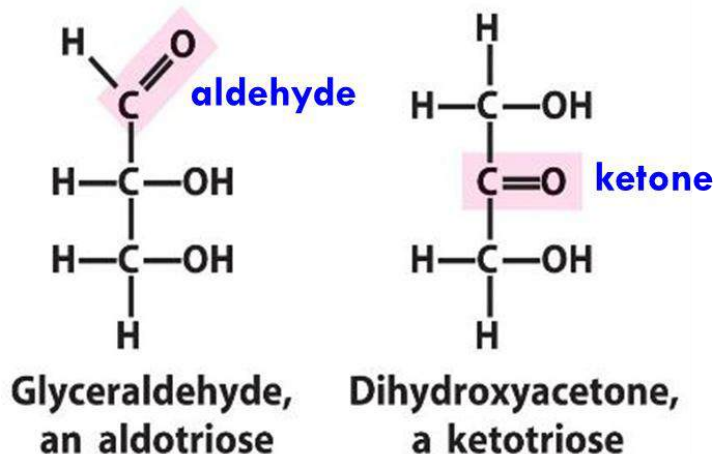
Aldose



Ketose



- Monosaccharides have common (trivial) names and systematic names.
  - ✓ **Systematic name** indicates both the number of carbon atoms present and functional group (aldehyde or ketone).
  - ✓ A sugar that bears an aldehyde group is called an aldose whereas a sugar with a ketone group is a ketose.
  - ✓ For example, glyceraldehyde is aldotriose and dihydroxyacetone is ketotriose





Depending on the number of carbons in the sugar, they also may be known as

| <i>No. of carbon atoms</i> | <i>Generic name</i> | <i>Aldoses (with aldehyde group)</i> | <i>Ketoses (with keto group)</i> |
|----------------------------|---------------------|--------------------------------------|----------------------------------|
| 3                          | Triose              | Glyceraldehyde                       | Dihydroxy-acetone                |
| 4                          | Tetrose             | Erythrose                            | Erythrulose                      |
| 5                          | Pentose             | Arabinose, Xylose, Ribose            | Xylulose, Ribulose               |
| 6                          | Hexose              | Glucose, Galactose, Mannose          | Fructose                         |
| 7                          | Heptose             |                                      | Sedoheptulose                    |



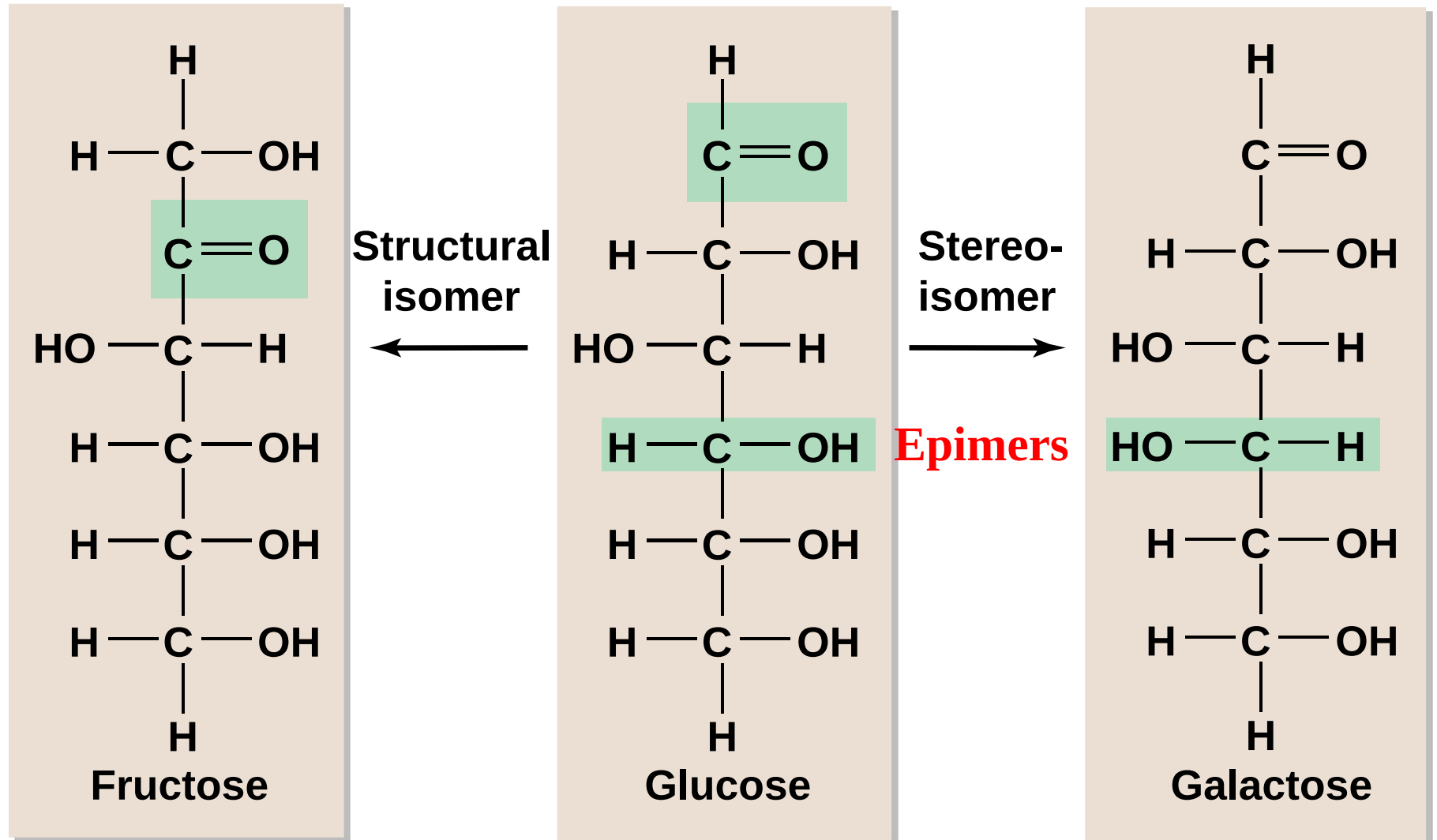
## Examples of Monosaccharides

- **Glucose:** the most predominant sugar in human body. It is the major source of energy
- **Galactose** is a constituent of lactose (milk sugar) and glycoproteins
- **Fructose** is the constituent of sucrose, the common sugar
- Although **glucose**, **galactose**, and **fructose** all have the **same chemical formula** ( $C_6H_{12}O_6$ ), **they differ structurally and chemically** (and are **isomers**)

**Isomers** are compounds that have the same chemical formula but different structures

**Stereoisomers** have the same chemical formula but differ in the position of the hydroxyl groups on one or more of their asymmetric carbons.

- ✓ **Epimers** are stereoisomers that differ in the position of the hydroxyl group at only one asymmetric carbon. For example, glucose and galactose are epimers that differ at carbon 4.



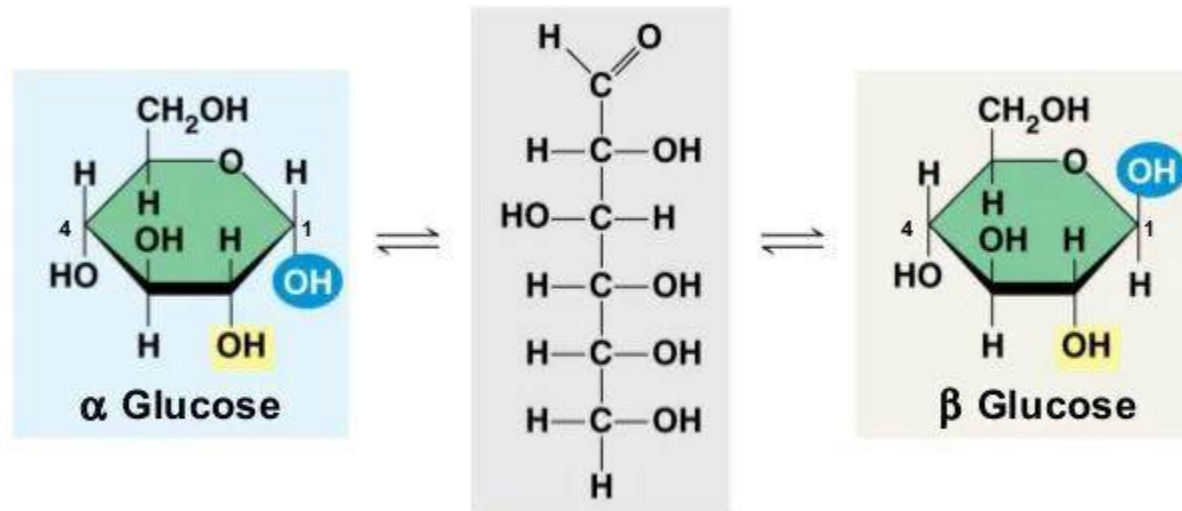


# Ring structure of carbohydrates

Monosaccharides with five or more carbons exist predominantly in a ring (cyclic) form

- In ring structures, the **aldehyde** or **ketone group reacts with a hydroxyl group** in the same molecule and results in  $\alpha$  and  $\beta$  forms are called **anomers**
- If the hydroxyl group is below carbon number 1 in the sugar, it is said to be in the alpha ( $\alpha$ ) position, and if it is above the plane, it is said to be in the beta ( $\beta$ ) position.

If the hydroxyl group is below carbon number 1 in the sugar, it is in the alpha ( $\alpha$ ) position, and if it is above the plane, it is in the beta ( $\beta$ ) position



(a)  $\alpha$  and  $\beta$  glucose ring structures

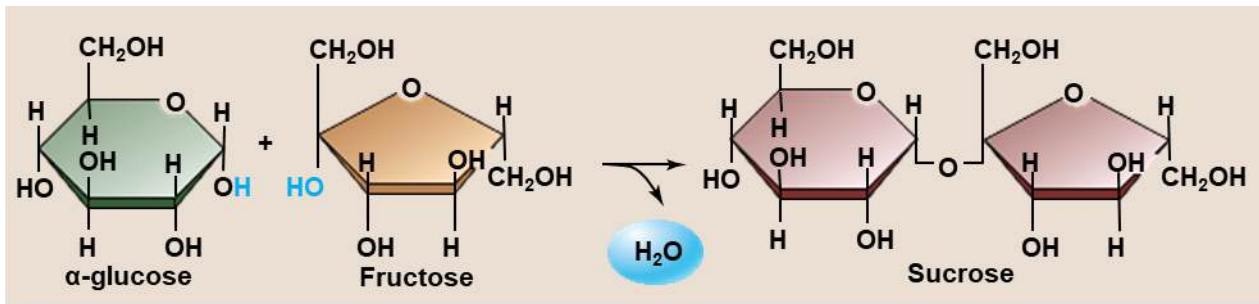


# Disaccharides

- **Disaccharides** (di- = “two”) form when two monosaccharides undergo a **dehydration reaction** (also known as a **condensation reaction** or dehydration synthesis).
- During this process, the hydroxyl group of one monosaccharide combines with the hydrogen of another monosaccharide, **releasing a molecule of water and forming a covalent bond**.
- A covalent bond formed between a carbohydrate molecule and another molecule (in this case, between two monosaccharides) is known as a **glycosidic bond**

**Common disaccharides include lactose, maltose and sucrose**

- **Lactose** = glucose + galactose (found in milk)
- **Maltose** = glucose + glucose (malt sugar)
- **Sucrose** = glucose + fructose (table sugar)



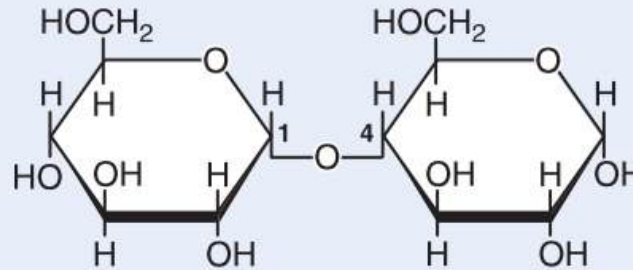




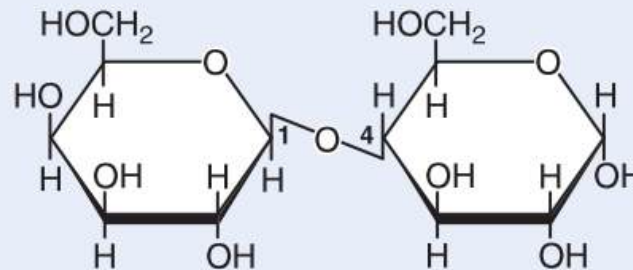
Glycosidic bonds (also called glycosidic linkages) can be of the alpha or the beta type.

An **alpha bond** is formed when the OH group on the carbon-1 of the first monosaccharide is below the ring plane, and

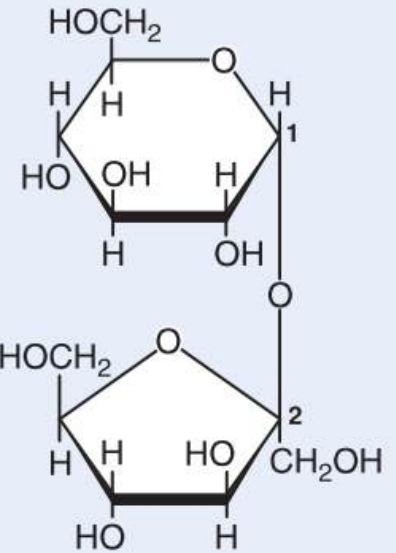
a **beta bond** is formed when the OH group on the carbon-1 is above the ring plane.



**Maltose**  
(Glucose- $\alpha$ (1 $\rightarrow$ 4)-glucose)



**Lactose**  
(Galactose- $\beta$ (1 $\rightarrow$ 4)-glucose)



**Sucrose**  
(Glucose- $\alpha$ (1 $\rightarrow$ 2)-fructose)





## Lactose intolerance

- Lactose is digestible by most humans by **lactase** enzyme present in the small intestine to galactose and glucose.
- 5% of people from Scandinavia and 90% of Asian adults suffer from lactose intolerance – a deficiency in the enzyme lactase.

For those with the deficiency:

- Lactose accumulates in the small intestine, Degraded by intestinal bacteria producing  $\text{CO}_2$ , hydrogen gas, and organic acid

### Symptoms

- Bloating, Nausea, Cramping, Diarrhea





# Polysaccharides

- A long chain of monosaccharides linked by glycosidic bonds is a polysaccharide (poly- = “many”) through **dehydration synthesis**  
**E.g. Starch, Glycogen, Cellulose.**

## Major classes of polysaccharides:

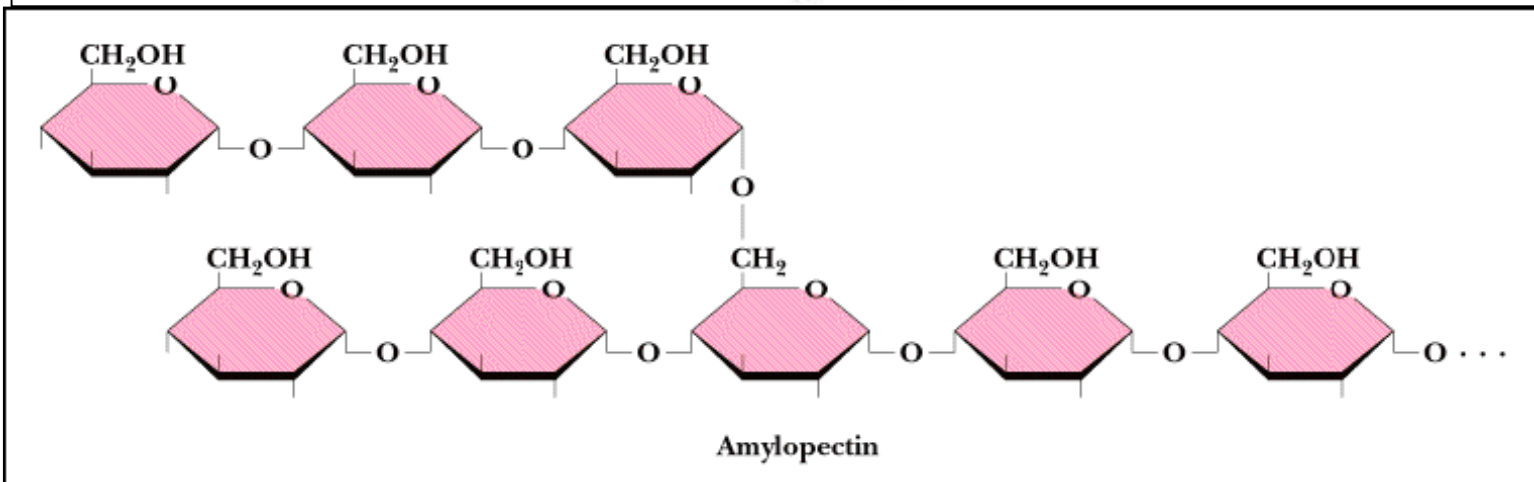
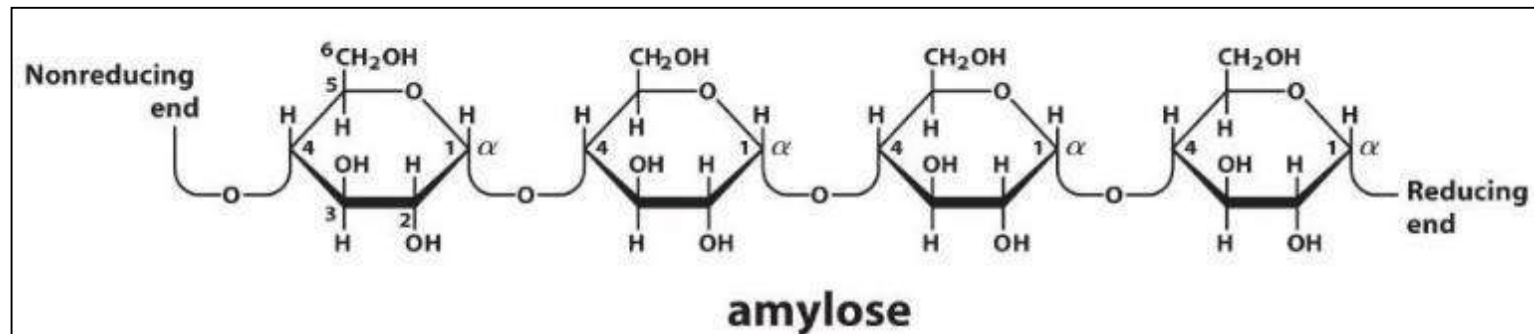
**Starch:** is the stored form of sugars in plants and is made up of a mixture of **amylose** and **amylopectin** (both polymers of glucose).

- Plants are able to synthesize glucose, and they store the excess glucose, beyond their immediate energy needs, as starch in different plant parts, including roots and seeds.
- The starch in the seeds provides food for the embryo as it germinates and can also act as a food source for humans and animals.

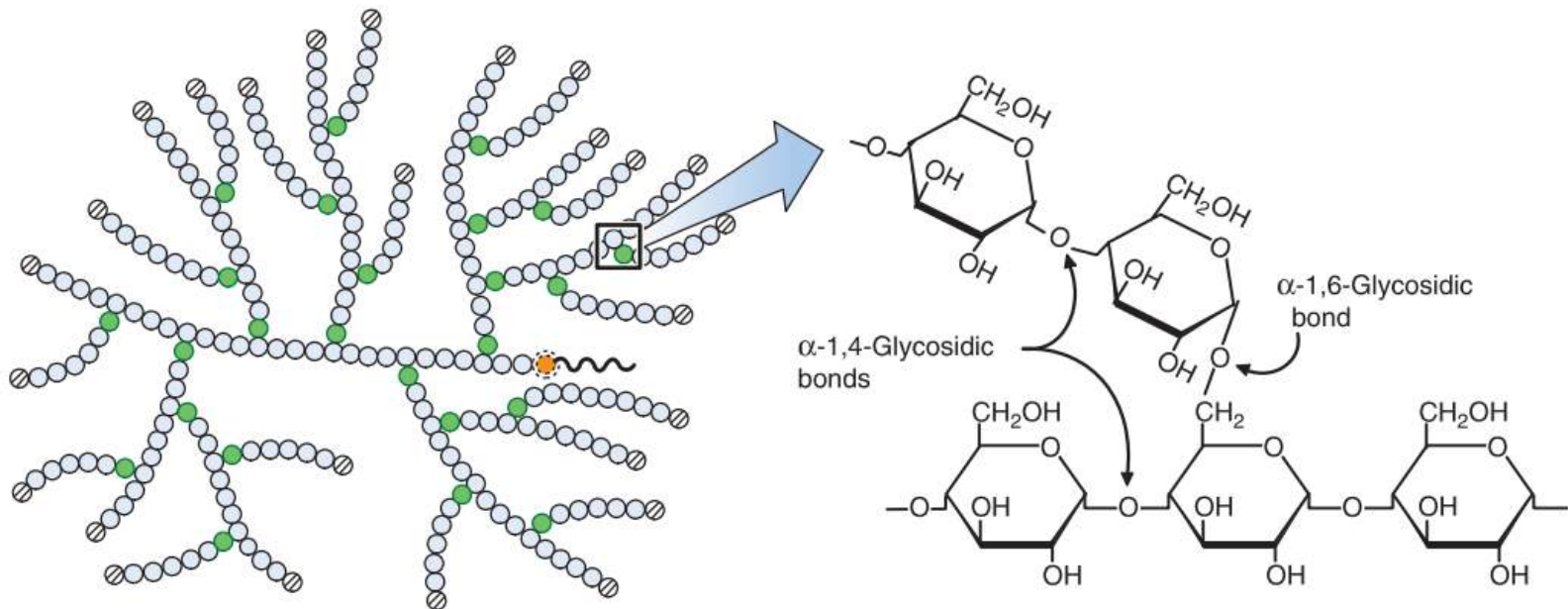


## Structure of starch

- In starch, glucose monomers are joined by  $\alpha$  1-4 or  $\alpha$  1-6 glycosidic bonds. The numbers 1-4 and 1-6 refer to the carbon number of the two residues that have joined to form the bond.



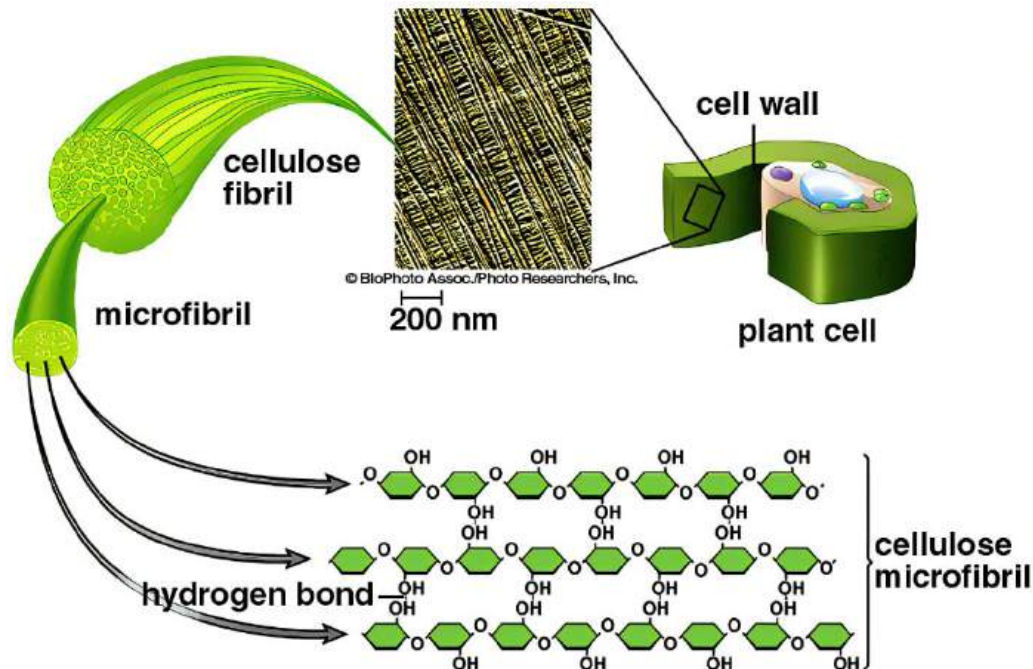
- 1) **Glycogen**: is the **storage form of glucose in humans and other vertebrates** and is made up of monomers of glucose.
- Glycogen is the animal equivalent of starch and is a highly branched molecule usually stored in liver and muscle cells.
  - Whenever blood glucose levels decrease, glycogen is broken down to release glucose in a process known as **glycogenolysis**.





## Cellulose:

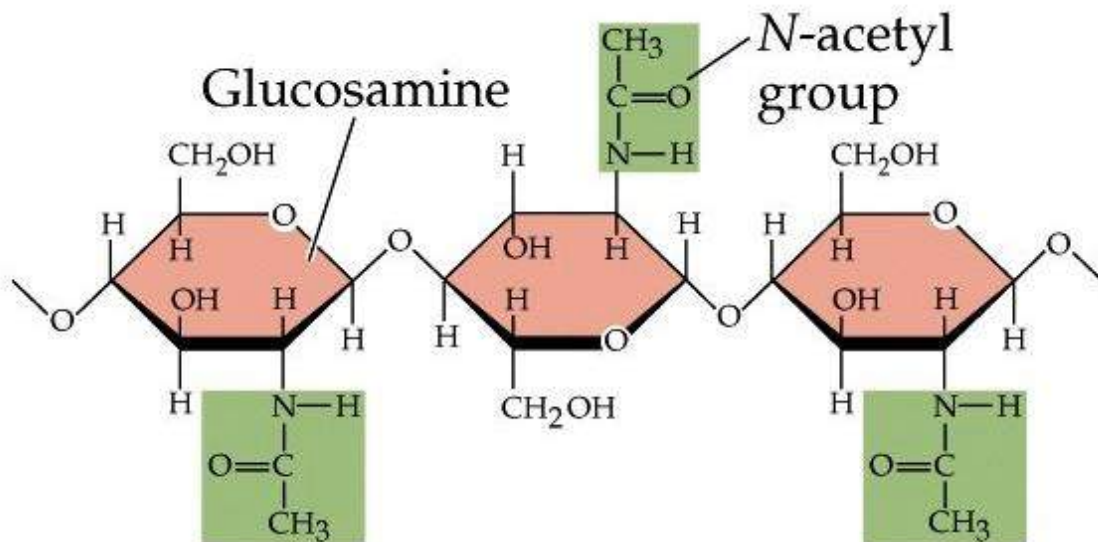
- Cellulose is made up of glucose monomers that are linked by  $\beta$  1-4 glycosidic bond
- make cell wall of plants providing structural support to the cell.
- Wood and paper are mostly cellulosic in nature.





## Chitin

- is a polysaccharide-containing nitrogen.
- It is made of repeating units of N-acetyl- $\beta$ -d-glucosamine, a modified sugar.
- Make the structure of fungal cell wall and insects exoskeleton







# Lipids

## Lipids

- include a diverse group of compounds that are largely nonpolar in nature (water-insoluble)
- They are esters of fatty acids and alcohol (glycerol or chains of alcohols).
- are highly soluble in organic solvents (ether, chloroform, and acetone)

## Functions of lipids

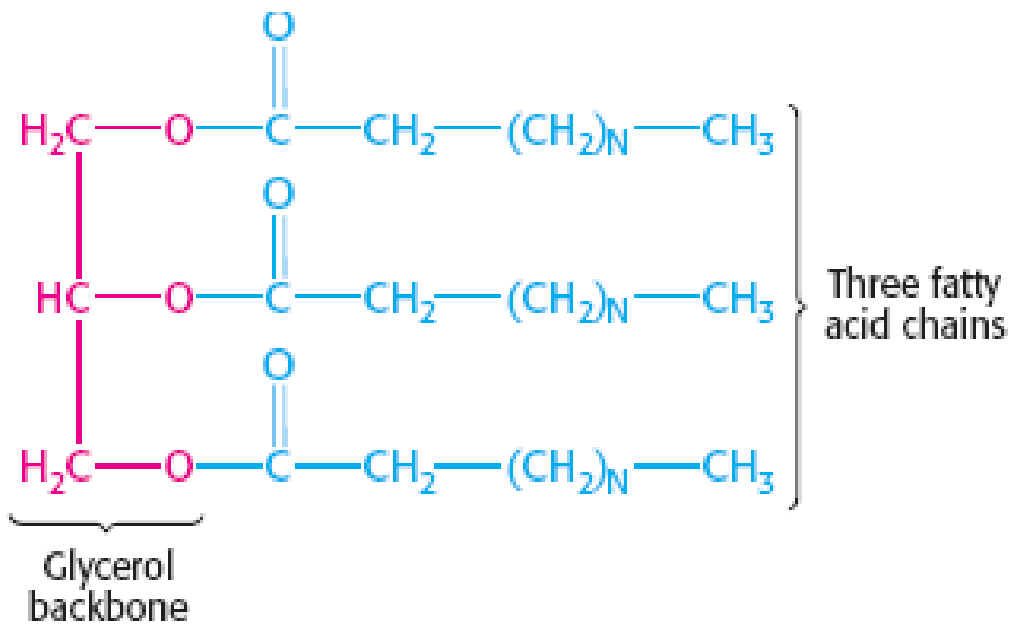
- Make structural components in cell membranes (e.g., phospholipids and sphingolipids)
- For long term storage of energy. e.g. fats and oils
- Serve as chemical signals (hormones), vitamins, or pigments.
- Form protective or waterproofing functions.
- provide insulation from the environment for plants and animals

***Lipids include fats, oils, waxes, phospholipids, and steroids.***



# Fats and Oils

- Also called Triacylglycerols
- consist of three fatty acid chains esterified to a glycerol backbone.







# Fatty acids

- are hydrocarbons, which terminate with carboxylic acid groups = **R-COOH**
- Most have an even number of carbon atoms
- Chain length = 12 to 24, most with 16 or 18 carbon atoms.
- **Amphipathic** – have both hydrophobic and hydrophilic parts

**R-COOH**

The diagram shows the text 'R-COOH' in blue. Two red curved arrows originate from the text 'Amphipathic' in the list above. One arrow points from the word 'Amphipathic' down to the 'R' in 'R-COOH', indicating the hydrophobic part. The other arrow points from the word 'Amphipathic' down to the 'COOH' in 'R-COOH', indicating the hydrophilic part.

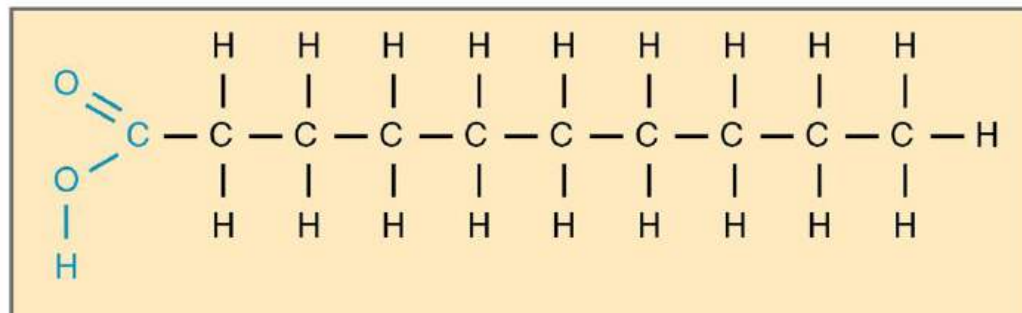
**Fatty acids may be saturated or unsaturated.**



- **Saturated fatty acids**

- ✓ no double bonds between carbon atoms. general formula  $\text{CH}_3(\text{CH}_2)_n\text{COOH}$ , where  $n$  is an even number
- ✓ are saturated with hydrogen, i.e. the number of hydrogen atoms attached to the carbon skeleton is maximized
- ✓ Have higher melting point and are solid at room temperature. E.g. fatty acids of animal origin
- ✓ saturated fats contribute to plaque formation in the arteries = Atherosclerosis- cause of heart attack, stroke, Gangrene

(a) Saturated

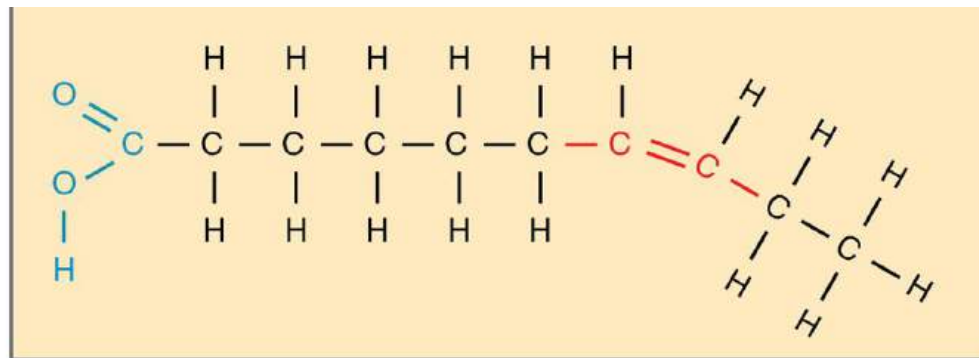




## Unsaturated fatty acids

- contain a double bond in their carbon-carbon chain
- Can be either
  - ✓ **Monounsaturated fatty acids:** have one double bond
  - ✓ **Polyunsaturated fatty acids:** have two or more double bonds.
- Have Lower melting points than saturated fatty acids and thus Liquid at room temperature. E.g. Fatty acids of plant origin

(b) Unsaturated





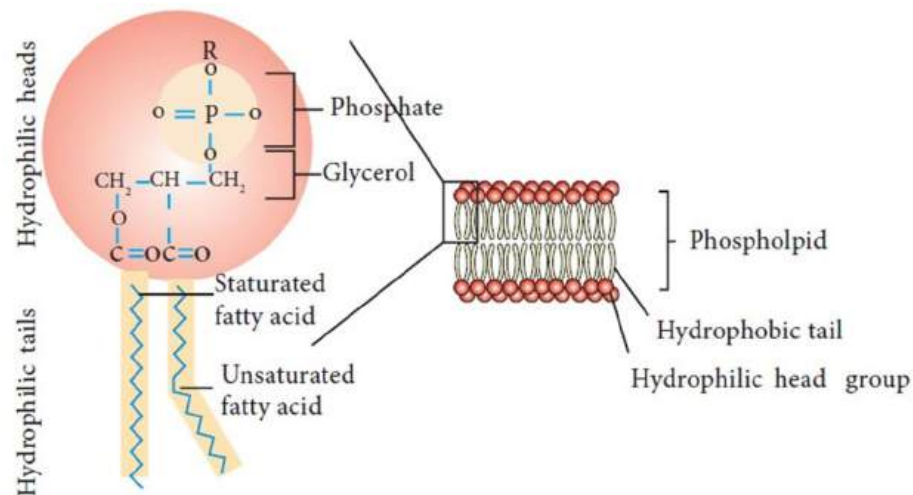
## Essential and Non-Essential fatty acids

- **Essential fatty acids** are fatty acids that are required but not synthesized by the human body; therefore they must be supplemented through the diet
  - ✓ There are two known essential fatty acids for humans: omega-3 fatty acid and omega-6 fatty acid
  - ✓ Omega-3 fatty acids are found in salmon, trout, and tuna; they are important in brain function, normal growth/development, and may prevent heart disease, and certain cancers
- **Non Essential Fatty acids** are fatty acids that can be synthesized in the body.
  - ✓ Organisms such as plants and bacteria can synthesize all the fatty acids they require



# Phospholipids

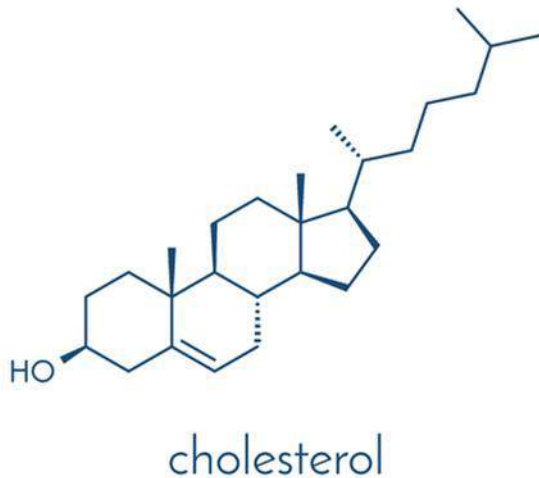
- Are structural components membranes
- Composed of glycerol + 2 FA (attached on C1 & C2 of glycerol) + phosphate (attached to C3 of glycerol)
- A phospholipid is an amphipathic molecule, meaning it has a hydrophobic and a hydrophilic part. The fatty acid chains are hydrophobic and cannot interact with water, whereas the phosphate-containing group is hydrophilic and interacts with water





## Steroids and Waxes

- Steroids have four carbon rings linked together and several of them (like cholesterol) have a short tail
- It is synthesized by the liver; and is a precursor of many hormones, of vitamins E and K, and of bile salts
- It is a key component of the plasma membrane of animal cells
- It is necessary for the proper functioning of the body





**Proteins** are the most abundant organic molecules in living systems

# Functions

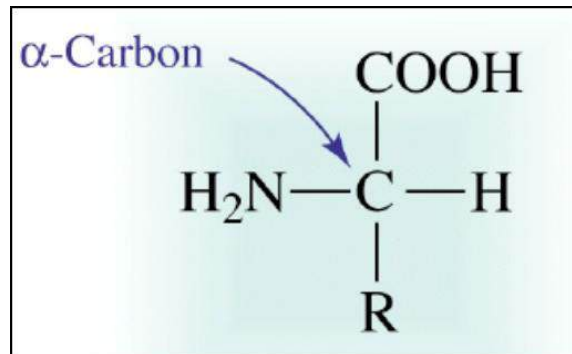
- Catalysis= enzymes
- Protection and support = collagen, Elastin
- Movement = actin, tubulin
- Defense = immunoglobulins (antibodies)
- Regulation = hormone, Growth factors
- Transport = hemoglobin, lipoproteins, Transferrin
- Storage = ovalbumin, casein
- Stress response = cytochrome P450, metallothionein, heat shock proteins







- All proteins are made up of different arrangements of the same 20 kinds of amino acids
- **Amino acids** are the monomers that make up proteins
- Each amino acid consists of a central carbon atom bonded to an amino group ( $-\text{NH}_2$ ), a carboxyl group ( $-\text{COOH}$ ) and a hydrogen atom



- Each amino acid has its own unique R group
- The R group is the only difference in structure between the 20 amino acid



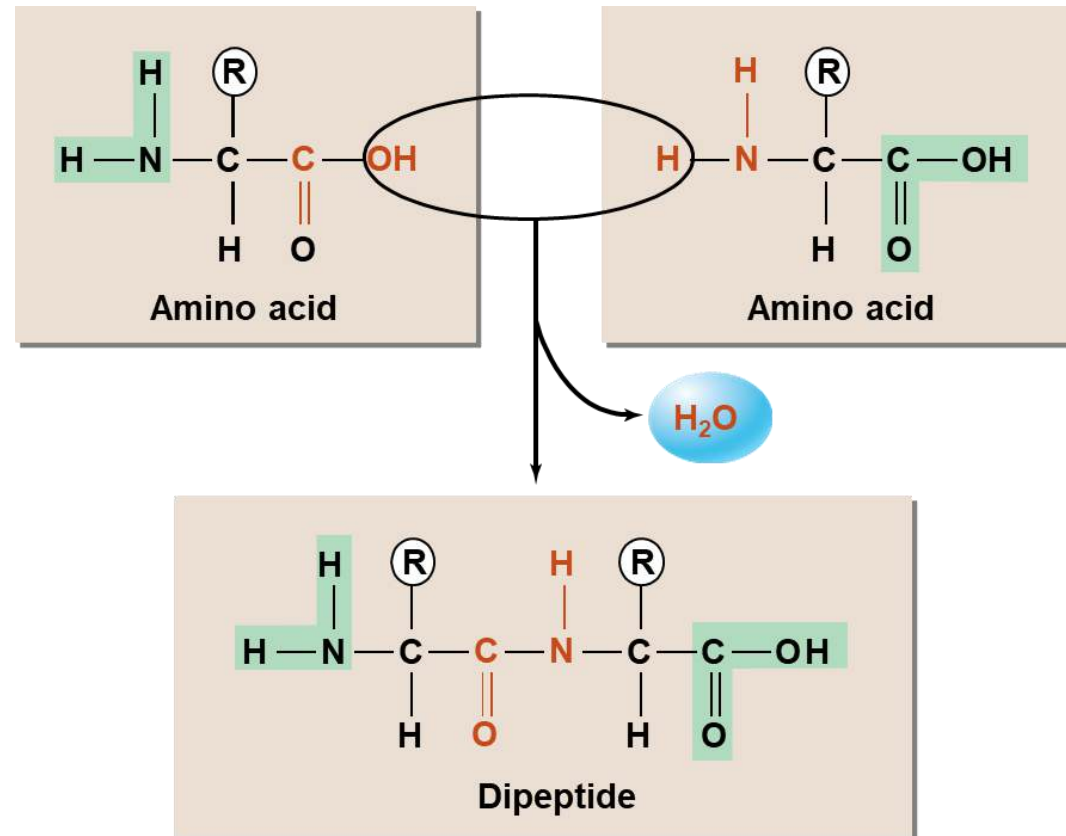
| AMINO ACID                   |                                                                                                                                                                                               |                                                                                                                                                                                           |                                                                                                                                                                                                          |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nonpolar, aliphatic R groups | $\begin{array}{c} \text{COO}^- \\   \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\   \\ \text{H} \end{array}$ <p>Glycine</p>                                                                    | $\begin{array}{c} \text{COO}^- \\   \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\   \\ \text{CH}_3 \end{array}$ <p>Alanine</p>                                                             | $\begin{array}{c} \text{COO}^- \\   \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\   \\ \text{CH} \\   \quad   \\ \text{CH}_3 \quad \text{CH}_3 \end{array}$ <p>Valine</p>                                 |
|                              | $\begin{array}{c} \text{COO}^- \\   \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\   \\ \text{CH}_2 \\   \\ \text{CH} \\   \quad   \\ \text{CH}_3 \quad \text{CH}_3 \end{array}$ <p>Leucine</p> | $\begin{array}{c} \text{COO}^- \\   \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\   \\ \text{CH}_2 \\   \\ \text{CH}_2 \\   \\ \text{S} \\   \\ \text{CH}_3 \end{array}$ <p>Methionine</p> | $\begin{array}{c} \text{COO}^- \\   \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\   \\ \text{H}-\text{C}-\text{CH}_3 \\   \\ \text{CH}_2 \\   \\ \text{CH}_3 \end{array}$ <p>Isoleucine</p>               |
|                              | $\begin{array}{c} \text{COO}^- \\   \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\   \\ \text{CH}_2\text{OH} \end{array}$ <p>Serine</p>                                                         | $\begin{array}{c} \text{COO}^- \\   \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\   \\ \text{H}-\text{C}-\text{OH} \\   \\ \text{CH}_3 \end{array}$ <p>Threonine</p>                       | $\begin{array}{c} \text{COO}^- \\   \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\   \\ \text{CH}_2 \\   \\ \text{SH} \end{array}$ <p>Cysteine</p>                                                         |
|                              | $\begin{array}{c} \text{COO}^- \\   \\ \text{H}_2\text{N}^+-\text{C}(\text{H})-\text{CH}_2-\text{CH}_2-\text{NH}_2 \end{array}$ <p>Proline</p>                                                | $\begin{array}{c} \text{COO}^- \\   \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\   \\ \text{CH}_2 \\   \\ \text{C}=\text{O} \\   \\ \text{H}_2\text{N} \end{array}$ <p>Asparagine</p>     | $\begin{array}{c} \text{COO}^- \\   \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\   \\ \text{CH}_2 \\   \\ \text{CH}_2 \\   \\ \text{C}=\text{O} \\   \\ \text{H}_2\text{N} \end{array}$ <p>Glutamine</p> |

| AMINO ACID                  |                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Positively charged R groups | $\begin{array}{c} \text{COO}^- \\   \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\   \\ \text{CH}_2 \\   \\ \text{CH}_2 \\   \\ \text{CH}_2 \\   \\ \text{CH}_2 \\   \\ \text{CH}_2 \\   \\ \text{NH}_3^+ \end{array}$ <p>Lysine</p> | $\begin{array}{c} \text{COO}^- \\   \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\   \\ \text{CH}_2 \\   \\ \text{CH}_2 \\   \\ \text{CH}_2 \\   \\ \text{CH}_2 \\   \\ \text{NH} \\   \\ \text{C}=\text{NH}_2^+ \\   \\ \text{NH}_2 \end{array}$ <p>Arginine</p> | $\begin{array}{c} \text{COO}^- \\   \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\   \\ \text{CH}_2 \\   \\ \text{C}-\text{NH}^+ \\   \quad   \\ \text{CH} \quad \text{N} \\    \quad   \\ \text{H} \quad \text{N} \end{array}$ <p>Histidine</p> |
|                             | $\begin{array}{c} \text{COO}^- \\   \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\   \\ \text{CH}_2 \\   \\ \text{COO}^- \end{array}$ <p>Aspartate</p>                                                                               | $\begin{array}{c} \text{COO}^- \\   \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\   \\ \text{CH}_2 \\   \\ \text{CH}_2 \\   \\ \text{COO}^- \end{array}$ <p>Glutamate</p>                                                                                        |                                                                                                                                                                                                                                                |
| Nonpolar, aromatic R groups | $\begin{array}{c} \text{COO}^- \\   \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\   \\ \text{CH}_2 \\   \\ \text{C}_6\text{H}_5 \end{array}$ <p>Phenylalanine</p>                                                                   | $\begin{array}{c} \text{COO}^- \\   \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\   \\ \text{CH}_2 \\   \\ \text{C}_6\text{H}_4\text{OH} \end{array}$ <p>Tyrosine</p>                                                                                            | $\begin{array}{c} \text{COO}^- \\   \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\   \\ \text{CH}_2 \\   \\ \text{Indole} \end{array}$ <p>Tryptophan</p>                                                                                         |



## Structure of Proteins

- In protein structure, each amino acid is attached to another amino acid by a covalent bond, known as a **peptide bond**, which is formed by a dehydration reaction
- The term **polypeptide** refers to a polymer of amino acids
- The sequence and number of amino acids ultimately determine a protein's shape, size, and function





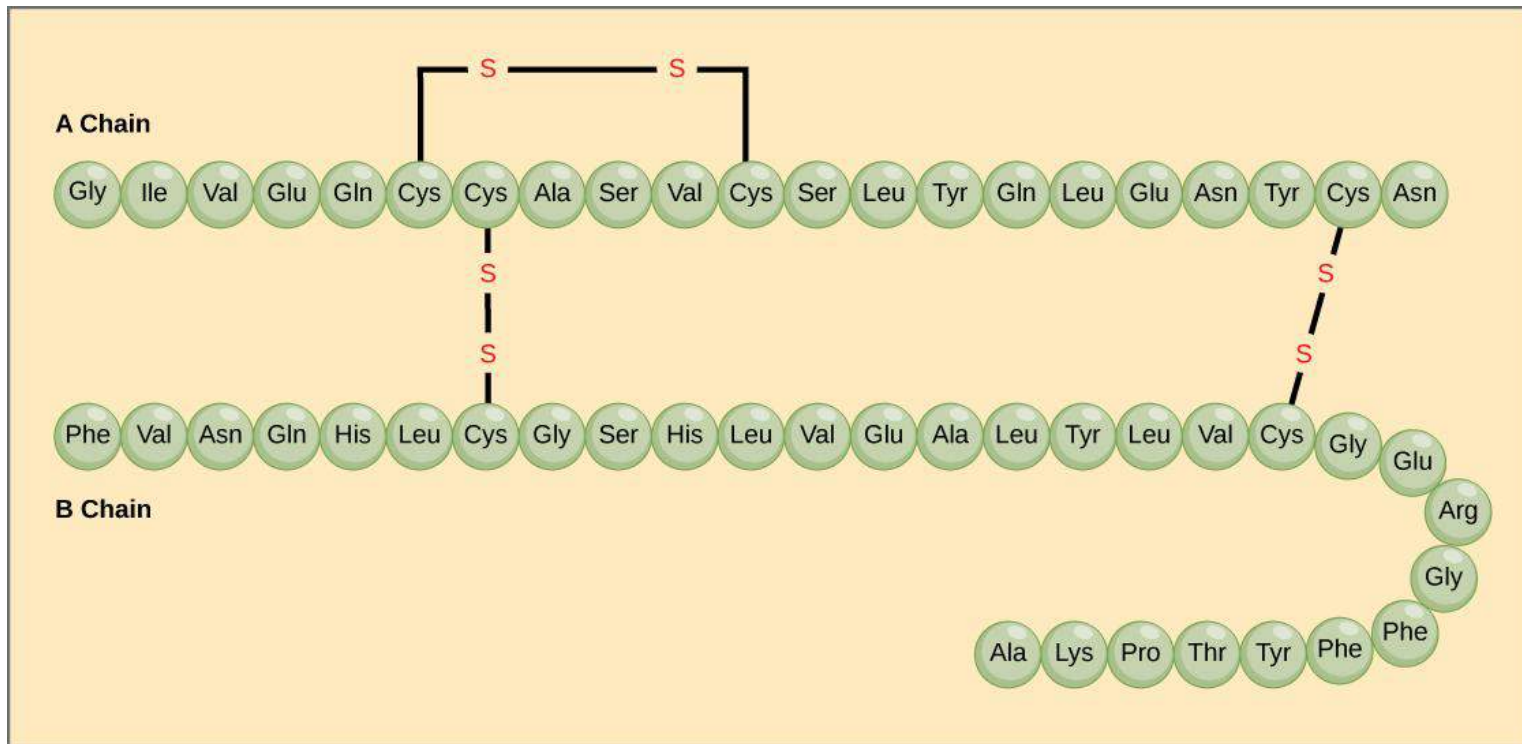
## Protein structure.....

- A protein's shape is critical to its function
- Each protein has its own unique sequence and shape
- **There are four levels to protein structure:**
  - ✓ Primary
  - ✓ Secondary
  - ✓ Tertiary
  - ✓ Quaternary



## 1) Primary structure:

- Refers to genetically determined sequence of amino acids
- Amino acids linked by peptide bonds
- Millions of different proteins made from the 20 standard amino acids





## Primar structure....

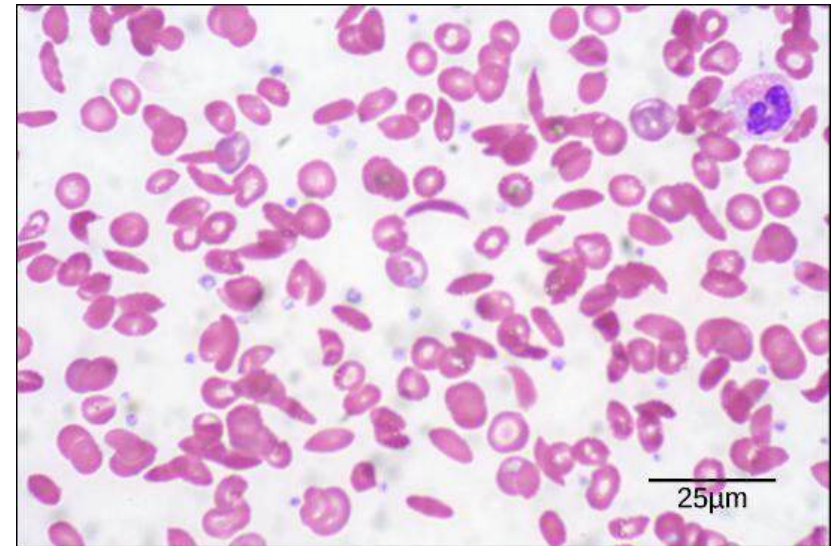
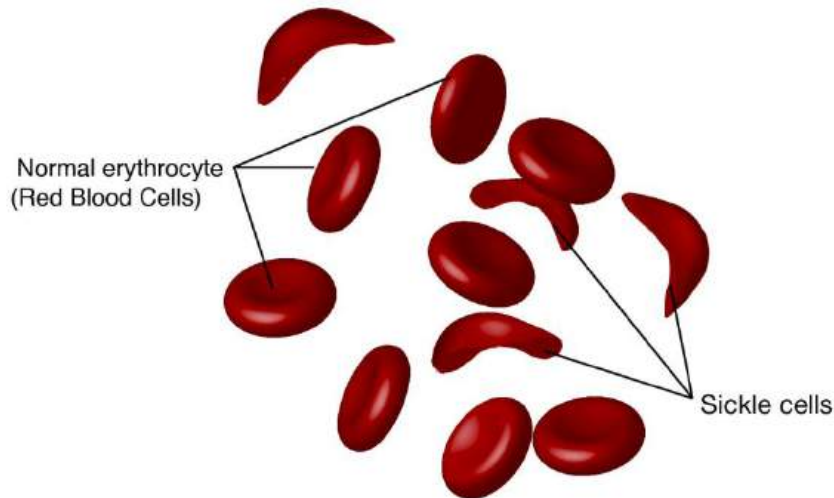
- A change in nucleotide sequence of the gene's coding region may lead to a different amino acid being added to the growing polypeptide chain, causing a change in protein structure and function. Example: **Sickle cell anemia**
  - ✓ In sickle cell anemia, the hemoglobin  $\beta$  chain has a single amino acid substitution, causing a change in protein structure and function. Specifically, the amino acid **glutamic acid** is **substituted by valine in the  $\beta$  chain**.
  - ✓ Because of this change of one amino acid in the chain, hemoglobin molecules form long fibers that distort the biconcave, or disc-shaped, red blood cells and assume a crescent or “sickle” shape, which clogs arteries.
- ✓ **sickle-cell anaemia** a condition caused by a mutation that affects the structure of the haemoglobin molecules in red blood cells causing the red blood cells to sickle under low oxygen tension.
- ✓ the haemoglobin molecule alters the triplet from GAG to GTG. This results in the amino acid valine replacing glutamate in the polypeptide chain.





## Cont'd

- ✓ This can lead to myriad serious health problems such as breathlessness, dizziness, headaches, and abdominal pain for those affected by this disease.



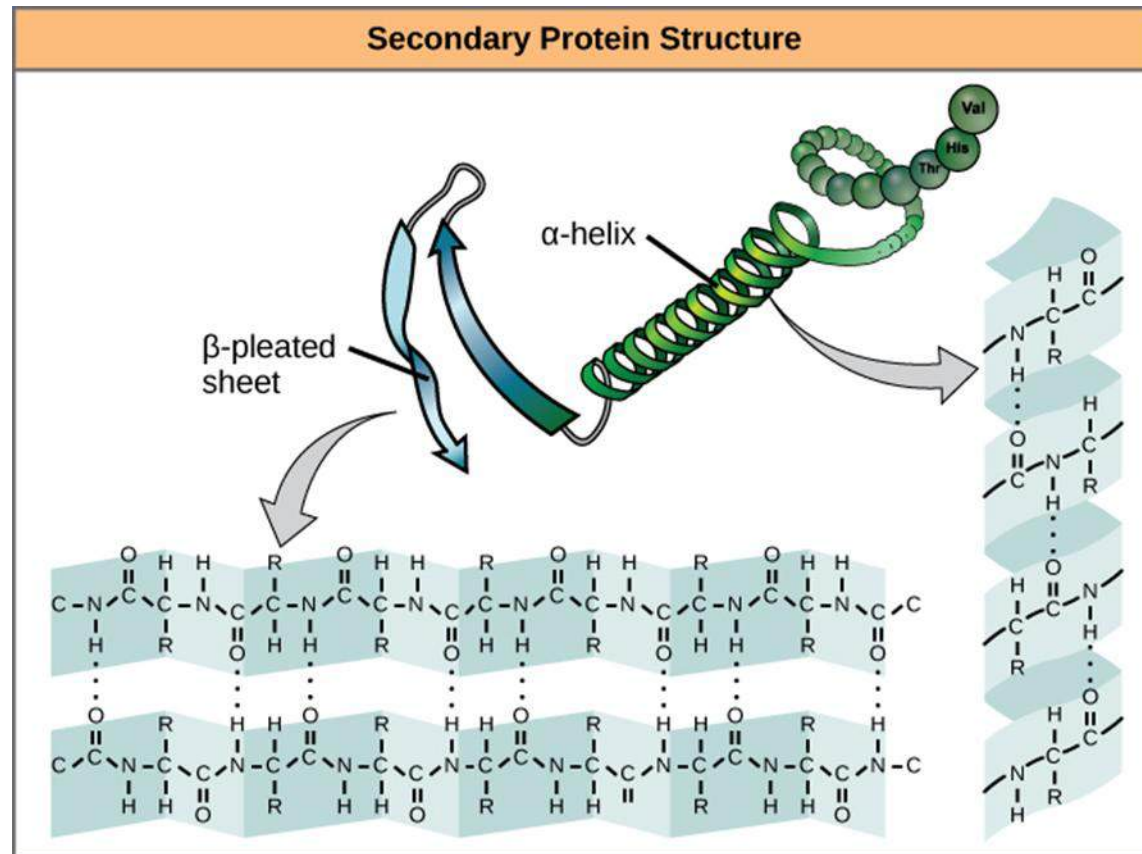


## 2) **Secondary structure:** formed due to interaction of groups in the peptide backbone of primary structure

- The structure is stabilized by **H-bonds** between AA

- (beta) sheet**
- (alpha) helix**

- $\alpha$ \_helix:** helical pattern is a result of hydrogen bonds between the coils of helix
- $\beta$ \_pleated sheets:-** hydrogen bond holds the peptide chain in parallel manner



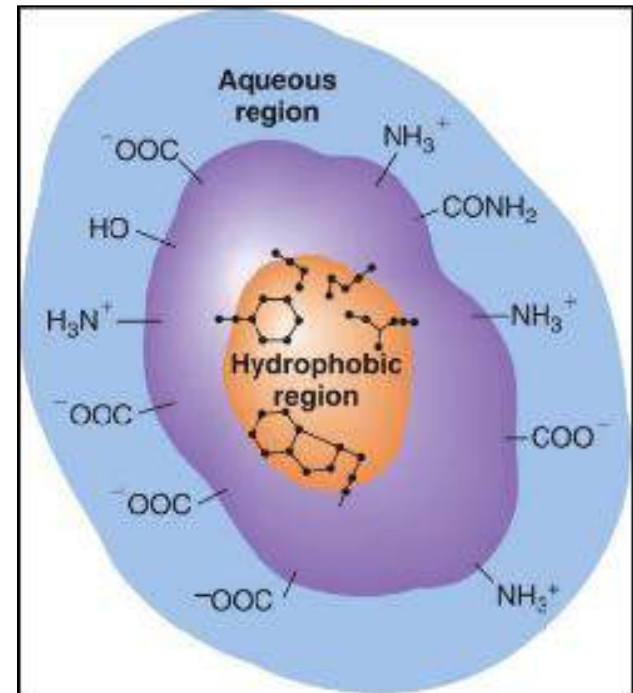
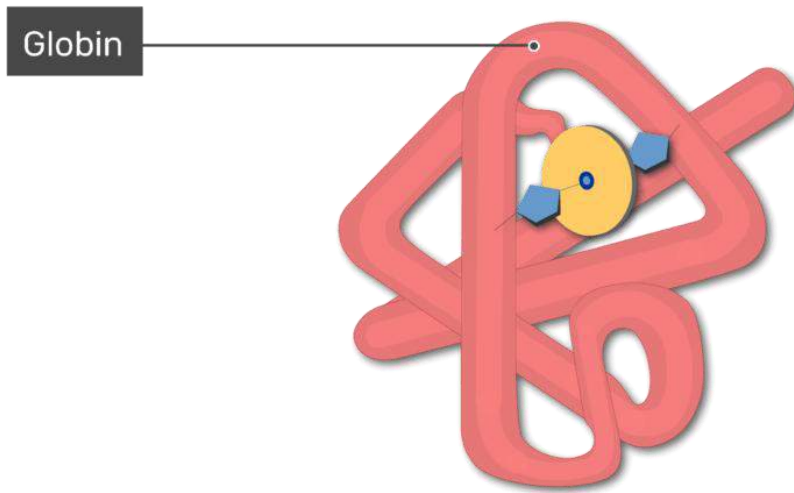


# Proteins cont'd

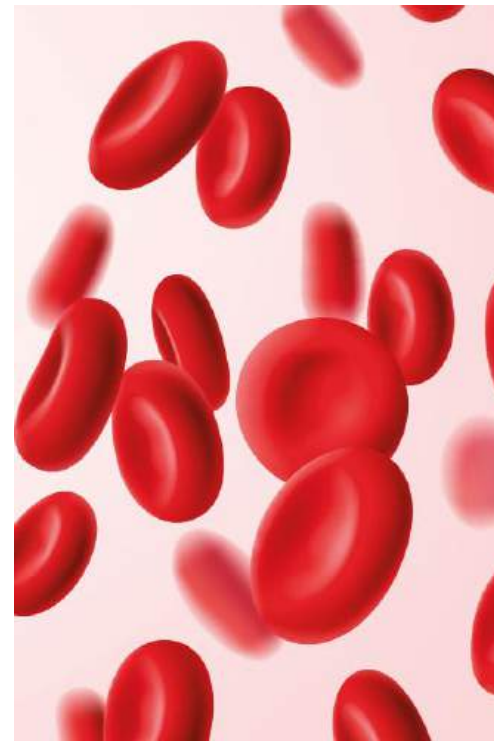
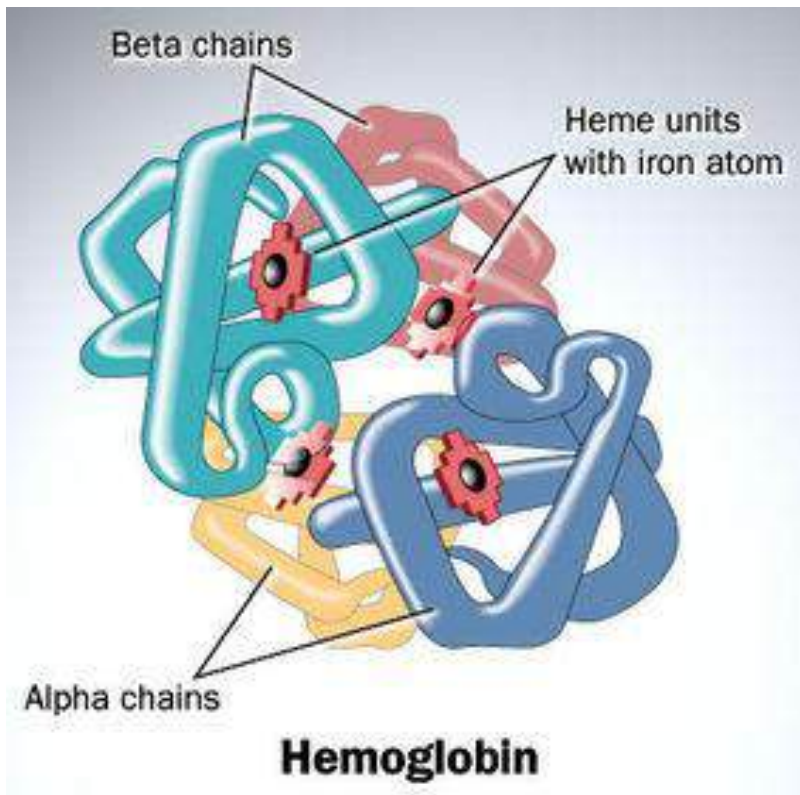
- ❑ Secondary structure- as the peptide grows longer, the chain begins to bend.
- ✓ When the amino group of one section on the peptide moves close to the carbonyl group of an other section, intermolecular bond called hydrogen bond develop.
- ✓ This intermolecular bond bend the linear section and giving two common types of secondary structure ( *$\alpha$ \_helix and  $\beta$ \_pleated sheets*)

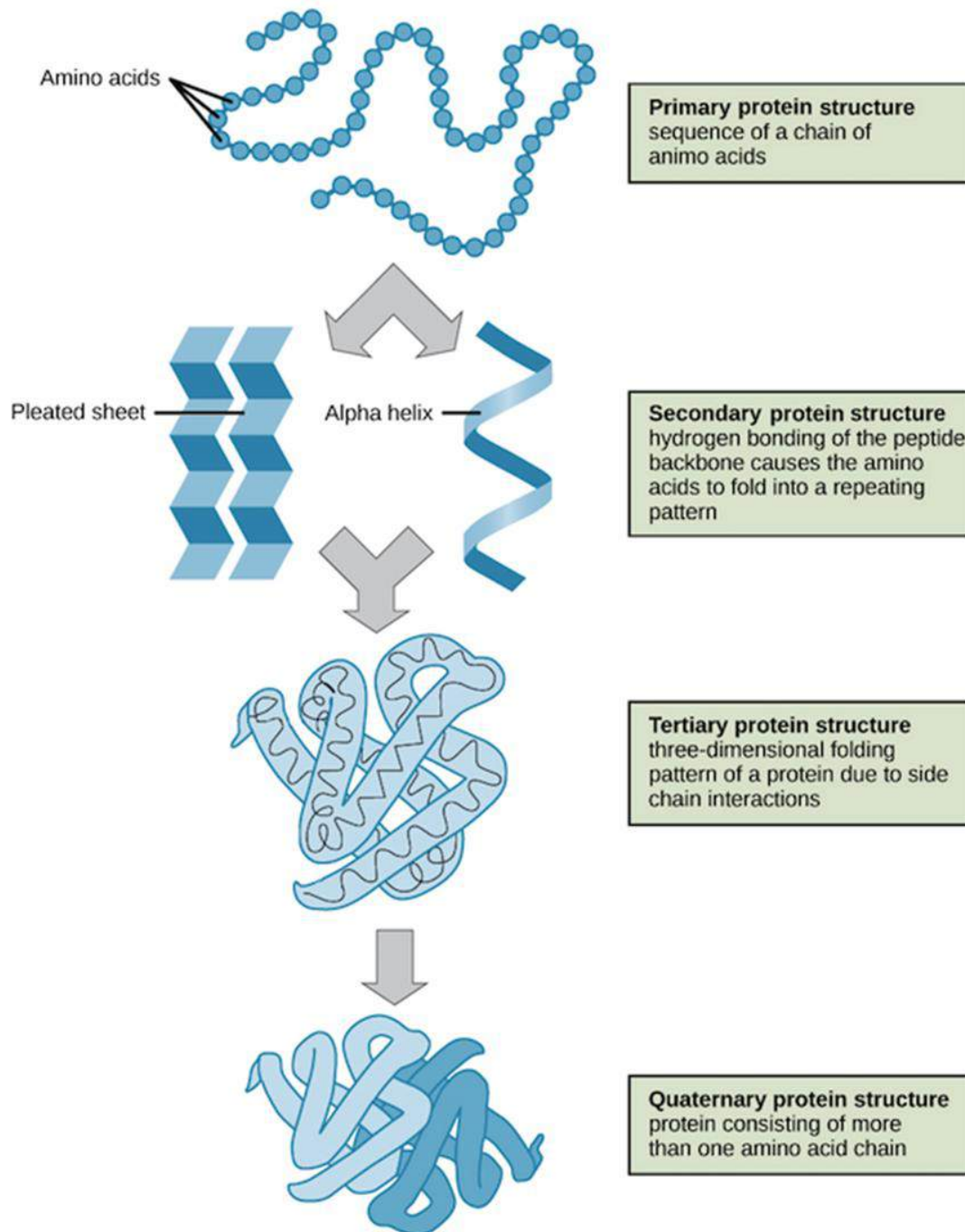
### 3) **Tertiary structure:** final folded shape of a globular protein

- Stabilized by a number of **forces** (H-bonds, covalent bonds, ionic bonds, hydrophobic interactions)
- The final level of structure for proteins consists of only a single polypeptide chain



- 4) **Quaternary structure:** arrangement of individual chains (subunits) in a protein with **2 or more** polypeptide chains









# Nucleic Acids

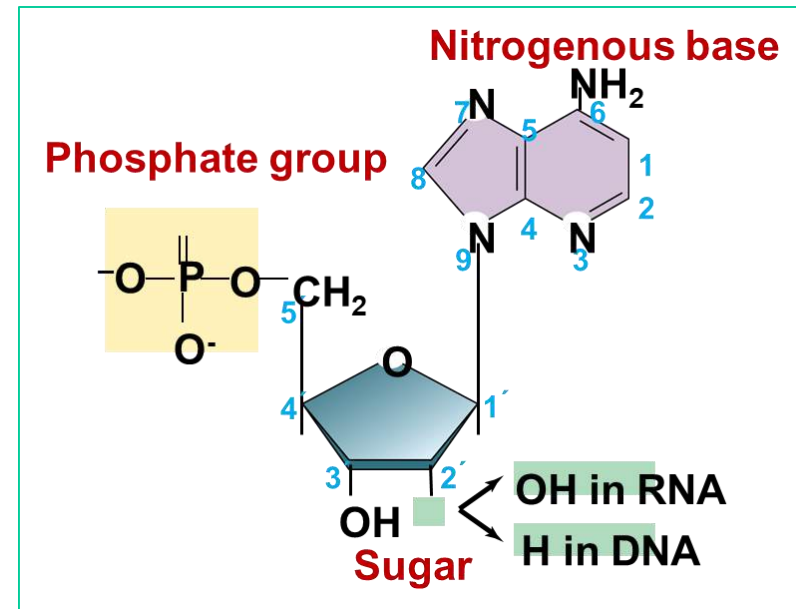
- Nucleic acids are **polymers of nucleotides that store and transmit genetic information.**

The two main types of nucleic acids are:

- **Deoxyribonucleic acid (DNA)** holds the genetic information in all cellular organisms and some viruses.
- **Ribonucleic acid (RNA)** is the genetic material in some viruses

Each nucleotide consists of three parts:

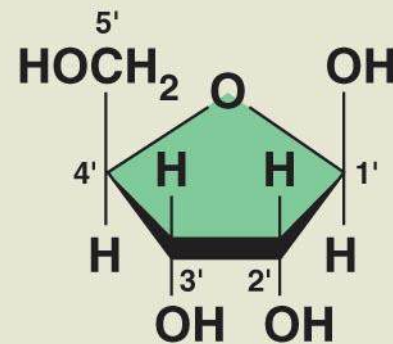
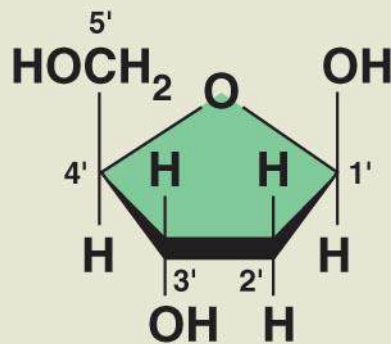
- ✓ **a five-carbon sugar,**
- ✓ **a phosphate group, and a**
- ✓ **nitrogenous base.**





## Differences in the structure of the sugars in DNA and RNA

- DNA's sugar, called **deoxyribose**, has a hydrogen atom at 2' position and contains one oxygen atom fewer overall whereas
- RNA's ribose sugar has a **hydroxyl group** attached to the 2'-carbon atom,

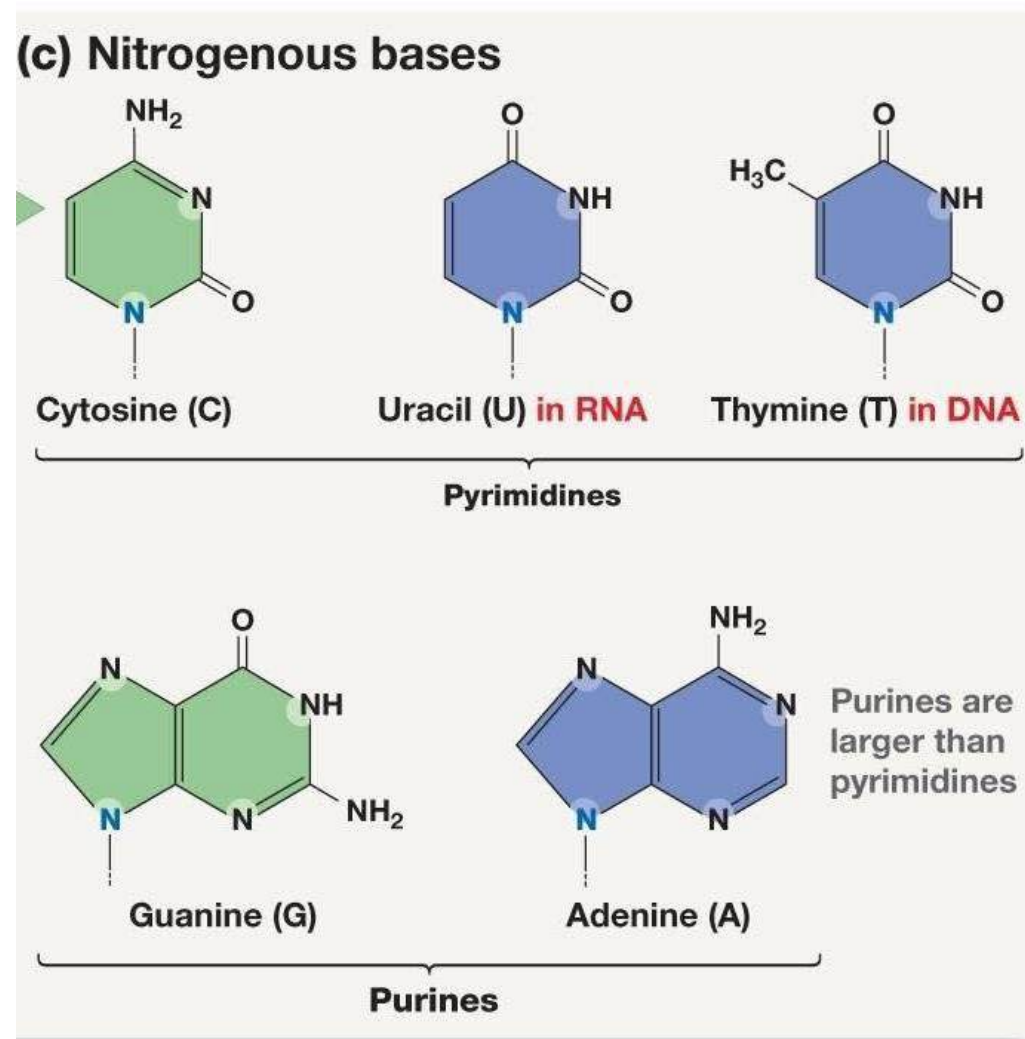






## There are two families of nitrogenous bases:

- **The purines:** adenine and guanine in both DNA and RNA.
- **The pyrimidines:** are cytosine and uracil in RNA; uracil is replaced by thymine in DNA.





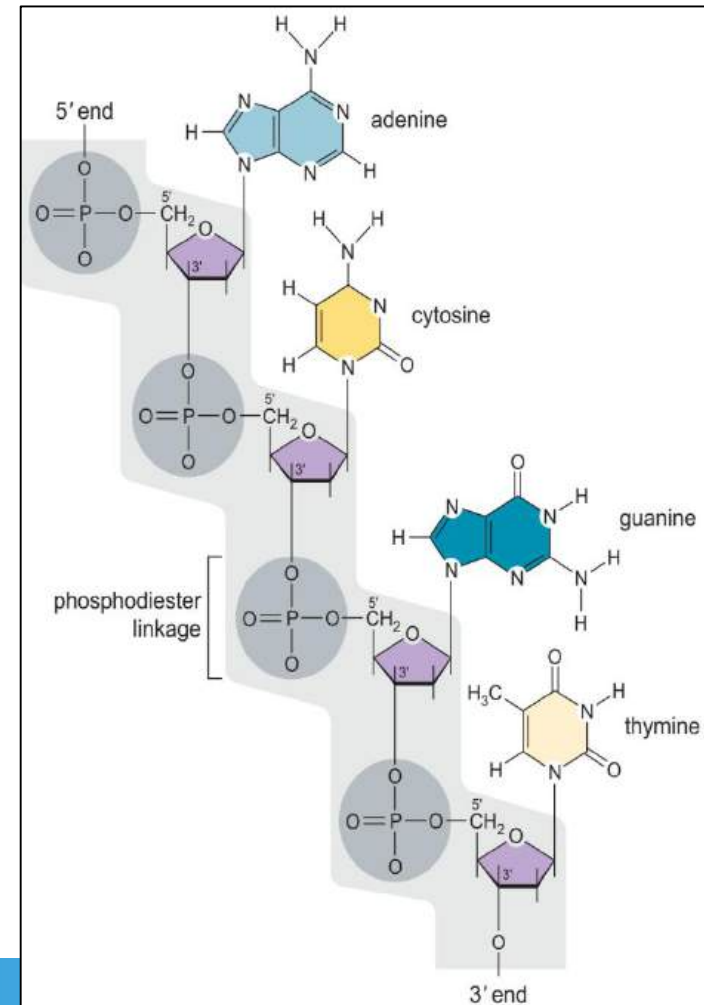
## Polynucleotide strands

- composed of many nucleotides connected by **phosphodiester linkages** between **5'-phosphate group** of one nucleotide and **3'-carbon** of the next nucleotide
- its **backbone** is composed of alternating sugars and phosphates and the bases project away from the long axis of the strand.

## Nucleic acid polarity

The two ends of the strands are different with

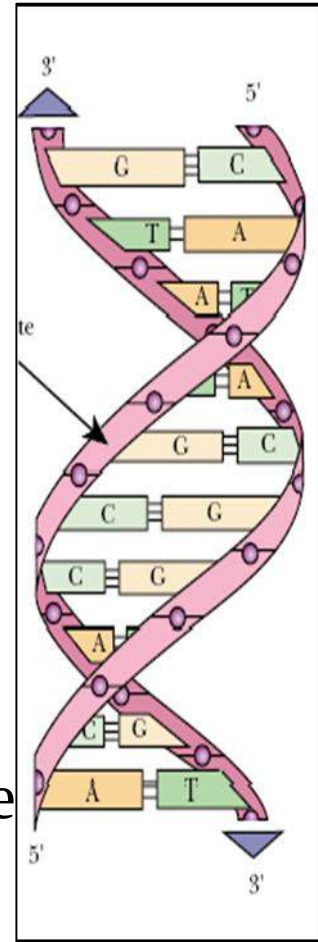
- a phosphate group attached to the 5'-carbon (5' end)
- OH attached to the 3'-carbon atom of the sugar (3' end)





## The DNA Double Helix

- deduced by **James Watson** and **Francis Crick** in 1953.
- A DNA molecule **has two polynucleotides** spiraling around an imaginary axis, forming a double helix
- In the DNA double helix, the two backbones run in opposite **5' → 3'** directions from each other, an arrangement referred to as **antiparallel**
- The strands are joined **non-covalently by hydrogen bonding**
  - ✓ **two hydrogen bonds** between Adenine thymine and
  - ✓ **three hydrogen bonds** between cytosine with guanine
- The two polynucleotide strands of a DNA molecules are **not identical but complementary**.





## Ribonucleic Acid (RNA)

- RNA similar to DNA except
  - Contains **ribose** *instead* of **deoxyribose**
  - Contains **uracil** *instead* of **thymine**
- Single stranded
- RNA plays many roles in the cell
  - **Messenger RNA (mRNA)**: DNA encodes the information to make proteins, mRNA relays this info to the structures that make proteins
  - **Ribosomal RNA (rRNA)**: make structure of Ribosome (Protein synthesizing machinery)
  - **Transfer RNA (tRNA)**: also involved in protein synthesis by decoding codons in the mRNA into amino acids



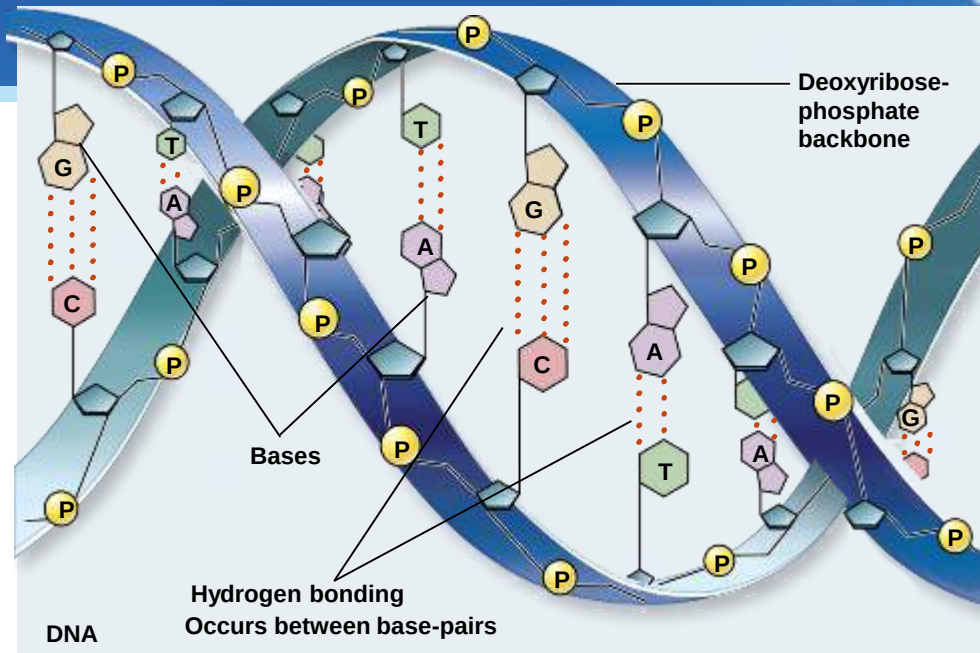
# The Differences between DNA and RNA

|                         | DNA                                                                | RNA                                                               |
|-------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------|
| Name                    | Deoxyribonucleic Acid                                              | Ribonucleic Acid                                                  |
| Number of strand        | Double-stranded                                                    | Single-stranded                                                   |
| Type of sugar           | Deoxyribose                                                        | Ribose                                                            |
| Types of nitrogen bases | Adenine (A), thymine (T), cytosine (C), and guanine (G)            | adenine, uracil (U), guanine, and cytosine                        |
| Base Pairing            | Adenine links to thymine (A-T) and cytosine links to guanine (C-G) | Adenine links to uracil (A-U) and cytosine links to guanine (C-G) |
| Location                | DNA is found in the nucleus, Mitochondria and chloroplast.         | RNA is found in the nucleus, cytoplasm, and ribosome              |

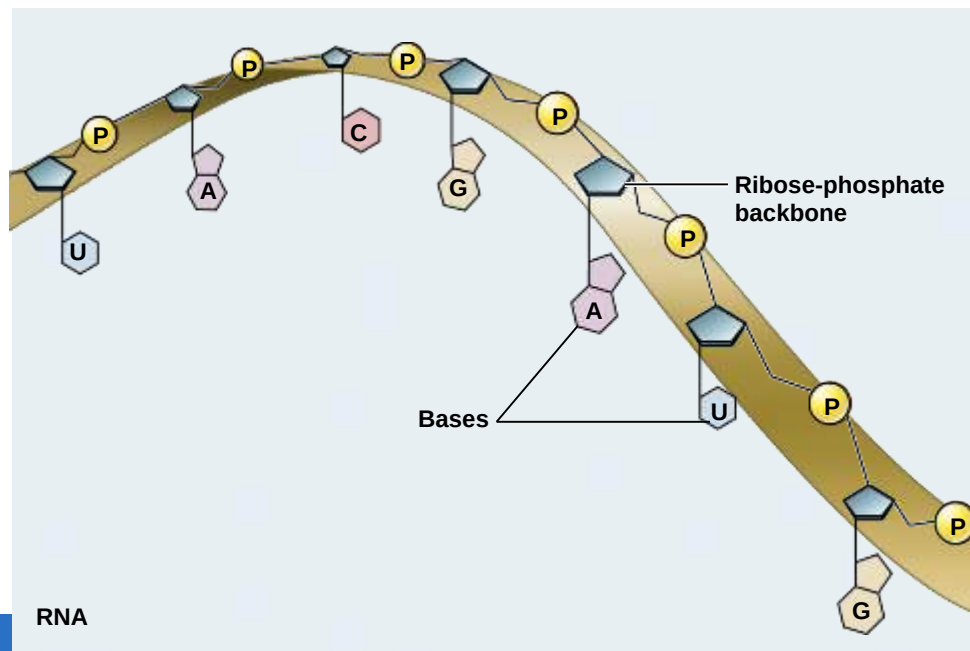




# DNA Double Stranded ATGC



# RNA Single Stranded AUGC







# Vitamins

- Are essential, organic nutrient needed in small amounts in the diet. They can not be synthesized by human body
- **needed for normal function, growth and maintenance.**
- are important enzyme cofactors
- Inadiduate supply of vitamins leads to deficiency diseases
- They are classified as **fat-soluble** and **water-soluble** vitamins.
- **Water-soluble includes: C and B group.**
  - ✓ The B complex vitamins may be sub divided into energy releasing (B1, B2, B6, biotin etc) and hematopoietic (folic acid and B12).
- **Fat-soluble includes: A, D, E and K**



**Table 41.1 Vitamin Requirements of Humans**

| Vitamin                                | Major Dietary Sources                                                                       | Major Functions in the Body                                                                                                   | Symptoms of Deficiency or Extreme Excess                                                                                                 |
|----------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Water-Soluble Vitamins</b>          |                                                                                             |                                                                                                                               |                                                                                                                                          |
| Vitamin B <sub>1</sub> (thiamine)      | Pork, legumes, peanuts, whole grains                                                        | Coenzyme used in removing CO <sub>2</sub> from organic compounds                                                              | Beriberi (nerve disorders, emaciation, anemia)                                                                                           |
| Vitamin B <sub>2</sub> (riboflavin)    | Dairy products, meats, enriched grains, vegetables                                          | Component of coenzymes FAD and FMN                                                                                            | Skin lesions such as cracks at corners of mouth                                                                                          |
| Niacin (B <sub>3</sub> )               | Nuts, meats, grains                                                                         | Component of coenzymes NAD <sup>+</sup> and NADP <sup>+</sup>                                                                 | Skin and gastrointestinal lesions, nervous disorders <b>Liver damage</b>                                                                 |
| Vitamin B <sub>6</sub> (pyridoxine)    | Meats, vegetables, whole grains                                                             | Coenzyme used in amino acid metabolism                                                                                        | Irritability, convulsions, muscular twitching, anemia <b>Unstable gait, numb feet, poor coordination</b>                                 |
| Pantothenic acid (B <sub>5</sub> )     | Most foods: meats, dairy products, whole grains, etc.                                       | Component of coenzyme A                                                                                                       | Fatigue, numbness, tingling of hands and feet                                                                                            |
| Folic acid (folacin) (B <sub>9</sub> ) | Green vegetables, oranges, nuts, legumes, whole grains                                      | Coenzyme in nucleic acid and amino acid metabolism                                                                            | Anemia, birth defects<br><b>May mask deficiency of vitamin B<sub>12</sub></b>                                                            |
| Vitamin B <sub>12</sub>                | Meats, eggs, dairy products                                                                 | Coenzyme in nucleic acid metabolism; maturation of red blood cells                                                            | Anemia, nervous system disorders                                                                                                         |
| Biotin                                 | Legumes, other vegetables, meats                                                            | Coenzyme in synthesis of fat, glycogen, and amino acids                                                                       | Scaly skin inflammation, neuromuscular disorders                                                                                         |
| Vitamin C (ascorbic acid)              | Fruits and vegetables, especially citrus fruits, broccoli, cabbage, tomatoes, green peppers | Used in collagen synthesis (such as for bone, cartilage, gums); antioxidant; aids in detoxification; improves iron absorption | Scurvy (degeneration of skin, teeth, blood vessels), weakness, delayed wound healing, impaired immunity<br><b>Gastrointestinal upset</b> |



**Table 41.1 Vitamin Requirements of Humans**

| Vitamin                     | Major Dietary Sources                                                                                  | Major Functions in the Body                                                                                          | Symptoms of Deficiency or Extreme Excess                                                                                        |
|-----------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Fat-Soluble Vitamins</b> |                                                                                                        |                                                                                                                      |                                                                                                                                 |
| Vitamin A (retinol)         | Provitamin A (beta-carotene) in deep green and orange vegetables and fruits; retinal in dairy products | Component of visual pigments; maintenance of epithelial tissues; antioxidant; helps prevent damage to cell membranes | Blindness and increased death rate<br><b>Headache, irritability, vomiting, hair loss, blurred vision, liver and bone damage</b> |
| Vitamin D                   | Dairy products, egg yolk; also made in human skin in presence of sunlight                              | Aids in absorption and use of calcium and phosphorus; promotes bone growth                                           | Rickets (bone deformities) in children, bone softening in adults <b>Brain, cardiovascular, and kidney damage</b>                |
| Vitamin E (tocopherol)      | Vegetable oils, nuts, seeds                                                                            | Antioxidant; helps prevent damage to cell membranes                                                                  | Degeneration of the nervous system                                                                                              |
| Vitamin K (phylloquinone)   | Green vegetables, tea; also made by colon bacteria                                                     | Important in blood clotting                                                                                          | Defective blood clotting<br><b>Liver damage and anemia</b>                                                                      |

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# Water: a vital compound for life

- Why do scientist spend so much time looking for water on other planets?
  - ✓ It is because water is essential to life
- Water is the most abundant molecule of life.
- Human beings are 70% water, tomatoes are 90% water,

## Functions of water

- serve as solvent of life
- transports compounds
- separates charged molecules
- dissipates heat
- Used as a reactant and as a product in chemical rxns



## Water is Polar compound

- The hydrogen and oxygen atoms within water molecules form polar covalent bonds
- Each hydrogen has a slight positive charge and the oxygen has a slight negative charge
- The opposite charges of water molecules are attracted to one another, and is also attracted to other polar molecules
- When a substance readily forms hydrogen bonds with water, it can dissolve in water and is referred to as **hydrophilic**
- When a substance does not readily form hydrogen bonds with water, it will not dissolve in water and is referred to as **hydrophobic**



# Properties of Water

Water has many important unique properties that result from water's polar nature

## 1. Water has a high heat of vaporization and stabilizes the temperature

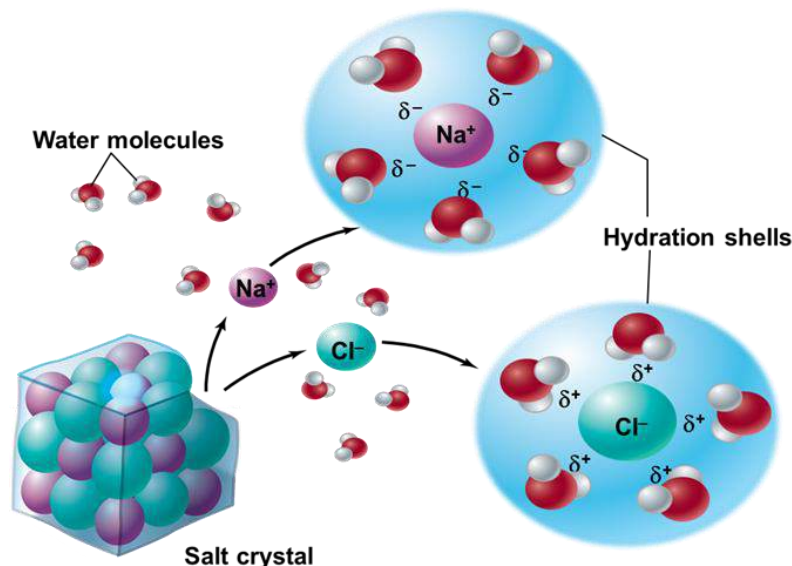
- The hydrogen bonds in water allow it to absorb and release heat energy more slowly than many other substances
- A large amount of energy is required to change the temperature of the water and evaporate it. This helps to keep body temperature constant
- The evaporation of water from a surface causes the cooling of that surface





## 2. Water is an Excellent Solvent

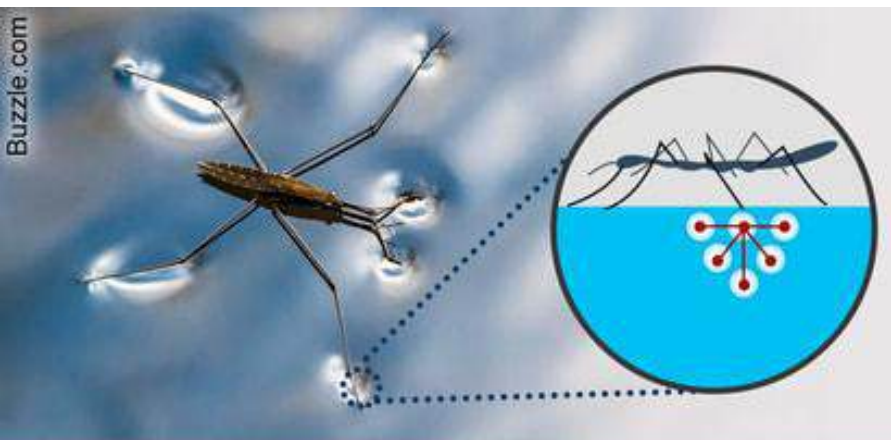
- Because water is polar, ionic compounds and polar molecules can readily dissolve in it
- Water is therefore referred to as a **solvent** - a substance capable of dissolving another substance
- A “sphere of hydration” serves to keep the charged particles separated or dispersed in the water





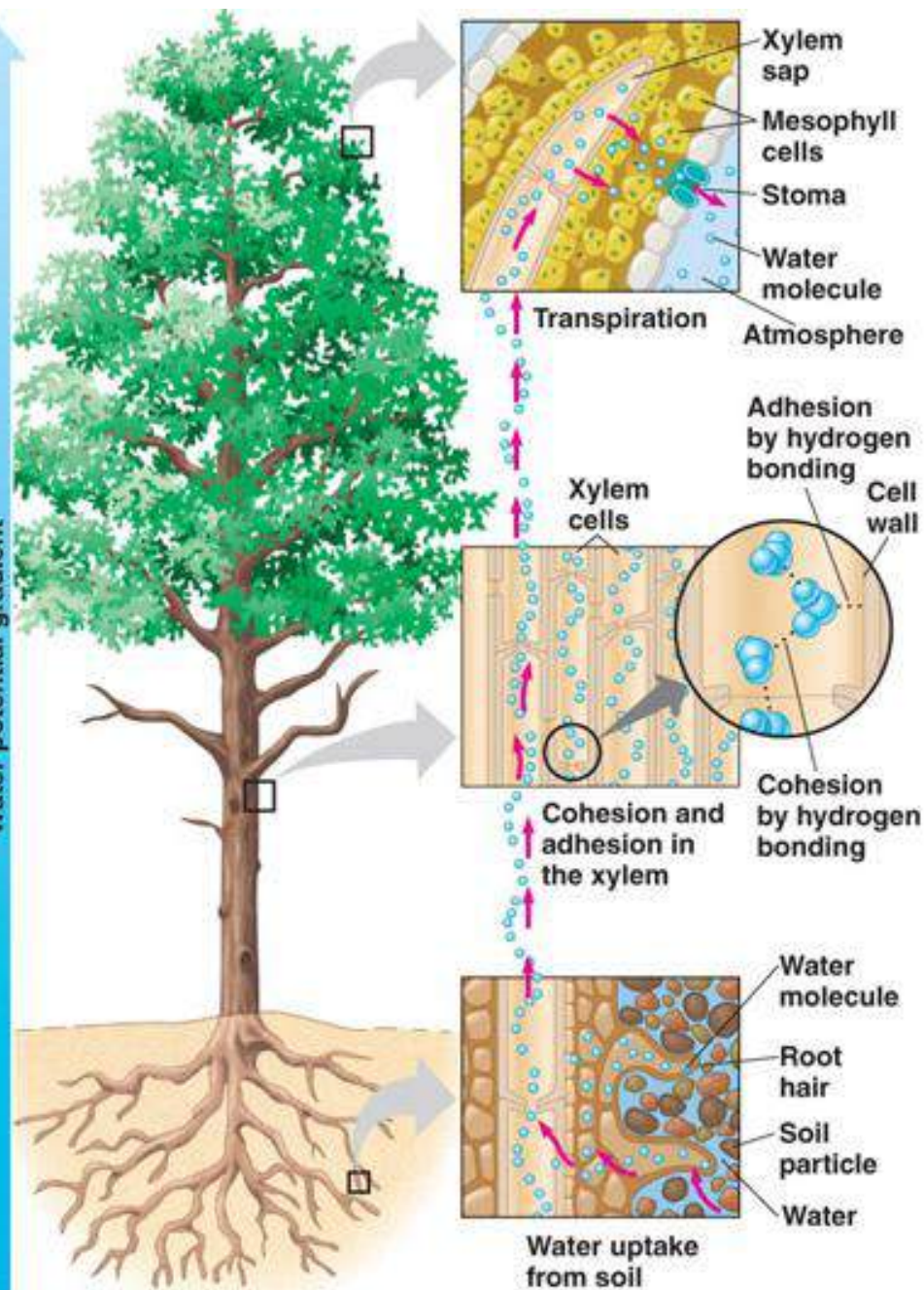
### 3. Water is Cohesive

- **Cohesion** - the attraction of water molecules to other water molecules
- Cohesion gives rise to **surface tension** - the capacity of a substance to withstand rupture when placed under tension or stress
  - ✓ Surface tension allows water to serve as transport medium
- **Adhesion** - the attraction between water molecules and other molecules
  - ✓ The adhesion of water molecules is responsible for transportation of water from the soil to tip of the tree



Water striders can walk on water because of the **SURFACE TENSION OF WATER.**

Water potential gradient





#### 4. **Water expands upon freezing**

- Solid water is less dense than liquid water
  - ✓ density of water decrease as it cools down
  - ✓ Water has the highest density at 4 degree centigrade
  - ✓ Bodies of water freeze from the top down. This allow organisms to live at the bottom of fresh water lakes, protected from freezing







# Minerals

Provide ions for:-Muscle Contraction -Nerve Impulse Conduction - transports substances into- out of cells

Below in the table are important minerals in human body

**Table 41.2 Mineral Requirements of Humans**

| Mineral                              | Major Dietary Sources                                      | Major Functions in the Body                                                            | Symptoms of Deficiency*                                                        |                                                     |
|--------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------|
| Greater than 200 mg per day required | Calcium (Ca)                                               | Dairy products, dark green vegetables, legumes                                         | Bone and tooth formation, blood clotting, nerve and muscle function            | Retarded growth, possibly loss of bone mass         |
|                                      | Phosphorus (P)                                             | Dairy products, meats, grains                                                          | Bone and tooth formation, acid-base balance, nucleotide synthesis              | Weakness, loss of minerals from bone, calcium loss  |
|                                      | Sulfur (S)                                                 | Proteins from many sources                                                             | Component of certain amino acids                                               | Symptoms of protein deficiency                      |
|                                      | Potassium (K)                                              | Meats, dairy products, many fruits and vegetables, grains                              | Acid-base balance, water balance, nerve function                               | Muscular weakness, paralysis, nausea, heart failure |
|                                      | Chlorine (Cl)                                              | Table salt                                                                             | Acid-base balance, formation of gastric juice, nerve function, osmotic balance | Muscle cramps, reduced appetite                     |
|                                      | Sodium (Na)                                                | Table salt                                                                             | Acid-base balance, water balance, nerve function                               | Muscle cramps, reduced appetite                     |
|                                      | Magnesium (Mg)                                             | Whole grains, green leafy vegetables                                                   | Cofactor; ATP bioenergetics                                                    | Nervous system disturbances                         |
| Iron (Fe)                            | Meats, eggs, legumes, whole grains, green leafy vegetables | Component of hemoglobin and of electron carriers in energy metabolism; enzyme cofactor | Iron-deficiency anemia, weakness, impaired immunity                            |                                                     |
| Fluorine (F)                         | Drinking water, tea, seafood                               | Maintenance of tooth (and probably bone) structure                                     | Higher frequency of tooth decay                                                |                                                     |



**Table 41.2 Mineral Requirements of Humans**

| Mineral         | Major Dietary Sources                                  | Major Functions in the Body                                                  | Symptoms of Deficiency*                                                     |
|-----------------|--------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Zinc (Zn)       | Meats, seafood, grains                                 | Component of certain digestive enzymes and other proteins                    | Growth failure, skin abnormalities, reproductive failure, impaired immunity |
| Copper (Cu)     | Seafood, nuts, legumes, organ meats                    | Enzyme cofactor in iron metabolism, melanin synthesis, electron transport    | Anemia, cardiovascular abnormalities                                        |
| Manganese (Mn)  | Nuts, grains, vegetables, fruits, tea                  | Enzyme cofactor                                                              | Abnormal bone and cartilage                                                 |
| Iodine (I)      | Seafood, dairy products, iodized salt                  | Component of thyroid hormones                                                | Goiter (enlarged thyroid)                                                   |
| Cobalt (Co)     | Meats and dairy products                               | Component of vitamin B <sub>12</sub>                                         | None, except as B <sub>12</sub> deficiency                                  |
| Selenium (Se)   | Seafood, meats, whole grains                           | Enzyme cofactor; antioxidant functioning in close association with vitamin E | Muscle pain, possibly heart muscle deterioration                            |
| Chromium (Cr)   | Brewer's yeast, liver, seafood, meats, some vegetables | Involved in glucose and energy metabolism                                    | Impaired glucose metabolism                                                 |
| Molybdenum (Mo) | Legumes, grains, some vegetables                       | Enzyme cofactor                                                              | Disorder in excretion of nitrogen-containing compounds                      |

\*All of these minerals are also harmful when consumed in excess.



# Chapter 3: The cellular basis of life

---

- Cells were discovered in 1665 by **Robert Hooke** after the invention of the microscope
    - ✓ He described the cells as tiny boxes or a honeycomb
    - ✓ He thought that cells only existed in plants and fungi
  - In 1673, **Anton van Leeuwenhoek**: first observed living cells, and called them animalcules (little animals)
  - 1838- German Botanist, **Matthias Schleiden**, reported that all plant parts are made of cells
  - 1839- German physiologist, **Theodor Schwann**, stated that all animal tissues are composed of cells.
-

# The cellular basis of life.....

---

- 1839- German doctor, **Rudolf Virchow**, stated All cells only arise from pre-existing cells

## The Cell Theory

- Based on the previous discovery, the cell theory was stated as
    1. All organisms are composed of cells
    2. Cells are the smallest living things
    3. Cells arise only from pre-existing cells
-

# Size and Shape of cells

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## Cell size

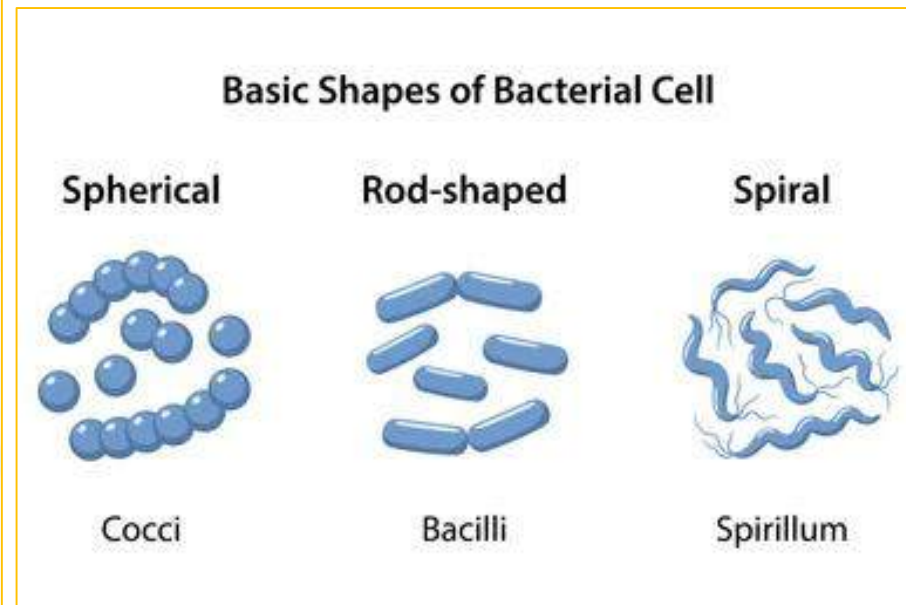
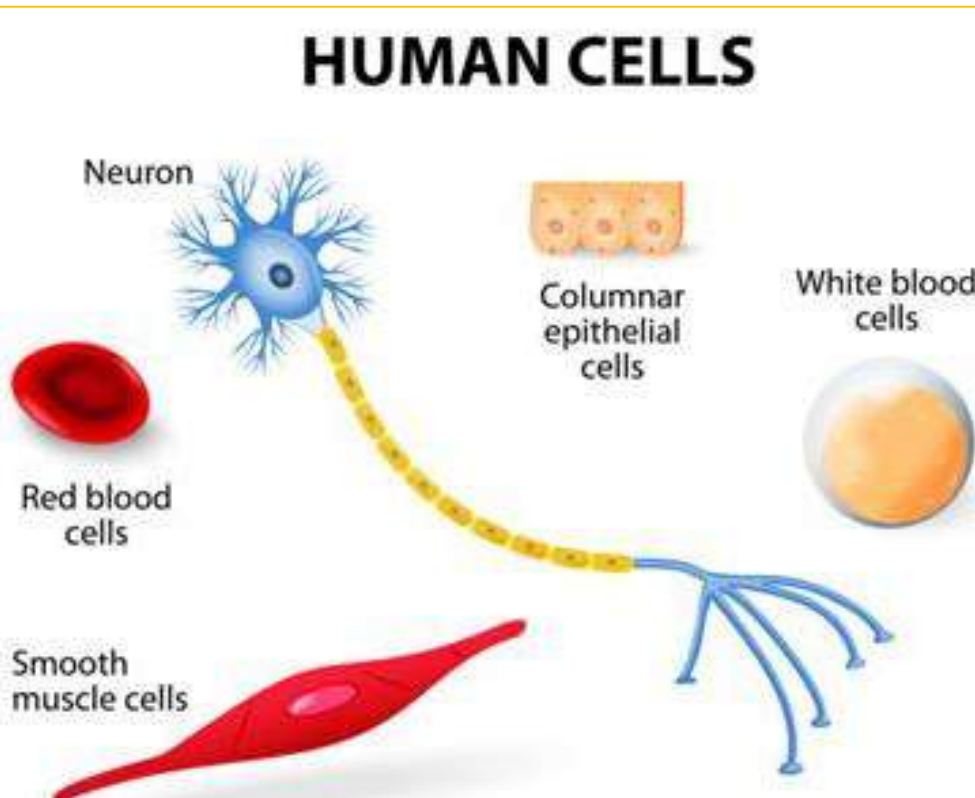
- Most cells are relatively small
- Cells size ranges from less than 1 micrometre - 200 micrometres.

## Why cell size is limited? What is the advantage of being small?

- The size of a cell is directly related to its level of activity and the rate that molecules move across its membranes
- Rate of **diffusion** of materials into and out of the cell is affected by surface area available
- Smaller cells have higher surface area to volume ratio so things can be moved in and out efficiently

# Cell shape

- Cells also show great variation in their shape.
- In human beings for example, there are about 200 types of cells that vary greatly in shape



# Methods to study cells

---

- Many cells are not visible to the naked eye
  - Most cells are  $< 50 \mu\text{m}$  in diameter
  - the study of cells thus depended heavily on the use of microscopes

## Microscopes

- are optical instruments that can **magnify** and **resolve** the image of an object
- **Magnification:** the process of enlarging the apparent size (Image) of an object.
- Microscopes are capable of magnifying image objects up to about thousands to millions of times



- **Resolution** – the ability of a microscope to distinguish objects separated by small distances—is even more important than magnification. Human eye resolution =  $100\ \mu\text{m}$

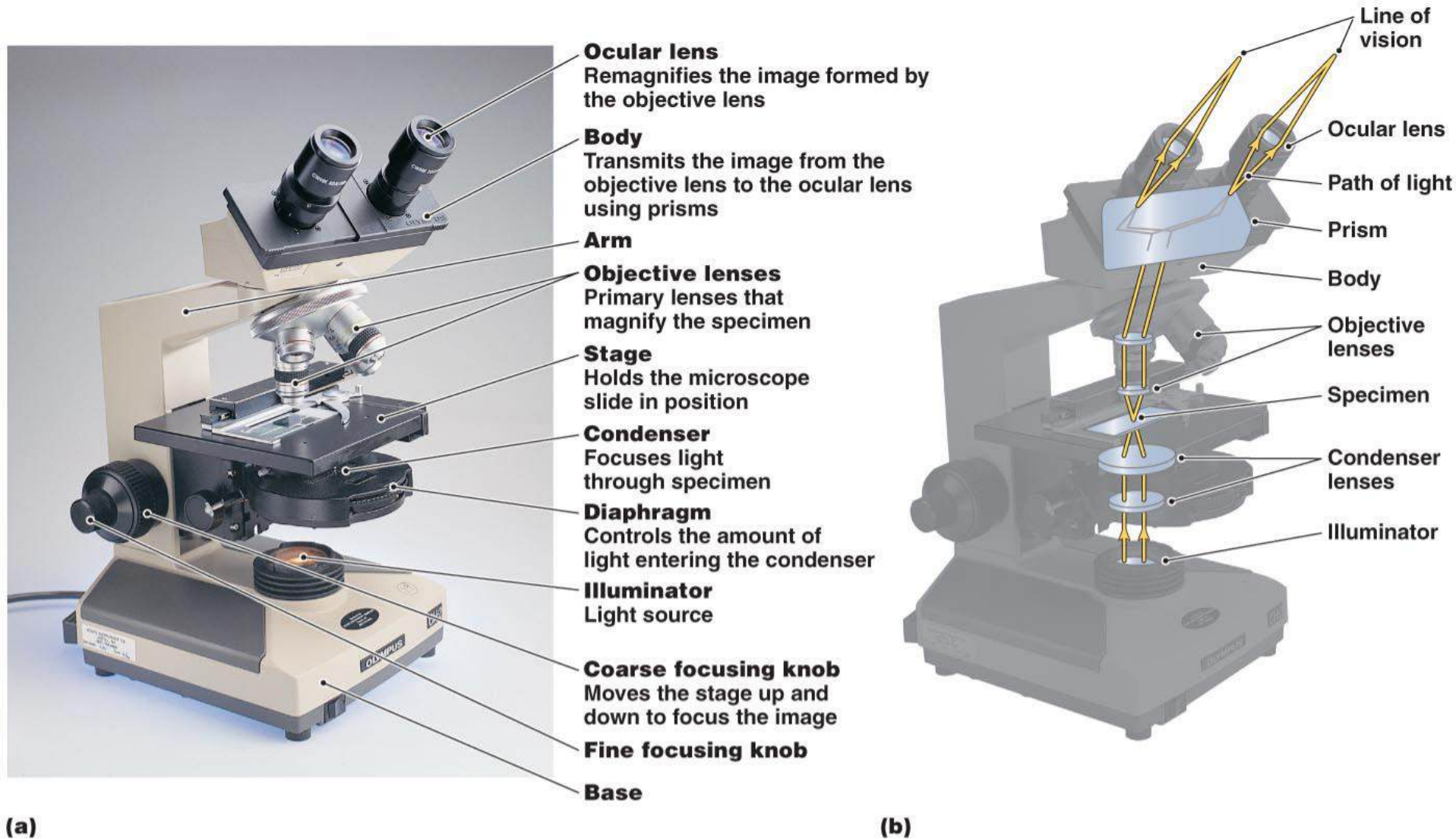
## Types of Microscopes

### 1. Light microscopes

- Use glass lenses to magnify the image of an object up to 1500X
- Use visible light to illuminate an object
- Resolve structures that are 200 nm apart
- Most subcellular structures, including organelles (membrane-enclosed compartments), are too small to be resolved by an LM



# Parts and functions of compound light microscope



# Electron microscopes

- have greater resolving power and magnification than light microscopes
- Use electromagnetic lense to magnify 10,000X to 500,000X
- use a beam of electrons to illuminate an object
- Resolve structures that are 0.2 nm apart
- Used for detailed views of bacteria, viruses, internal cellular structures(organelles), molecules, and large atoms
- Two types: Transmission and Scanning electron microscopes



# Electron microscopes:

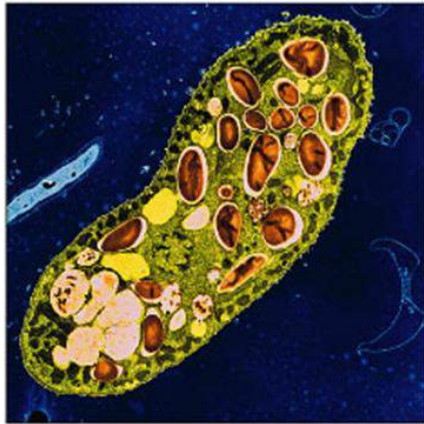
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## Transmission electron microscopes (TEM)

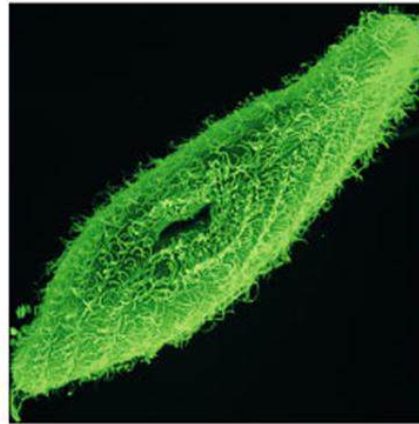
- transmit electrons through the material
- used to study internal cellular ultrastructure.

## Scanning electron microscopes (SEM)

- beam electrons onto the specimen surface
- used to provide a three-dimensional image of cells.



TEM



SEM

# Levels of cellular organization

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- **Unicellular organisms:** organisms with only one cell (e.g., Bacteria, Archea, Some algae, Protozoa, etc.).
- **Multicellular organisms:** organisms having many cells in their body. (e.g., most plants, animals and fungi).

Any cellular organism may contain either prokaryotic or eukaryotic cell types

- Only organisms of the domains Bacteria and Archaea consist of prokaryotic cells
- Protists, fungi, animals, and plants all consist of eukaryotic cells

# Basic features of all cells

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All cells (prokaryotic and eukaryotic) possess

- Plasma membrane (Cell membrane)
- Semifluid substance called **cytosol (cytoplasm)**
- Chromosomes (carry genes)
- Ribosomes (synthesize proteins)

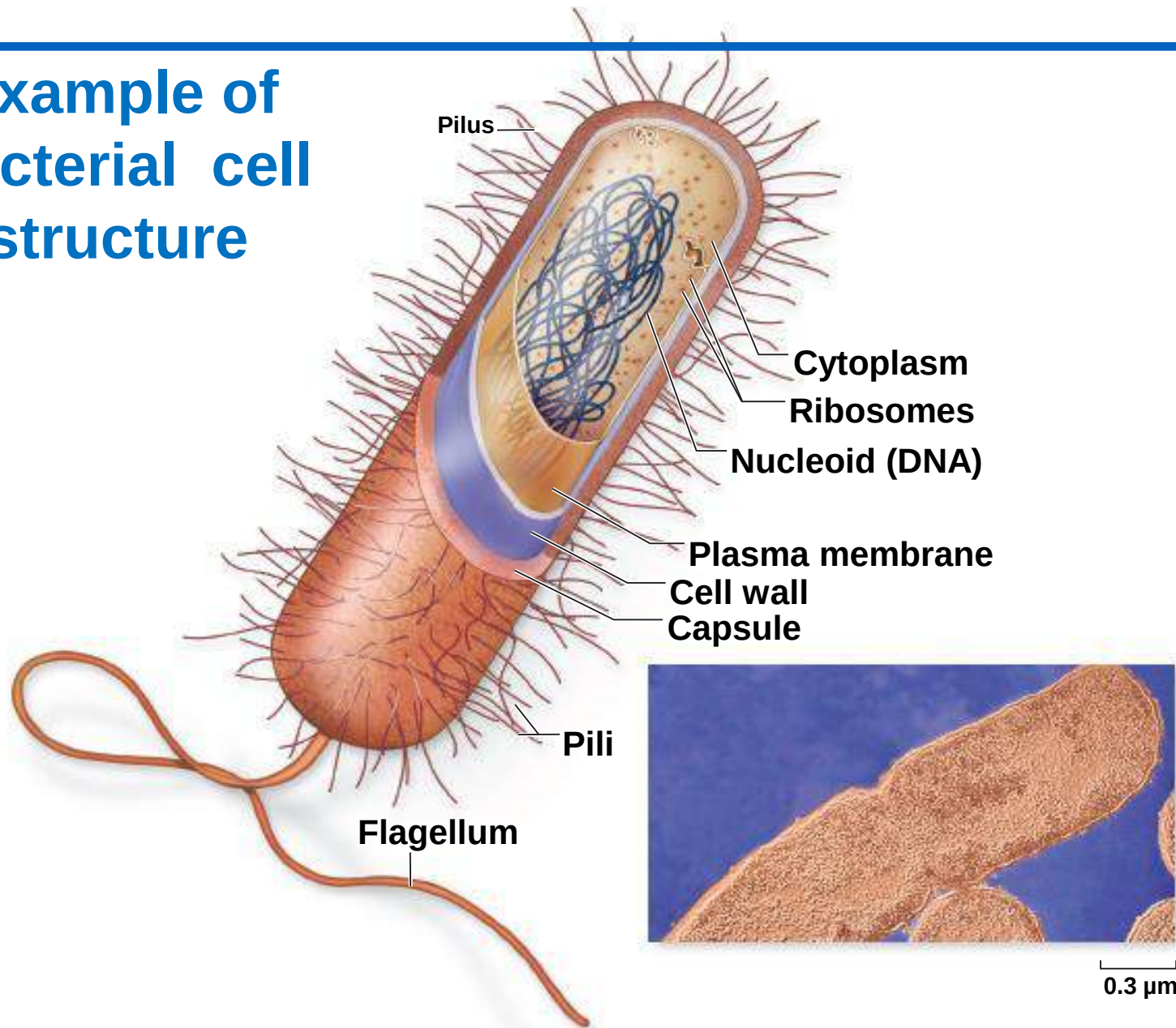


# Prokaryotic Cells

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- The prokaryotic (Gr., pro = primitive or before; karyon = nucleus) are small, simple and most primitive.
- Lack a membrane-bound nucleus
  - DNA in the **nucleoid**
- Lacks internal membrane structures
- Contain **Cell wall** outside of **plasma membrane**
- Possess **ribosomes**: machinery for protein synthesis
- **Constitute two Domains** : Archaea and Bacteria

# Example of Bacterial cell structure



# Prokaryotic Cells

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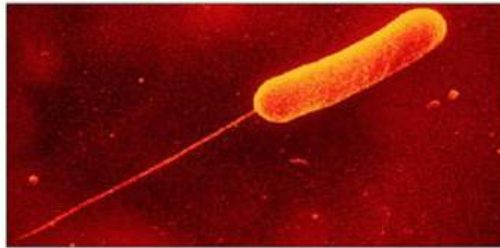
## Bacterial Cell Walls:

- Most bacterial cells possess a strong cell wall
  - composed of **peptidoglycan**
- ***Functions*** include...
  - Protection
  - Maintains cell shape
  - Prevents excessive H<sub>2</sub>O uptake
- **Archaea lack peptidoglycan**
- **Capsule:** gelatinous covering external to the cell wall (not always present)

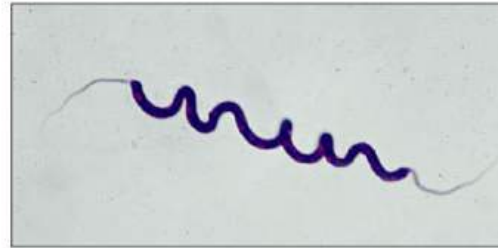
# Prokaryotic Cells

## Flagella:

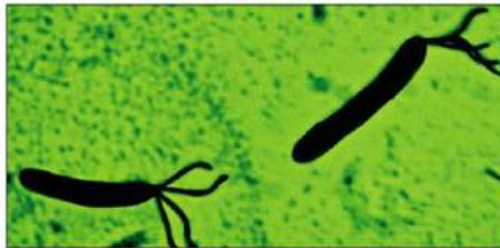
- Present in some prokaryotic cells
  - May be one or more or none
- Used for **locomotion**



(a) Monotrichous



(b) Amphitrichous



(c) Lophotrichous



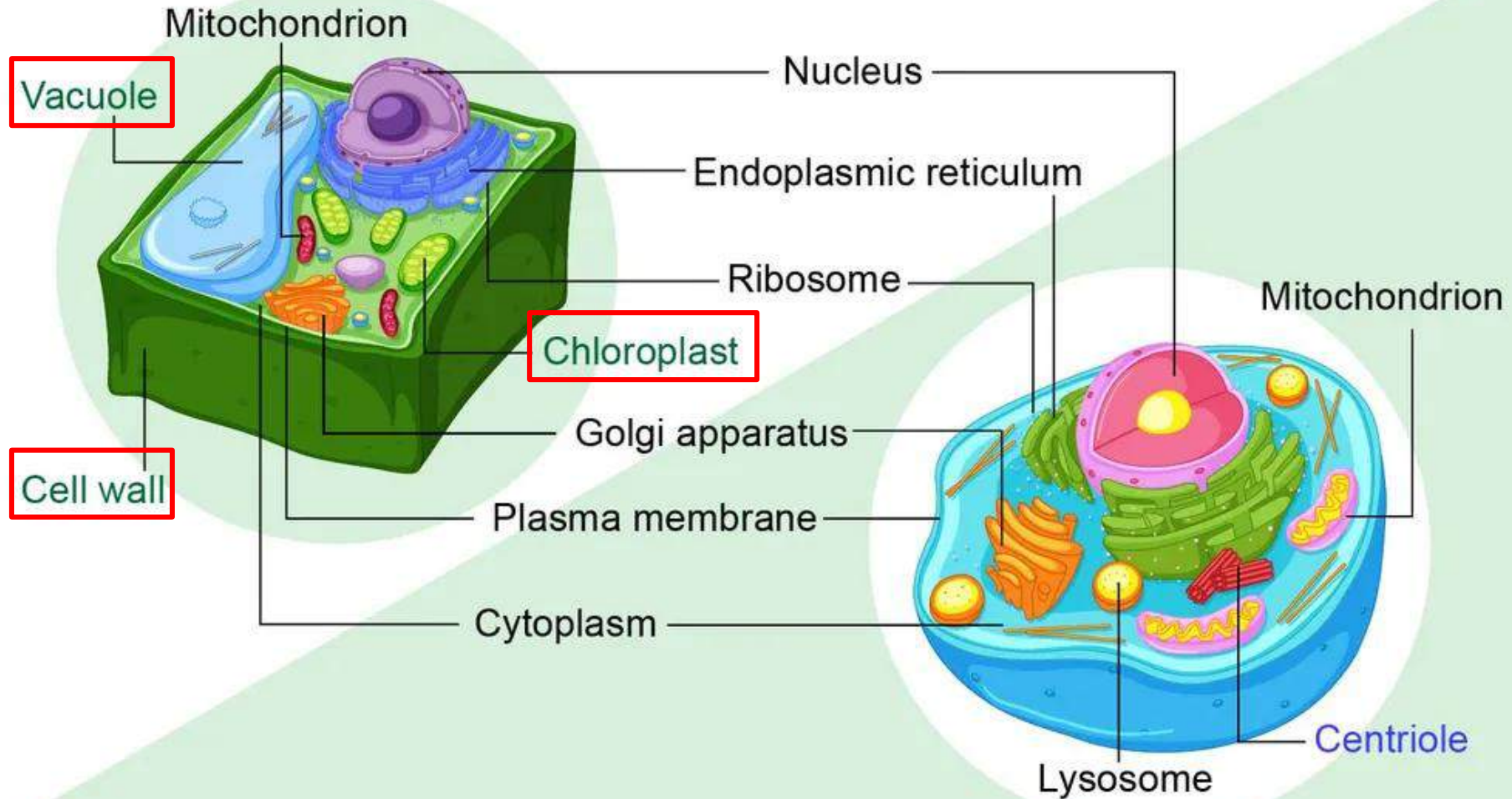
(d) Peritrichous

# Eukaryotic Cells

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- The eukaryotic cells (Gr., eu=true, karyon=nucleus). Possess a membrane-bound **nucleus**
- More *complex* than prokaryotic cells
- Show cytoplasmic compartmentalization (subdivided into functionally distinct, membrane-bound compartments called organelles)
- organelles within cells are specialized for one or more functions
- Also possess a **cytoskeleton** for
  - support
  - maintain cell structure

# PLANT CELL



# ANIMAL CELL



# Organelles

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## Nucleus

- Storage of cell's genetic information (DNA)
- is the largest organelle surrounded by two Nuclear envelopes
  - **Possess Nuclear pores** – control transport in and out
- Contain **Nucleolus** (one, two or four per nucleus)– a region where **ribosomal RNA (rRNA)** synthesis takes place
- The space between the nuclear envelope and the nucleolus is filled by nucleoplasm
- cells contain a single nucleus but the number of the nucleus may vary from cell to cell.

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# Ribosomes

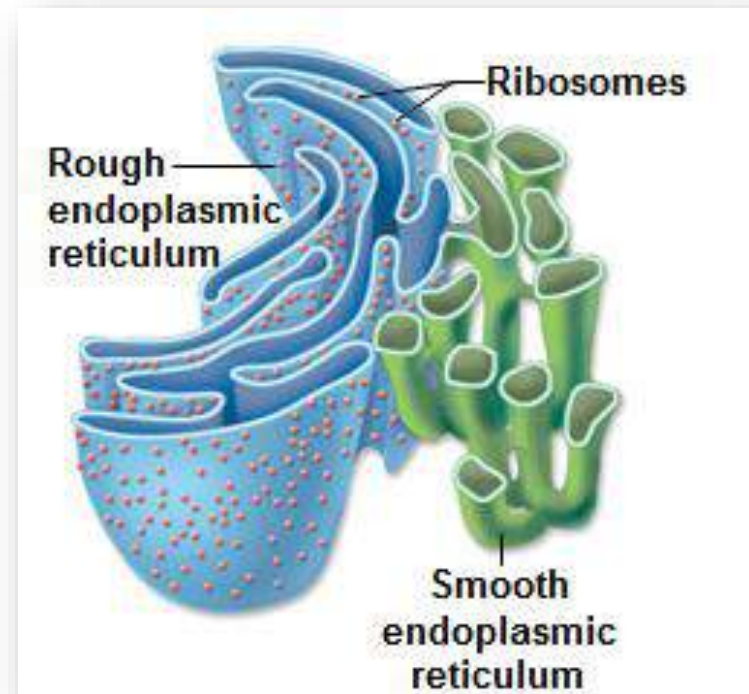
- non membrane bounded cytoplasmic organelle found in all Organisms
- Cell's **protein synthesis** machinery
- Composed of: Ribosomal RNA (rRNA) and protein complex
- function either free in the cytosol or bound to endoplasmic reticulum.
- two types of ribosomes have been recognized
  - ✓ 70s in prokaryotes
  - ✓ 80s in eukaryotes

# Endoplasmic Reticulum

- Endoplasmic reticulum (ER) = (Endoplasmic = within the cytoplasm; reticulum = network); extensive membranous network of membranes in the cell
- Two types

## 1. Rough endoplasmic reticulum (RER)

- Attachment of ribosomes to the membrane gives a rough appearance
- Involved in the Synthesis of proteins and membranes



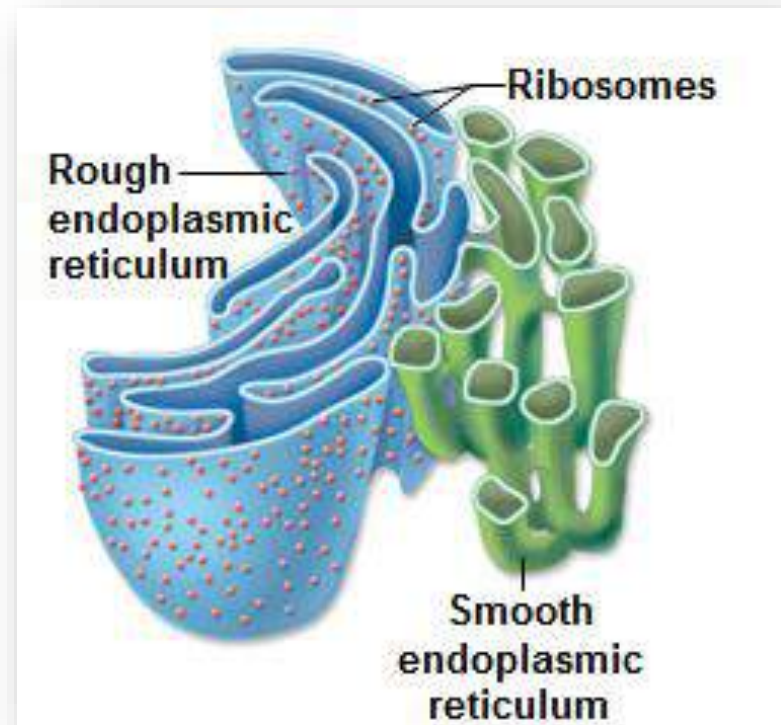
# Endoplasmic Reticulum

## 2. Smooth endoplasmic reticulum (SER)

- Appears smooth in the electron microscope because its cytoplasmic surface lacks ribosomes.

Have a variety of functions:

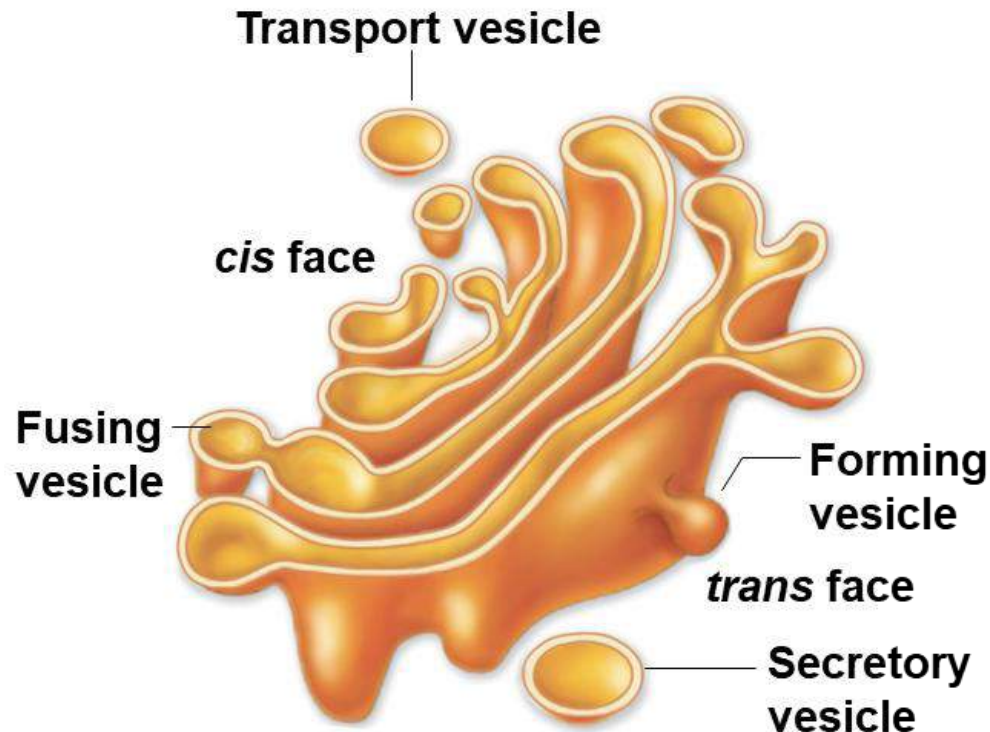
- ✓ synthesize lipids,
- ✓ Metabolizes carbohydrates
- ✓ store  $\text{Ca}^{2+}$ ,
- ✓ detoxify drugs and poisons



# Golgi Apparatus (Golgi bodies)

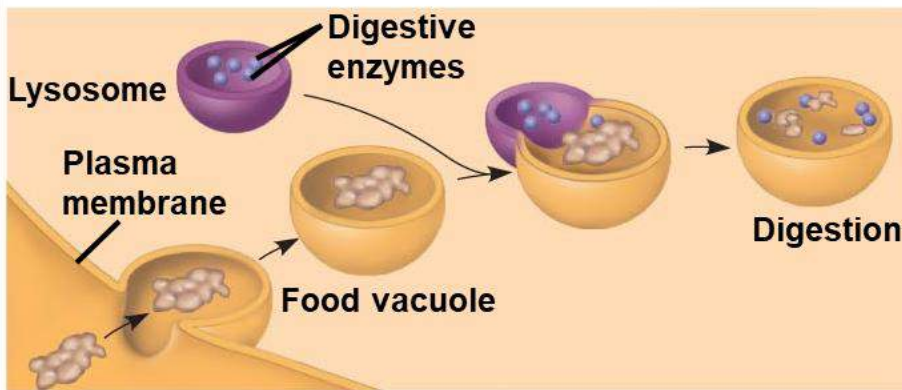
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- Flattened stacks of interconnected membranes
- Functions in **packaging** and **distribution** of molecules
- **Vesicles** transport molecules to the destination

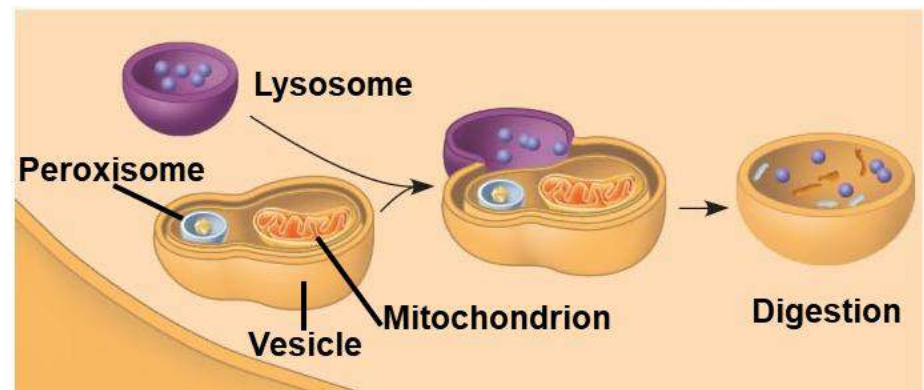


# Lysosomes (Digestive Compartments)

- Membrane-bounded **digestive vesicles**
- **Arise from Golgi apparatus**
- ***Store Enzymes*** that hydrolyze proteins, fats, polysaccharides, and nucleic acids
- destroy cells or foreign matter taken in by **phagocytosis**
- recycle the cell's own organelles and macromolecules, a process called **autophagy**



(a) Phagocytosis

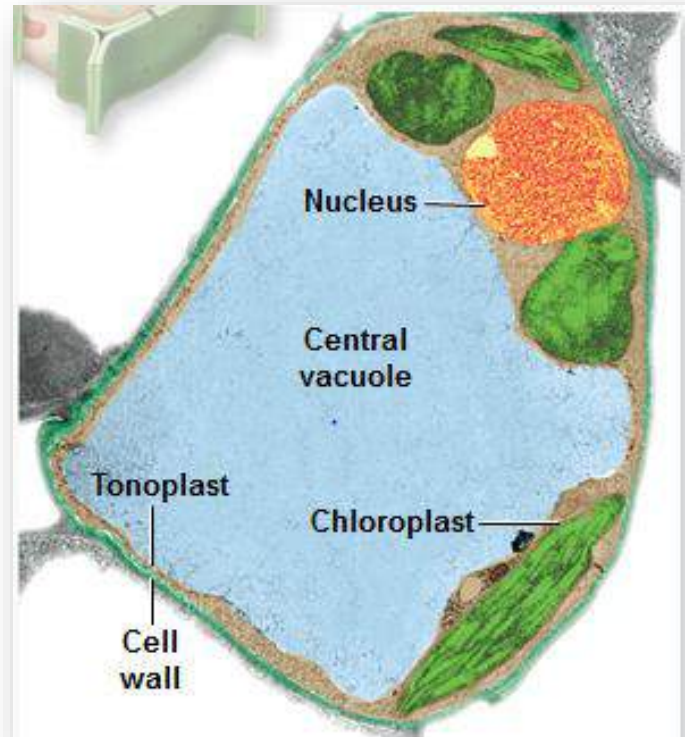


(b) Autophagy



# Vacuoles

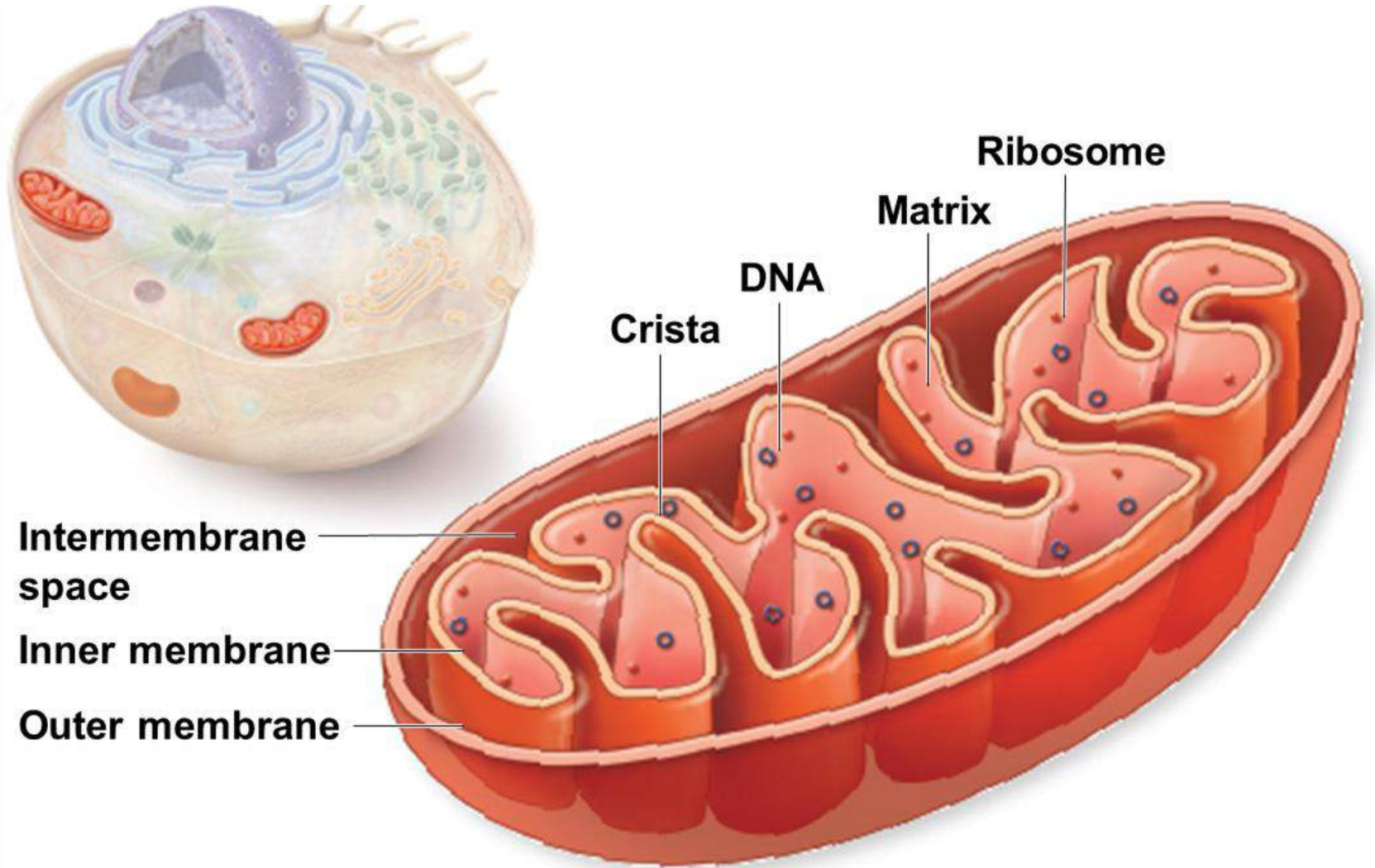
- Membrane-bounded structures in plants, fungi and protists
- Various functions depending on the cell type
  1. **Central vacuole: in** plant cells hold organic compounds and water (maintain turgor pressure)
  2. **Contractile vacuole: in** some fungi and protists pump excess water out of cells
  3. **Storage vacuoles**



# Mitochondria

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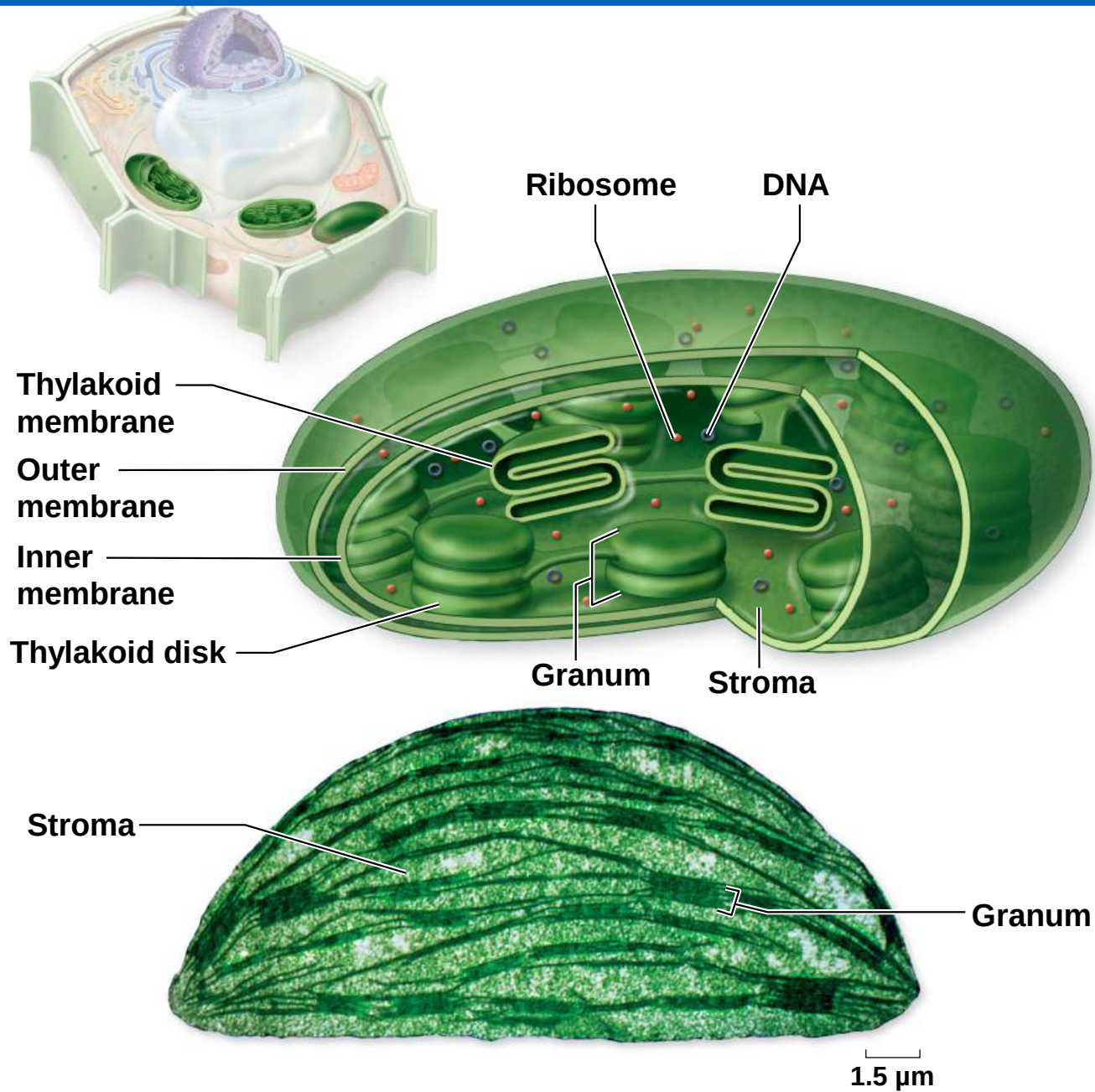
- Found in all types of eukaryotic cells
- Used for cell respiration (ATP energy production = powerhouse of the cell)
- Bound by 2 membranes
  - Outer membrane and inner membrane separated by Intermembrane space
  - The inner membrane has folds known as cristae (site for electron transport system)
  - Space inside the inner membrane is Matrix (site for Krebs cycle)
- *Have their own DNA*



# Chloroplasts

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- Organelles present in cells of plants and some other eukaryotes
- Contain chlorophyll for photosynthesis
- Surrounded by 2 membranes
- membranous sacs within the inner membrane are called Thylakoids (contain chlorophyll and are the site light reaction):
  - stacks of thylakoids are known as Grana
- the fluid matrix inside chloroplast is called Stroma (site of dark reaction of CO<sub>2</sub> fixation)
- Have their own DNA



(inset): © Dr. Jeremy Burgess/Photo Researchers, Inc.

# Peroxisomes:

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- Peroxisomes are specialized metabolic compartments bounded by a single membrane
  - Peroxisomes produce hydrogen peroxide and convert it to water
  - Oxygen is used to break down different types of molecules
-



# Endosymbiosis

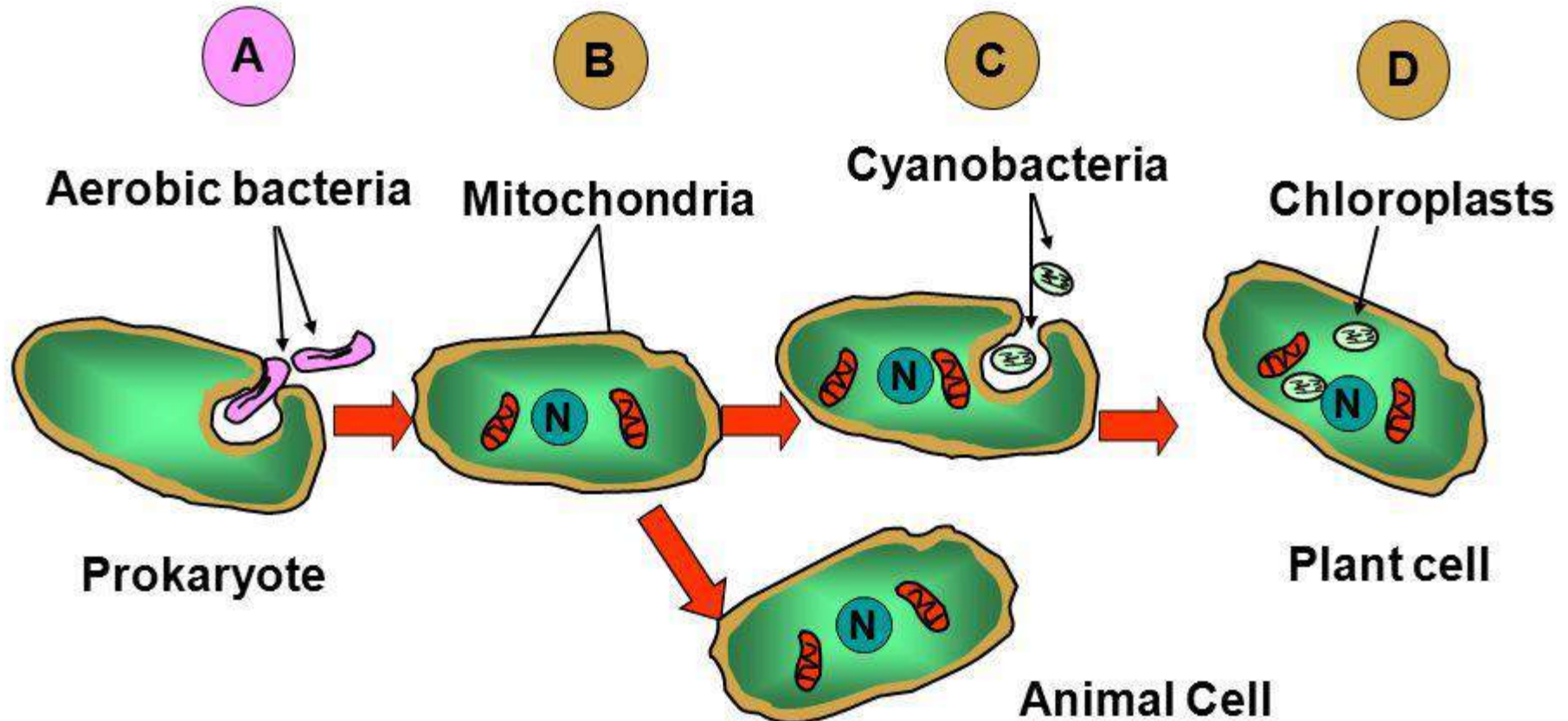
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- The endosymbiotic theory is the idea that a long time ago, prokaryotic cells engulfed other prokaryotic cells by endocytosis. This resulted in the first eukaryotic cells.
- Explains the evolution of eukaryotic cells from simple prokaryotic organisms.
- Focuses mainly on the origins of chloroplasts and mitochondria

# Endosymbiosis Hypothesis

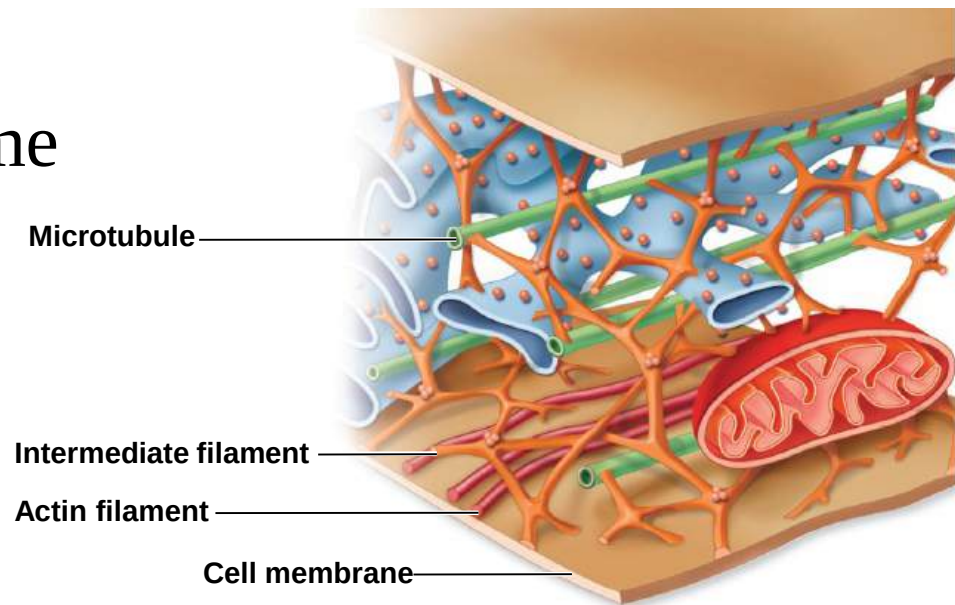
A

A prokaryote ingested some aerobic bacteria. The aerobes were protected and produced energy for the prokaryote



# Cytoskeleton

- Network of protein fibers found in all eukaryotic cells
  - Supports the shape of the cell
  - Keeps organelles in fixed locations
- It is composed of three types of molecular structures:
  - ✓ Microfilaments
  - ✓ Microtubules
  - ✓ Intermediate filaments

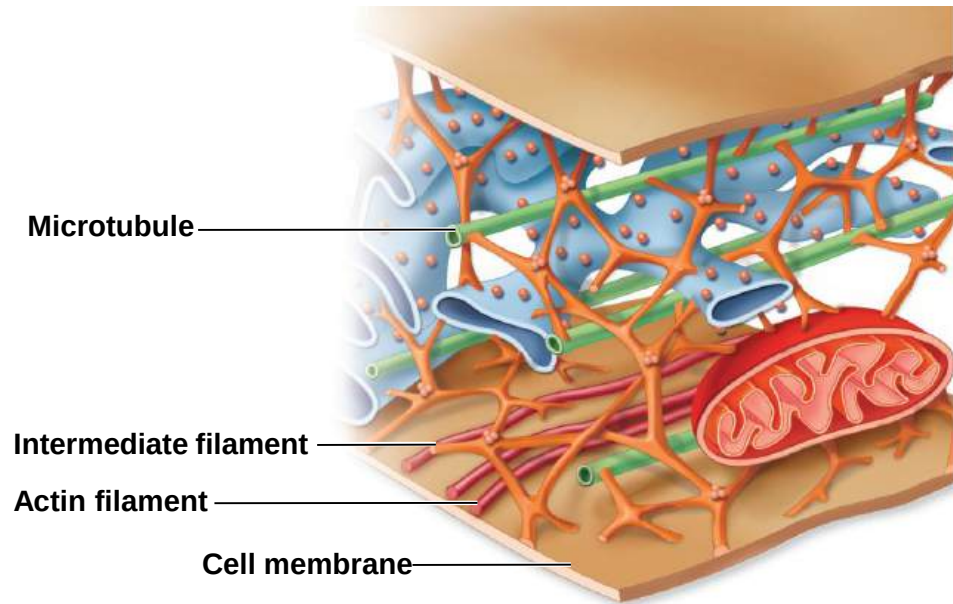
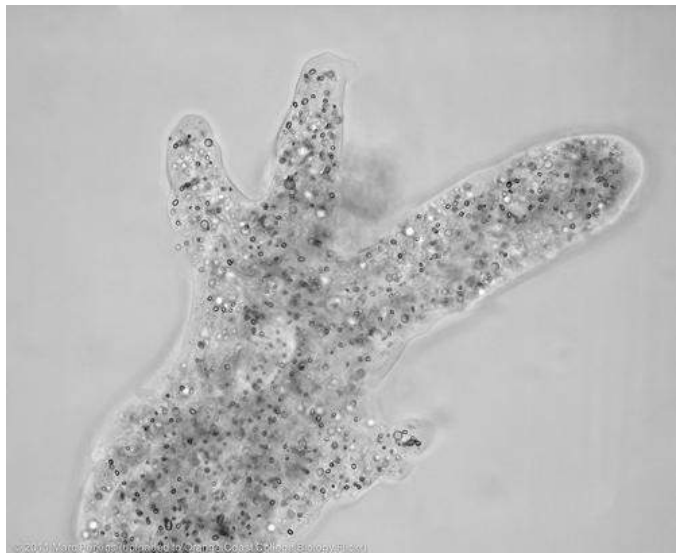


# Cytoskeleton: 3 types of fibers

## 1. Microfilaments (actin filaments)

- Two protein chains loosely twined together
- Used for movements like contraction, crawling, “pinching”

### Ameboid Movement



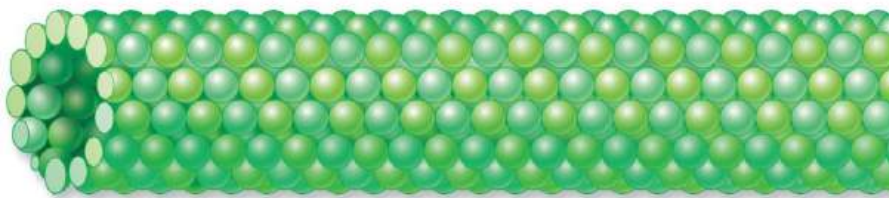
a. Actin filaments

## 2. Microtubules

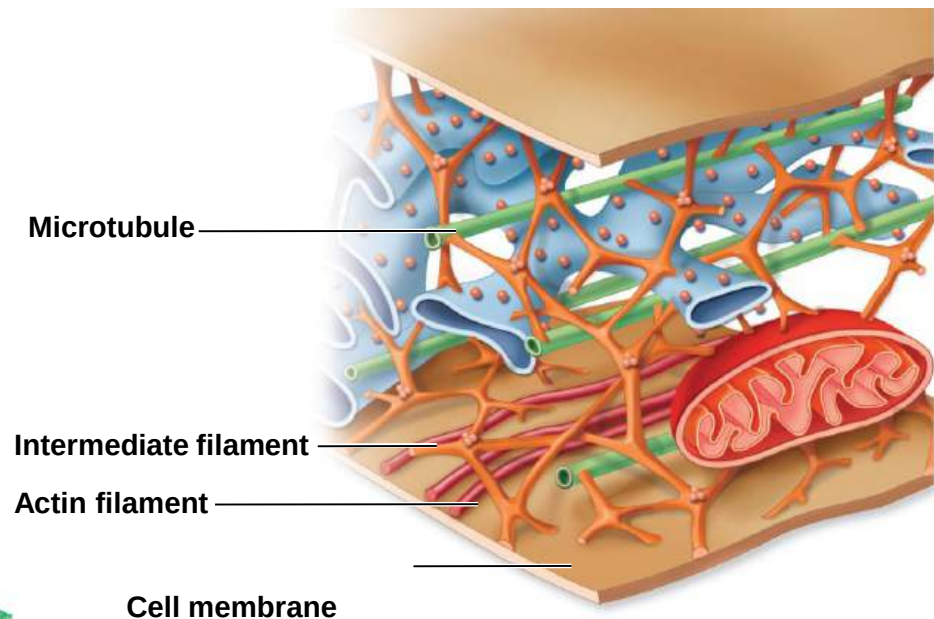
- *Largest* of the cytoskeletal elements
- Dimers of  $\alpha$ - and  $\beta$ -tubulin subunits

### Functions

- Shaping the cell
- Guiding movement of organelles
- Separating chromosomes during cell division



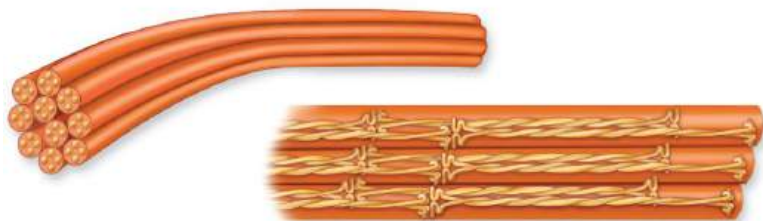
b. Microtubules



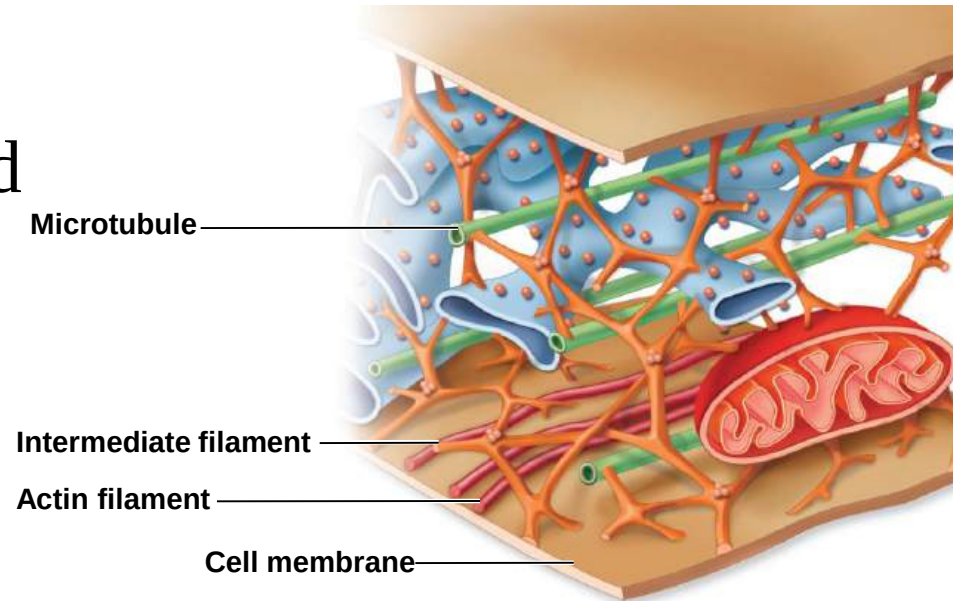


### 3. Intermediate filaments

- Between the size of actin filaments and microtubules
  - **Very stable** – usually not broken down
  - Example: Keratin
- They support cell shape and fix organelles in place



c. Intermediate filament





# Centrosomes

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- Located near the nucleus
- Is Microtubule-organizing center
  - Can nucleate the assembly of microtubules
- Animal cells and most protists have centrioles – pair of organelles
- Organize microtubules for *cell division*
- Plants and fungi usually lack centrioles

# Eukaryotic Cell Walls

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- Presents in plants, fungi, and many protists
- Different from prokaryote
- **Cell wall of Plants** and **protists** is composed of *cellulose* while that of **Fungi** is composed of *chitin*
- The cell wall
  - ✓ protects the cell,
  - ✓ maintains its shape, and
  - ✓ prevents excessive uptake of water

# Extracellular Matrix (ECM)

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- Animal cells lack cell walls but are covered by an elaborate extracellular matrix (ECM)
- The ECM is made up of glycoproteins such as collagen, proteoglycans, and fibronectin
- ECM proteins bind to receptor proteins in the plasma membrane called integrins

## Functions of the ECM:

- ✓ Support
- ✓ Adhesion
- ✓ Movement
- ✓ Regulation

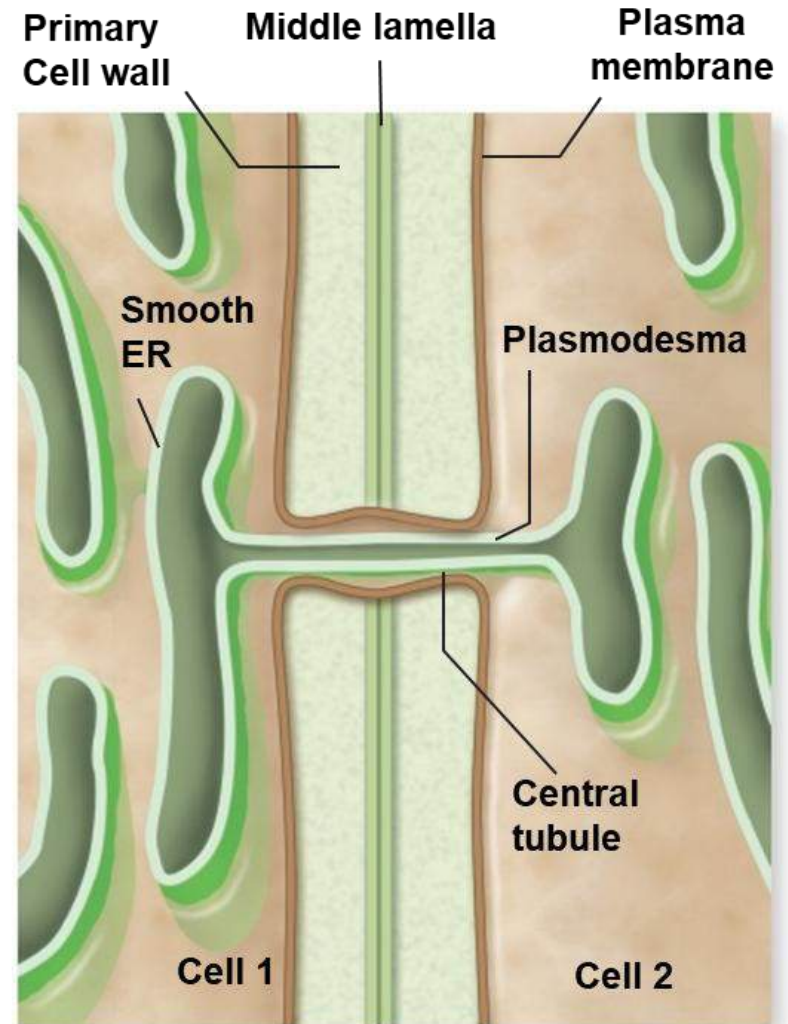
# Intercellular Junctions

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- Neighboring cells in tissues, organs, or organ systems often adhere, interact, and communicate through direct physical contact
  - Intercellular junctions facilitate this contact
  - There are several types of intercellular junctions
    - Plasmodesmata
    - Tight junctions
    - Desmosomes
    - Gap junctions
-

# *Plasmodesmata in Plant Cells*

- **Plasmodesmata** are channels that perforate plant cell walls
- Through plasmodesmata, water and small solutes (and sometimes proteins and RNA) can pass from cell to cell



# ***Tight Junctions, Desmosomes, and Gap Junctions in Animal Cells***

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- At **tight junctions**, membranes of neighboring cells are pressed together, preventing leakage of extracellular fluid
  - **Desmosomes** (anchoring junctions) fasten cells together into strong sheets
  - **Gap junctions** (communicating junctions) provide cytoplasmic channels between adjacent cells
-



# Structure Functions of cell membrane

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- is an extremely thin layer of lipids and proteins forming the outermost boundary of the living cell and enclosing the intracellular fluid (ICF).

## Functions of membranes

- encloses the cell & defines cell boundaries
  - determining the composition of cell by selective permeability of substances to pass between the cell and its ECF environment
  - Establish ion gradients across membranes
  - Can be used to synthesize ATP
  - Control movement of selected solutes or, in nerve and muscle cells, to produce and transmit electrical signals.
  - contains proteins that act as sensors of external signals
-

# Cell membrane...

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## Membrane structure

Despite their differing functions, all biological membranes have a common general structure:

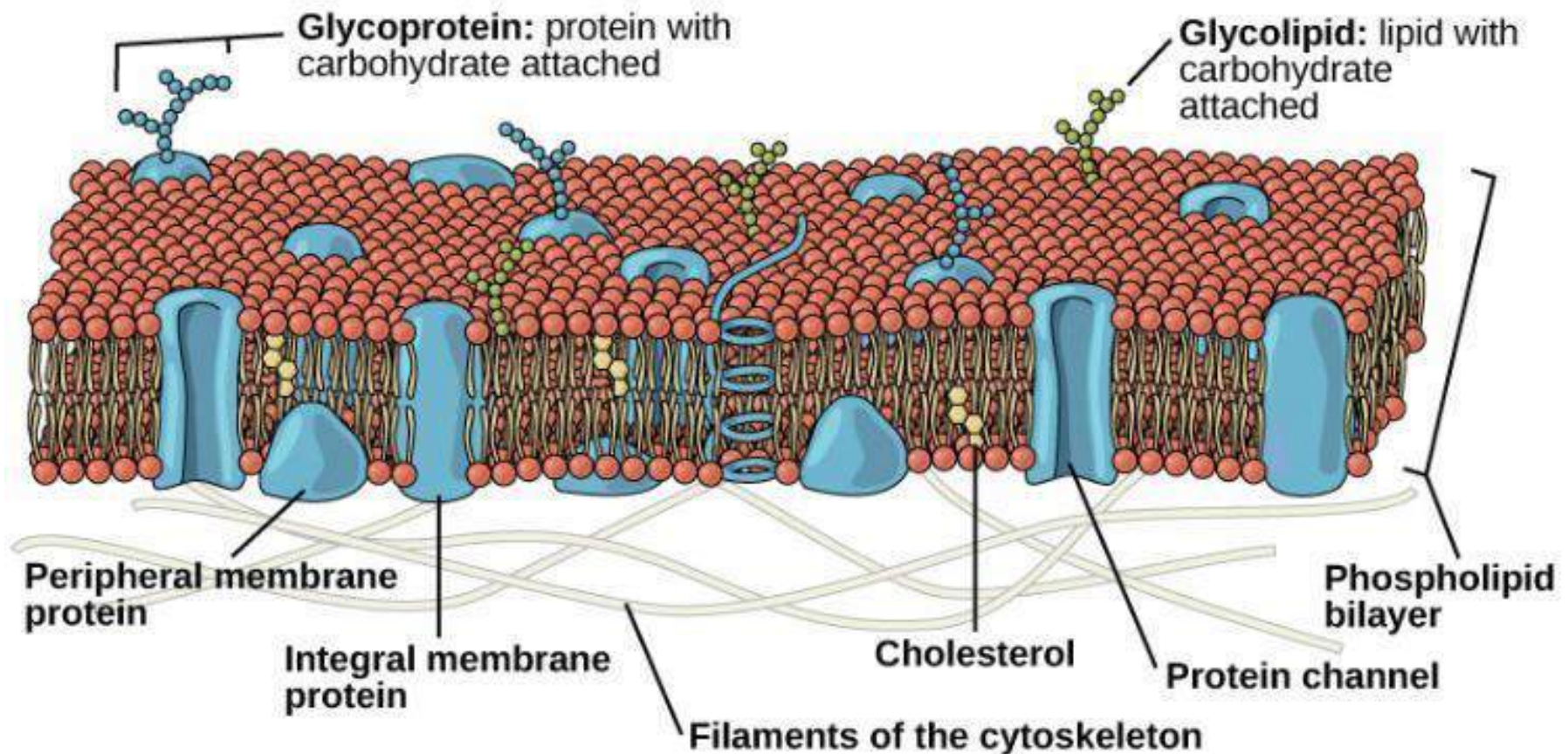
- each is a very thin film of lipid bilayer and protein molecules, held together mainly by non-covalent interactions.

## Fluid mosaic model of membrane

- states that a membrane is a fluid structure with a “mosaic” of various proteins embedded in it
  - The components include phospholipids, cholesterol, proteins and carbohydrates in which the components are able to flow and change position
-

## Importance of membrane fluidity

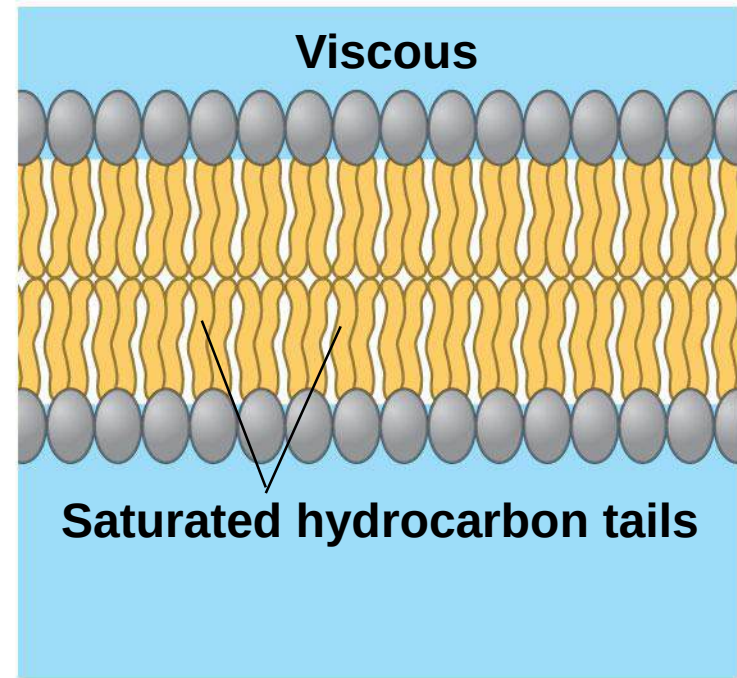
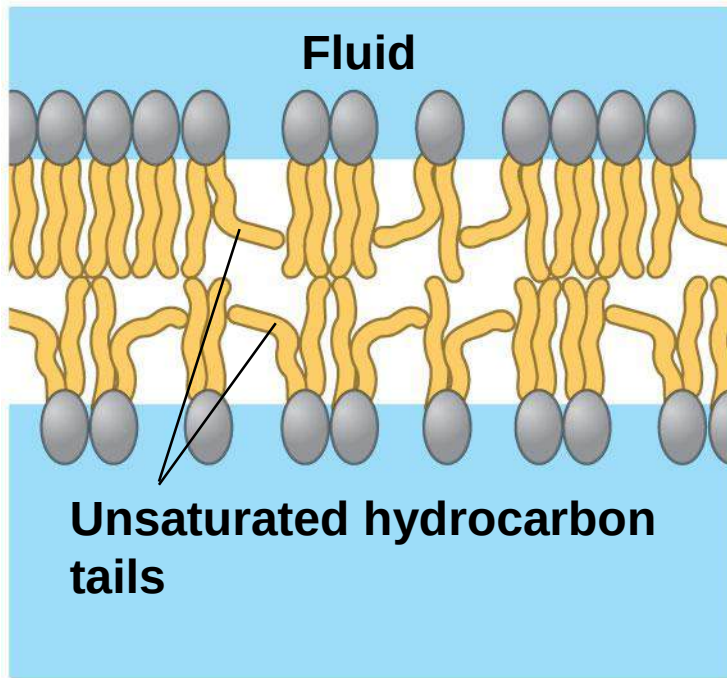
- Fluidity of the membrane gives flexibility to the cell to change its shape; transport processes are also dependent on the fluidity of the lipid bilayer.



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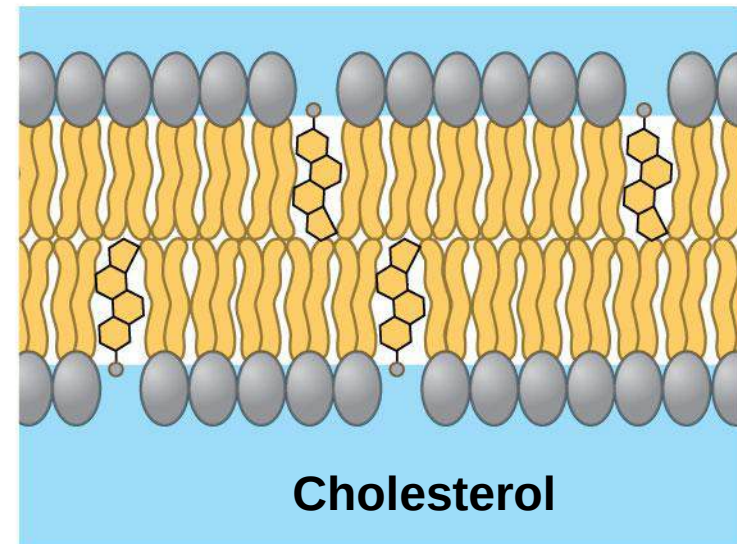
## Factors affecting Membrane fluidity

- The fluidity of a lipid bilayer depends on both its composition (hydrocarbon chain length, degree of saturation, and amount of cholesterol) and temperature
- The membrane becomes more fluid if the hydrocarbon chains are short or have double bonds.
- Membranes rich in unsaturated fatty acids are more fluid than those rich in saturated fatty acids
- Cholesterol because of its shape prevents long-chain fatty acids from packing close to each other at low temperature and increase membrane fluidity.
- Fluidity increases with increased temperature



**(a) Unsaturated versus saturated hydrocarbon tails**

**(b) Cholesterol within the animal cell membrane**



**Fig: Factors that affect membrane fluidity**



# Composition of Lipid bilayer

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- The lipid bilayer of many cell membranes is composed of Phospholipids, Cholesterol, Carbohydrates (Glycolipids and glycoprotein) and proteins

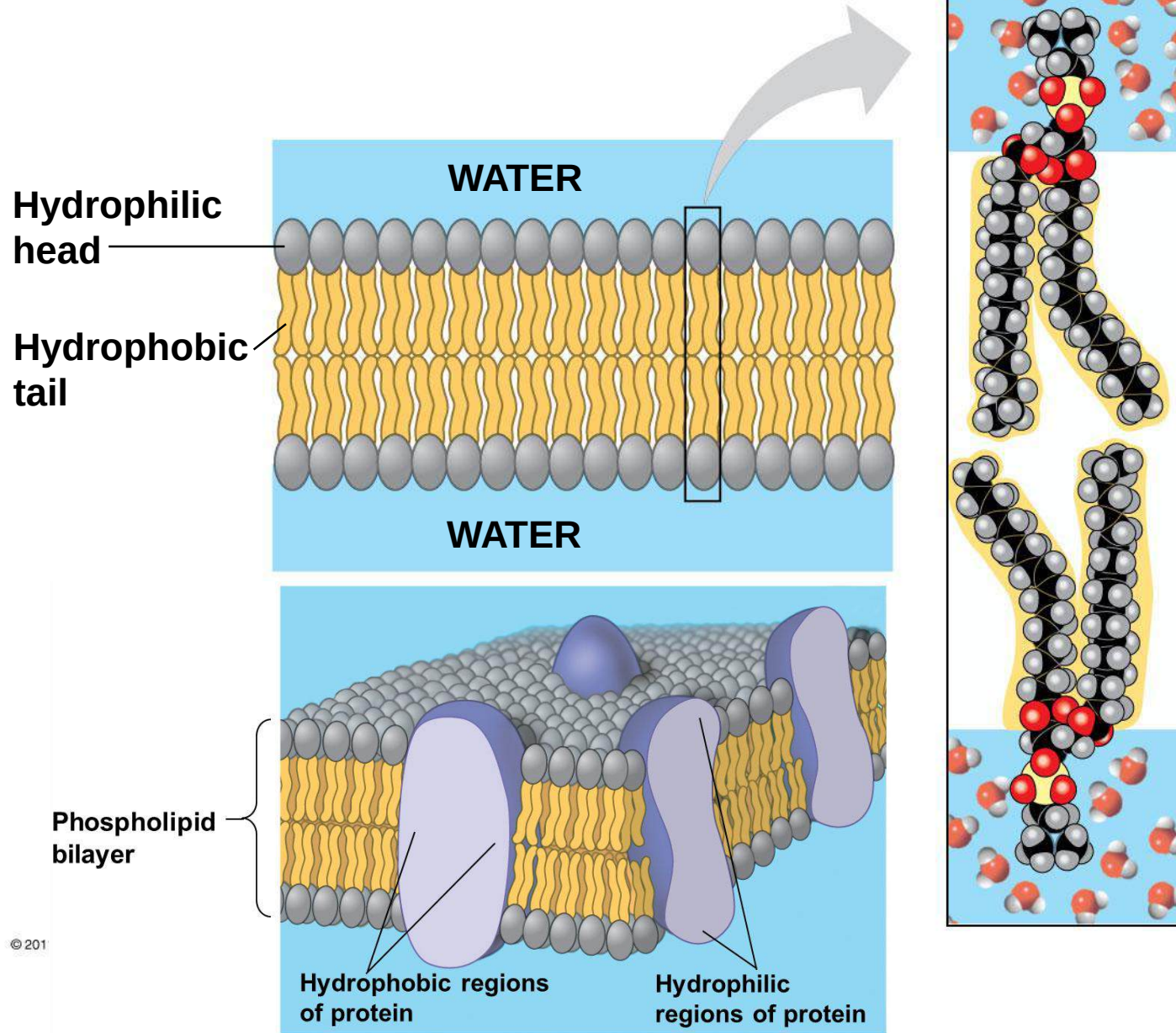
## Phospholipids

- Two layers of phospholipids molecules make the main fabric of the membrane
- are amphipathic molecules, containing hydrophobic and hydrophilic regions
  - ✓ **The hydrophilic (polar) ends** of the phospholipids are in contact with the aqueous fluids inside and outside the cell
  - ✓ **The hydrophobic tails** bury themselves in the center away from the water, while the hydrophilic heads line up on both sides in contact with water.



- 
- The lipid bilayer serves as a relatively impermeable barrier to the passage of most water- soluble molecules.
  - Protein molecules that span the lipid bilayer mediate nearly all of the other functions of the membrane, transporting specific molecules across it, or catalyzing membrane-associated reactions, such as ATP synthesis

# Phospholipid bilayer (cross section).



# Carbohydrates in the membrane

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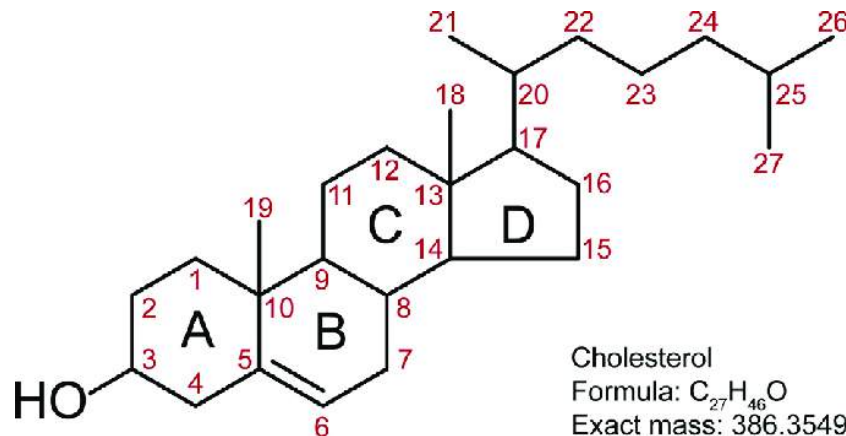
- Short-chain carbohydrates are bound primarily to membrane proteins and to a lesser extent to lipids, forming **glycoproteins** and **glycolipids**.
- Found only on the outer surface of the plasma membrane. This makes the cell membrane asymmetrical; i.e the two surfaces are not the same
- vary among species, individuals, and even cell types in an individual

## Functions

- They act as antennae receiving chemical messages from other cells
- They also serve as cell surface markers. Identify self cell from non-self

# Cholesterol

- It is a steroid, built from four linked hydrocarbon rings. A hydrocarbon tail is linked to the steroid at one end, and a hydroxyl group is attached at the other end.
- Cholesterol is absent from prokaryotes but is found to varying degrees in virtually all animal membranes.
- It constitutes almost 25% of the membrane lipids in certain nerve cells but is essentially absent from some intracellular membranes.



# Membrane proteins

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- membrane proteins perform most of the specific functions of membranes

- There are two classes of membrane proteins.

## 1. **Peripheral proteins (or extrinsic proteins):**

- Do not penetrate the bilayer to any significant degree. It is associated with the membrane on its surface

## 2. Integral proteins

- Also known as intrinsic proteins or transmembrane proteins
- penetrate the lipid bilayer

# Membrane proteins...

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## Functions of membrane proteins

- **Transport Proteins:** transport lipid insoluble molecules and ions across the membrane
- **Receptor Proteins:** Bind to chemical messengers (Ex. hormones) which send a message into the cell
- **Enzymatic Proteins:** Carry out enzymatic reactions right at the membrane when a substrate binds to the active site
- **Cell Recognition Proteins:** Glycoproteins (and glycolipids) on extracellular surface serve as ID tags (which species, type of cell, individual)
- **Attachment Proteins:** Attach to the cytoskeleton to maintain cell shape
- **Intercellular Junction Proteins:** Bind cells together



# Membrane Transport

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- The internal composition of the cell is maintained because the plasma membrane is selectively permeable to small molecules.
- Plasma membrane forms a barrier that blocks the free exchange of charged molecules between the cytoplasm and the external environment of the cell.
- In general small non-polar molecules, such as  $O_2$  and  $N_2$  and small uncharged polar molecules, such as water readily dissolve in lipid bilayers and therefore diffuse rapidly across them.
- By contrast, lipid bilayers are highly impermeable to charged molecules (ions) such as  $Na^+$  and  $K^+$   $Cl^-$ ,  $HCO_3^-$ , small hydrophilic molecules like glucose, macromolecules like proteins and RNA can not freely diffuse through the plasma membrane

# Membrane Transport

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## Types of Membrane Transport

- Generally, there are two membrane transport systems based on energy requirements

### 1. Passive transport

- ✓ Simple Diffusion
- ✓ Facilitated Diffusion

### 2. Active transport

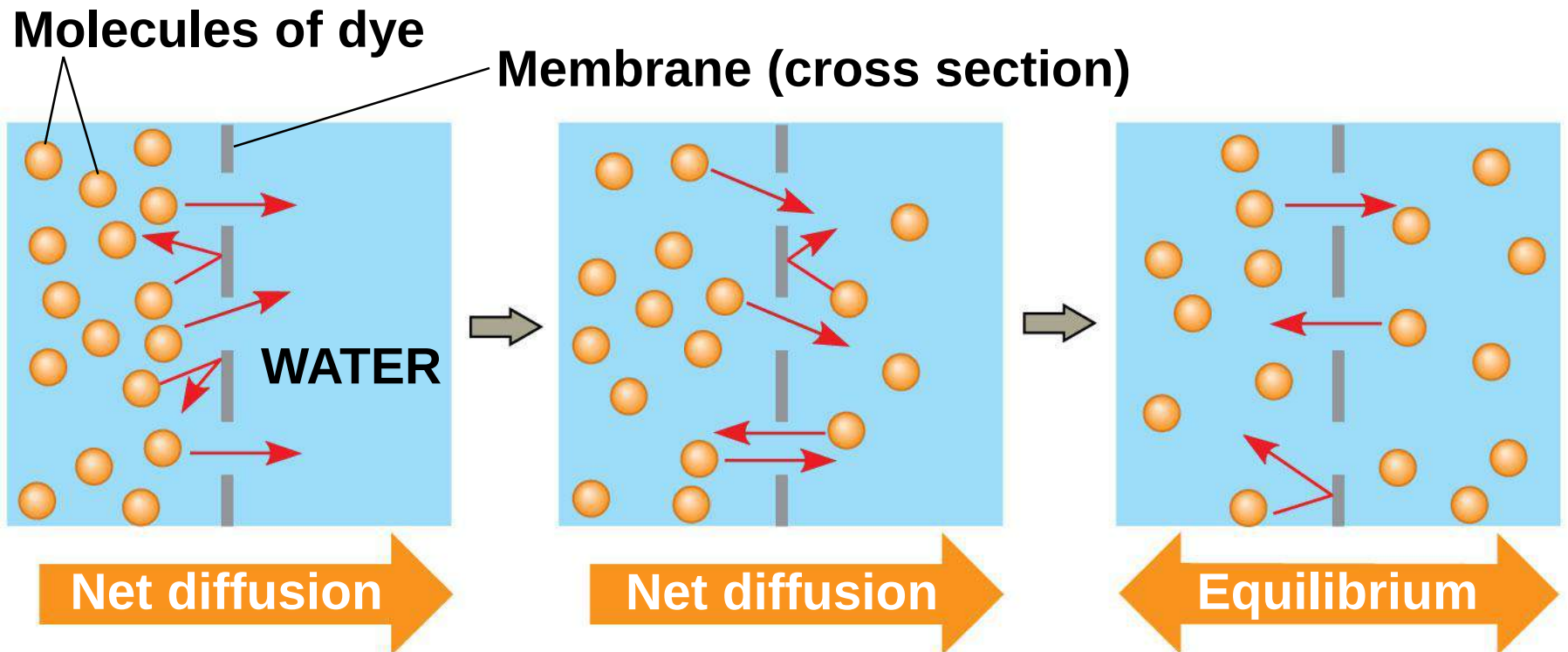
- ✓ Direct active transport
- ✓ Indirect Active transport

# Membrane Transport

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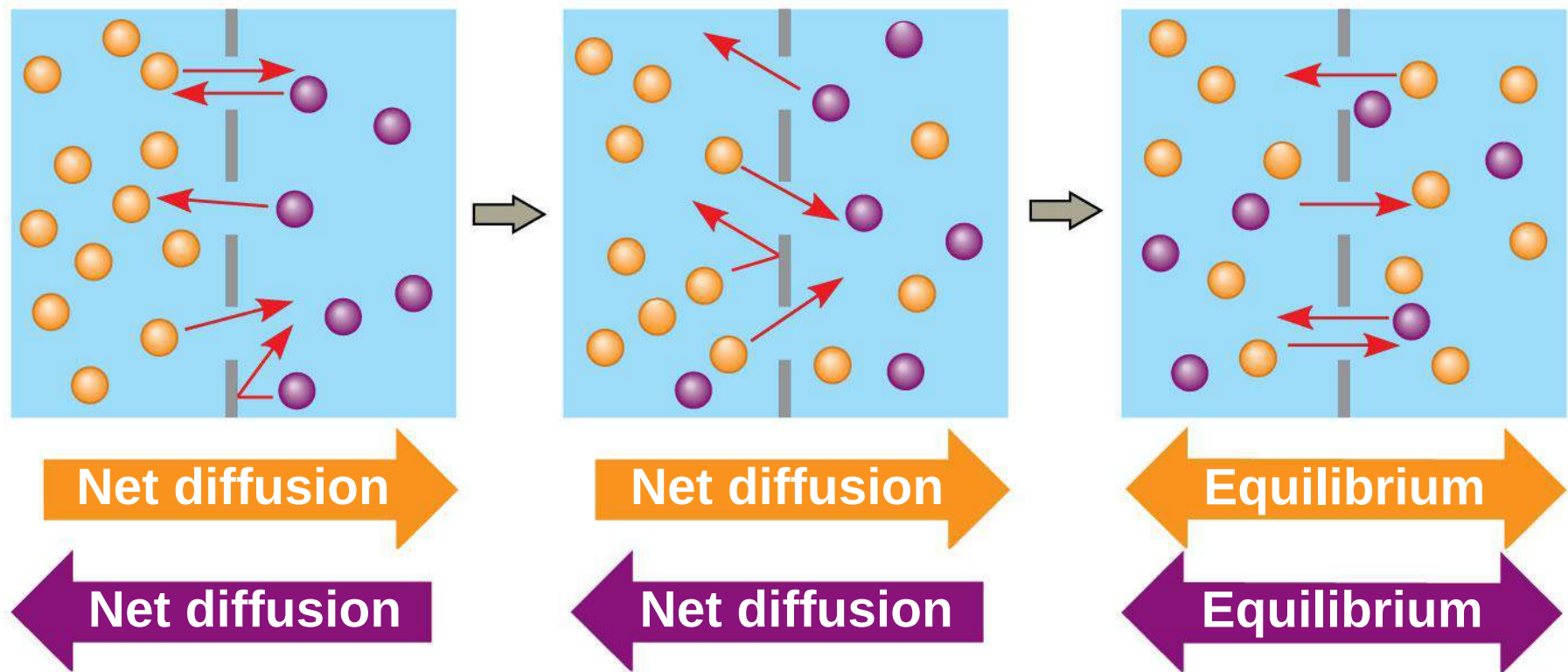
## Passive transport

- a naturally occurring phenomenon in which substances move from an area of higher concentration to an area of lower concentration in a process called diffusion
- **Simple Diffusion** is the movement of molecules from a region of high concentration to a region of low concentration (down” its concentration gradient)
  - ✓ does not need energy for the process.
  - ✓ No membrane proteins are involved



**(a) Diffusion of one solute**

Figure 7.13b

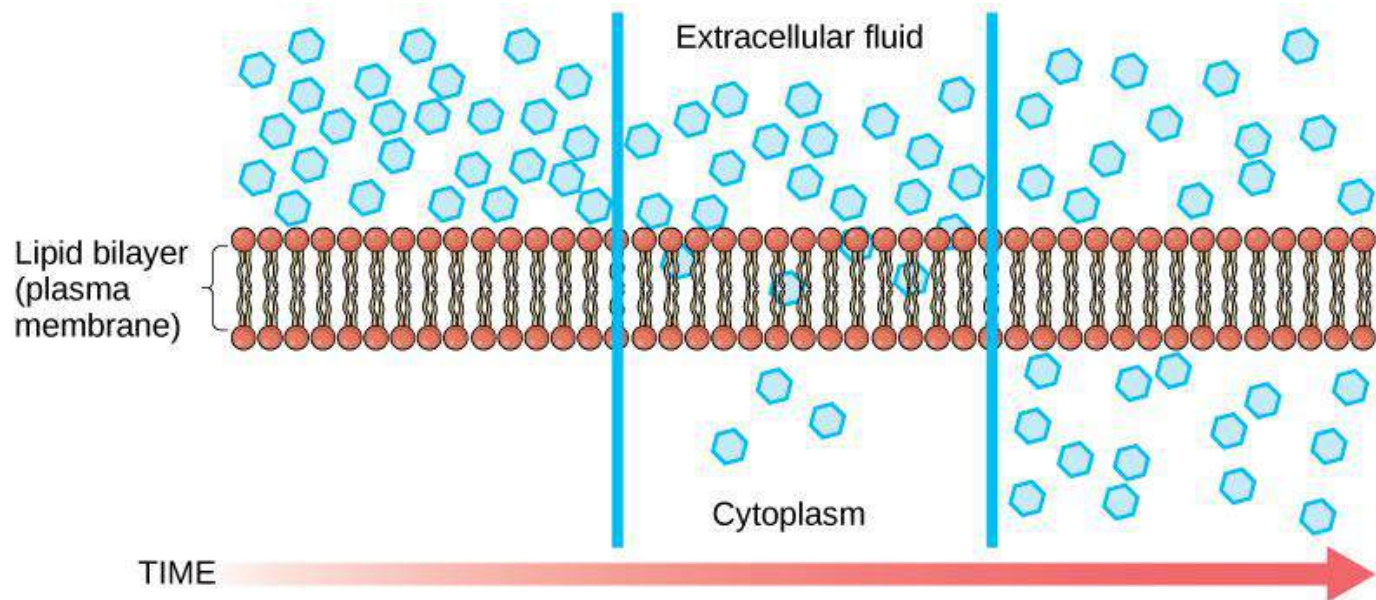


**(b) Diffusion of two solutes**

# Membrane Transport

Molecules that can pass through plasma membrane by simple diffusion are

- Gases (such as  $O_2$  and  $CO_2$ )
- Hydrophobic molecules (such as benzene)
- Small polar but uncharged molecules (such as  $H_2O$  and ethanol) are able to diffuse across the plasma membrane.





# Membrane Transport

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- Several factors affect the rate of diffusion:
  - ✓ **Concentration gradient** – the greater the difference in concentration, the more rapid the diffusion
  - ✓ **Mass of the molecules diffusing** – the more massive a molecule is the slower it will diffuse
  - ✓ **Temperature** – higher temperatures increase the rate of diffusion
  - ✓ **Distance travelled** – as the distance increases, the rate of diffusion decreases

# Membrane Transport

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## Facilitated Diffusion

- ✓ moves materials across the plasma membrane with the assistance of transmembrane proteins
- ✓ The transport is down the concentration gradient (from high to low concentration)
- ✓ Needs no expenditure of cellular energy
- ✓ It allows the movement of polar and charged molecules, such as carbohydrates, amino acids, nucleosides, and ions, to cross the plasma membrane.
- ✓ The direction of their transport is determined both by the concentration and voltage(electrical) differences-  
Electrochemical gradient

# Membrane Transport

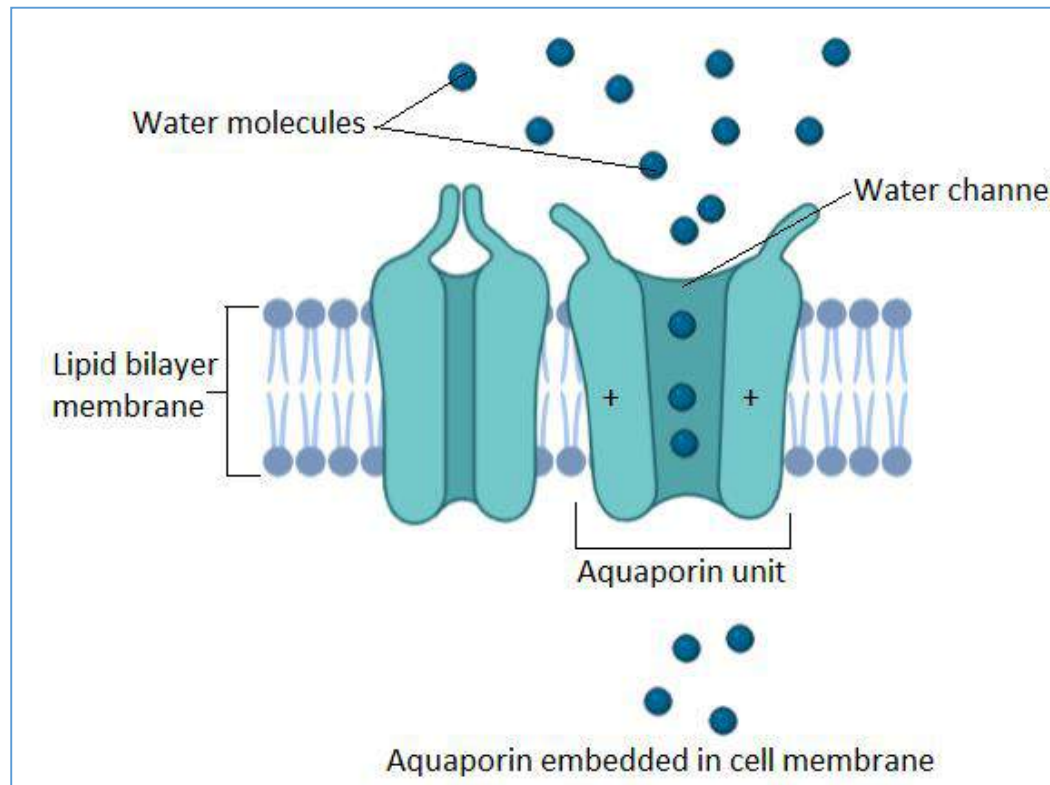
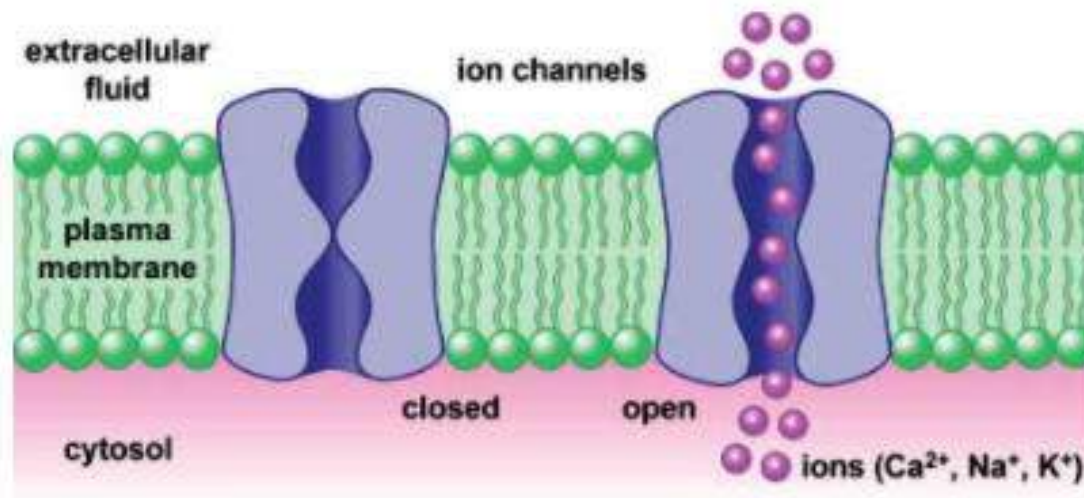
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Types of proteins that mediate facilitated diffusion

1. Channel proteins.
2. carrier proteins

## Channel proteins

- form a hydrophilic passageway across the membrane through which polar molecules or ions move. include
  1. Aquaporins, for facilitated diffusion of water
  2. Ion channels that open or close in response to a stimulus (gated channels)

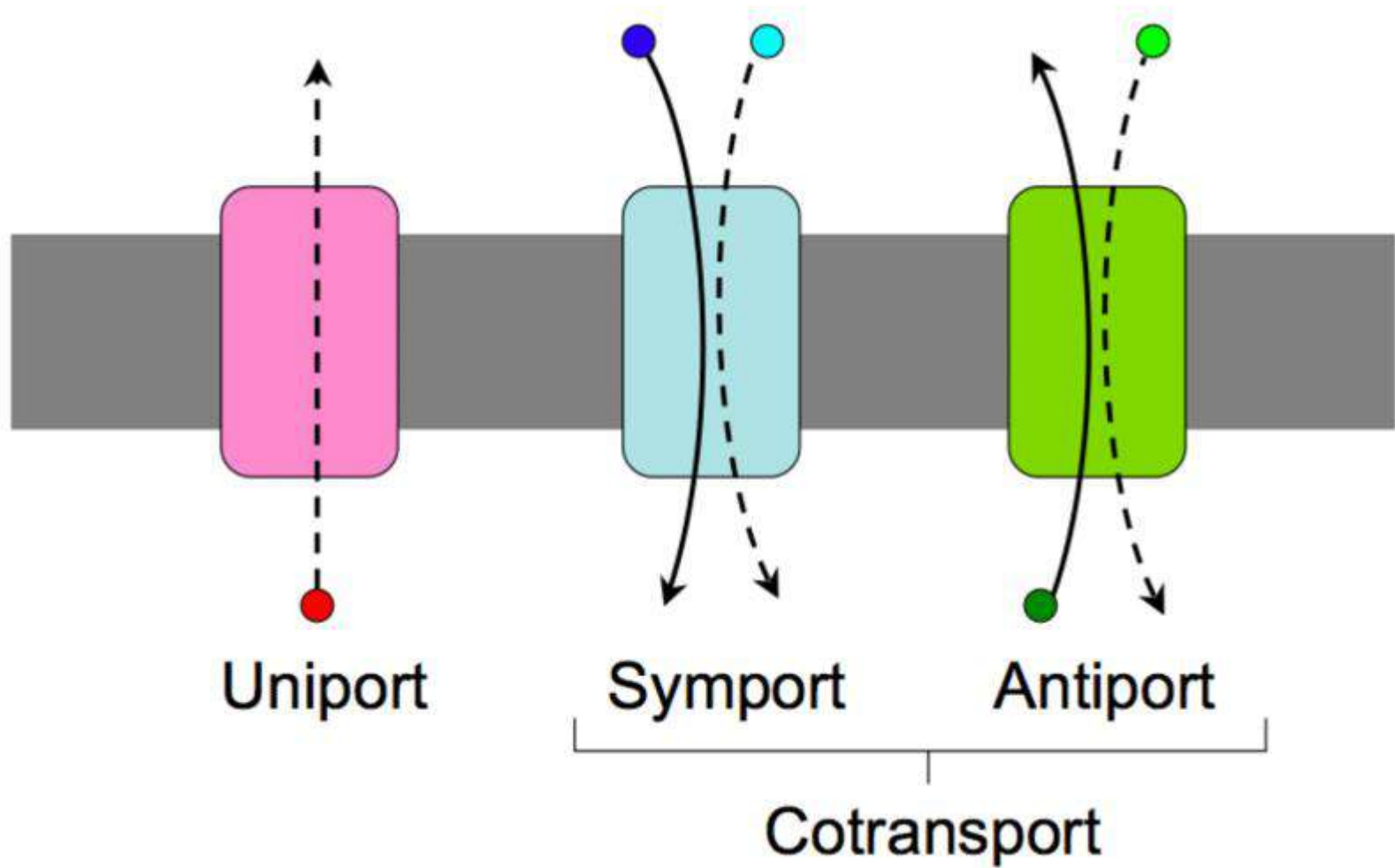


# Membrane Transport

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**Carrier proteins** (are also known as transporters)

- Carrier proteins transport a wide variety of ions and molecules across cell membranes.
- Three types of transporters have been identified
  1. **Uniporters**: transport a single type of molecule down its concentration gradient via facilitated diffusion
  2. **Symporters** are also known as Co-transporters: They transport two different solutes simultaneously in the same direction
  3. **Antiporters**: are also known as counter transporters. They also transport two different solutes simultaneously but in the opposite direction

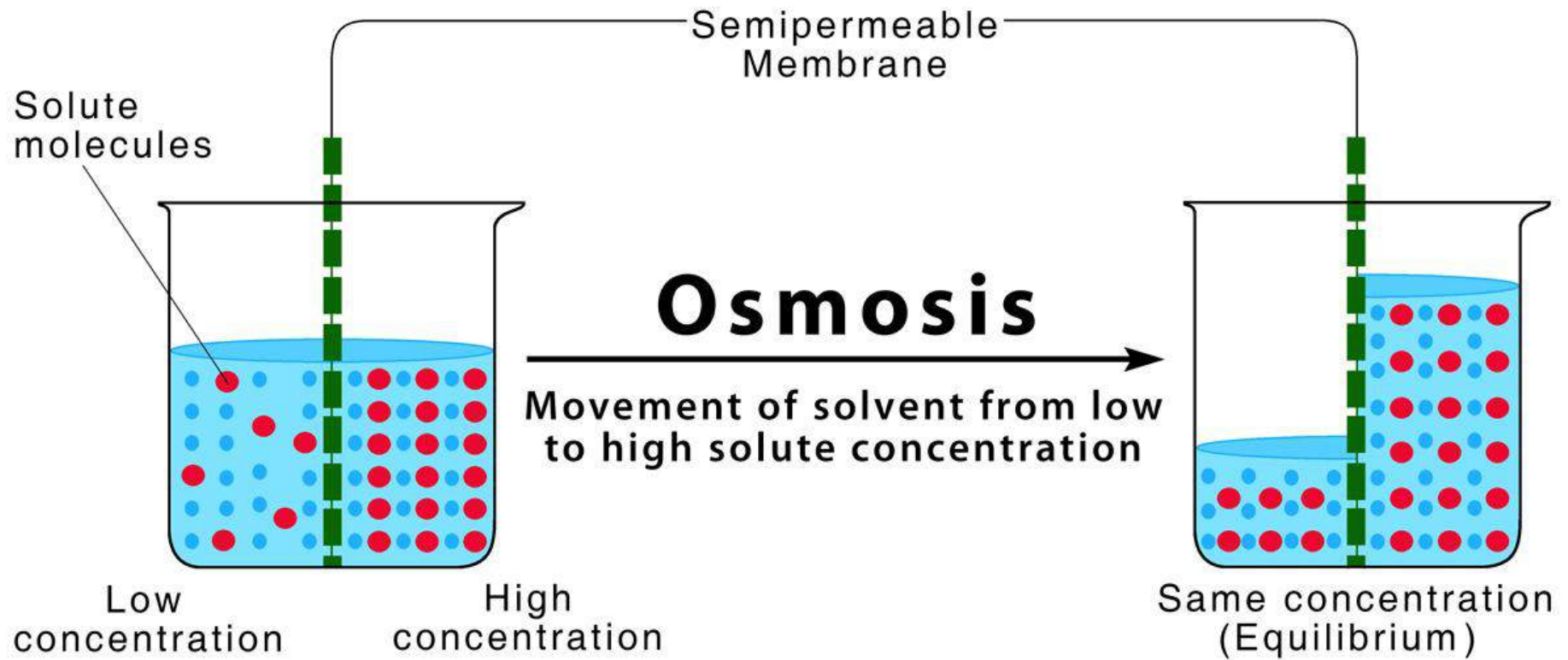




# Membrane Transport

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- **Osmosis** - the diffusion of water through a semipermeable membrane
- It is the movement of water from an area of high WATER concentration to an area of low WATER concentration
- Water is the **solvent** (what does the dissolving), and the dissolved substance is called the **solute**
- If the solute can not cross the membrane, the solvent (water) will



In osmosis, water always moves from an area of higher concentration of water (weak solution) to a lower concentration of water (strong solution). In this system, the solute cannot pass through the selectively permeable membrane.

# Membrane Transport

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## Tonicity and Osmolarity

- **Tonicity** describes the amount of solute in a solution
- **Osmolarity** – the total amount of solutes dissolved in a specific amount of solution

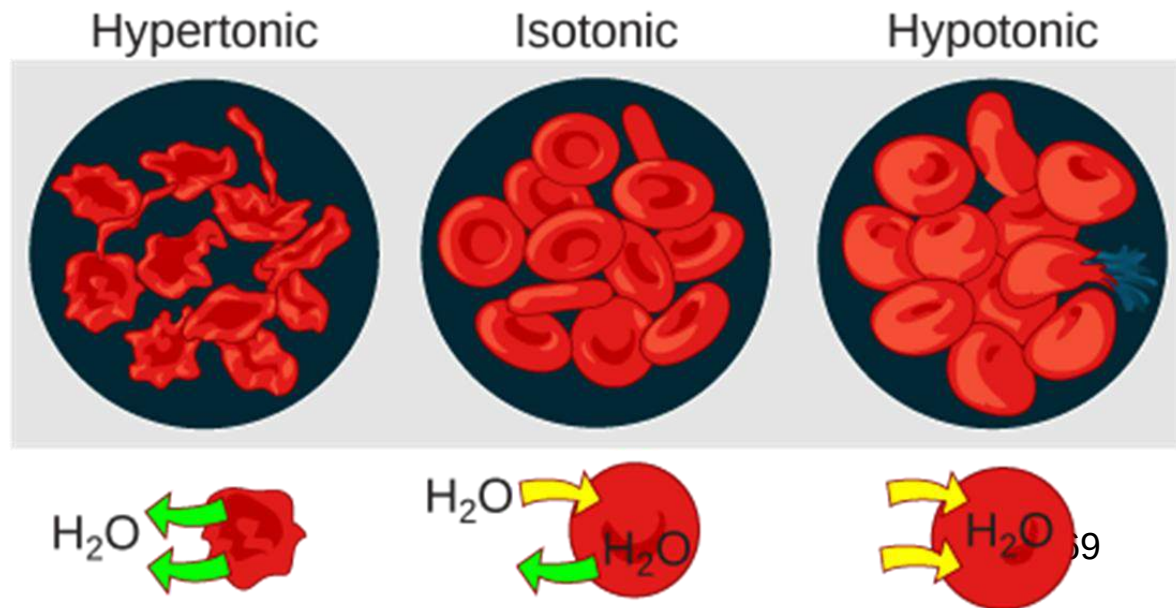
Three terms are used to relate osmolarity of a cell to the osmolarity of the extracellular fluid:

- Hypotonic – the extracellular fluid has a lower concentration of solute than the fluid inside the cell. This situation causes water to enter the cells and animal cells may burst, or lyse

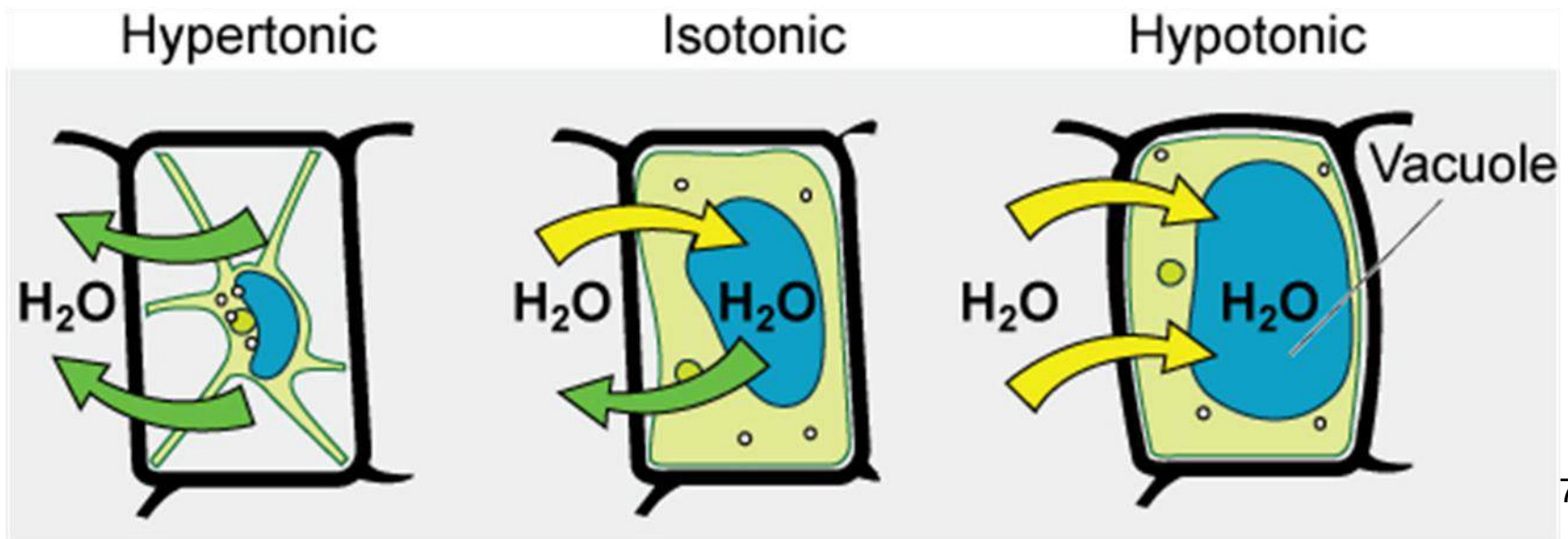
# Membrane Transport

- Hypertonic – the extracellular fluid has a higher concentration of solute than the fluid inside the cell. This situation causes water to leave the cells. This may cause animal cells to shrivel or crenate
- Isotonic – the extracellular and intracellular fluid have equal concentrations of solute, water moves into and out of the cell at the same rate, so the cell remains normal

Osmotic pressure changes the shape of red blood cells in hypertonic, isotonic, and hypotonic solutions. (credit: modification of work by Mariana Ruiz Villarreal)



- in cells that have a cell walls, such as plants, fungi, bacteria, and some protist
- ✓ In a hypotonic solution, the plasma membrane can only expand to the limit of the cell wall, it won't lyse
- ✓ In a hypertonic solution, water will leave the cell, the plant will lose turgor pressure and wilt (Plasmolysis)



# Active Transport

---

Active transport – the movement of materials AGAINST the concentration gradient from low concentration to the high concentration

- Requires the cell to use energy (ATP) is performed by specific proteins embedded in the membranes
- Active transport mechanisms, collectively called pumps or carrier proteins, work against electrochemical gradients
- Example: **sodium-potassium pump**



# Active .....

---

## Types

- Primary active transport
- Secondary active transport
- Endocytosis
  - Pinocytosis
  - Phagocytosis
  - Receptor mediated endocytosis
- Exocytosis

# Active .....

---

## I. Primary active transport

- Examples -  $\text{Na}^+$  -  $\text{K}^+$  pump,  $\text{Ca}^{++}$  pump,  $\text{H}^+$ - $\text{K}^+$  pump
- Inner surface of carrier mol. has ATPase which is activated by attachment of specific ions and causes hydrolysis of ATP molecule
- Energy released from ATP causes conformational change in the carrier which transports ions to the opposite side.

## a) $\text{Na}^+ - \text{K}^+$ pump- electrogenic pump

- Attachment of  $2\text{K}^+$  on outer side &  $3\text{Na}^+$  on inner side

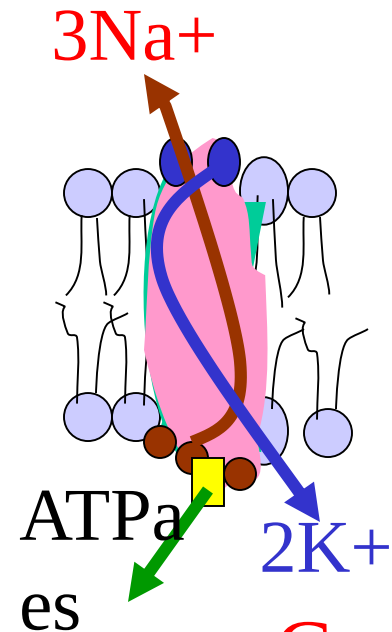
↓  
Activation of ATPase

↓  
Conformational change

↓  
Efflux of  $3\text{Na}^+$  & influx of  $2\text{K}^+$

↓  
Creates high  $\text{K}^+$  conc. & - vity inside the cell

Helps in maintaining cell volume



# Active .....

---

## Secondary Active Transport

- uses the energy of the electrochemical gradient created by the primary active transport system to bring in other substances such as amino acids and glucose
  - ✓ Active transport depending upon conc. gradient of  $\text{Na}^+$  from ECF to ICF created by utilization of energy-carrier, does not have ATPase activity
  - ✓ Another substance (e.g. Glucose) is transported along with  $\text{Na}^+$
  - ✓  $\text{Na}^+$  is transported only when glucose mol. is attached

# Active .....

---

## Bulk Transport: Exocytosis and Endocytosis

- Transports of large polar molecules
- multimolecular material may leave or enter the cell, such as hormone secretion or ingestion of invading microbe by leukocytes
- transport substance is enclosed in - vesicles
- Transport into the cell is termed **endocytosis**, whereas transport out of the cell is called **exocytosis**

# Active .....

---

## Endocytosis

- It is an active processes involving the bulk transport of materials through membranes, in to cells
- It occurs by an in folding or extension of cell surface membrane to form a vesicle or vacuole

### Endocytosis can occur in three ways

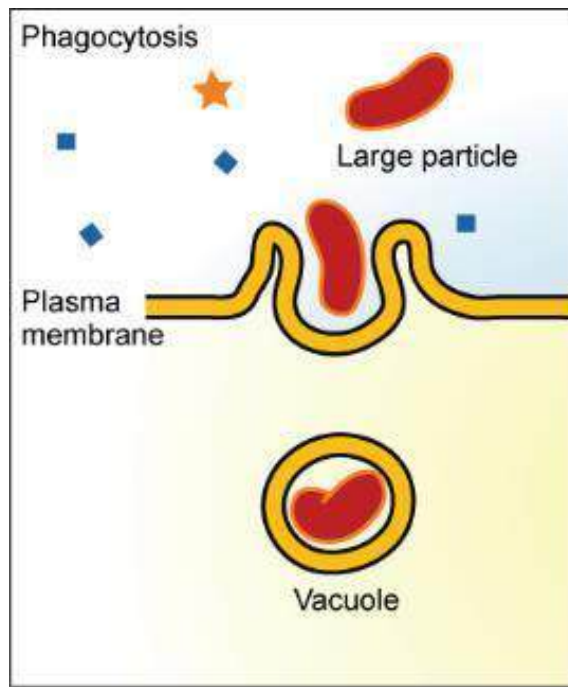
- ✓ Phagocytosis ("cell eating")
- ✓ Pinocytosis ("cell drinking")
- ✓ Receptor-mediated endocytosis



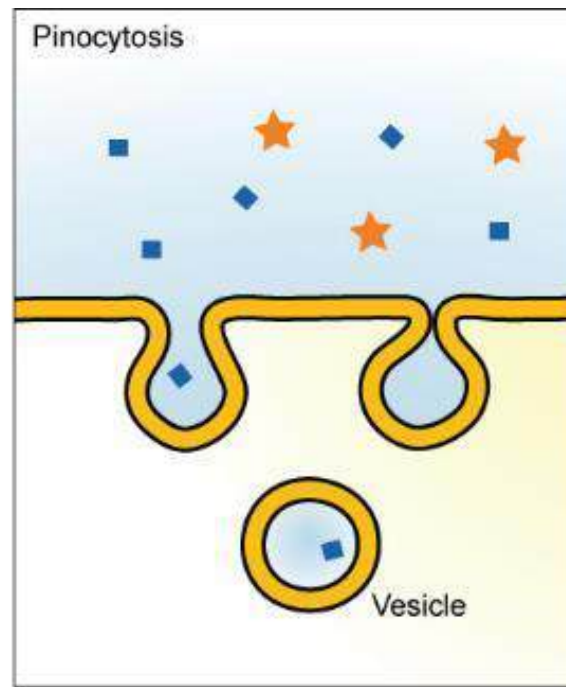
# Active Transport

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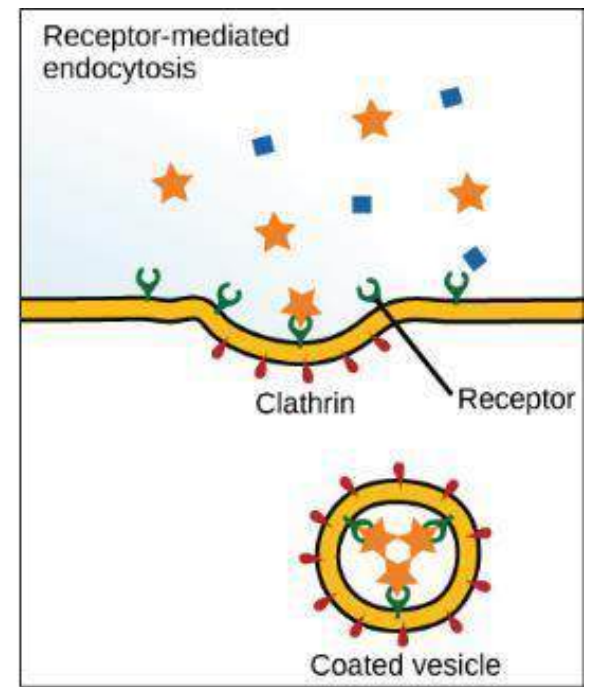
- ✓ **Phagocytosis** – the process by which large particles, such as cells, are taken in by a cell
- ✓ **Pinocytosis** – the process by which fluids are taken in by a cell
- ✓ **Receptor-mediated endocytosis** – the process by which particles bind to the receptor proteins, causing the plasma membrane to invaginate, bringing the particle and the proteins into the cell



(a)



(b)



(c)

- Three variations of endocytosis are shown.
- In one form of endocytosis, phagocytosis, the cell membrane surrounds the particle and pinches off to form an intracellular vacuole.
  - In another type of endocytosis, pinocytosis, the cell membrane surrounds a small volume of fluid and pinches off, forming a vesicle.
  - In receptor-mediated endocytosis, uptake of substances by the cell is targeted to a single type of substance that binds at the receptor on the external cell membrane.

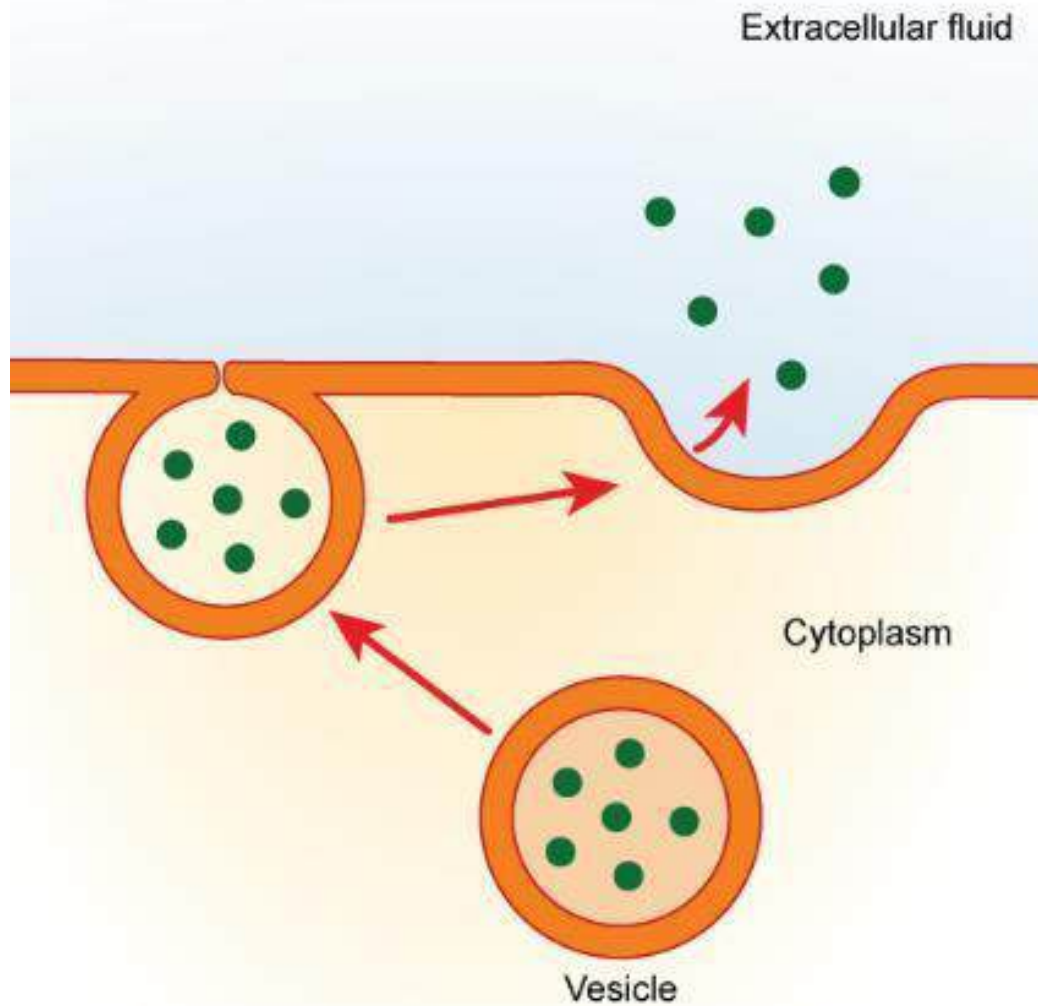
# Active .....

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## Exocytosis

- It is an active processes involving the bulk transport of materials through membranes (out of cells)
- Exocytosis is the reverse process of endocytosis
- Waste materials such as solid and undigested remains removed from cells
- Secretion of enzymes from the pancreas is achieved in this way
- Plant cells use exocytosis to export the materials needed to form cell wall

## Exocytosis



In exocytosis, a vesicle migrates to the plasma membrane, binds, and releases its contents to the outside of the cell.



# Chapter 4: Cellular metabolism

Virtually every task performed by living organisms requires energy extracted from environments (nutrients and/or solar energy).

Examples

- ✓ Assemblage & breakdown of macromolecules
- ✓ Transport of substances across cell membranes
- ✓ Transcription & translation of Genetic instructions
- ✓ Movement, growth, development, reproduction & maintenance of cell/body part

*Metabolism refers to all of the chemical reactions that take place inside cells including those that use and release energy*

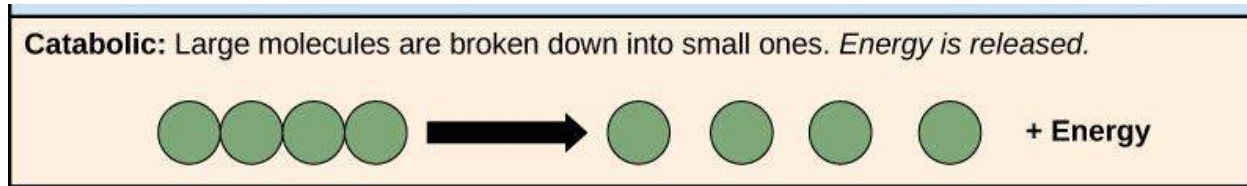


# Cellular...

Divided into 2: Anabolism & Catabolism reactions

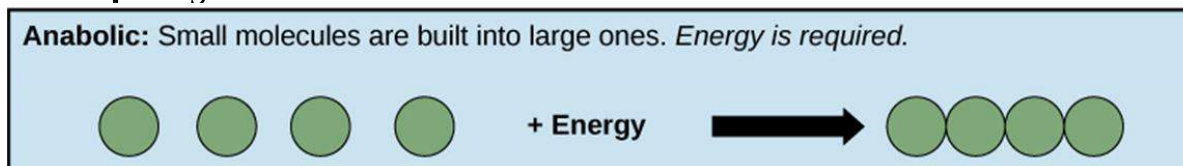
## Catabolic reactions: Degradation reactions

- are series of chemical reactions that release energy as polymers are broken down into their monomers



## Anabolic reactions: Synthesis Reactions

- are series of chemical reactions that require the input of energy to build polymers



*Each reaction step in these pathways is catalyzed by a specific enzyme*





# Enzymes and their role in Metabolism

- Enzymes are biological catalysts that **increase the rate of chemical reactions without being used up in the process.**
- Without enzymes, most cellular reactions would take place too slowly to support life

## Chemical Nature of Enzymes (Properties)

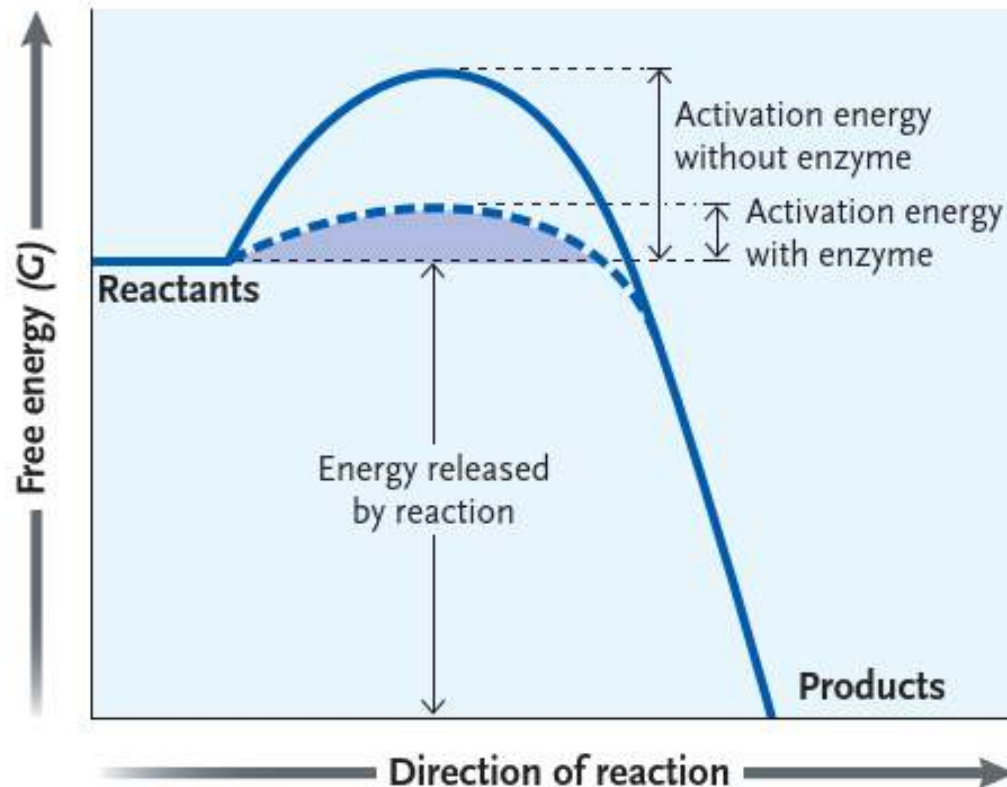
- Most are globular proteins
- Enzymes are well suited to their roles in three major ways:
  - they have **enormous catalytic power**, they are **highly specific** in the reactions they catalyze, and their activity as catalysts can be regulated
- acts upon reactants called substrates
- do not alter the equilibrium constant ( $K_{eq}$ ).



# Mechanisms of enzyme action

## How enzymes work?

- *Enzymes accelerates the rate of reaction by decreasing the energy of activation allowing it to achieve equilibrium more rapidly*
- **Activation Energy** is the energy required to start a reaction.





## Enzyme Classification and Nomenclature

- Some of the earliest enzymes to be discovered were given names ending in –*in* to indicate that they were proteins (e.g. the digestive enzymes pepsin, trypsin, chymotrypsin).
- Because of the large number of enzymes that are now known, a systematic nomenclature called the Enzyme Commission (EC) system is used to name them. [*International Union of Biochemistry and Molecular Biology*]



## Enzyme Classification and....

- EC is a four-digit number. The **first digit indicates major class**, **second digit indicates sub class**, **third digit denotes sub-sub class** and **final digit indicates specific enzyme**
- Enzymes are grouped into **six major classes** on the basis of the reaction which they catalyze.
- Each enzyme has an unambiguous (and often long) systematic name that specifies
  - ✓ *the substrate of the enzyme (the substance acted on)*
  - ✓ *the functional group acted on, and*
  - ✓ *the type of reaction catalyzed.*
- All EC names end in **ase**



## Example of systematic name

- The enzyme commonly called “hexokinase” is designated “ATP: D-hexose-6-phospho-transferase E.C. 2.7.1.1.” This identifies hexokinase as
  - ✓ a member of class 2 (transferases)
  - ✓ subclass 7 (transfer of aphosphoryl group)
  - ✓ sub-subclass 1 (alcohol is the phos-phoryl acceptor). and
  - ✓ the term “hexose-6” indicates that the alcohol phosphorylated is that of carbon six of a hexose
- Enzymes are also assigned common names derived by adding -ase to the name of the substrate or to a combination of substrate name and type of reaction



## EC Classification of Enzymes

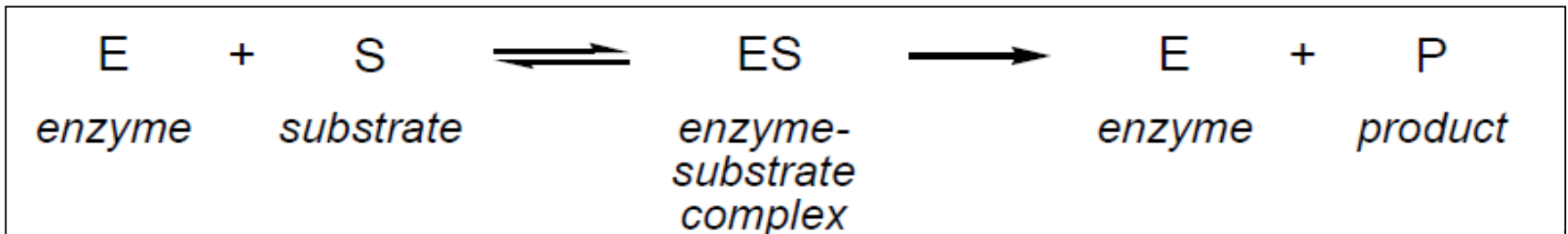
| EC Code | Group Name      | Type of Reaction Catalyzed                               |
|---------|-----------------|----------------------------------------------------------|
| EC 1    | Oxidoreductases | Oxidation-reduction reactions                            |
| EC 2    | Transferases    | Transfer of functional groups                            |
| EC 3    | Hydrolases      | Hydrolysis reactions                                     |
| EC 4    | Lyases          | Addition to double bonds or the reverse of that reaction |
| EC 5    | Isomerases      | Isomerization reactions                                  |
| EC 6    | Ligases         | Formation of bonds with ATP cleavage                     |





## Enzyme-substrate interaction

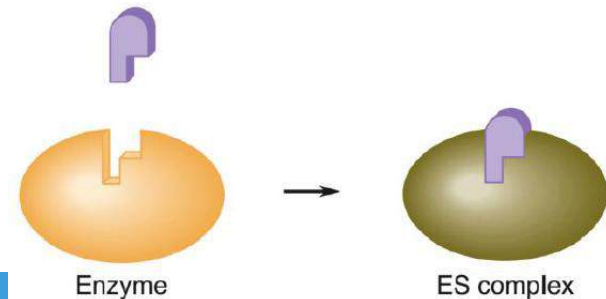
- Substrate interact with enzyme at its **active site**
- The active site
  - ✓ Is a pocket or cleft that is specialized to recognize specific substrates and catalyze chemical transformations.
- Substrates bind in active site by weak non-covalent interactions (*hydrogen bonding, hydrophobic interactions, ionic interactions*)
- The interactions hold the substrate in the proper orientation for most effective catalysis



# Two models for enzyme/substrate interactions:

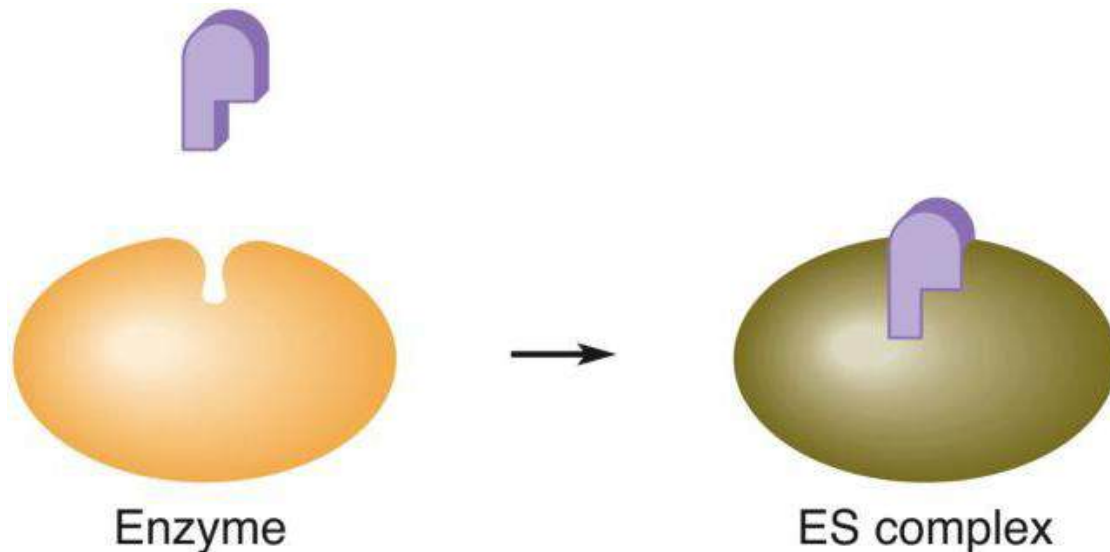
## 1. Lock and Key Model:

- Proposed by Emil Fischer in 1894
- Explains the high specificity of enzyme activity. Enzyme surfaces accommodate substrates having specific shapes and sizes, so only specific substances —fit in an active site to form an ES complex
- A limitation of this theory is that it requires enzymes conformations to be rigid. Research suggests that instead enzymes are at least somewhat flexible.
- This model fails to take into account proteins conformational changes to accommodate a substrate molecule.



## 2. In the induced-fit model

- Proposed by Daniel E. Koshland in 1958
- Proposes that enzymes have **flexible conformations** that may adapt to incoming substrates.
- The active site adopts a shape that is complementary to the substrate only after the substrate is bound





## Enzyme Cofactors

- **Cofactors** are non-protein molecules required for activity of some enzymes
- There are two types of cofactors:
  - ✓ **Organic cofactors(co-enzymes)** such as  $\text{NAD}^+$ ,  $\text{FAD}^+$  and
  - ✓ **Inorganic cofactors(metal Ions)** such as  $\text{Fe}^{2+}$ ,  $\text{Mg}^{2+}$ ,  $\text{Mn}^{2+}$ , or  $\text{Zn}^{2+}$
- A coenzyme or metal ion that is very tightly or even covalently bound to the enzyme protein is called **a prosthetic group**.
- **Co-enzyme + Apoenzyme = holoenzyme.**



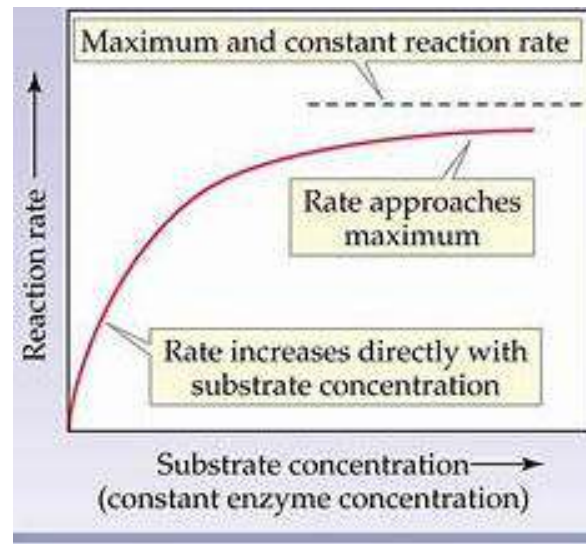
## Enzyme Kinetics

- Refers to the rate of an enzyme-catalyzed reaction
- The velocity of all enzymes is dependent on
  - ✓ Substrate concentration
  - ✓ Enzyme concentration
  - ✓ Temperature, and
  - ✓ pH

# 1. Effect of substrate and enzyme concentration

## Substrate Concentration

- Increasing  $[S]$  increases the rate of the reaction, but eventually, the rate reaches a maximum ( $V_{max}$ ), and remains constant after that.
- The maximum rate is reached when the enzyme is saturated with substrate, and cannot react any faster under those conditions
- The shape of the resulting graph when  $V_0$  is plotted against  $[S]$  is called a hyperbolic curve



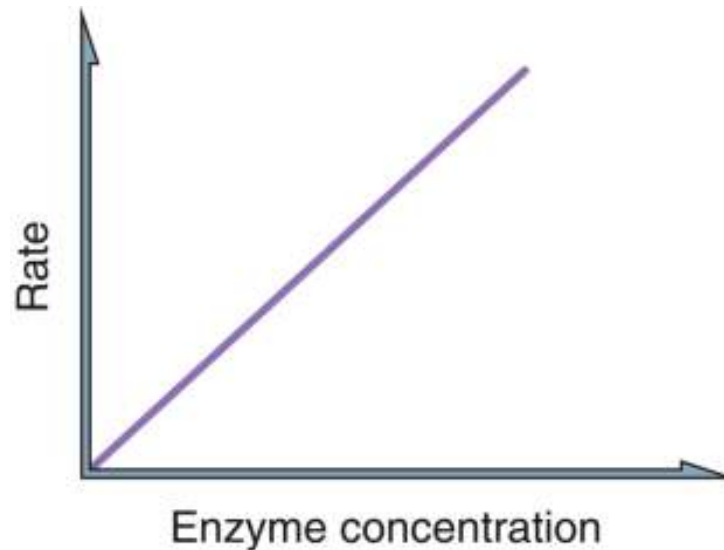




## Effect of substrate and enzyme concentration....

### Enzyme Concentration

- At saturating substrate concentration (**i.e. all the enzyme molecules are bound to substrate**), a doubling of the enzyme concentration will lead to a doubling of  $V_0$ . This gives a straight line graph when  $V_0$  is plotted against enzyme concentration.



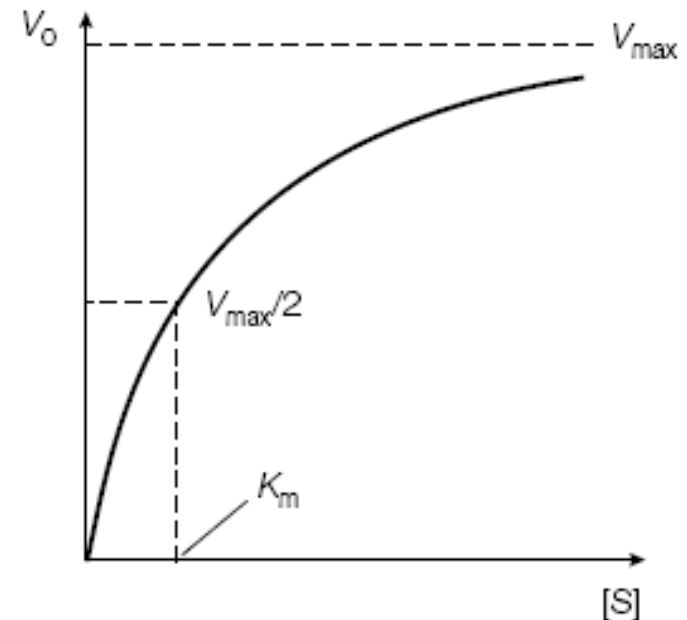


## Effect of substrate and enzyme concentration....

- The dependence of enzyme rate (activity) on substrate concentration has been provided in a quantitative way.
- The simplest of these equations, the Michaelis-Menten equation, relates the **initial velocity ( $V_0$ )** to the concentration of **substrate  $[S]$**  and the two parameters  **$K_m$**  and  **$V_{max}$** .

Michaelis–Menten equation:

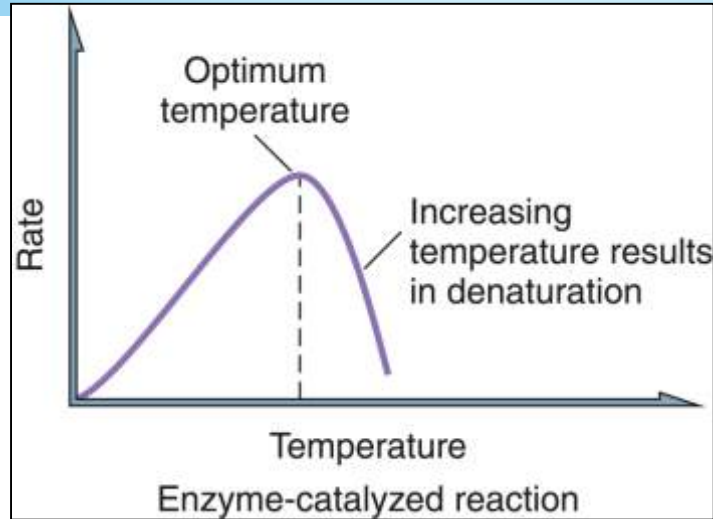
$$V_0 = \frac{V_{max} \cdot [S]}{K_m + [S]}$$





## 2. Temperature

- Temperature affects the rate of enzyme-catalyzed reactions in two ways.
- ✓ **First, arise in temperature increases the thermal energy of the substrate molecules.**
  - *This raises the proportion of substrate molecules with sufficient energy to overcome energy of activation, and hence increases the rate of the reaction.*
- ✓ **a second effect comes into play at higher temperatures.**
  - *Increasing the thermal energy of the molecules increase the chances of breaking the multiple weak, non-covalent interactions which hold the three-dimensional structure of the enzyme together. Ultimately this will lead to the denaturation (unfolding) of the enzyme, decreasing enzyme activity*



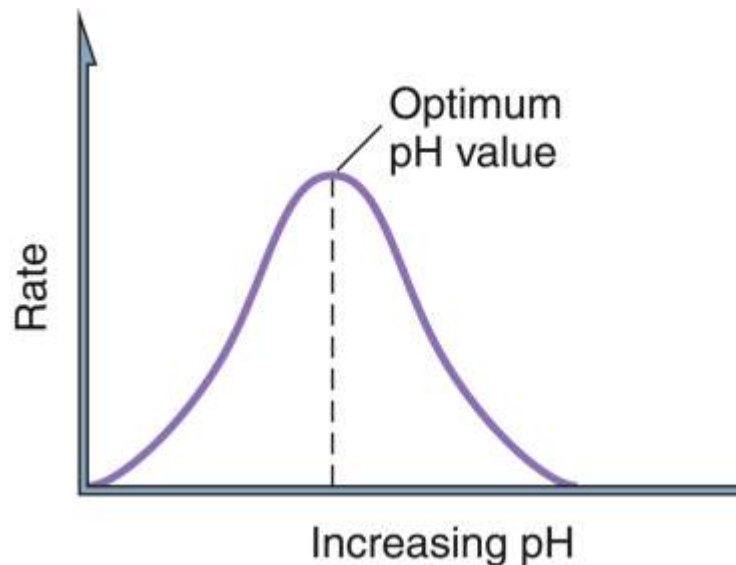
- **3. pH**

Each enzyme has an **optimum pH** at which the rate of the reaction that it catalyzes is at its maximum.

- Slight deviations in the pH from the optimum lead to a decrease in the reaction rate.
- *Larger deviations in pH lead to denaturation of the enzyme due to changes in the ionization of amino acid residues and the disruption of non covalent interactions.*



- Many enzymes have a pH optimum of around 6.8, but there is great diversity in the pH optima of enzymes, due to the different environments in which they are adapted to work. **For example**, the digestive enzyme **pepsin** is adapted to work at the acidic pH of the stomach (around pH 2.0)



**Table 20.4** Examples of Optimum pH for Enzyme Activity

| Enzyme                  | Source         | Optimum pH |
|-------------------------|----------------|------------|
| pepsin                  | Gastric mucosa | 1.5        |
| $\beta$ -glucosidase    | Almond         | 4.5        |
| sucrase                 | Intestine      | 6.2        |
| urease                  | Soybean        | 6.8        |
| catalase                | Liver          | 7.3        |
| succinate dehydrogenase | Beef heart     | 7.6        |
| arginase                | Beef liver     | 9.0        |
| alkaline phosphatase    | Bone           | 9.5        |



## Enzyme inhibition

- Any substance that can diminish the velocity of an enzyme-catalyzed reaction is called an inhibitor
- Many poisons and medicines inhibit one or more enzymes and thereby decrease the rate of the reactions they carry out.
- Some substances normally found in cells inhibit specific enzymes, providing a means for internal regulation of cell metabolism.
- **Enzyme inhibition can be of two main types:**
  - ✓ irreversible or
  - ✓ reversible.
- Reversible inhibition can be subdivided into
  - ✓ competitive and
  - ✓ noncompetitive



# 1. Irreversible Inhibitors

- Inhibitor forms a covalent bond with a specific functional group of an enzyme, thereby inactivating it. Also called suicide inhibitors
  - ✓ e.g. Cyanide ion,  $\text{CN}^-$ , is a rapidly-acting, highly toxic inhibitor, which interferes with the iron-containing enzyme cytochrome oxidase: Cell respiration stops, and death occurs within minutes
- Antibiotics are enzyme inhibitors that act on life processes that are essential to certain strains of bacteria
  - ✓ Penicillin interfere with transpeptidase, which is used by bacteria for the construction of cell walls.



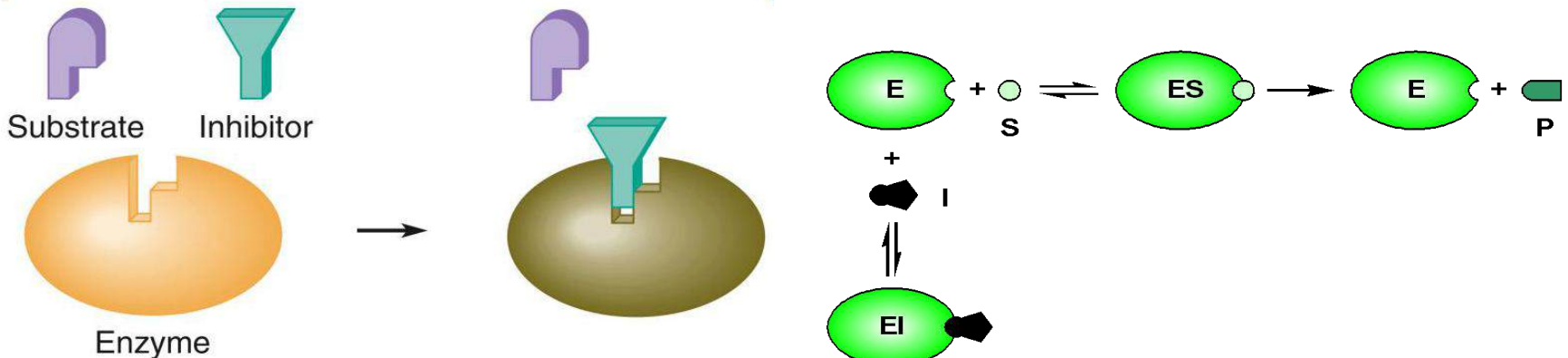
## 2. Reversible Inhibitors

- Once the inhibitor combines with the enzyme by non-covalent interaction, the active site is blocked, and no further catalysis takes place.



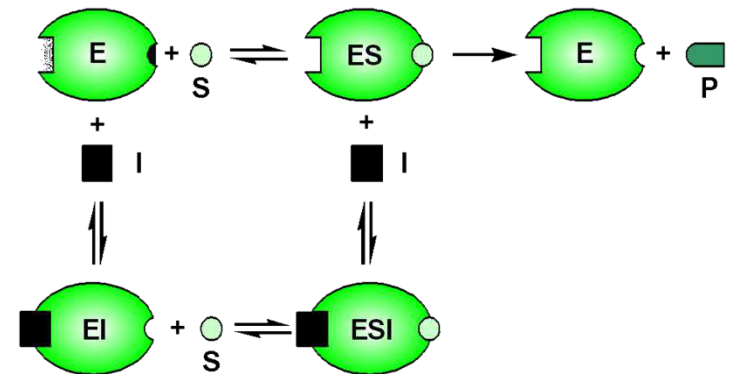
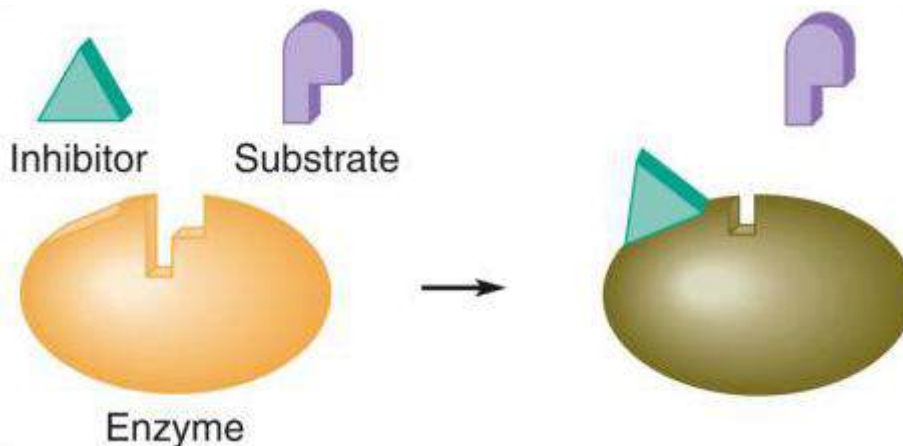
### a. Competitive inhibitors

- Competitive inhibitors bind to the enzyme's active site and compete with the normal substrate molecules. They often have structures that are similar to those of the normal substrate
- Can be overcome by increasing [S]



## b. Non-competitive inhibitors

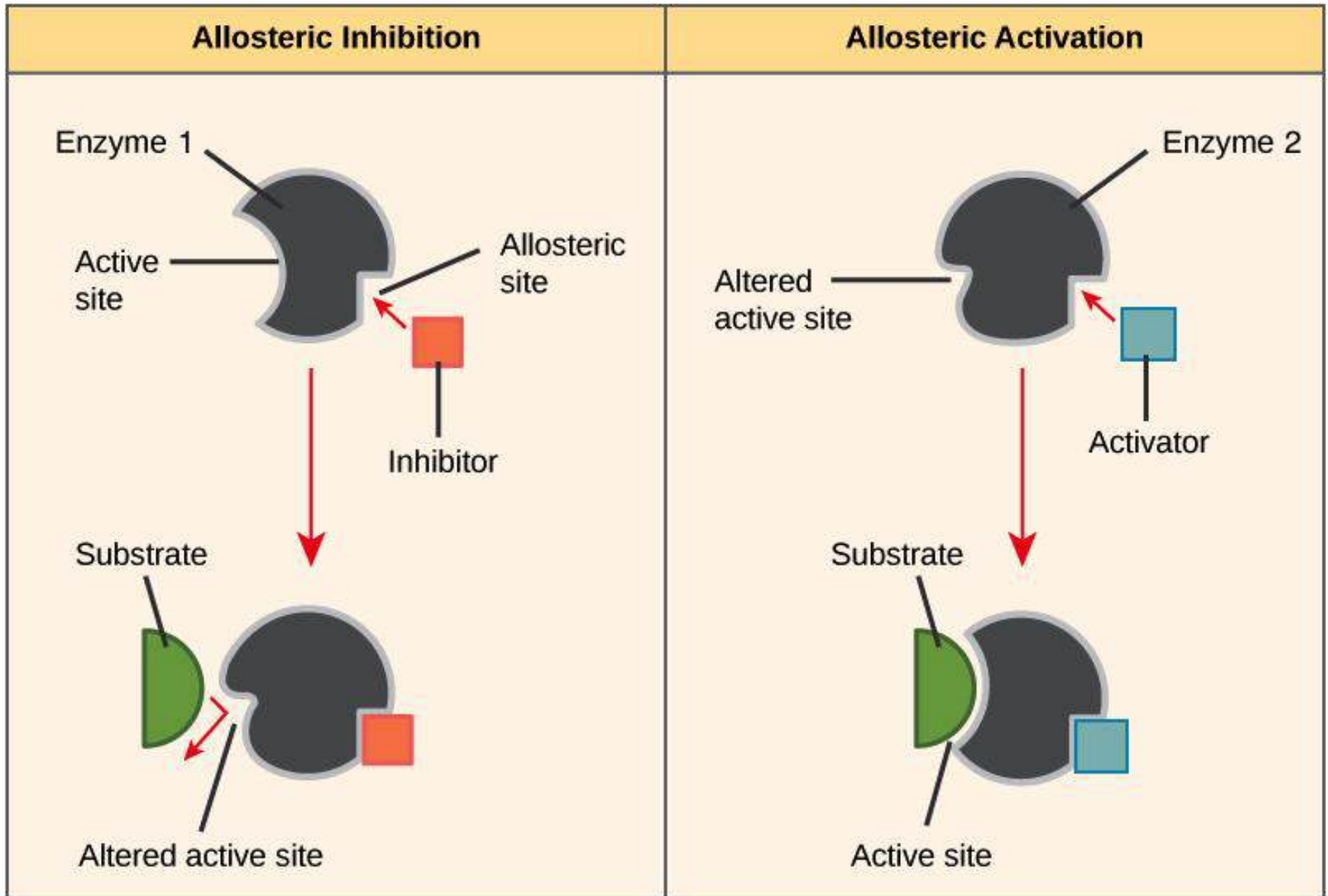
- Noncompetitive inhibitors bind reversibly to the enzyme at a site other than the active site, changing the 3D shape of the enzyme and the active site, so that the normal substrate no longer fits correctly.
- Noncompetitive inhibitors do not look like the enzyme substrates.
- Increasing the substrate concentration does not affect noncompetitive inhibition because it can't bind to the site occupied by the inhibitor





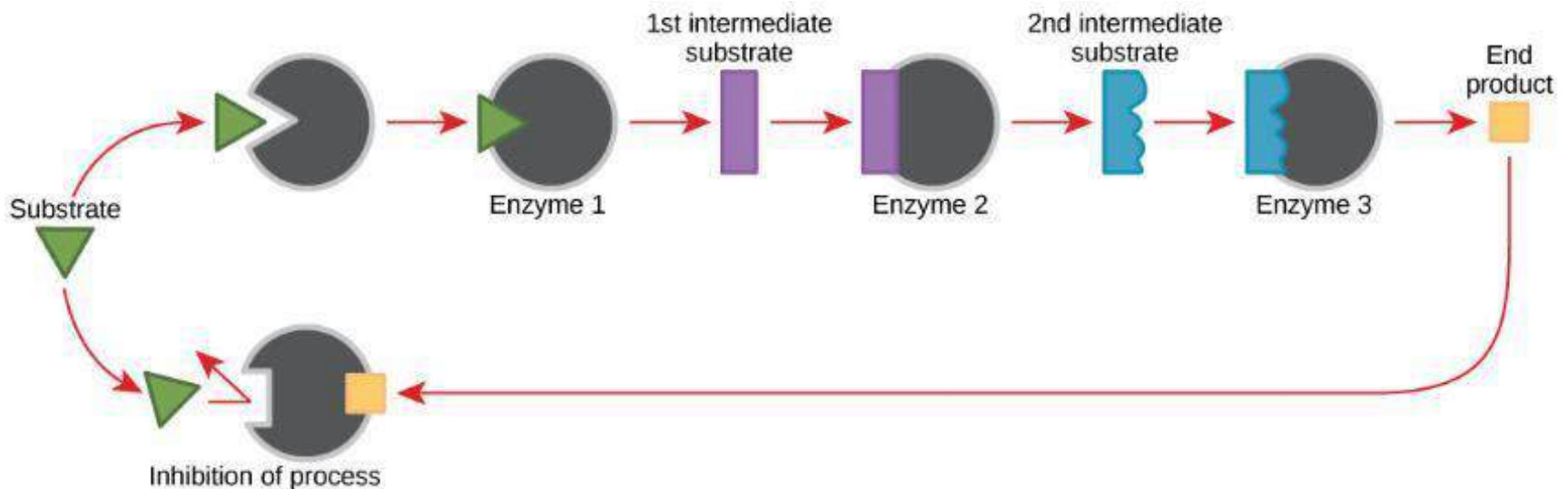
# Enzyme regulation

- The rates of enzyme-catalyzed reactions are altered by **activators** and **inhibitors** (a.k.a. effector molecules).
- **Allosteric enzymes:** have more than one site, where effector binding at one site induces a conformational change in the enzyme, altering its affinity for a substrate.
  - ✓ An allosteric activator increases enzyme rate of activity,
  - ✓ An allosteric inhibitor decreases its activity.



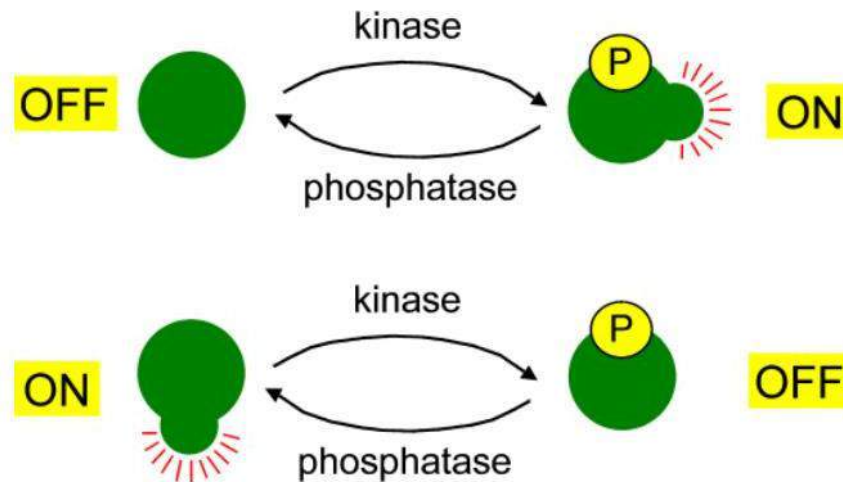
## 2. Feedback regulation

- Feedback inhibition is the inhibition of an enzyme by the reaction product.
- an enzyme early on in the pathway is inhibited by an end-product of the metabolic pathway in which it is involved



## Reversible covalent modification

- The activity of many enzymes is altered by the reversible making and breaking of a covalent bond between the enzyme and a small non-protein group (Phosphorylation, Methylation, Glycosylation...etc)



- Some enzymes are activated by phosphorylation others are inactivated.



## Regulation of enzyme activity....

### 3. Proteolytic cleavage

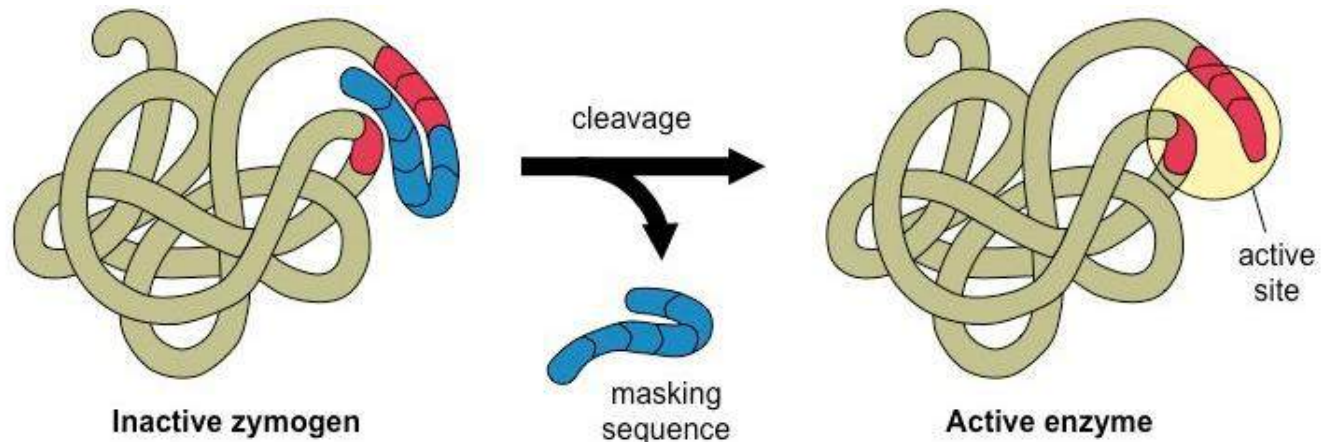
- Zymogens or proenzymes are inactive precursors of an enzyme.
- Some enzymes in their active form would degrade the internal structures of the cell. These enzymes are synthesized and stored as inactive precursors, and when the enzyme is needed, the zymogen is released and activated where it is needed.
- Activation usually requires the cleavage of one or more peptide bonds.
- The digestive enzymes pepsin, trypsin, and chymotrypsin, as well as enzymes involved in blood clotting, are activated this way.



## Regulation of enzyme activity....

### Example: Trypsin and Chymotrypsin

- Initially synthesized as trypsinogen and chymotrypsinogen which are both inactive
- Formed in the pancreas where they would do damage if active
- In the small intestine, where their digestive properties are needed, they are activated by cleavage of specific peptide bonds by enteropeptidase





# Cellular Respiration

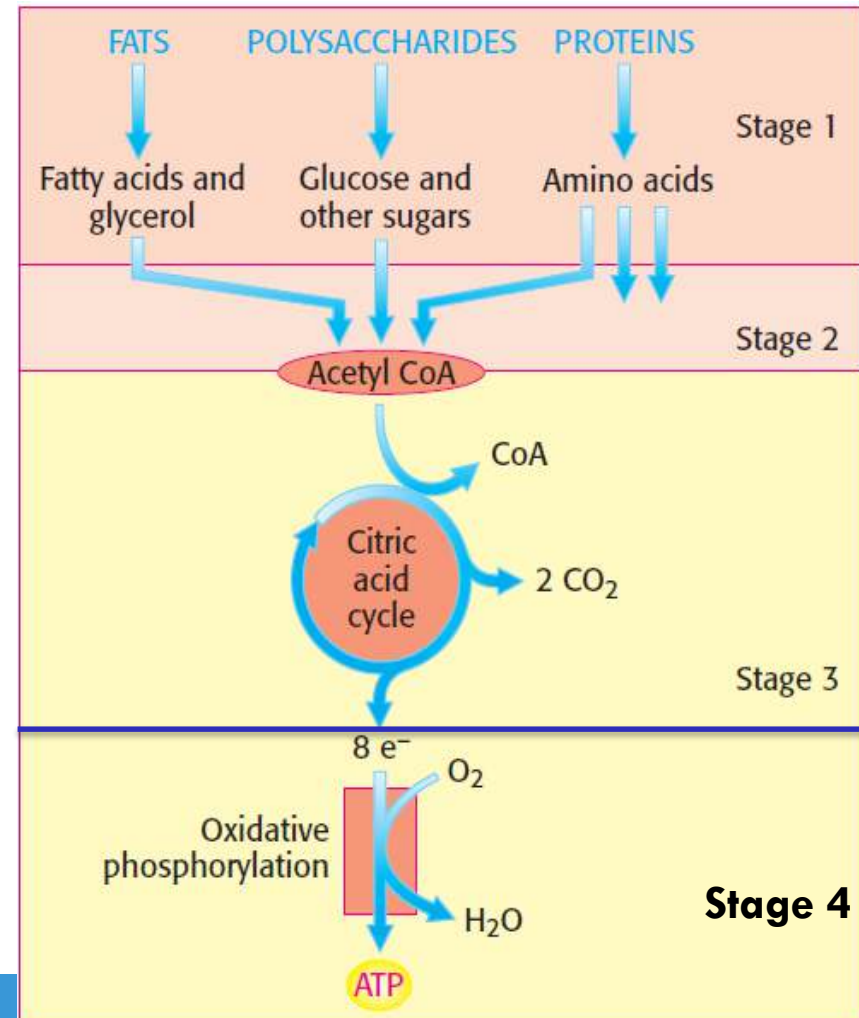
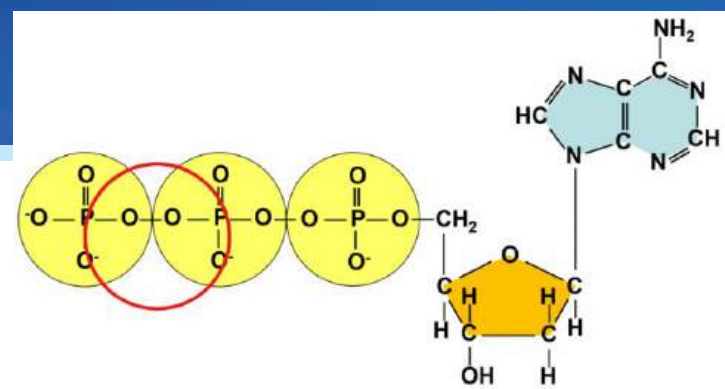
Is a series of chemical reactions that breakdown biomolecules to release energy (ATP)

**Stage 1:** Digestion of larger molecules to smaller molecules

**Stage 2:** Conversion of products of digestion to acetyl CoA

**Stage 3:** the citric acid cycle

**Stage 4:** Electron transport chain and oxidative phosphorylation produce ATP





# Three major mechanisms of ATP biosynthesis.

## **Substrate level phosphorylation** as in Glycolysis

- ✓ Chemical reaction coupled to ATP synthesis

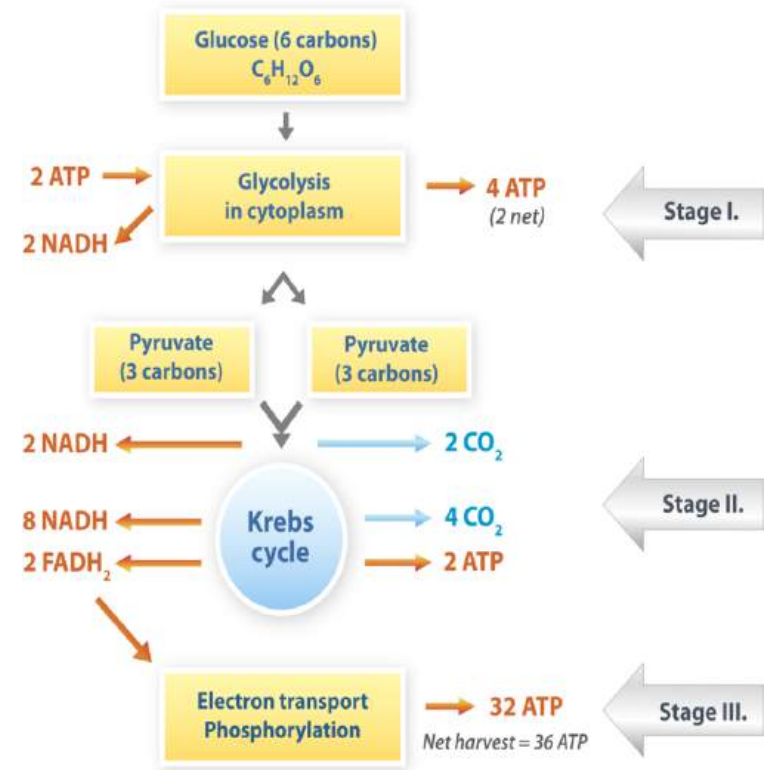
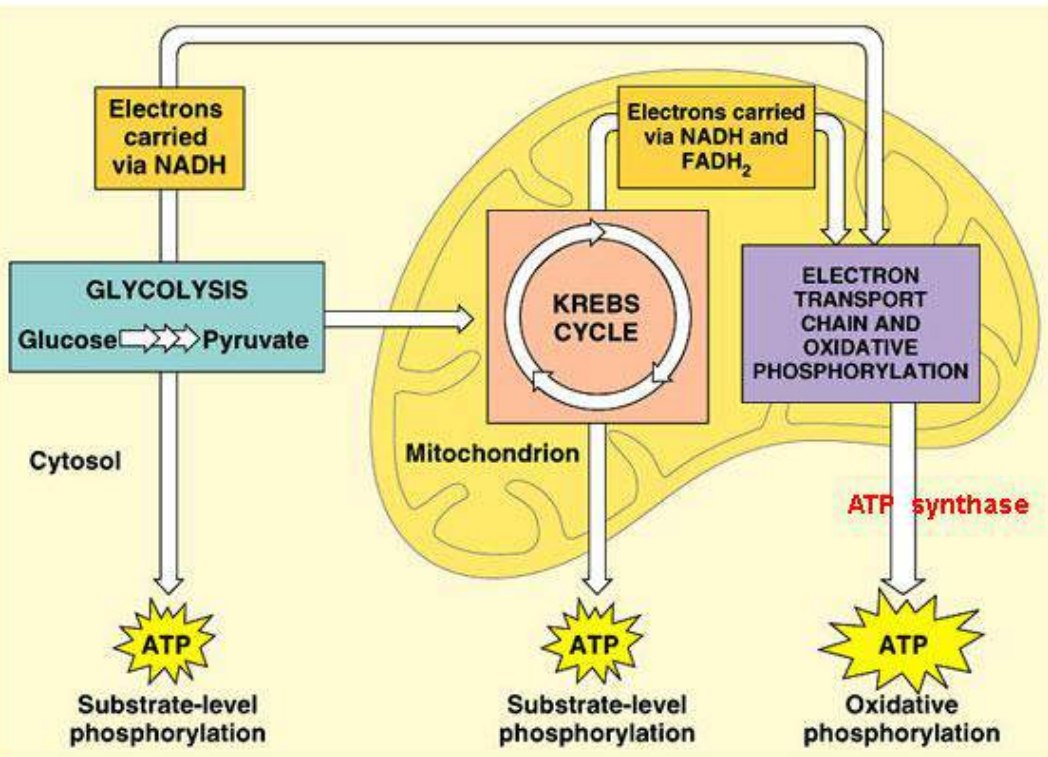
## **Oxidative phosphorylation** in cellular respiration and

- ✓ Pumping of protons powered by electron transport with oxygen as terminal electron acceptor yields ATP

## **Photophosphorylation** in photosynthesis

- ✓ Pumping of protons powered by absorption of light.

# Cellular respiration = Glycolysis + Krebs Cycle + Electron Transport Chain





## Cellular respiration.....

**Glycolysis-** Embden-Meyerhof-Parnas pathway)

- **converts one molecule of glucose to two molecules of pyruvate while generating ATP.**
- the major source of energy for anaerobic microorganisms
- takes place in the cytosol of cells
- some tissues, such as the brain and red blood cells, rely solely on glucose as a fuel; consequently, glycolysis is especially important in these tissues.

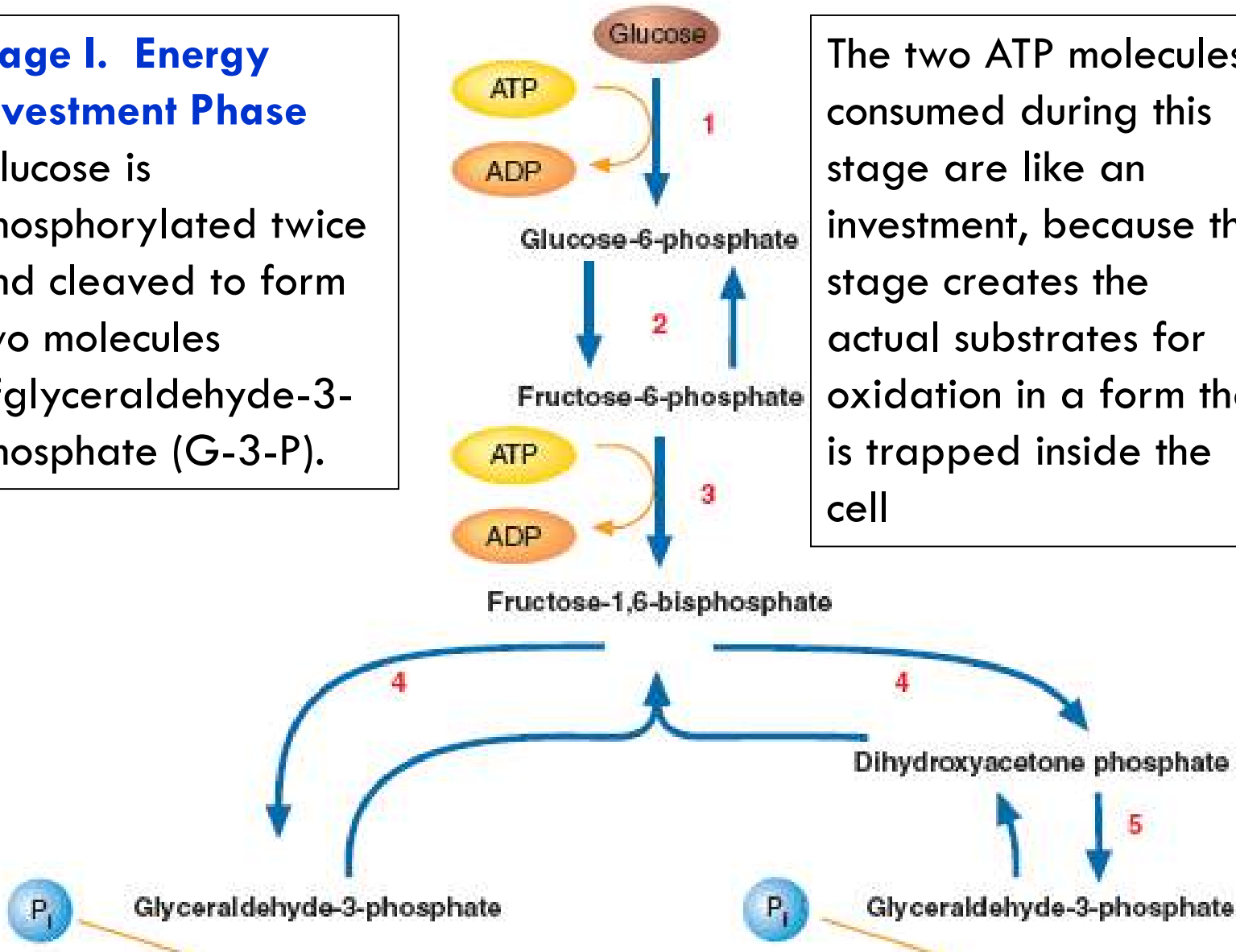


# Glycolysis: divided into two stages (each with 5 steps)

## Stage I. Energy Investment Phase

Glucose is phosphorylated twice and cleaved to form two molecules of glyceraldehyde-3-phosphate (G-3-P).

The two ATP molecules consumed during this stage are like an investment, because this stage creates the actual substrates for oxidation in a form that is trapped inside the cell



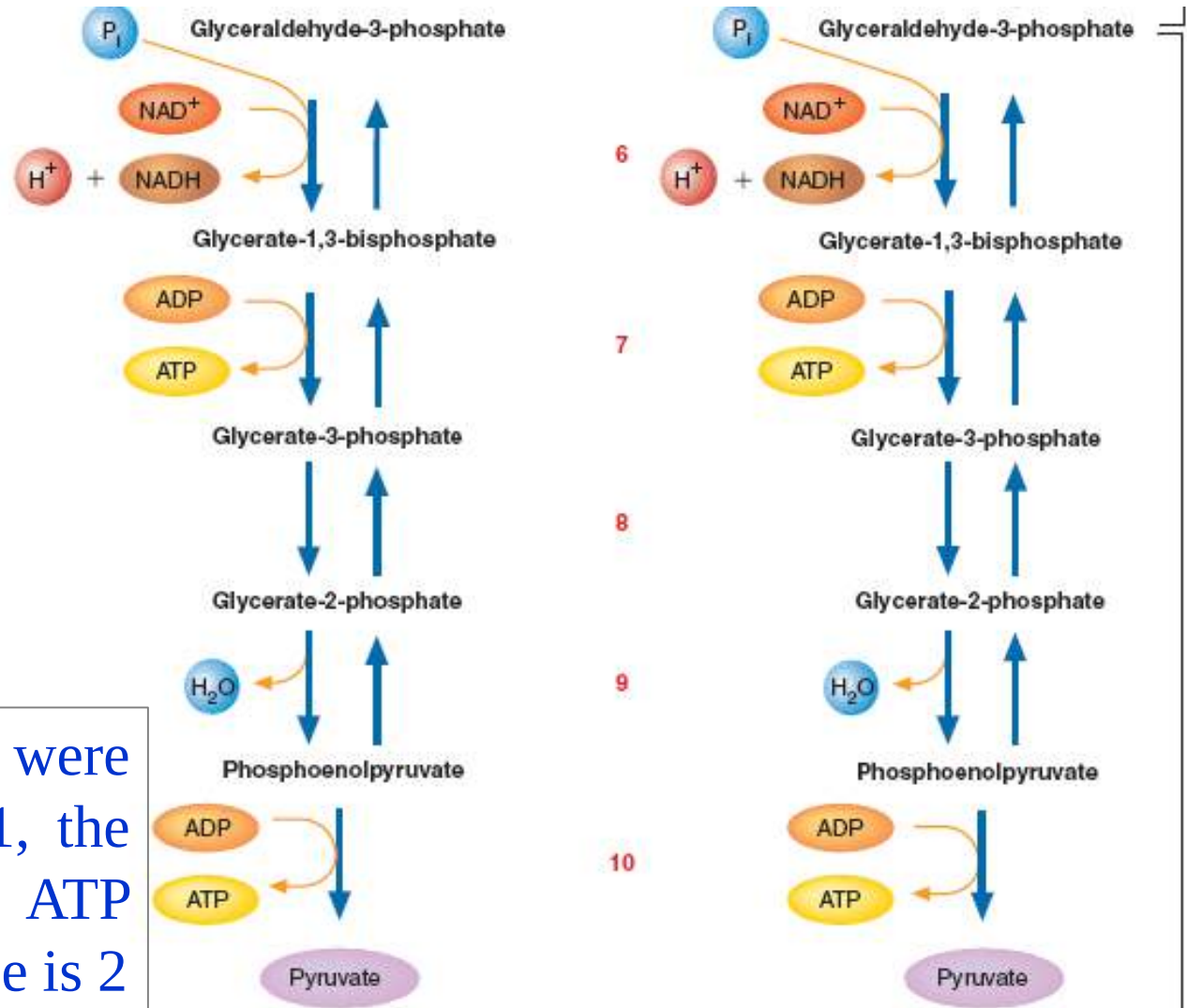
STAGE  
1





## Stage II: Energy payoff phase

- Convert Glyceraldehyde-3 phosphate to pyruvate.
- Generate four ATP and two NADH molecules



STAGE  
2

Because two ATP were consumed in stage1, the net production of ATP per glucose molecule is 2

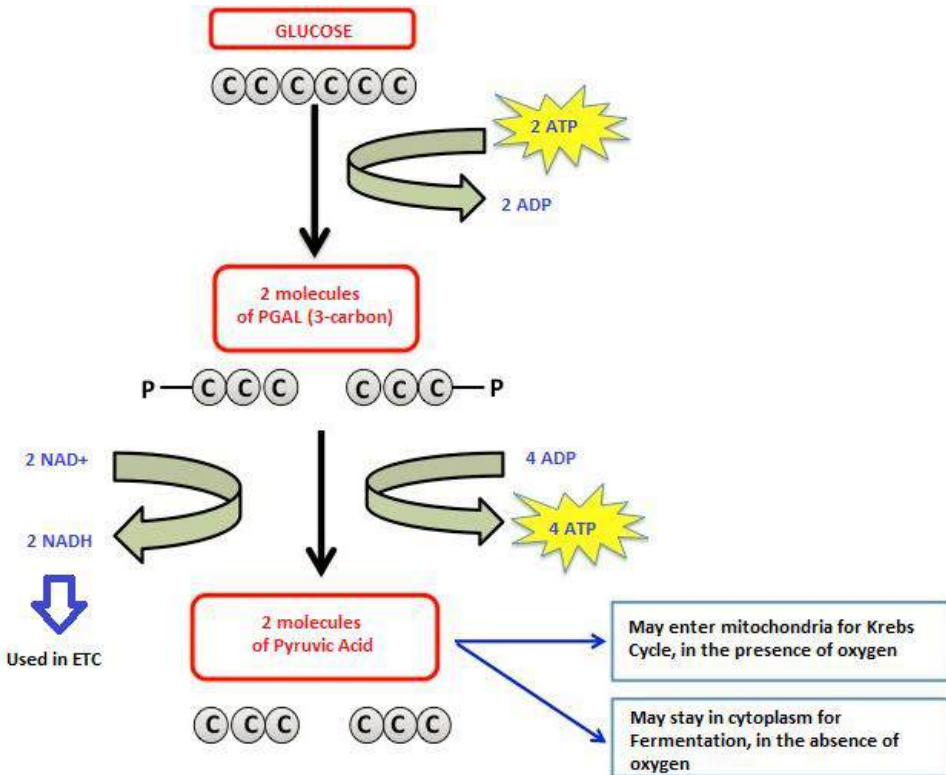




# Glycolysis....

Overall, glycolysis

- generates two molecules of NADH (nicotinamide adenine dinucleotide) and two molecules of ATP
- produce intermediates that act as precursors for a number of biosynthetic pathways .
- For example acetyl CoA is the precursor for fatty acid synthesis



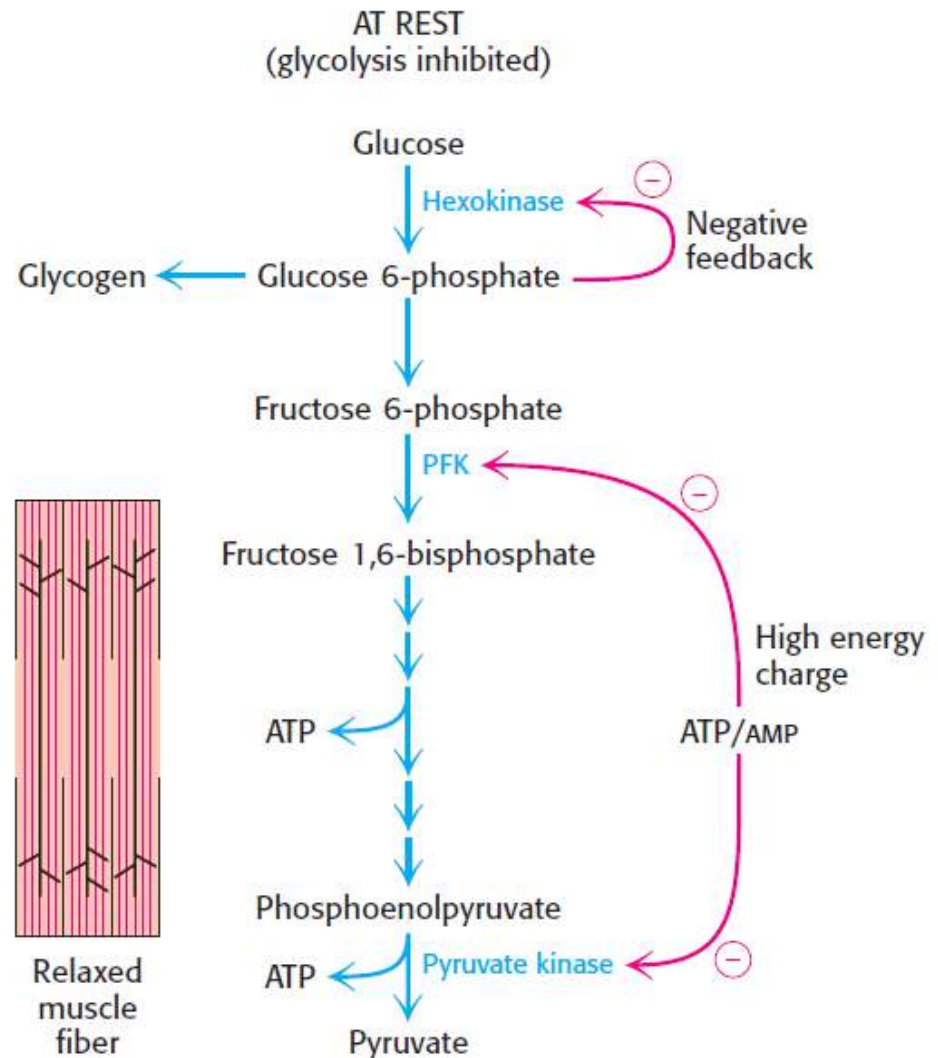
The glycolytic pathway can be summed up in the following equation:



## Regulation of glycolysis

Glycolysis is regulated by,  
feedback inhibition

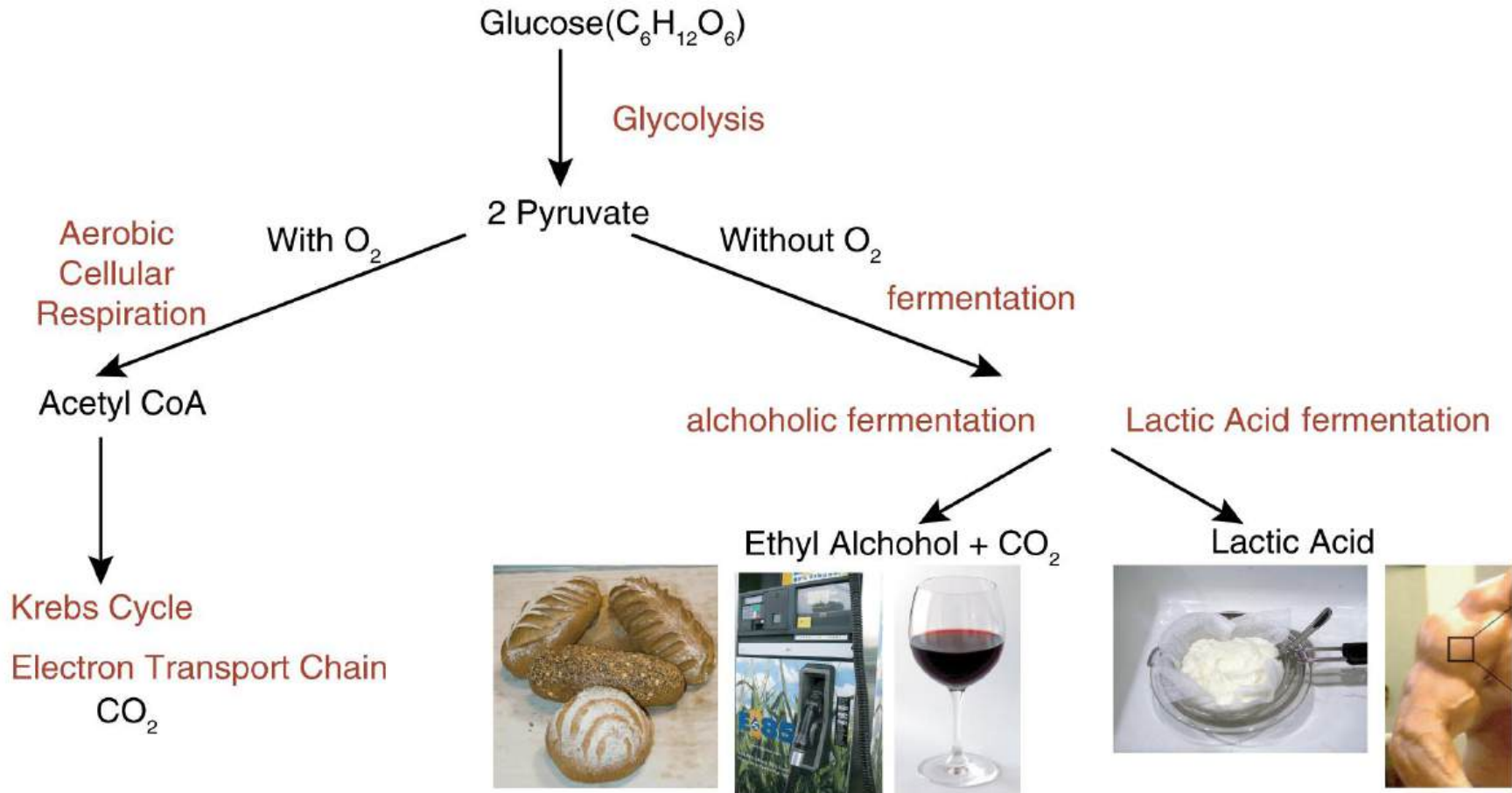
- high levels of ATP inhibit the enzyme
- phosphofructokinase early in the pathway
- **As more G-6-P produced, the rxn rate slow down**
- **Hexokinase governs rate limiting steps in**





# Fates of pyruvate

## Major Respiratory Pathways



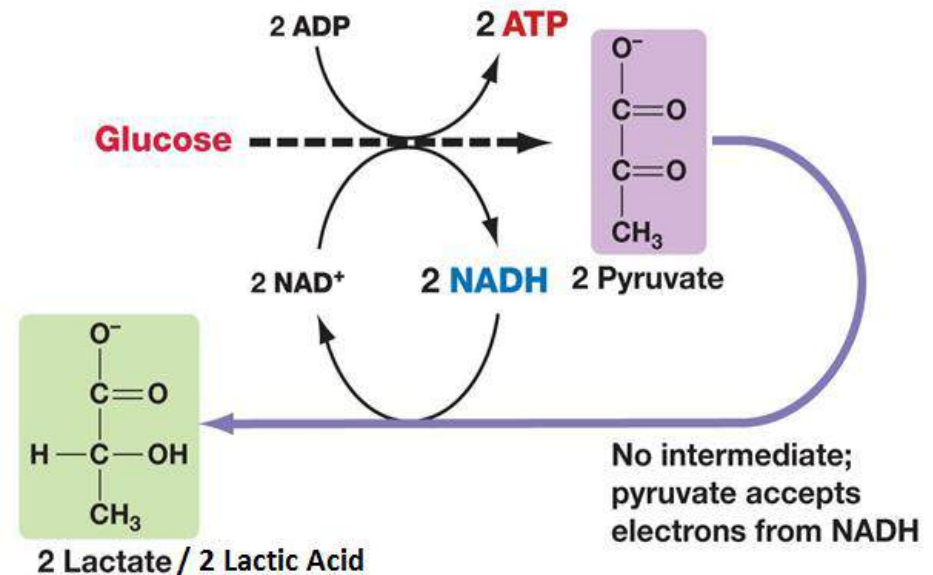


# Lactic acid fermentation

- When oxygen is limiting in muscle during vigorous exercise, pyruvate is converted into lactate by **lactate dehydrogenase**
- Regenerates  $\text{NAD}^+$ , Ensures the continuation of the process of glycolysis

- Under aerobic conditions,  $\text{NAD}^+$  is regenerated by re-oxidation of  $\text{NADH}$  via the electron transport chain.
- Skeletal muscles ferment glucose to lactate during exercise= Results in muscle fatigue
- Lactate released to the blood may be taken up by other tissues, or by skeletal muscle after exercise, and converted via Lactate Dehydrogenase back to pyruvate

(a) Lactic acid fermentation occurs in humans.

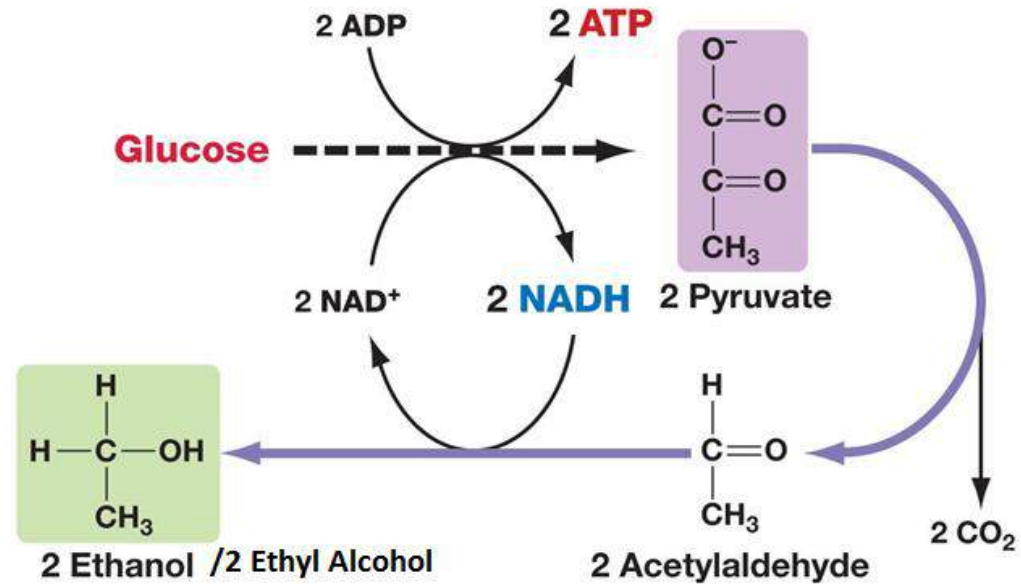


# Alcoholic fermentation

- pyruvate is converted to acetaldehyde (by pyruvate decarboxylase) and then to ethanol (by alcohol dehydrogenase)
- Produce  $\text{CO}_2$  as byproduct
- Also regenerates  $\text{NAD}^+$

Example: yeast

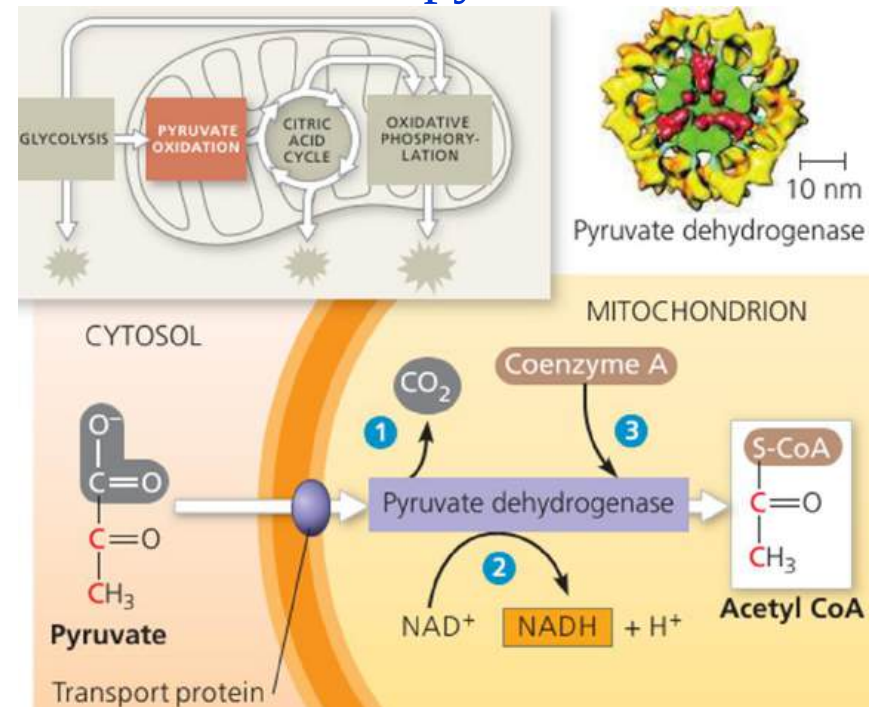
## (b) Alcohol fermentation occurs in yeast.





# The Krebs Cycle/tricarboxylic acid cycle/ citric acid cycle

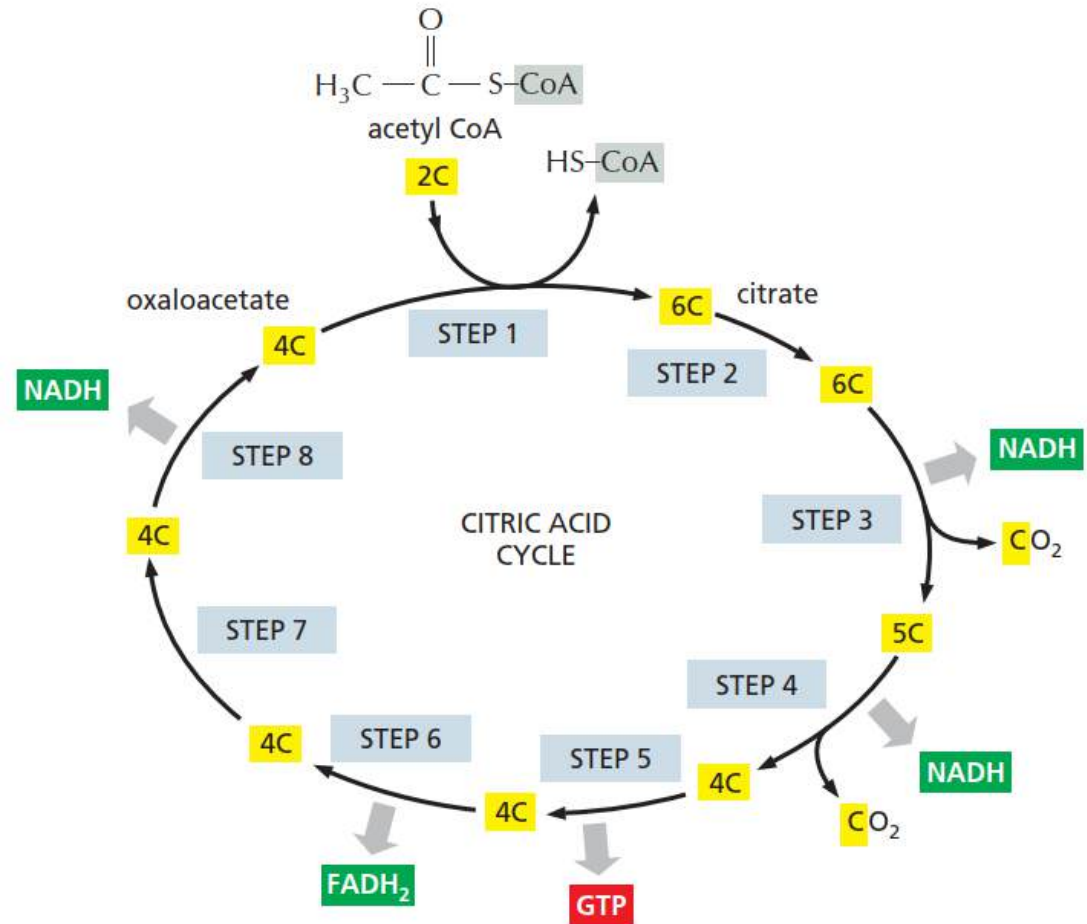
- named after Hans Krebs
- Consists of series of eight steps of reactions
- take place in mitochondria
- If oxygen is available, the pyruvate molecules produced at the end of glycolysis are transported into the mitochondria **via a pyruvate/H<sup>+</sup> symport.**
- Within the mitochondria, **pyruvate is oxidatively decarboxylated to acetyl-CoA.** This reaction is catalyzed by several different enzymes working sequentially in a multi-enzyme complex collectively known as the pyruvate dehydrogenase complex.





## Kreb cycle....

- The acetyl groups are fed into the citric acid cycle, which enzymatically oxidizes them to  $\text{CO}_2$ .
- The energy released by oxidation is conserved in the reduced electron carriers NADH and  $\text{FADH}_2$
- ✓ The reduced coenzymes are then oxidized to protons and electrons.
- ✓ The electrons are transferred, eventually, to  $\text{O}_2$ , via a chain Electron transport system.

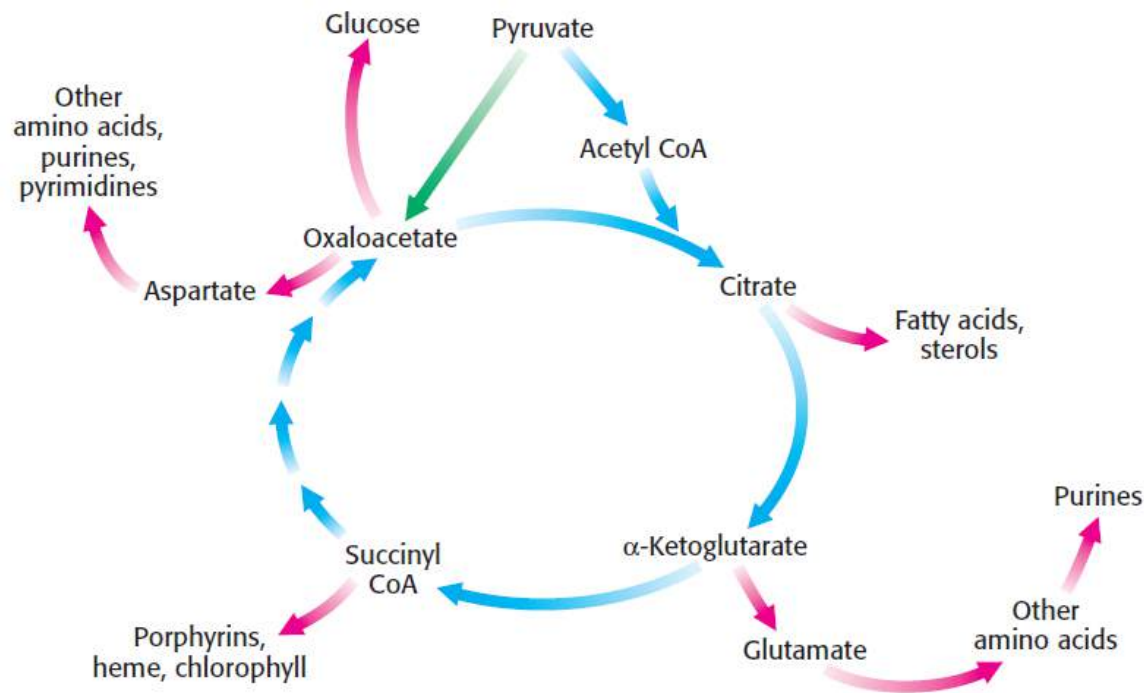






## Role of TCA cycle

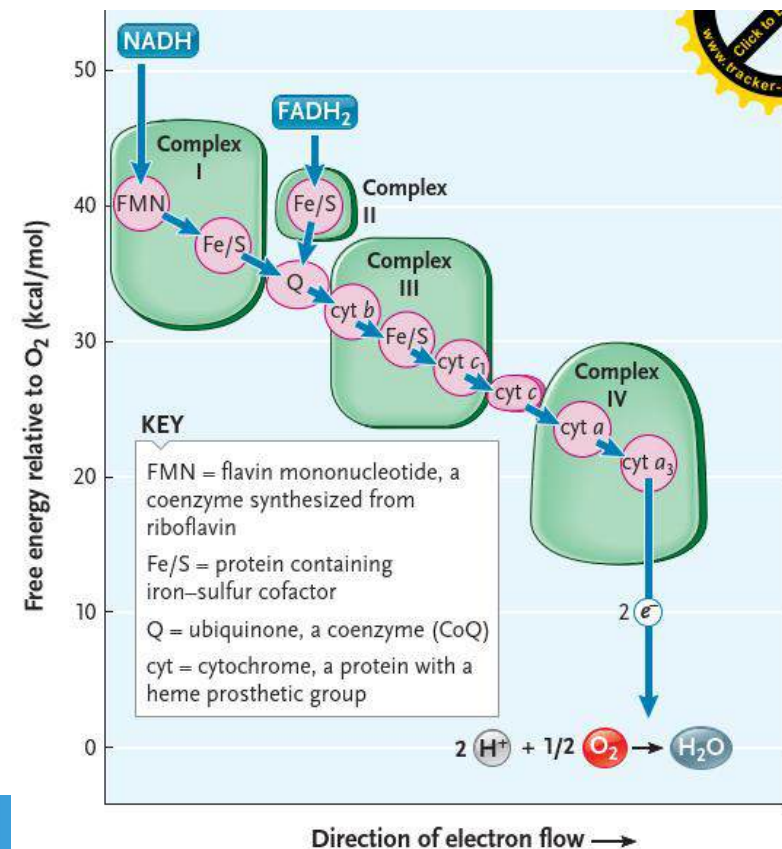
- acts as a final common pathway for the oxidation of carbohydrate, lipids, and protein
  - ✓ glucose, fatty acids, and many amino acids can all be metabolized to acetyl-CoA or intermediates in the cycle.





# The Electron Transport Chain (ETC)

- Series of electron carriers that transfer electrons captured in the form of NADH and  $\text{FADH}_2$  molecules during glycolysis, pyruvate oxidation and TCA cycle to  $\text{O}_2$  (the final electron acceptor).
- components of ETC are arranged according to their increasing redox potential (affinity to gain electron)
- Electrons flow down an energy gradient from NADH and  $\text{FADH}_2$  to  $\text{O}_2$  catalyzed by four protein complexes (Complex I - IV)
- occurs in the inner mitochondrial membrane
- is the major source of ATP in aerobic organisms
- ATP is formed as a result of the transfer of electrons to  $\text{O}_2$  through oxidative phosphorylation.





## Mechanisms of ATP synthesis (chemiosmotic coupling)

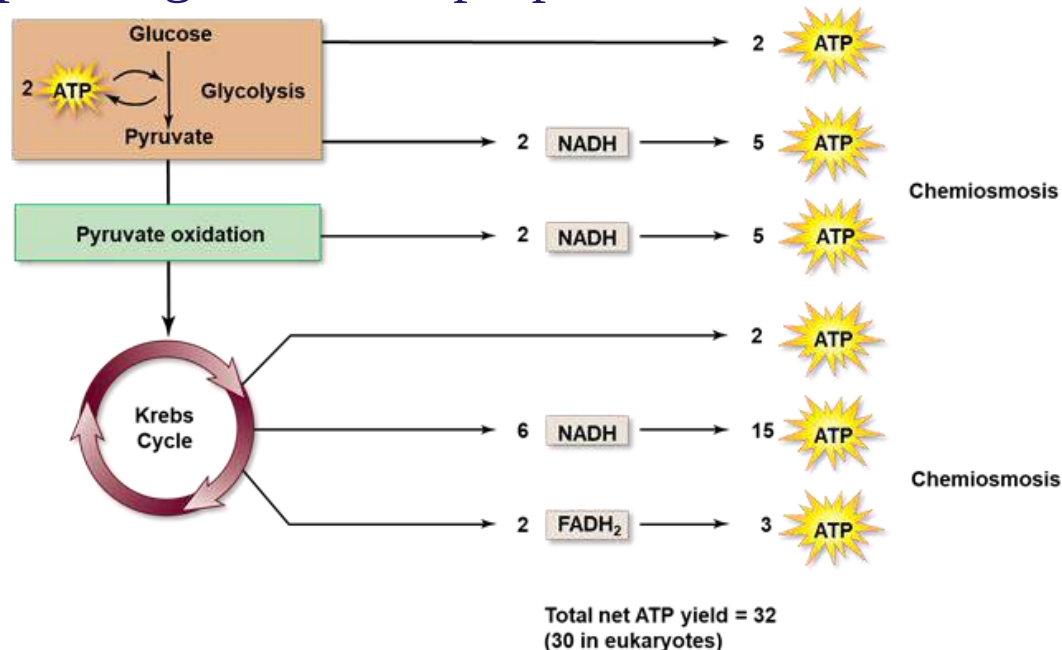
- The process of electron transfer is coupled to ATP synthesis
- Most of the free energy from the transfer of electrons from NADH or  $\text{FADH}_2$  to  $\text{O}_2$  through ETC pump protons out of the matrix across the inner mitochondrial membrane into the intermembrane space generating a charge separation (Proton motive force (PMF)) across the membrane.
- The PMF in the intermembrane space drives the synthesis of ATP as protons flow back from the intermembrane space to the matrix of the mitochondria
- These protons are channeled through an enzyme called *ATP synthase* which catalyzes the formation of ATP from ADP and  $\text{P}_i$  (a process known as chemiosmosis).





## Energy Yield of Respiration

- Theoretical energy yield (these values vary from book to book); 36-38 ATP per glucose
- Actual energy yield: 30 ATP per glucose for eukaryotes, 32 ATP per glucose for prokaryotes.
- Reduced yield is due to “Leaky” inner membrane
- Use of the proton gradient for purposes other than ATP synthesis

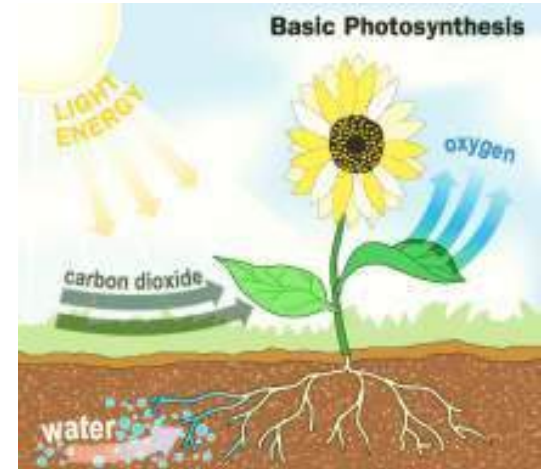




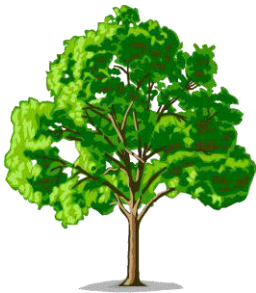


# Photosynthesis

Sunlight plays a bigger role in our lives than you may think. All the food we eat and the fossil fuels we burn are products of photosynthesis.



In photosynthesis, plants turn solar energy into chemical bond energy ( food)



**Photosynthesis** is the process that converts solar energy into chemical energy that is used by biological systems (that means us).





**Photosynthesis** has 3 major events:

1. Sunlight is converted into chemical energy
2. Water ( $\text{H}_2\text{O}$ ) is split into oxygen ( $\text{O}_2$ )
3. Carbon dioxide ( $\text{CO}_2$ ) is fixed into sugars ( $\text{C}_6\text{H}_{12}\text{O}_6$ )

The photosynthesis reaction:



6 Carbon  
dioxide  
molecules

12 Water  
molecules

1 Sugar  
(glucose)  
molecule

6 Water  
molecules

6 Oxygen  
molecules



Photosynthesis is carried out by:  
certain bacteria      plants



These organisms are known as photoautotrophs or **producers** meaning they make their own food and energy from the sun.

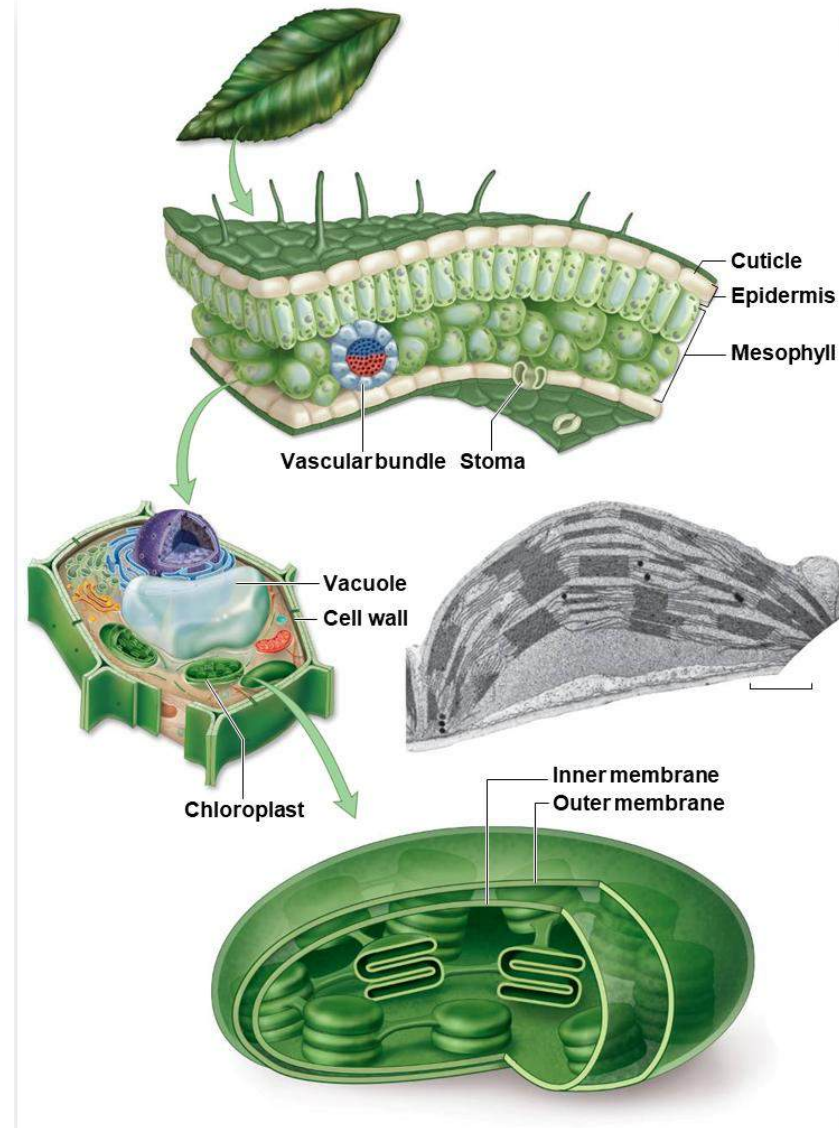


most algae      cyanobacteria      phytoplankton

Consumers such as herbivores and carnivores depend on the products of photosynthesis that producers make to live.



- Photosynthesis takes place in:
  - Thylakoid membrane– light-dependent phase
  - Stroma – Calvin cycle phase
- The main structures needed for photosynthesis are
  - ✓ pigments called chlorophylls that are present in the thylakoid membrane of the chloroplast. These pigments are clustered into photosystems.
  - ✓ Stomata: the structure of the leaf that allows gas exchange ( $\text{CO}_2$  enters,  $\text{O}_2$  exits during photosynthesis) and water evaporation





# Reactions of Photosynthesis

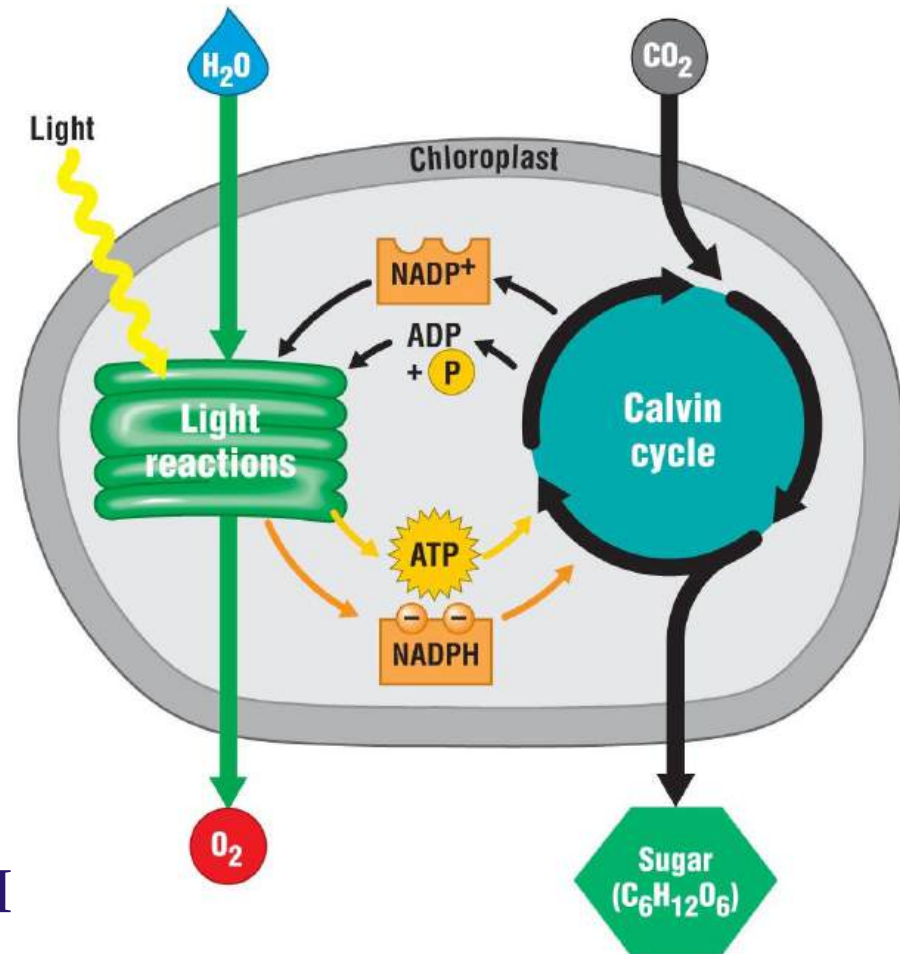
Photosynthesis is composed of two processes:

## 1. The light reactions:

- ✓ Occur in the grana
- ✓ Split water and releases oxygen
- ✓ convert solar energy to chemical energy (ATP & NADPH).

## 2. The Calvin cycle: CO<sub>2</sub> fixation

- makes sugar from carbon dioxide.
- Occurs in the stroma
- Forms sugar from carbon dioxide, using ATP for energy, and NADPH as reducing power





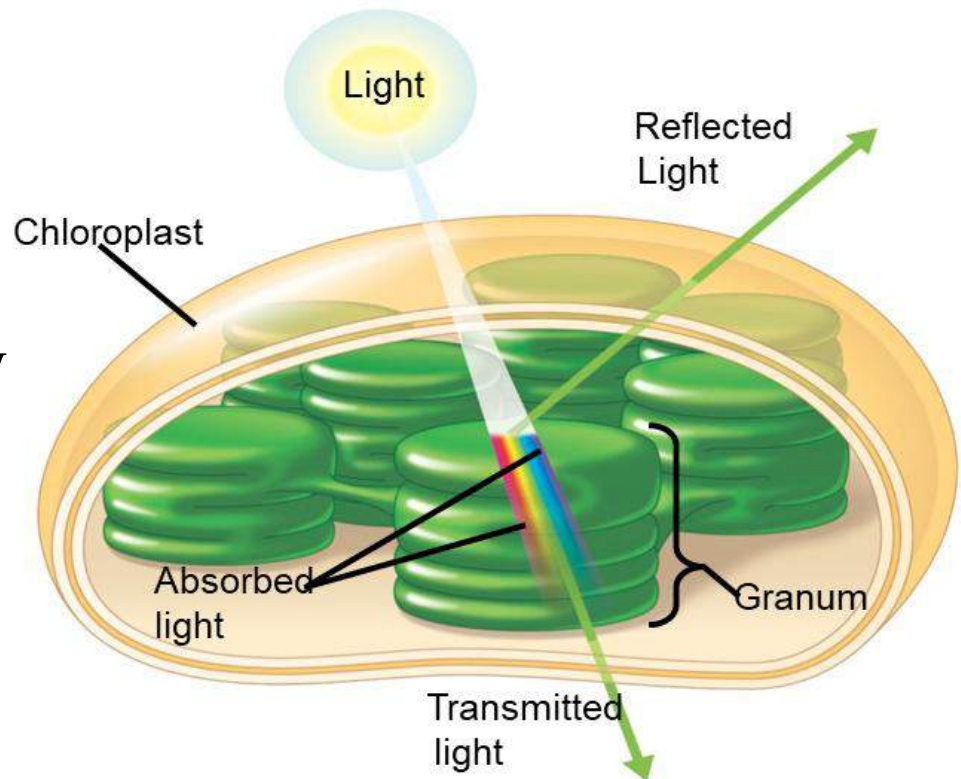


## Photosynthetic Pigments: The Light Receptors

- Are substances that absorb visible light
- Reflect light, which includes the colors we see

### Chloroplasts contain several pigments:

- **Chlorophyll a:** Is the main photosynthetic pigment
- **Chlorophyll b:** Is an accessory pigment
- **Carotenoids:** An accessory pigment (protective)





# Photosystems

- The pigments are organized into groups called Photosystems.
- Composed of a protein complex called a reaction centre complex (Light-harvesting complex)
- Light harvesting complex: Consists of various pigments molecules (chlorophyll a, b, & carotenoid)

## Two photosystems:

- ✓ Water-splitting photosystem (P680 or Photosystem II)
- ✓ NADPH-producing photosystem (P700 or Photosystem I)

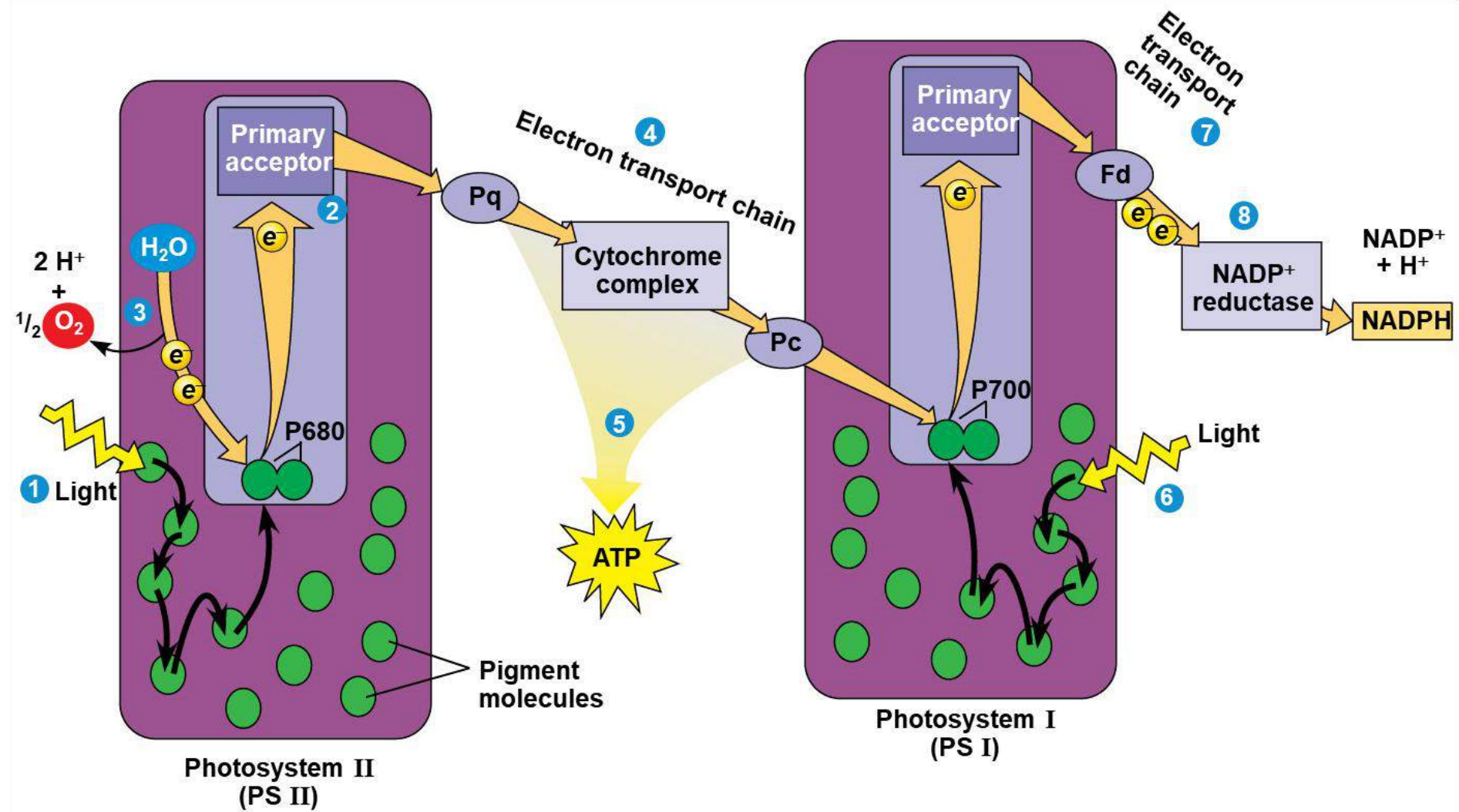




## Photosystem II

- The energy absorbed by the chlorophyll during the light reactions is used to make electrons “jump” out of the pigment’s atom and break the bonds of water absorbed through the plant’s roots.
  - ✓ Water is here to help and donate its own electron to the pigment; as a result  $H_2O$  molecule is split into electrons, protons ( $H^+$ ) and  $O_2$ ;
  - ✓  $O_2$  is a waste that will be released into the atmosphere.
  - ✓ The “lost” electron has to be replaced in the pigment’s atom by another electron from water.

*Then, the electron moves through a chain of proteins called electron transport chain (ETC).*





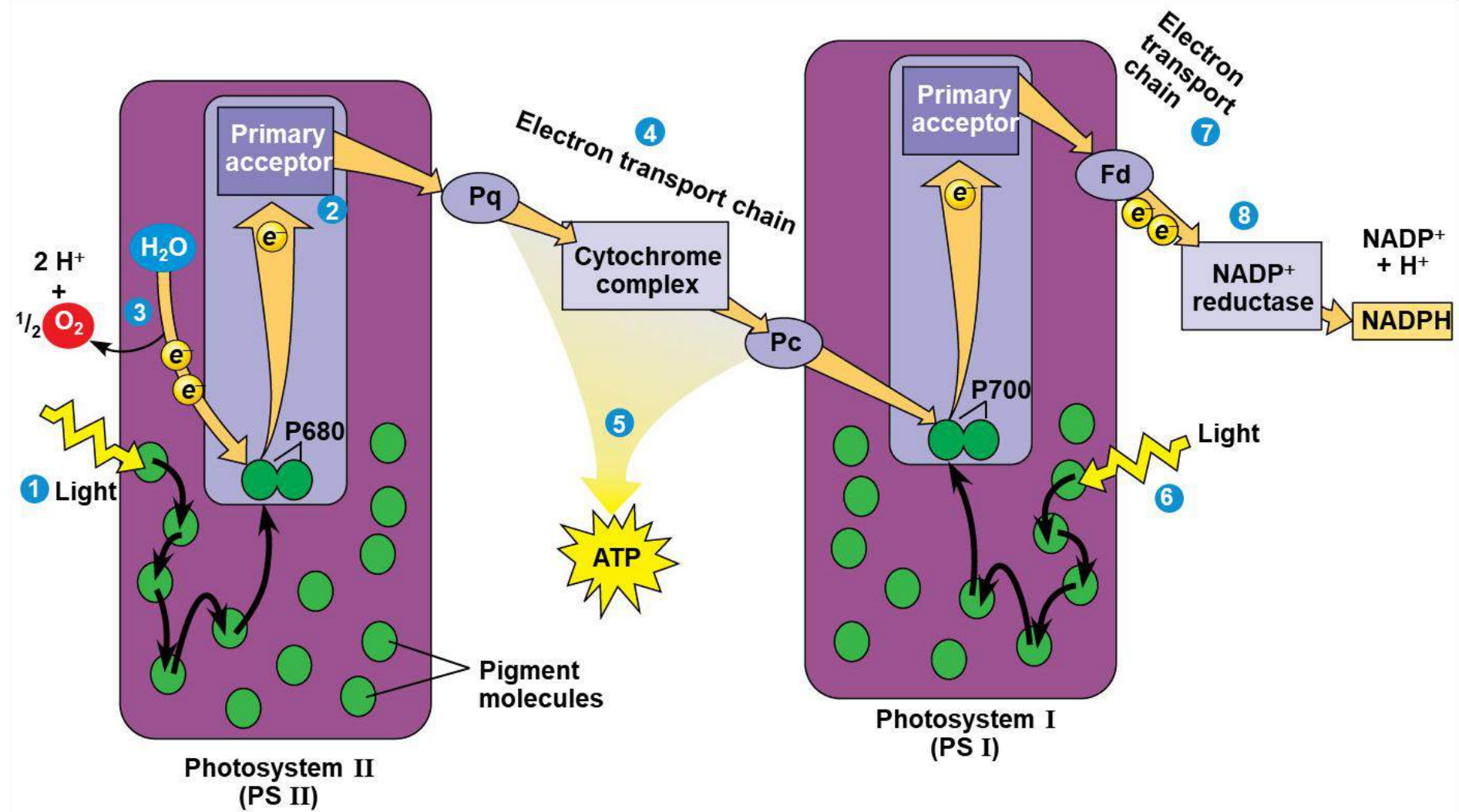
## ATP synthesis by PSII

- As the electron moves, its energy is used by the proteins in the ETC to transport  $H^+$  from the chloroplast inside the thylakoid.
- As a result there will be more  $H^+$  inside the thylakoid space than outside of it;
- This difference in concentration triggers  $H^+$  movement through diffusion from the HIGH concentration (inside the thylakoid) to LOW (stroma).
- $H^+$  will move back into the chloroplast stroma through the ATP-synthase protein which will make ATP (!) = chemical energy



## PSI and production of NADPH

- As the electron moves, its energy is used by the proteins in the ETC to transport  $H^+$  from the chloroplast inside the thylakoid.
- PSI absorbs energy and loses an electron in a similar way with PSII.
- Its lost electron is replaced by the electron from the PSII coming down from the ETC
- PSI electron will eventually end up in a molecule called NADPH (!) = chemical energy.
- Light-dependent reactions harness energy from the sun to produce ATP and NADPH. These energy-carrying molecules travel into the stroma where the Calvin cycle reactions take place





## Calvin cycle (CO<sub>2</sub> fixation)

- Named after Melvin Calvin (1911–1997)
- Takes place in the stroma of the chloroplast
- The cycle gets its energy to conduct the reaction via high-energy electron from NADPH and ATP
- The NADPH and ATP are two of the products of the light-dependent reactions
- In the Calvin reaction, 6 CO<sub>2</sub> molecules enter the cycle from the atmosphere. The CO<sub>2</sub> molecules combine with 6, 5-carbon molecules. Result: 12, 3-carbon molecules.
- Key step is the attachment of CO<sub>2</sub> to **RuBP** (ribulose 1, 5 biphosphate) to form **PGA** (3-phosphoglycerate)
- Uses enzyme ribulose biphosphate carboxylase/oxygenase or **rubisco**



## Three Phases of Calvin Cycle

### Carbon fixation

- $\text{RuBP} + \text{CO}_2 \rightarrow \text{PGA}$  (3-phosphoglycerate)

### 1. Reduction

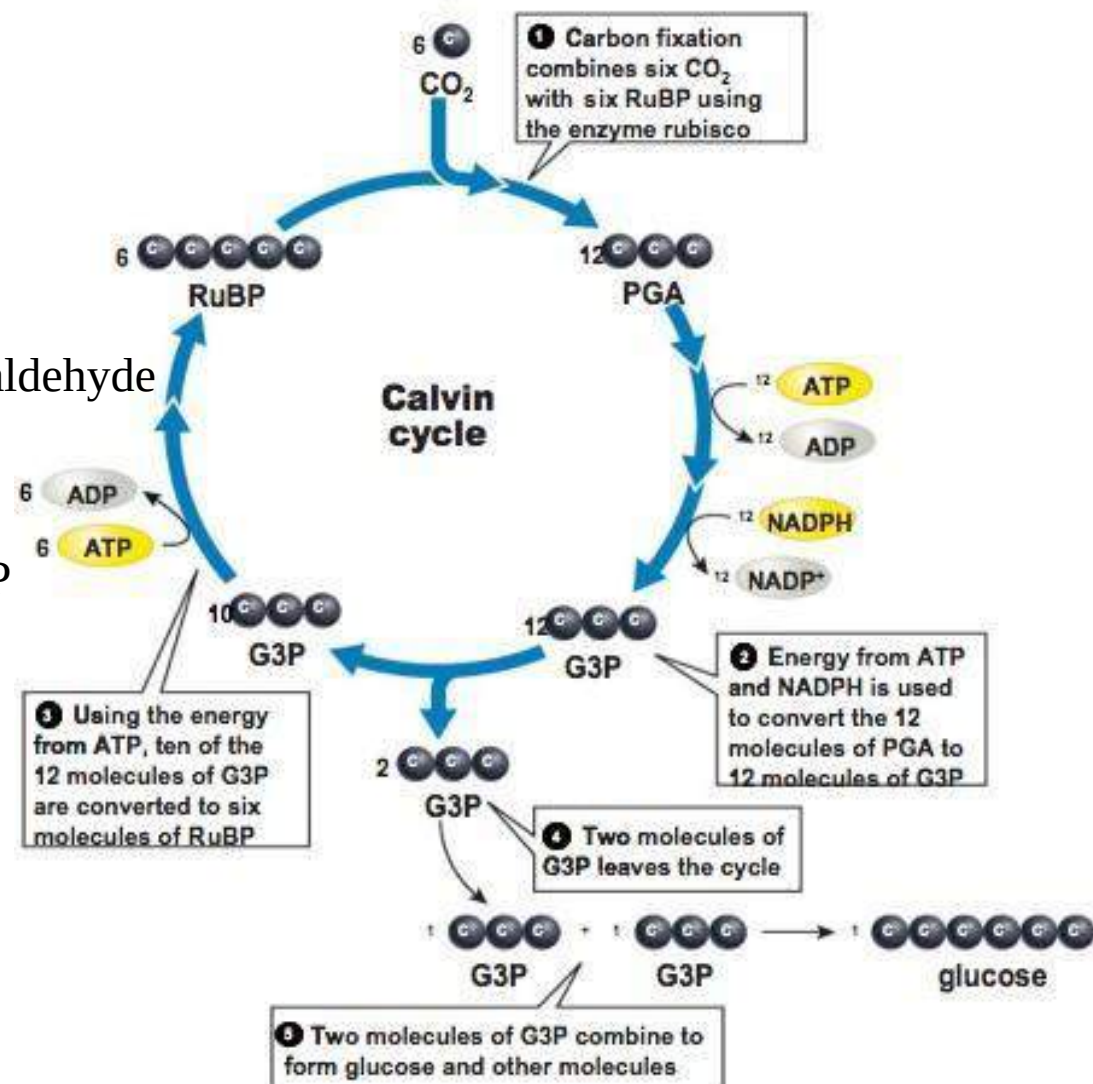
- PGA is reduced to **G3P** (glyceraldehyde 3-phosphates)

### 2. Regeneration of RuBP

- PGA is used to regenerate RuBP

### Output of Calvin Cycle

- Glucose is not a direct product of the cycle
- G3P a 3-carbon sugar is the direct product of the cycle
- Used to form sucrose, a major transport sugar in plants





# **General Biology (Biol1012)**

## **Lecture Slides**

### **Chapter 5: Genetics & Evolution**



# What is Genetics

- Genetics is a field of biology that studies how **traits** (hereditary or genetic information) are passed from parents to their offspring.
- Genetic information is carried from generation to generation through **DNA** molecules
- A particular section(segment) of a DNA molecule that tells a cell to perform a specific task is known as a **gene** (encodes for RNA or functional protein)
- The place on a chromosome where a specific gene resides – is the **locus**
- Each Locus contains an alternative form of a gene called **allele** (e.g. A / a)



- When an organism inherits two identical alleles for a trait, the organism is said to be **homozygous** for the trait (e.g. AA).
  - When an organism inherits two different alleles for one trait, the organism is called **heterozygous** for the trait (e.g. Aa).
  - **Genotype:** genetic makeup of an organism or alleles carried by an individual or Arrangement of genes that produces the phenotype
  - **Phenotype:** physical characteristic or appearance of an individual
- ❖ **The first quantitative studies of inheritance were carried out by Gregor Mendel.**



# Mendelian Genetics

- Mendel was an Austrian monk. Born in 1822 to peasant parents, Mendel was educated in a monastery and went on to study science and mathematics
- He returned to the monastery and spent the rest of his life there, eventually becoming abbot.
- In the garden of the monastery, Mendel initiated a series of experiments on plant hybridization.
- The results of these experiments would ultimately change our views of heredity irrevocably.
- Considered the FATHER OF GENETICS!!!!!!





# What did Mendel do?

- Mendel spent his spare time breeding pea plants.
- He did this over & over & over again, and noticed patterns to the inheritance of traits, from one set of pea plants to the next.
- By carefully analyzing his pea plant numbers, he discovered three laws of inheritance.
  1. Law of Dominance
  2. Law of Segregation
  3. Law of Independent Assortment





# Why Mendel succeeded?

- Choice of the plant: Pea plant had discrete traits
  - ✓ It had seven contrasting traits
  - ✓ Produce pure plants when allowed to self-pollinate (A large number of true-breeding varieties of peas were available)
  - ✓ Can be artificially cross-pollinated
  - ✓ Pea plants are small and easy to grow, and they have a relatively short generation time.
- Used mathematical analysis in his studies

## The seven pea characteristics studied by Mendel

**Flower color**



Purple



White

**Flower position**



Axial



Terminal

**Seed color**



Yellow



Green

**Seed shape**



Round

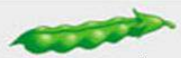


Wrinkled

**Pod shape**



Inflated



Constricted

**Pod color**



Green



Yellow

**Stem length**



Tall



Dwarf

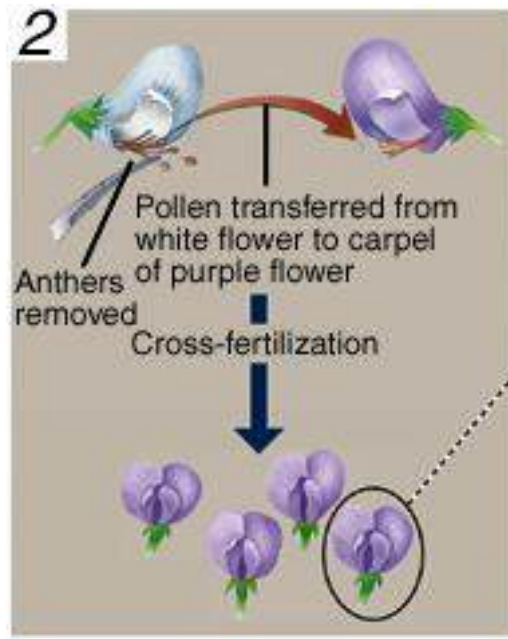


# How Mendel Began

- Mendel produced pure strains by allowing the plants to self-pollinate for several generations before doing the cross pollination

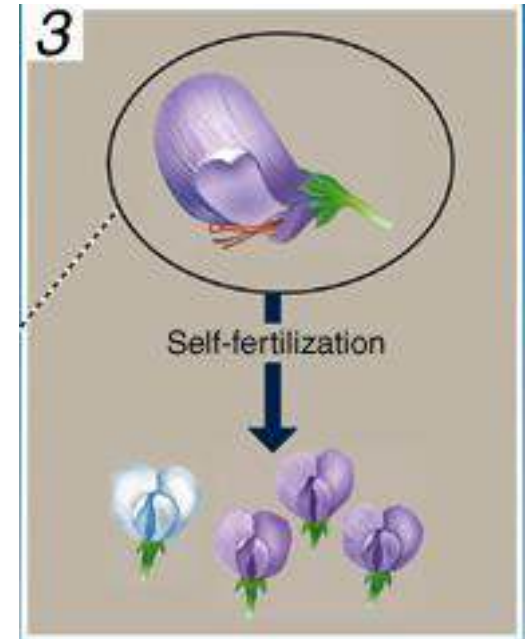


1. Self-Fertilize & Produce a True-Breeding Generation = **pure white** and **pure purple**



2. Produce first filial generation (F1) = all were **purple**

Transfer pollen from one parent to the other.



3. Let F1 generation self-fertilize to produce a second filial generation (F2) =  $\frac{3}{4}$  purple and  $\frac{1}{4}$  white



# Interpretation

- $F_1$  all heterozygous:  $Pp$
- $\frac{1}{4}$  of  $F_2$  Homozygous purple flower –  $PP$
- $\frac{1}{2}$  of  $F_2$  Heterozygous –  $\frac{1}{4} Pp$  and  $\frac{1}{4} pP$ 
  - Both  $PP$  and  $pP$  are tall
- $\frac{1}{4}$  of  $F_2$  Homozygous recessive –  $pp$ 
  - These are white
- One allele, when present, masks the effect of the other
  - In this case,  $P$  masks the effect of  $p$

## *Mendel's Law of Dominance*

### Why were all the flowers in the $F_1$ generation purple?

- A **dominant allele** (upper case  $P$ ) masked the effect of the **recessive allele** (lower case  $p$ ), thus dominant phenotype (**purple**) was expressed in both homozygous dominant ( $PP$ ) and heterozygous ( $Pp$ ) genotypes
- *A recessive allele was masked by the dominant allele, thus recessive phenotype (**White**) was expressed only in homozygous recessive genotypes ( $pp$ )*



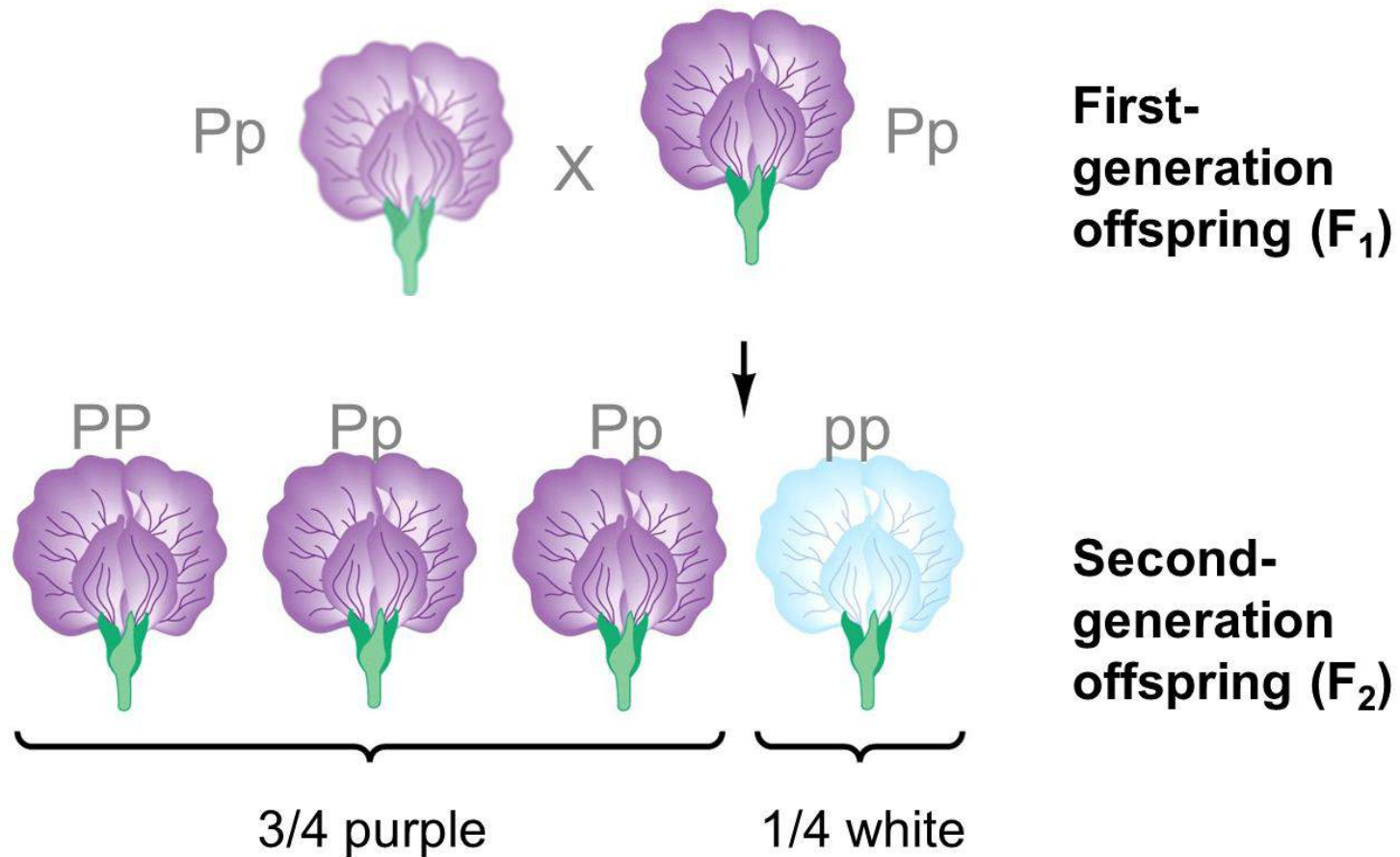
# Mendel's law of Dominance

- Mendel learned that for each pair of alternative forms of a character, one alternative was not expressed in the F1 hybrids, although it reappeared in some F2 individuals.
  - ✓ Mendel's law of dominance states that in a heterozygote, one trait will conceal the presence of another trait for the same characteristic.
  - ✓ The recessive trait will only be expressed by offspring that have two copies of this allele, and these offspring will breed true when self-crossed.



# Monohybrid and dihybrid crosses

- A **monohybrid cross** is defined as the cross happening in the F<sub>1</sub> generation offspring of parents differing in one trait only. E.g Flower colour







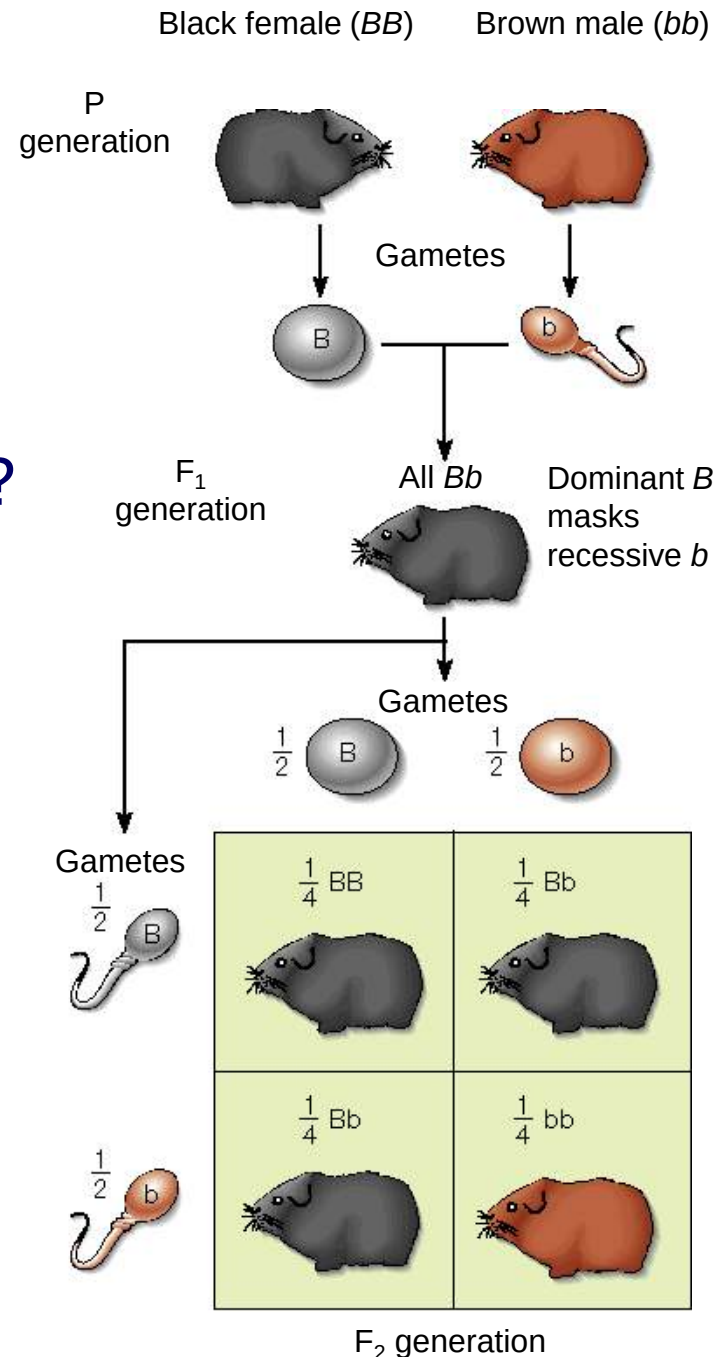
## The Results of Mendel's Garden Pea Hybridizations

| Characteristic  | Contrasting P <sub>0</sub> Traits | F <sub>1</sub> Offspring Traits | F <sub>2</sub> Offspring Traits | F <sub>2</sub> Trait Ratios |
|-----------------|-----------------------------------|---------------------------------|---------------------------------|-----------------------------|
| Flower color    | Violet vs. white                  | 100 percent violet              | 705 violet<br>224 white         | 3.15:1                      |
| Flower position | Axial vs. terminal                | 100 percent axial               | 651 axial<br>207 terminal       | 3.14:1                      |
| Plant height    | Tall vs. dwarf                    | 100 percent tall                | 787 tall<br>277 dwarf           | 2.84:1                      |
| Seed texture    | Round vs. wrinkled                | 100 percent round               | 5,474 round<br>1,850 wrinkled   | 2.96:1                      |
| Seed color      | Yellow vs. green                  | 100 percent yellow              | 6,022 yellow<br>2,001 green     | 3.01:1                      |
| Pea pod texture | Inflated vs. constricted          | 100 percent inflated            | 882 inflated<br>299 constricted | 2.95:1                      |
| Pea pod color   | Green vs. yellow                  | 100 percent green               | 428 green<br>152 yellow         | 2.82:1                      |



# Example: Monohybrid cross

- Start with true breeding parents
  - What types of gametes can each produce?
    - Black  $\Rightarrow$  only black
    - Brown  $\Rightarrow$  only brown
  - What are the genotypes in the  $F_1$ ?
    - All  $Bb$        $B$  - black  
                     **$b$  - brown**
  - $F_1$  tells us what?
    - Black is dominant to brown
  - What are the **phenotypic** ratios in the  $F_2$ ?
    - 3 black:1 brown
  - What are the **genotypic** ratios in the  $F_2$ ?
    - 1  $BB$ :2  $Bb$ :1  $bb$





# Mendel's Law (Principle) of segregation of alleles

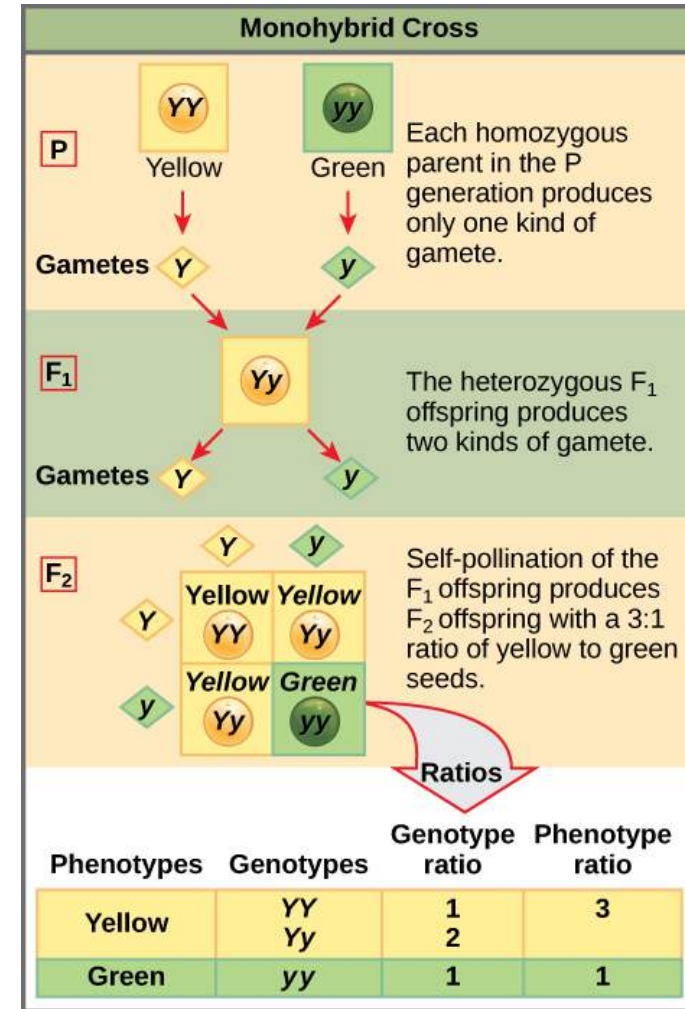
Observing that true-breeding pea plants with contrasting traits gave rise to F1 generations that all expressed the dominant trait and F2 generations that expressed the dominant and recessive traits in a 3:1 ratio, Mendel proposed the law of segregation.

- This law states that paired unit factors (genes) must segregate equally into gametes such that offspring have an equal likelihood of inheriting either factor.
- For the F2 generation of a monohybrid cross, the following three possible combinations of genotypes could result: homozygous dominant, heterozygous, or homozygous recessive



# Punnett Squares

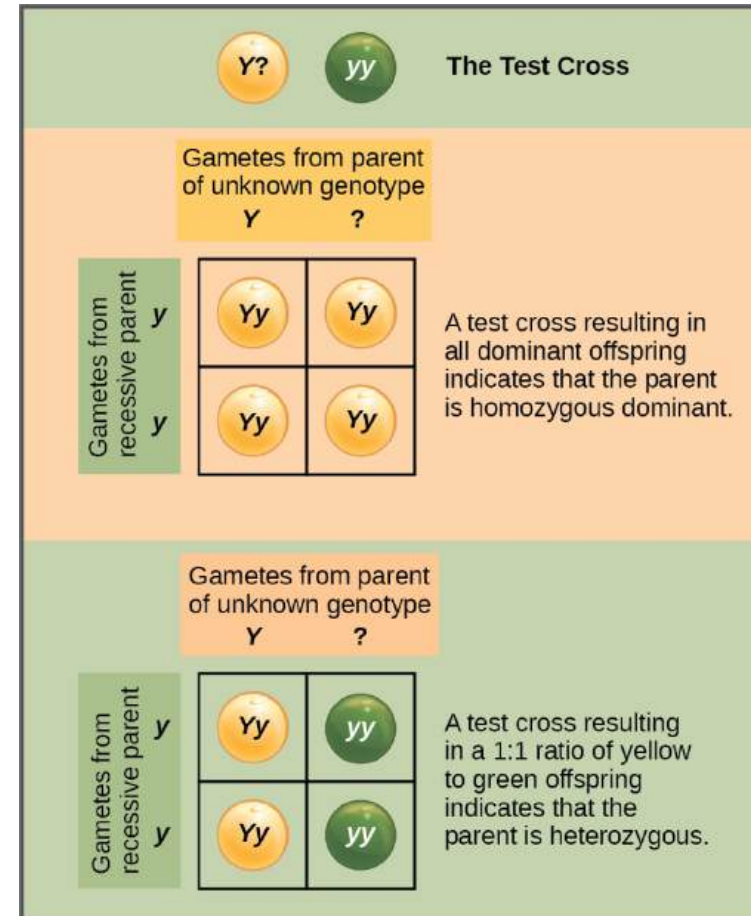
- Simplest tool for predicting the outcome of a genetic cross is a *Punnet Square*
- Mendel postulated that each parent in the monohybrid cross contributed one of two paired unit factors(alleles) to each offspring, and every possible combination of allele was equally likely.*
- Example:  $F_1$  Hybrid cross:  $Tt$  male X  $Tt$  female
  - ✓ Place possible alleles from one of parents on top; the other set from the other parent on the side. Location doesn't matter.
  - ✓ Fill across from sides and down from top to find genotypic combinations in offspring





# The Test Cross

- Distinguishes the Dominant Phenotype
- Beyond predicting the offspring of a cross between known homozygous or heterozygous parents, Mendel also developed a way to determine whether an organism that expressed a dominant trait was a heterozygote or a homozygote. This test is called the test cross
- In a test cross, the dominant-expressing organism is crossed with an organism that is homozygous recessive for the same characteristic
  - ✓ If the dominant-expressing organism is a homozygote, then all F1 offspring will be heterozygotes expressing the dominant trait
  - ✓ Alternatively, if the dominant expressing organism is a heterozygote, the F1 offspring will exhibit a 1:1 ratio of heterozygotes and recessive homozygotes

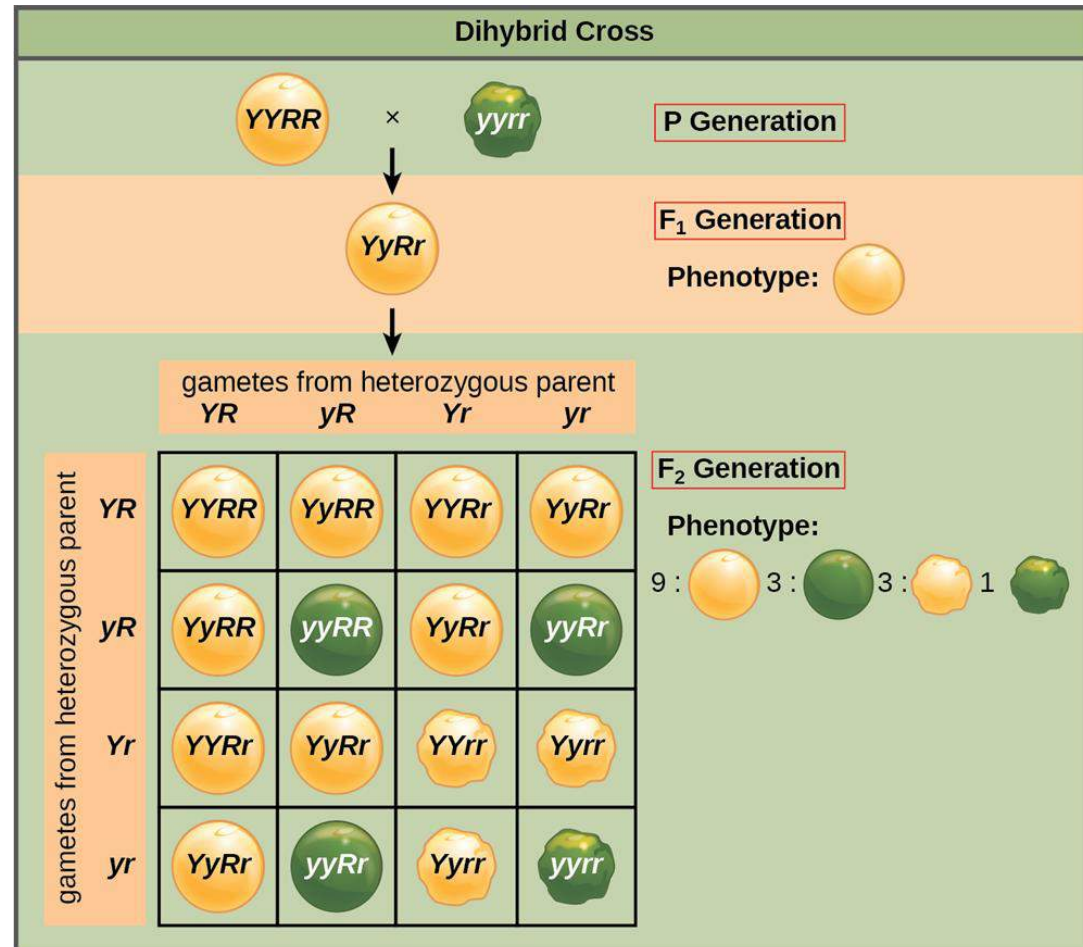






# Mendel's Law of Independent Assortment

- Mendel's law of independent assortment states that genes do not influence each other with regard to the sorting of alleles into gametes, and every possible combination of alleles for every gene is equally likely to occur.
- Dihybrid cross:** a cross between two true-breeding parents that express different traits for two characteristics illustrates the law of independent assortment. For example; consider the characteristics of seed color and seed texture for two pea plants





# Beyond Mendelian Genetics

## Alternatives to Dominance and Recessiveness

Mendel's experiments with pea plants suggested that:

- two “units” or alleles exist for every gene;
- alleles maintain their integrity in each generation (no blending); and
- in the presence of the dominant allele, the recessive allele is hidden and makes no contribution to the phenotype. Therefore, recessive alleles can be “carried” and not expressed by individuals.

However, further genetic studies in other plants and animals have shown that much more complexity exists such as

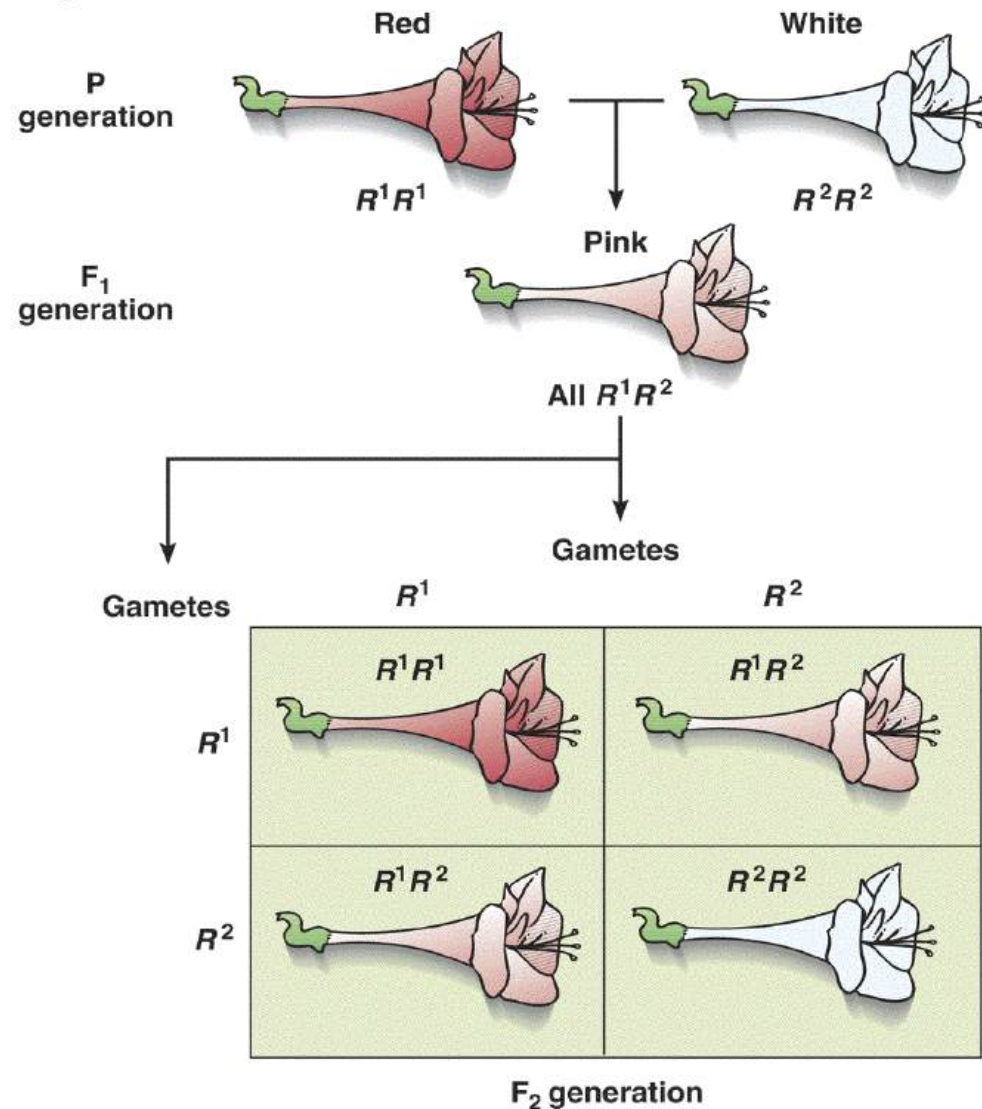
- ✓ Incomplete dominance
- ✓ Co-dominance
- ✓ Multiple allelism ...etc





## Incomplete Dominance

- Also known as partial dominance
- Inheritance where offspring exhibit a blend of their parents' traits. i.e. the individual displays an intermediate phenotype of the two parents
- For example: the allele for red flowers is incompletely dominant over the allele for white flowers resulting in Pink flower in F<sub>1</sub>.
- The phenotypic ratio would be 1:2:1 for red:pink:white in F<sub>2</sub>

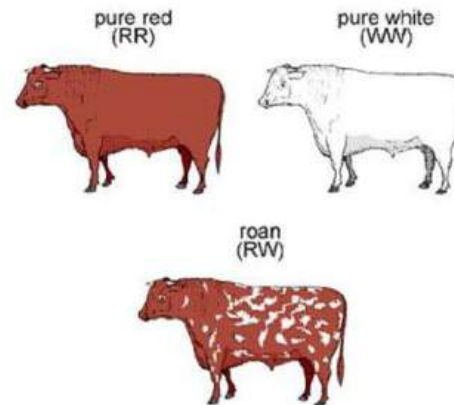




## Codominance

- both alleles for the same characteristic are simultaneously expressed in the heterozygote. An example of codominance is the skin color of cattle.
- In a self-cross between heterozygotes expressing a codominant trait, the three possible offspring genotypes are phenotypically distinct. However, the 1:2:1 genotypic ratio characteristic of a Mendelian monohybrid cross still applies.

- Red Cattle = RR (homozygous)
- White Cattle = WW (homozygous)
- Roan (Red & White Hair) = RW (Heterozygous)





# Multiple alleles

- Mendel implied that only two alleles, one dominant and one recessive, could exist for a given gene. We now know that this is an oversimplification.
- A gene may have more than two alleles in a population, and most genes possess several different alleles. Example ABO blood group.
- A person's blood type is determined by which allele he/she inherits from each parent

3 Alleles determine your blood group:

- $I^A$   $\Rightarrow$  possess A antigen on its surface and produce anti B antibody
- $I^B$   $\Rightarrow$  possess B antigen on its surface and produce anti A antibody
- $i^o$   $\Rightarrow$  possess no antigen on its surface and produce anti A and anti B antibodies
- $I^A$  &  $I^B$  are codominant while  $i^o$  is recessive

## Note:

- Blood types A and B have two possible genotypes – homozygous and heterozygous.
- Blood types AB and O only have one genotype each.

| Phenotype | Genotype  | Antigen present | Antibody produced |
|-----------|-----------|-----------------|-------------------|
| A         | $I^A I^A$ | A               | Anti-B            |
| A         | $I^A i$   | A               | Anti-B            |
| B         | $I^B I^B$ | B               | Anti-A            |
| B         | $I^B i$   | B               | Anti-A            |
| AB        | $I^A I^B$ | A & B           | None              |
| O         | $ii$      | None            | Anti-A & Anti-B   |



# Blood typing

Blood type is determined by mixing a drop of blood with a drop of anti-A or anti-B antiserum. Clumping indicates that the antibody has detected the corresponding antigen in the blood.

- The ABO system is one of several blood types that must be tested before blood transfusion to ensure the safety of the procedure.
- The general rule for safe blood transfusion is that the recipient blood must not contain an antibody that reacts with an antigen in the donated blood. When such a reaction occurs, blood clots produced by clumping blood cells form at the site of transfusion and could be lethal

| Blood type | Response to                                                                                        |                                                                                                    | Possible genotypes   |
|------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------|
|            | Anti-A                                                                                             | Anti-B                                                                                             |                      |
| A          | <br>Clumping      | <br>No clumping   | $I^A I^A$ or $I^A i$ |
| B          | <br>No clumping   | <br>Clumping      | $I^B I^B$ or $I^B i$ |
| AB         | <br>Clumping     | <br>Clumping     | $I^A I^B$            |
| O          | <br>No clumping | <br>No clumping | $ii$                 |

Table 4.1

Donor-Recipient Compatibility for ABO Blood Types

| Donor blood type | Recipient blood type |   |    |   |
|------------------|----------------------|---|----|---|
|                  | A                    | B | AB | O |
| A                | ✓                    | ✗ | ✓  | ✗ |
| B                | ✗                    | ✓ | ✓  | ✗ |
| AB               | ✗                    | ✗ | ✓  | ✗ |
| O                | ✓                    | ✓ | ✓  | ✓ |

✓ = Safe transfusion

✗ = Clumping, unsafe transfusion





## Gene, DNA, Chromosomes and Cell division

- At its core, hereditary transmission is the process of dispersing genetic information from parents to offspring.
- In sexually reproducing organisms, this process is accomplished by the generation of reproductive sex cells in males (the sperm or pollen) and females (the egg), followed by the union of egg and sperm (animals) or pollen (plants) or spores (yeast) at fertilization, with the subsequent development of an organism.
- DNA is the hereditary molecule that replicate during cell division to ensures that successive generations of cells are identical.
- When a cell divides, the two new cells contain all the same DNA that the original cell had
- DNA molecule contains a code that can be used by a cell to express their genes into functional proteins to direct the development of many different types of traits.

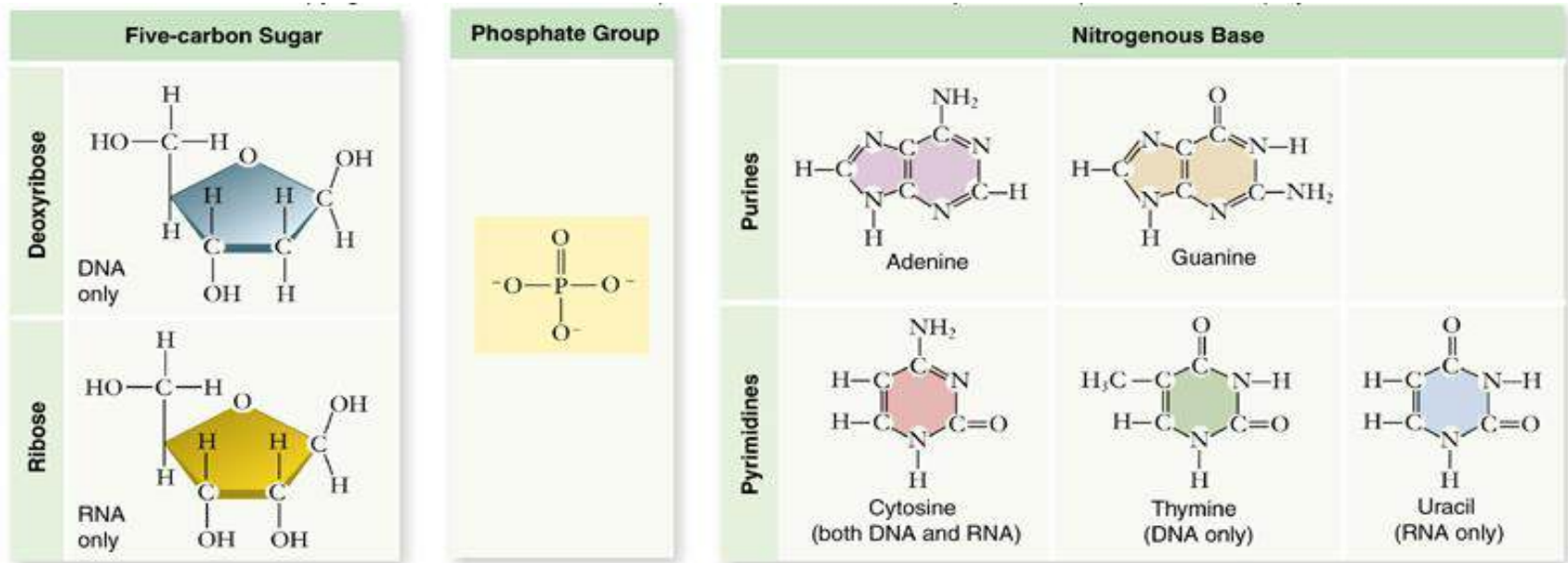


# DNA Structure: made up of polynucleotides

The building blocks of DNA are nucleotides

The nucleotide structure consists of

- The nitrogenous base attached to the 1' carbon of deoxyribose
- The phosphate group attached to the 5' carbon of deoxyribose
- a free hydroxyl group (-OH) at the 3' carbon of deoxyribose

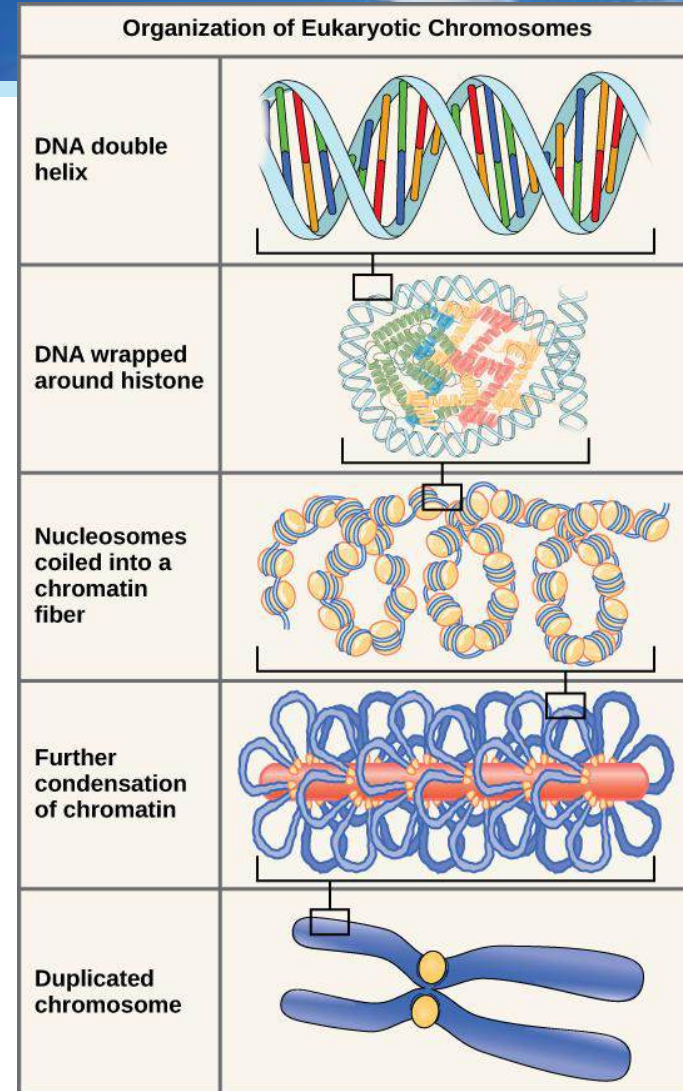






# Structure of DNA and chromosome

- In Eukaryotic cells, the linear DNA molecule is packed to fit inside the nucleus with the help of proteins known as Histones.
- At the most basic level, DNA is wrapped around proteins known as histones to form structures called nucleosomes.
- The nucleosomes are further compacted by passing through different compacting stages to form a chromosome.
- In non dividing eukaryotic cell, chromosomes have two distinct regions that can be distinguished by staining.
- The tightly packaged region is known as heterochromatin, and the less dense region is known as euchromatin.

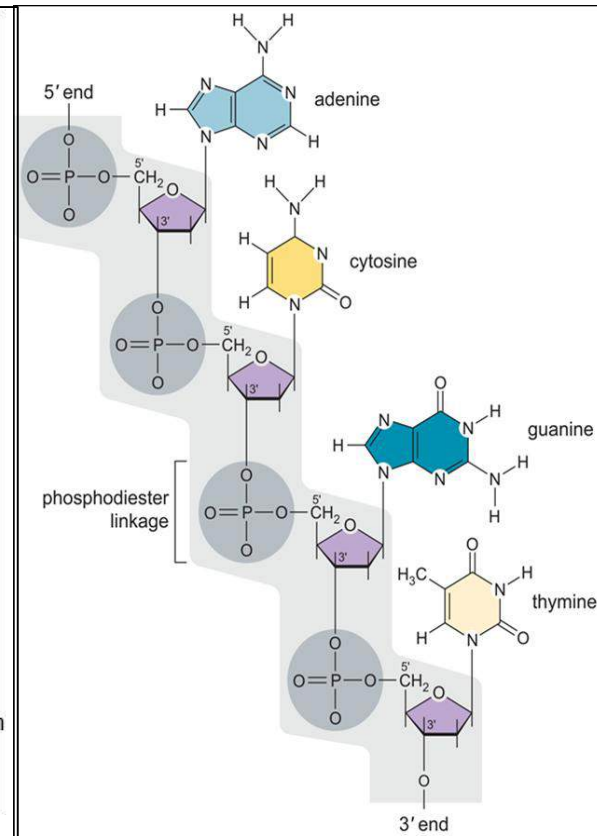
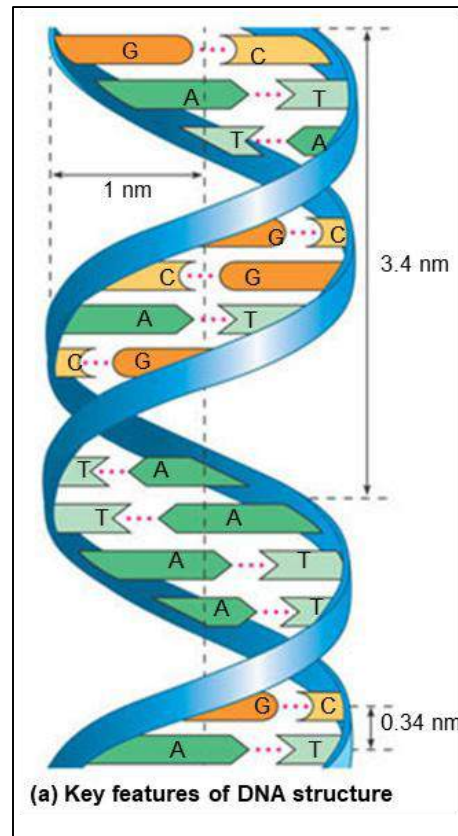


✓ *Heterochromatin usually contains genes that are not expressed, and is found in the regions of the centromere and telomeres. The euchromatin usually contains genes that are transcribed, with DNA packaged around nucleosomes but not further compacted..*



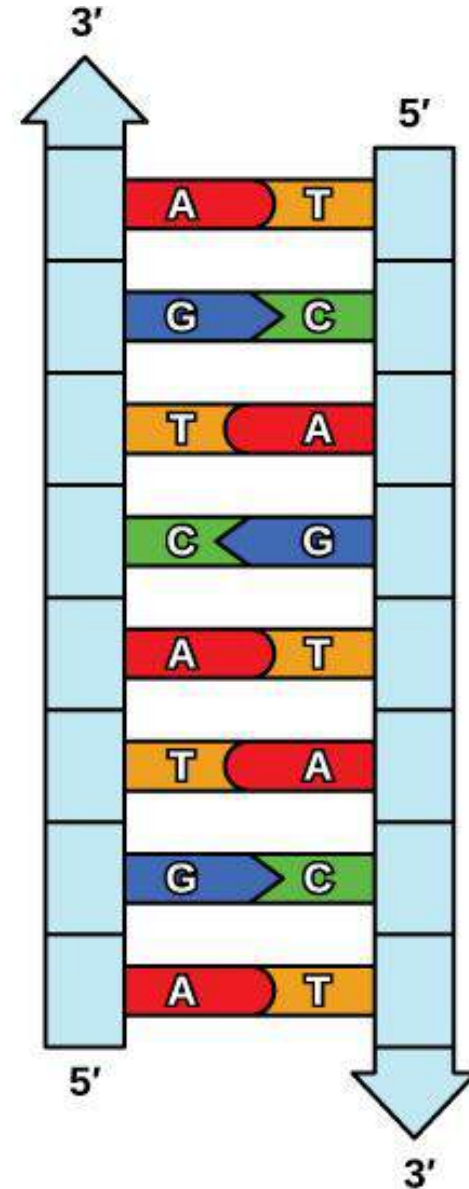
# Double Helical structure of DNA

- The structure of DNA was determined in the 1950s, by Francis Crick and James Watson
- They proposed that DNA is made up of two strands twisted around each other to form a right-handed helix.
- Polynucleotide chains are formed when the 3'-hydroxyl of one nucleotide and the 5'-phosphate are attached through a phosphodiester linkage
- Phosphodiester linkages create the repeating, sugar-phosphate backbone of the polynucleotide chain





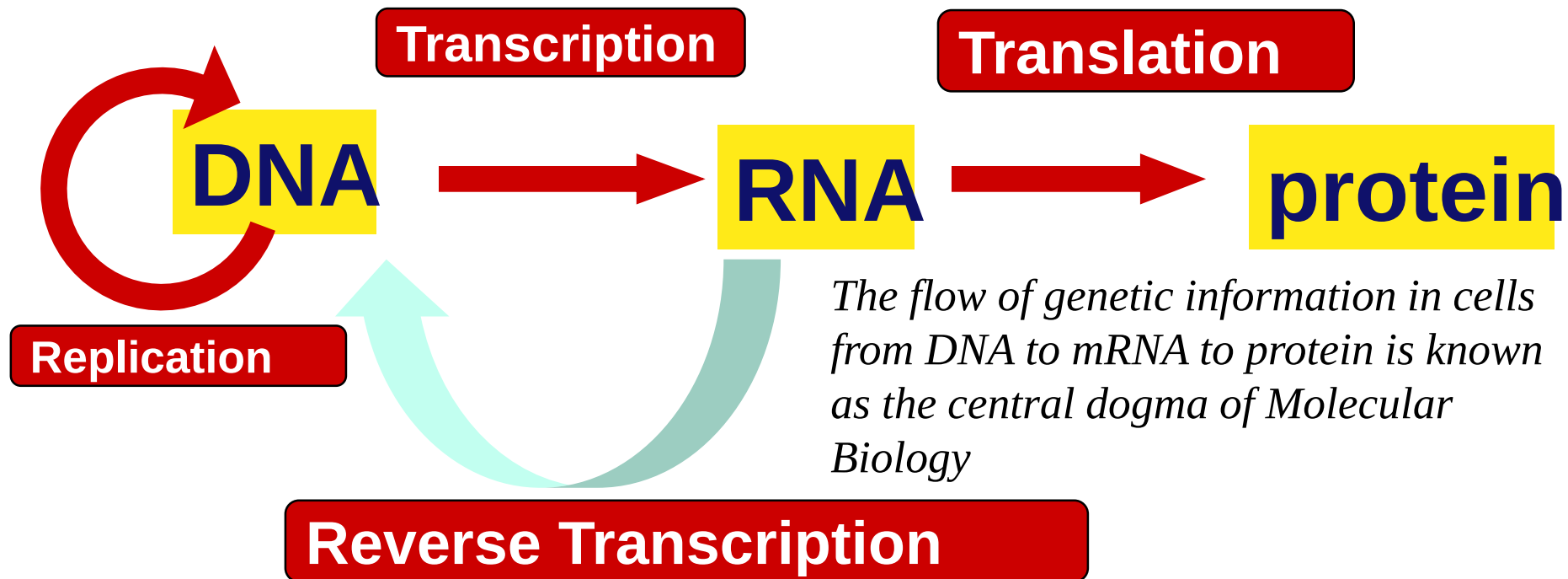
- The two polynucleotide strands are connected by hydrogen bonds between the complementary bases (A-T, G-C).
- The two strands of DNA are complementary, meaning the sequence of bases in one strand can be used to create the correct sequence of bases in the other strand.
- The two strands are anti-parallel in nature; that is, one strand will have the 3' carbon of the sugar in the “upward” position, whereas the other strand will have the 5' carbon in the upward position.
- The phosphate group is attached to the 5' carbon of one nucleotide and the 3' carbon of the next nucleotide





# DNA replication and cell division

- Replication is the process by which the DNA of the ancestral cell is duplicated, prior to cell division. Upon cell division, each of the descendants will get one complete copy of the DNA that is identical to its predecessor
- The ultimate goal of DNA replication is to maintain the constancy of genetic material throughout generations

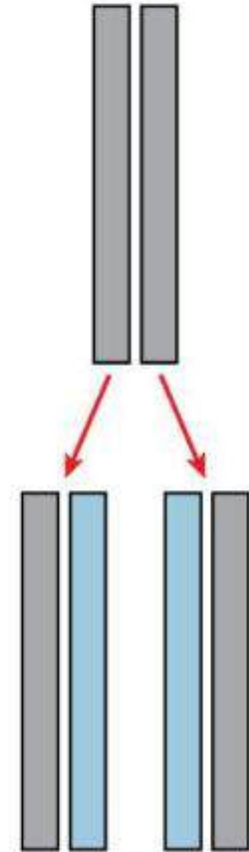




# DNA replication and cell division

- ✓ The replication of DNA occurs during the synthesis phase, or S phase, of the cell cycle, before the cell enters mitosis or meiosis
- ✓ during DNA replication, the two parental strands separate, and each strand serves as a template to synthesize a complementary daughter strand
- ✓ Each daughter cell gets one old strand and one new strand. i.e one of the parental strands is conserved and thus DNA replication is semi-conservative
- ✓ DNA replication is catalyzed by enzyme called **DNA polymerase**

Semiconservative Model of DNA Replication







# The DNA replication process

It occurs in three main stages:

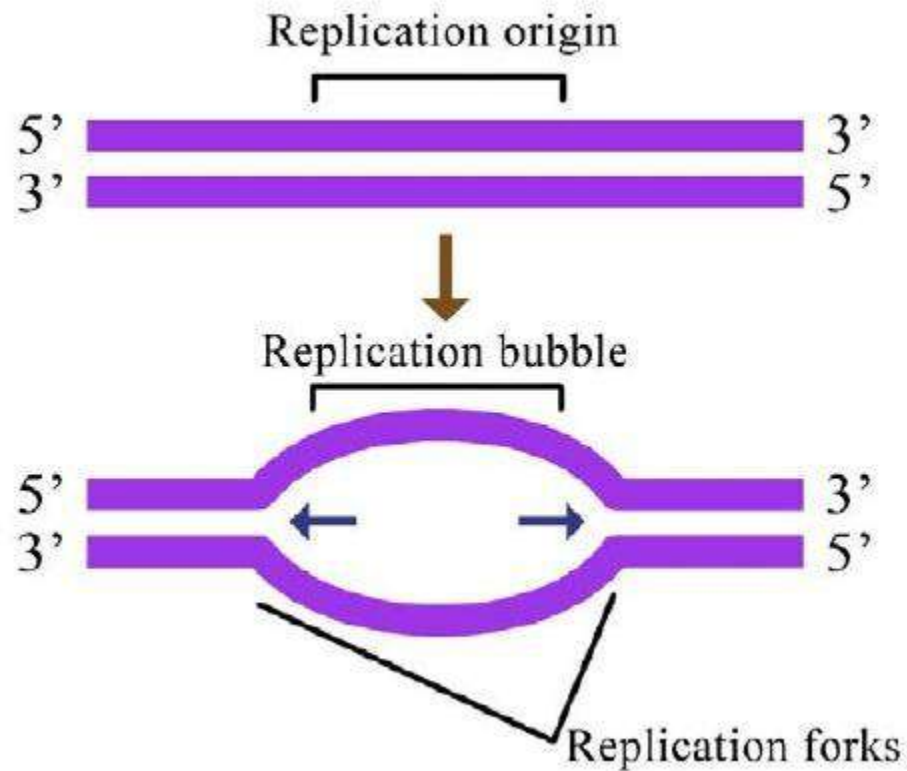
- initiation
- elongation
- termination

## Initiation of DNA replication

### 1. Unwinding and opening DNA double strand

- There are specific nucleotide sequences called **origins of replication** at which replication begins.
- At the origin of replication, the double helix unwound and the base pair bonds are broken (“opening” the DNA molecule) by the **DNA helicase** enzyme
- As the DNA opens up, Y-shaped structures called **replication forks** are formed



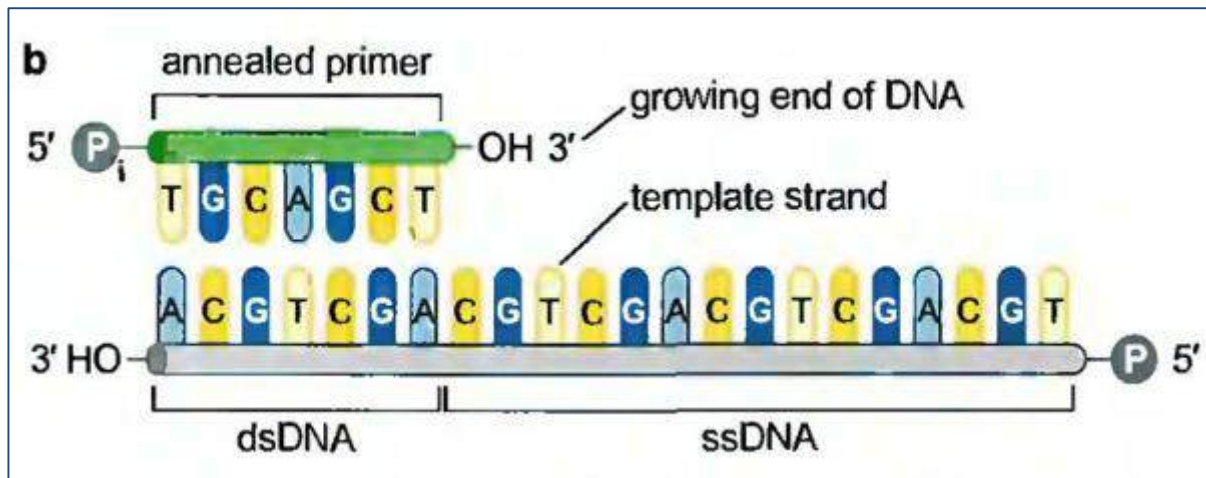


- A replication fork is formed by the opening of the origin of replication, and helicase separates the DNA strands.



## 2. Priming new strands

- DNA polymerase can only add nucleotide onto an existing strand during DNA replication. For this, an enzyme known as **primase** synthesizes a **short ribonucleotide** (usually ~10bp) called **RNA primer**.
- DNA polymerase can then add nucleotides starting at the end of the RNA primer
- the RNA primer is later degraded and (usually) replaced with DNA

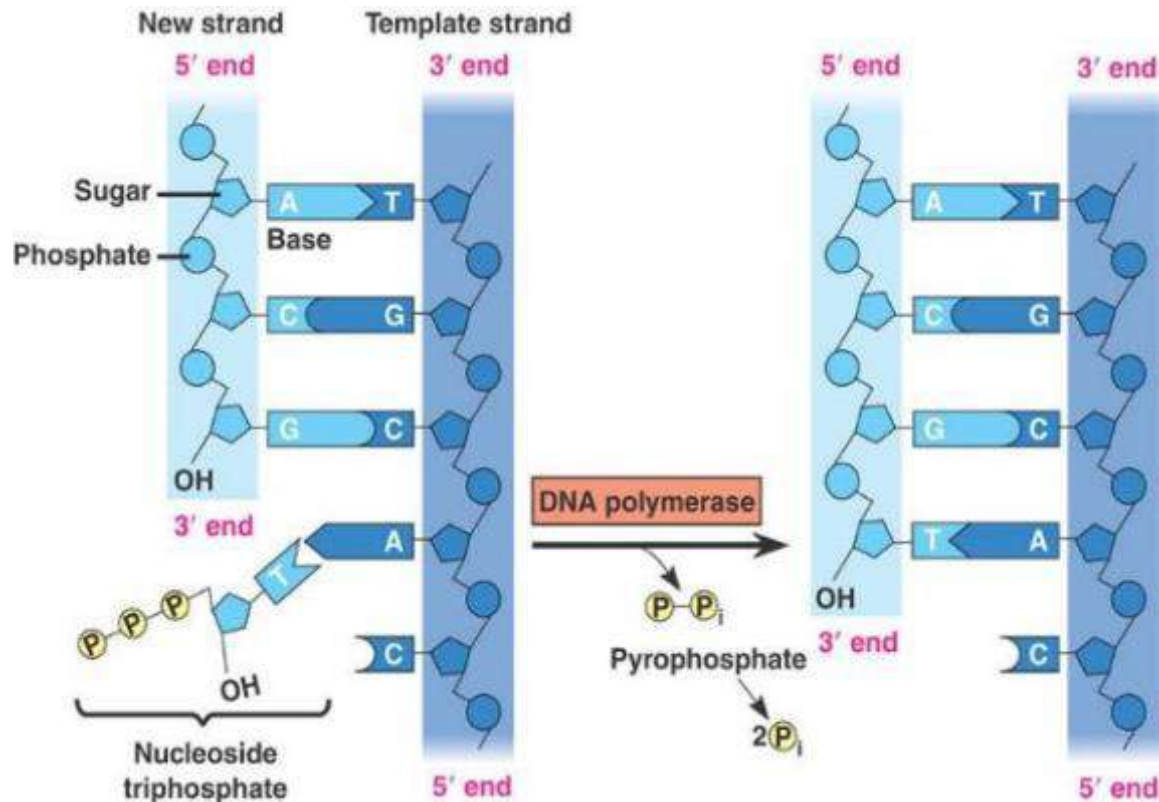




# DNA replication...

## DNA synthesis

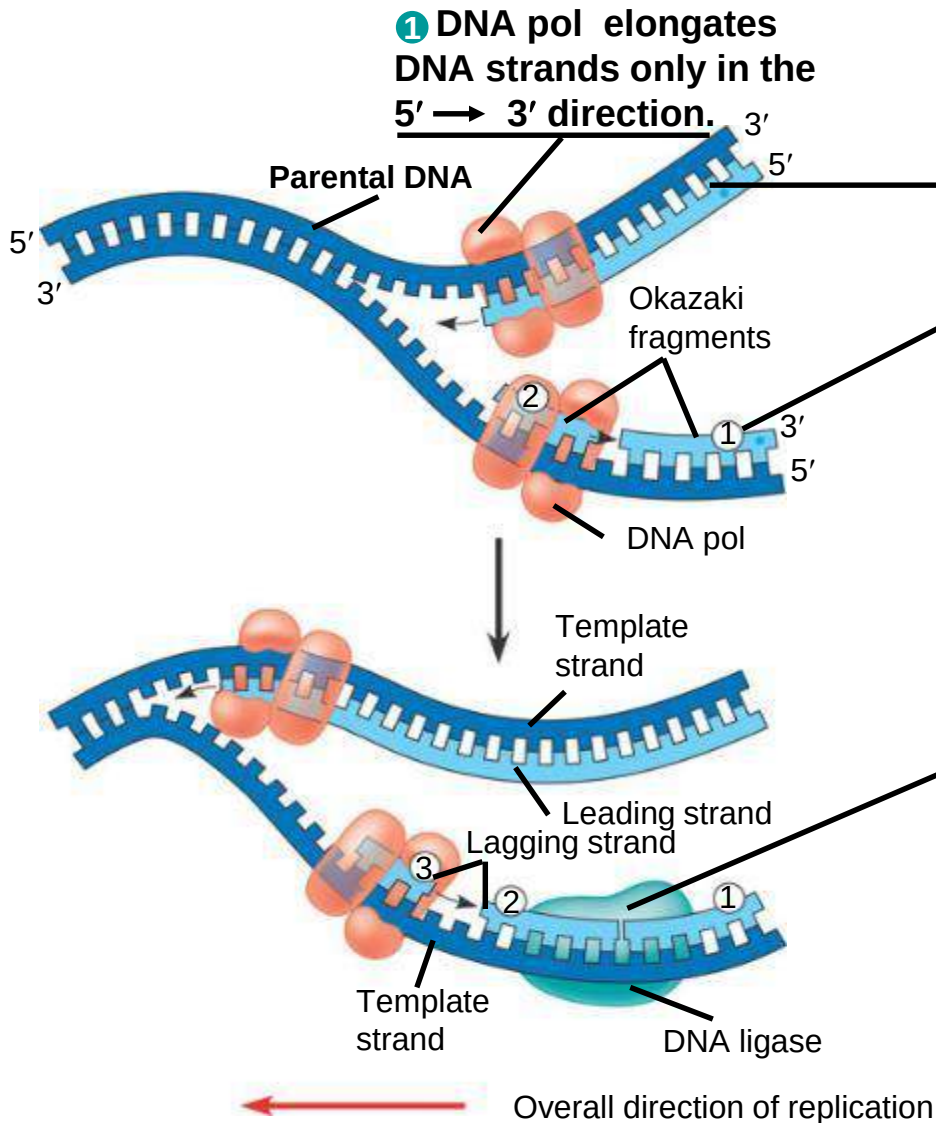
- **DNA polymerases** catalyze the synthesis of new strands.
- The synthesis proceeds by adding nucleotides onto the 3' end of a strand thus, synthesis can only proceed in the **5' → 3'** direction





## Elongation

- On one template strand, the newly synthesized strand grows in the same direction of replication fork movement and is called the “**leading strand**”. The leading strand is synthesized continuously from the beginning to the end.
- On the other template strand, the daughter strand is synthesized in the opposite direction and is called the “**lagging strand**”.
  - ✓ Synthesis of the lagging strand is discontinuous and must be re-initiated each time the replication fork moves and exposes a segment of the template. The lagging strand is composed of **Okazaki fragments**
  - ✓ These Okazaki fragments are then joined to form an intact strand



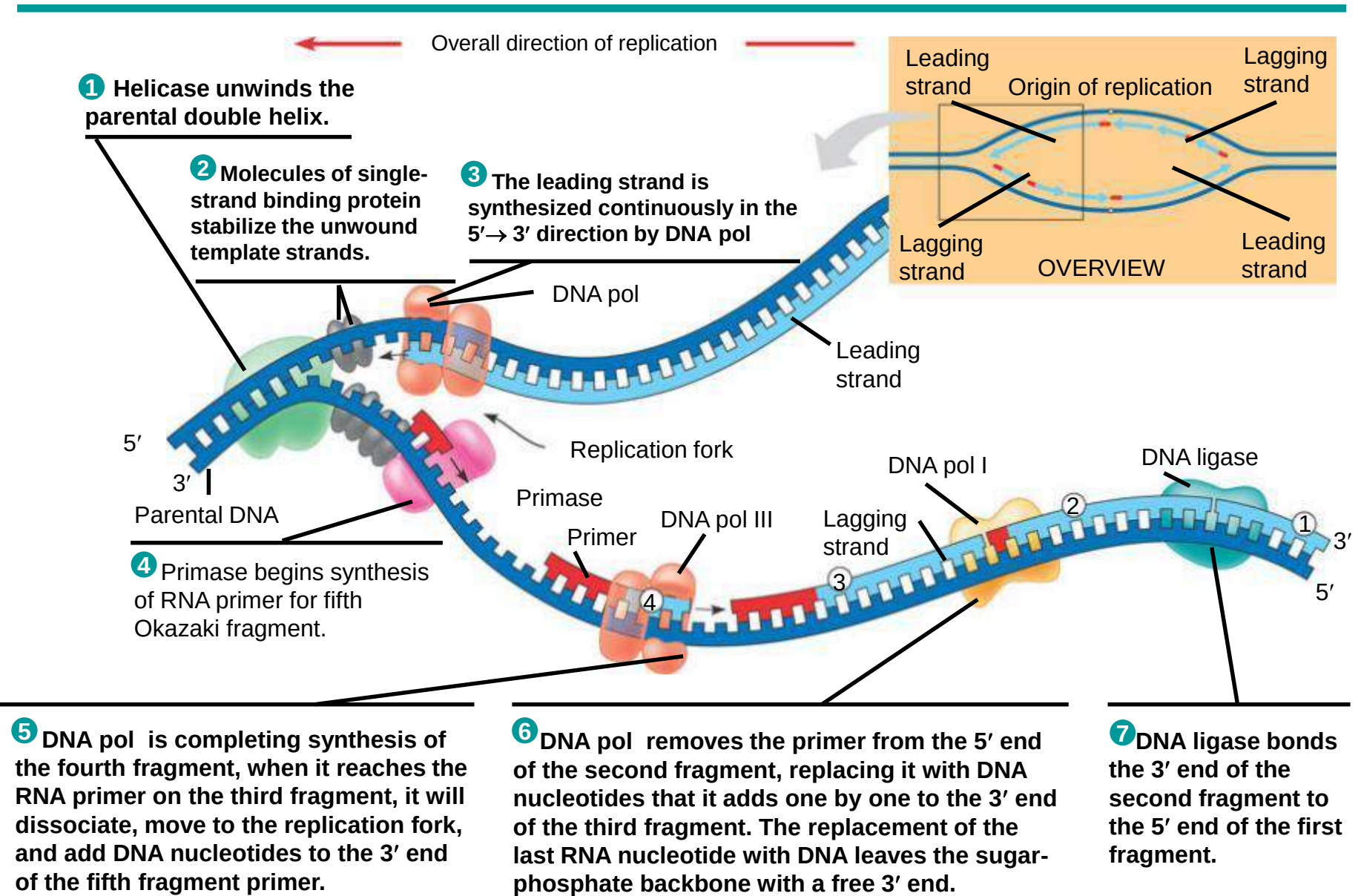
① DNA pol elongates DNA strands only in the 5' → 3' direction.

② One new strand, the leading strand, can elongate continuously 5' → 3' as the replication fork progresses.

③ The other new strand, the lagging strand must grow in an overall 3' → 5' direction by addition of short segments, Okazaki fragments, that grow 5' → 3' (numbered here in the order they were made).

④ DNA ligase joins Okazaki fragments by forming a bond between their free ends. This results in a continuous strand.









# Cell division

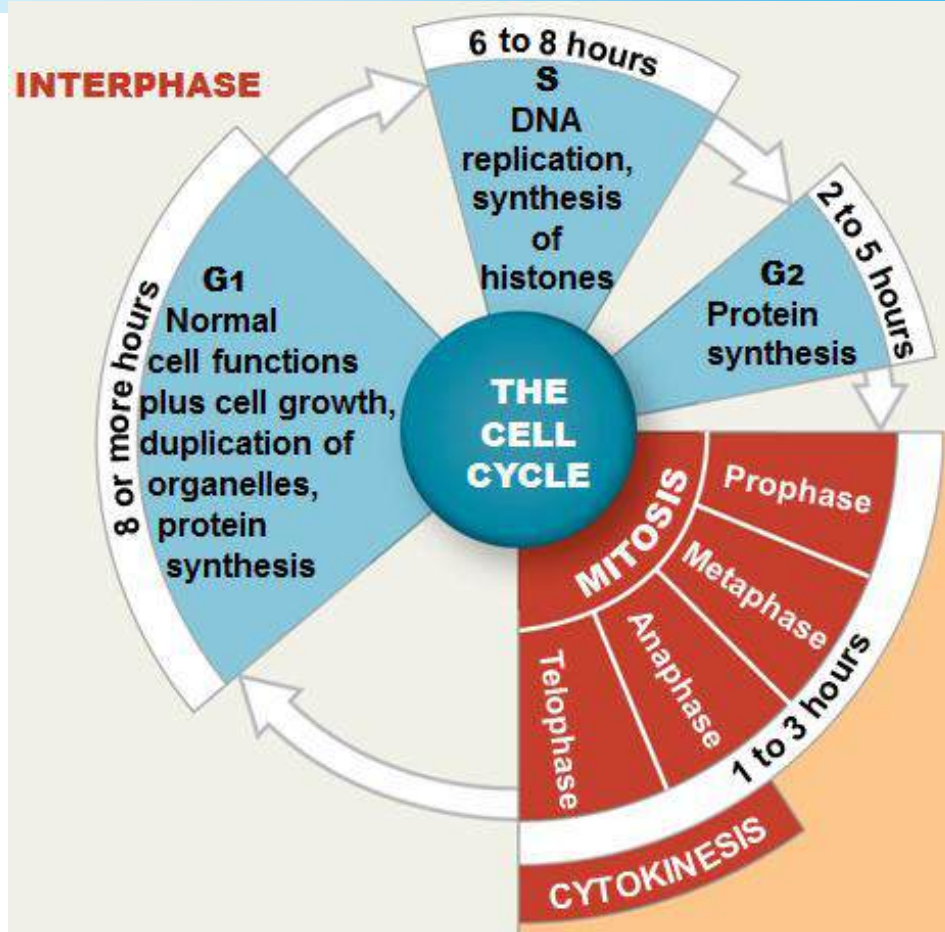
- The ability of organisms to reproduce best distinguishes living things from nonliving matter. The continuity of life is based on the reproduction of cells, or cell division
- Cell division is the process by which a parent cell divides into two or more daughter cells.
- In unicellular organisms, the division of one cell reproduces the entire organism
- Multicellular organisms depend on cell division for:
  - ✓ Development from a fertilized cell
  - ✓ Growth
  - ✓ Repair



Repeating sequence of cellular growth and division throughout the life of an organism is known as the **cell cycle**

The cell cycle consists of

1. **Interphase** (cell growth and copying of chromosomes in preparation for cell division)
2. **Mitotic (M) phase** (mitosis and cytokinesis)



*In multicellular organisms, mitosis is primarily used for tissue growth, regeneration, and repair. In many unicellular organisms, mitosis is the primary form of reproduction.*



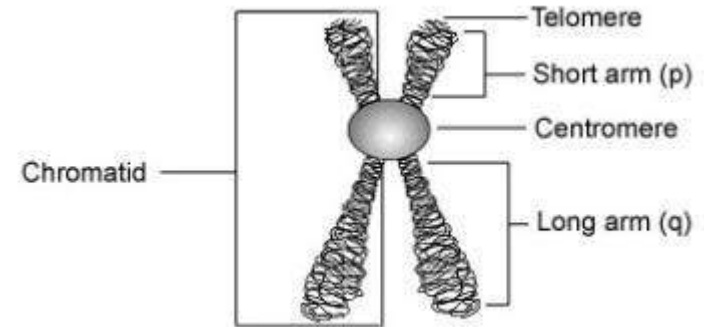
## Interphase

- Interphase covers about 90% of the cell cycle and can be divided into subphases:
  - ✓ G1 phase (“first gap”)
  - ✓ S phase (“synthesis”)
  - ✓ G2 phase (“second gap”)
- During the G1 phase, the cell replicates its organelles and cytosol so there is enough for two functional cells.
- During the S phase, the DNA of the cell is completely duplicated.
- During the G2 phase, any additional growth needed by the cell before division takes place.



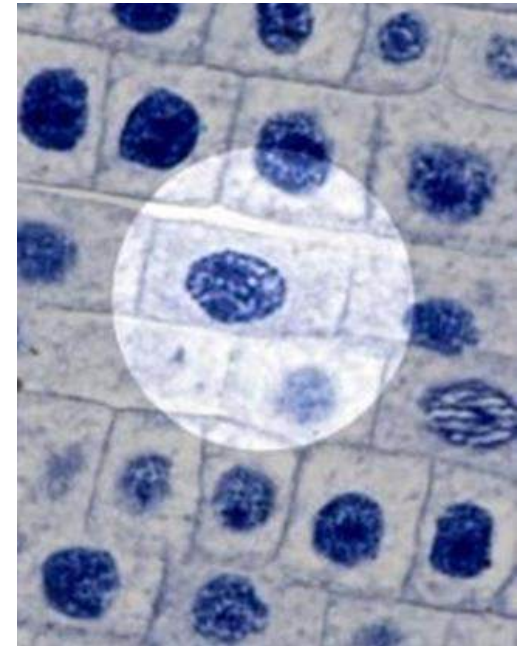
# Mitosis

- There are two joined copies of each chromosome before mitosis, called chromatids. The chromatids are connected by a single point, called the centromere.
- Mitosis is part of the cell cycle where the nucleus physically divides into two, each containing an exact copy of the original DNA.
- Includes prophase, metaphase, anaphase and telophase.



## Prophase

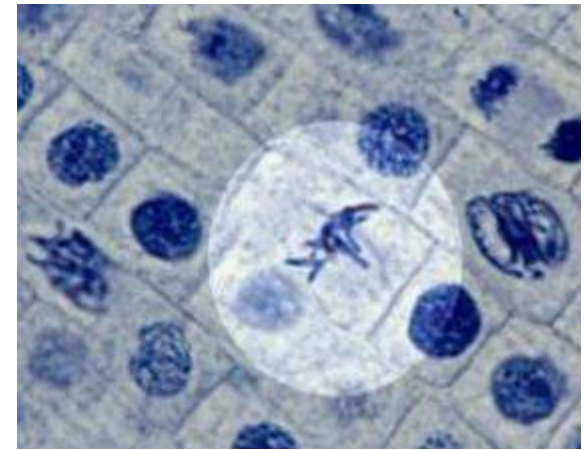
- During prophase, the DNA begins to condense.
- The nucleolus disappears and spindle fibers formed by centrioles attach to the centromeres.





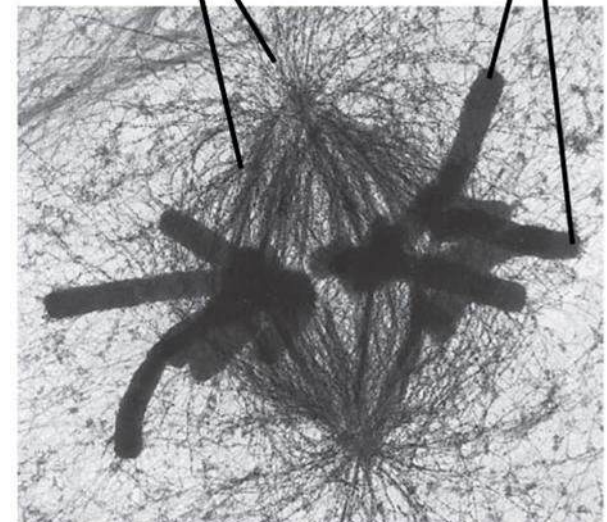
## Metaphase

- Metaphase begins as the chromosomes move and align themselves in a narrow central zone in the cell (Metaphase plate).
- a network of protein microtubules called the spindle apparatus produced by the centrioles attach to centromere of the sister chromatids



Microtubules

Chromosomes



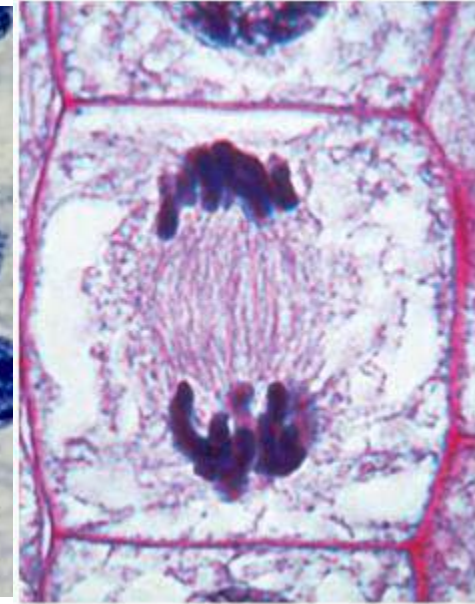
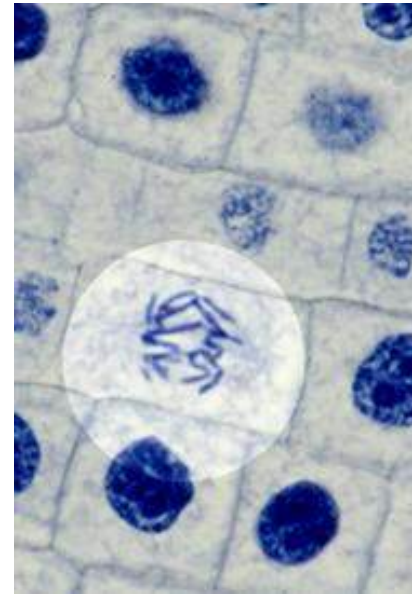
Centrosome





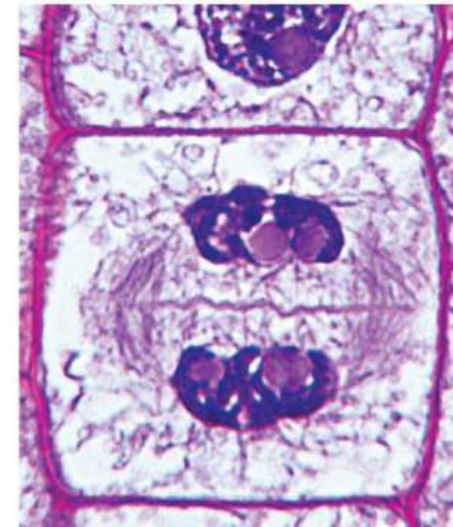
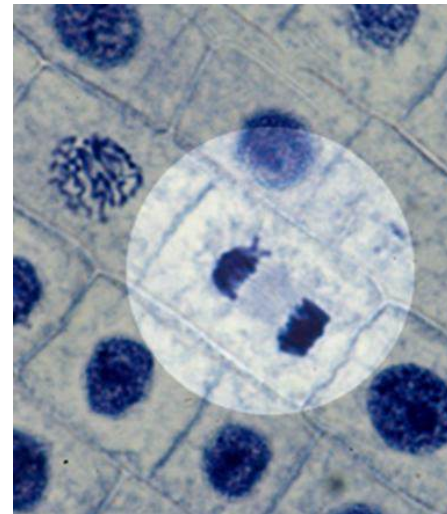
## Anaphase

- In anaphase, sister chromatids separate and move along the microtubules toward opposite ends of the cell
- The microtubules shorten by depolymerizing at their kinetochore ends



## Telophase

- With the chromosomes now fully divided, the nuclear membranes form during telophase.
- The DNA begins returning to its uncoiled chromatin form.
- The nucleolus becomes visible again.





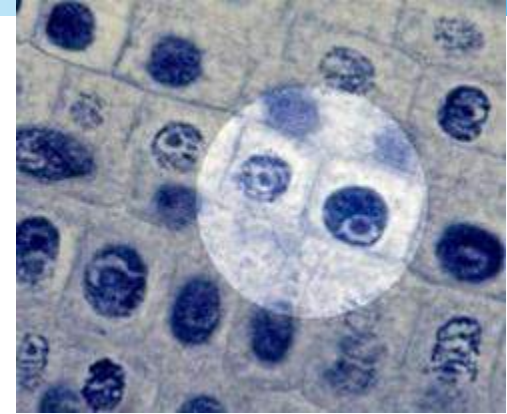


# Cytokinesis

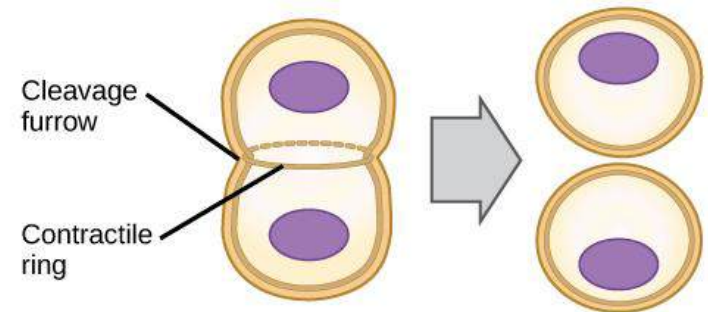
- The process of mitosis only separated the nucleus.
- The cytosol, organelles, and cell membrane are physically divided during cytokinesis. The cells return to interphase at this point.

Cytokinesis proceeds much differently in animal and plant cells.

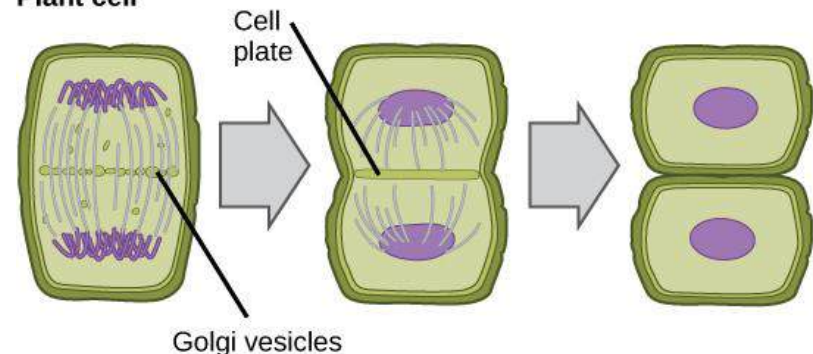
- In animal cells, cytokinesis causes the formation of a shallow indentation across the cell membrane called a cleavage furrow.
- The furrow tightens, eventually pinching shut as the cells separate.



Animal cell



Plant cell



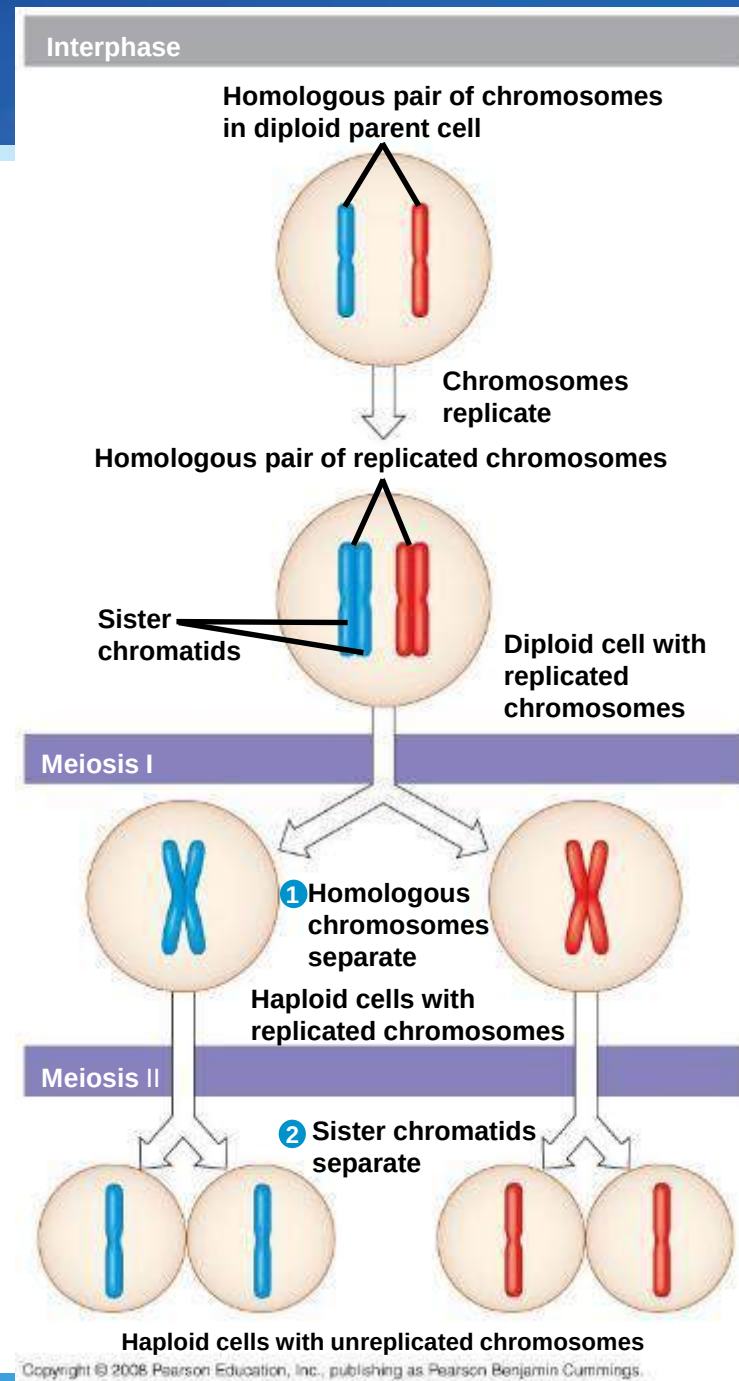


# Meiosis

- Sexually reproducing organisms require the union of two specialized cells, called gametes,
- A gamete (sperm or egg) contains a single set of chromosomes and is haploid (n)
- When gametes unite, they form a zygote, or fertilized egg that contains two sets of Chromosomes (diploid=  $2n$ )
- Male and female gametes are produced through the process of Meiosis; the nuclear division that forms haploid cells from diploid cells,
- Fertilization and meiosis alternate in sexual life cycles to maintain chromosome number
- Meiosis employs many of the same cellular mechanisms as mitosis. However, meiosis involves two separate sets of cell divisions labelled as Meiosis I and Meiosis II.



- Like mitosis, meiosis is preceded by the replication of chromosomes
- Meiosis takes place in two sets of cell divisions, called **meiosis I** and **meiosis II**
- The two cell divisions result in four daughter cells, rather than the two daughter cells in mitosis
- Each daughter cell has only half as many chromosomes as the parent cell





# Meiosis I

- In the first cell division (meiosis I), homologous chromosomes separate
- Meiosis I results in two haploid daughter cells with replicated chromosomes; it is called the reductional division
- Division in meiosis I occur in four phases:
  - Prophase I
  - Metaphase I
  - Anaphase I
  - Telophase I and cytokinesis



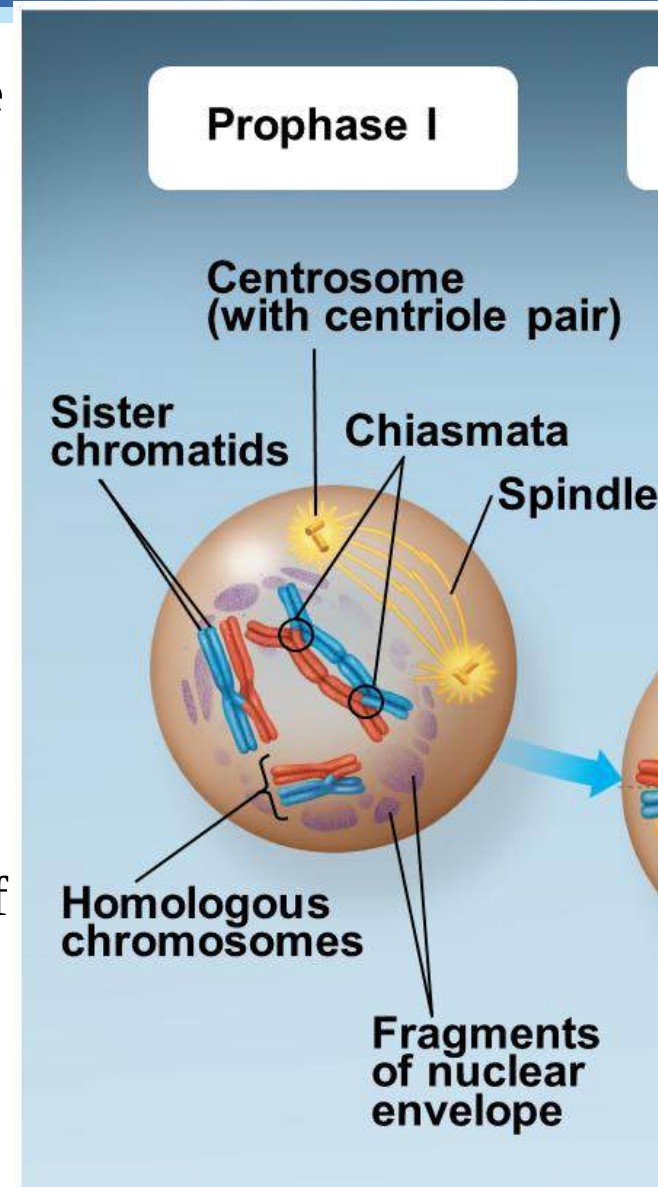
# Origins of Genetic Variation Among Offsprings

- The behavior of chromosomes during meiosis and fertilization is responsible for most of the variation that arises in each generation
- Three mechanisms contribute to genetic variation:
  - Independent assortment of chromosomes
  - Crossing over
  - Random fertilization



# Prophase I

- Prophase I typically occupies more than 90% of the time required for meiosis
- During this phase, chromosomes begin to condense
- Homologous chromosomes loosely pair up. The tight pairing of the homologous chromosomes is called **synapsis**. In synapsis, the genes on the chromatids of the homologous chromosomes are aligned precisely with each other.
- The homologous non-sister chromatids exchange segments through a process called **crossing over**, producing combinations of genes unique to those of the original stem cell.
- Each tetrad usually has one or more chiasmata, X-shaped regions where crossing over occurred

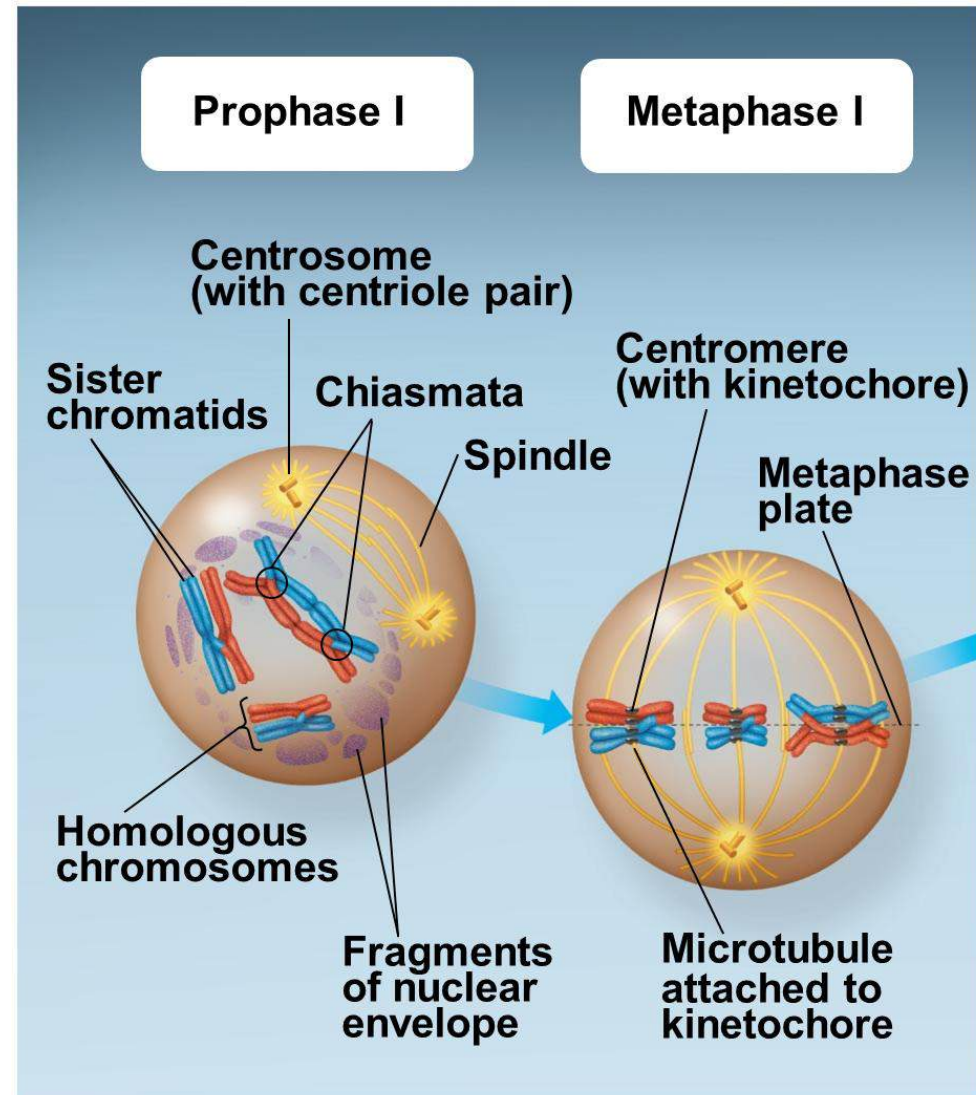


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## Metaphase I

- In metaphase I, tetrads line up at the metaphase plate, with one chromosome facing each pole
- Microtubules from one pole are attached to the kinetochore of one chromosome of each tetrad
- Microtubules from the other pole are attached to the kinetochore of the other chromosome

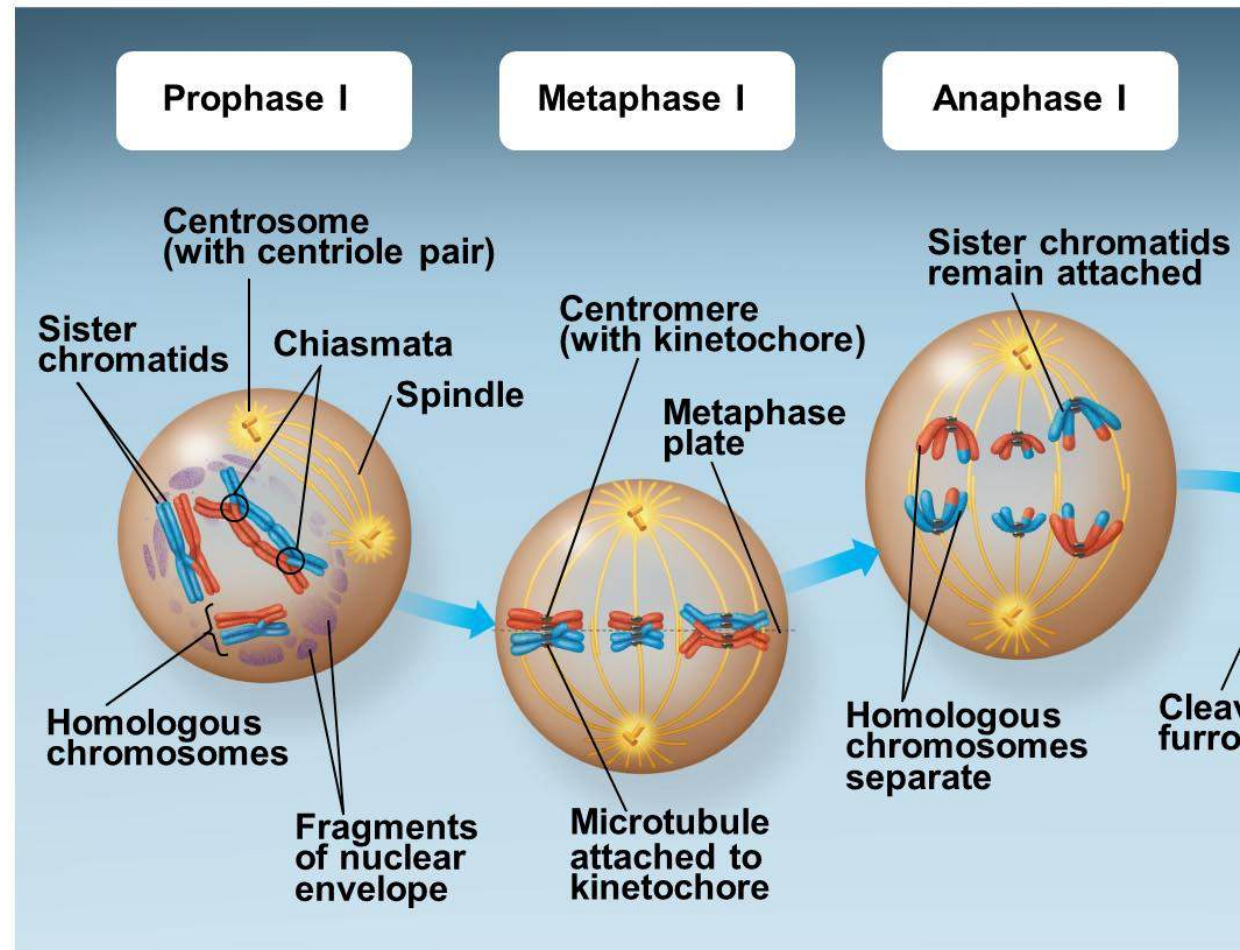


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# Anaphase I

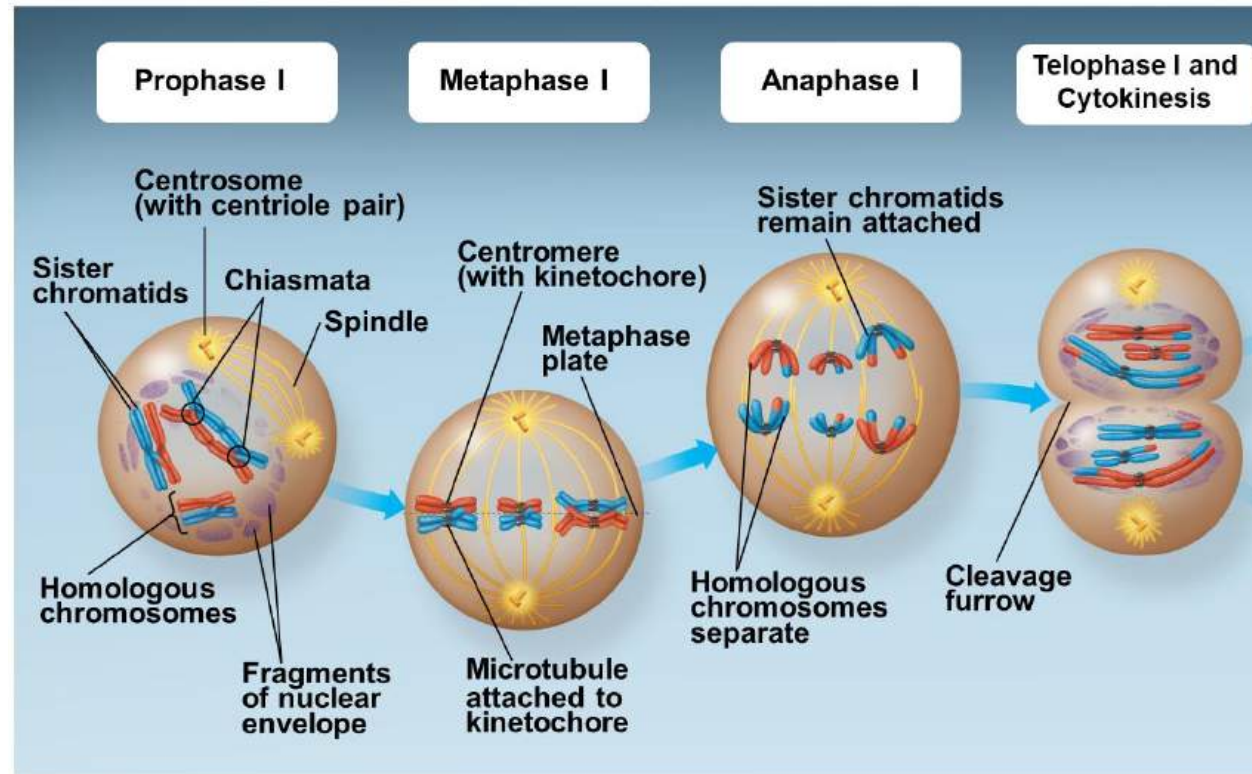
- In anaphase I, pairs of homologous chromosomes separate
- One chromosome moves toward each pole, guided by the spindle apparatus
- Sister chromatids remain attached at the centromere and move as one unit toward the pole



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# Telophase I and Cytokinesis

- In the beginning of telophase I, each half of the cell has a haploid set of chromosomes; each chromosome still consists of two sister chromatids
- Cytokinesis usually occurs simultaneously, forming two haploid daughter cells

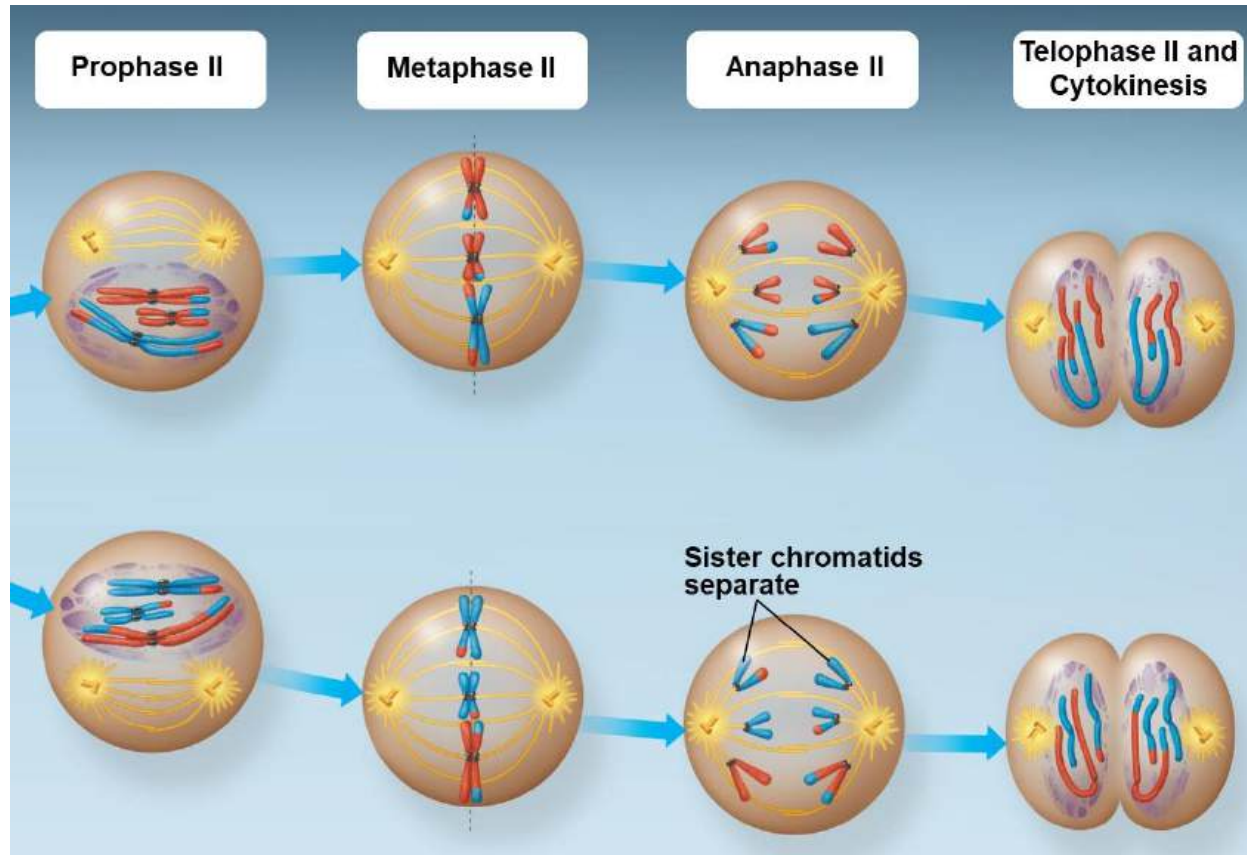






# Meiosis II

- Division in meiosis II also occurs in four phases:
  - Prophase II
  - Metaphase II
  - Anaphase II
  - Telophase II and cytokinesis
- Meiosis II is very similar to mitosis





# A Comparison of Mitosis and Meiosis

## SUMMARY

### Property

### Mitosis

**DNA replication**

Occurs during interphase before mitosis begins

**Number of divisions**

One, including prophase, metaphase, anaphase, and telophase

**Synapsis of homologous chromosomes**

Does not occur

**Number of daughter cells and genetic composition**

Two, each diploid ( $2n$ ) and genetically identical to the parent cell

**Role in the animal body**

Enables multicellular adult to arise from zygote; produces cells for growth, repair, and, in some species, asexual reproduction

### Meiosis

Occurs during interphase before meiosis I begins

Two, each including prophase, metaphase, anaphase, and telophase

Occurs during prophase I along with crossing over between nonsister chromatids; resulting chiasmata hold pairs together due to sister chromatid cohesion

Four, each haploid ( $n$ ), containing half as many chromosomes as the parent cell; genetically different from the parent cell and from each other

Produces gametes; reduces number of chromosomes by half and introduces genetic variability among the gametes

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# Transcription

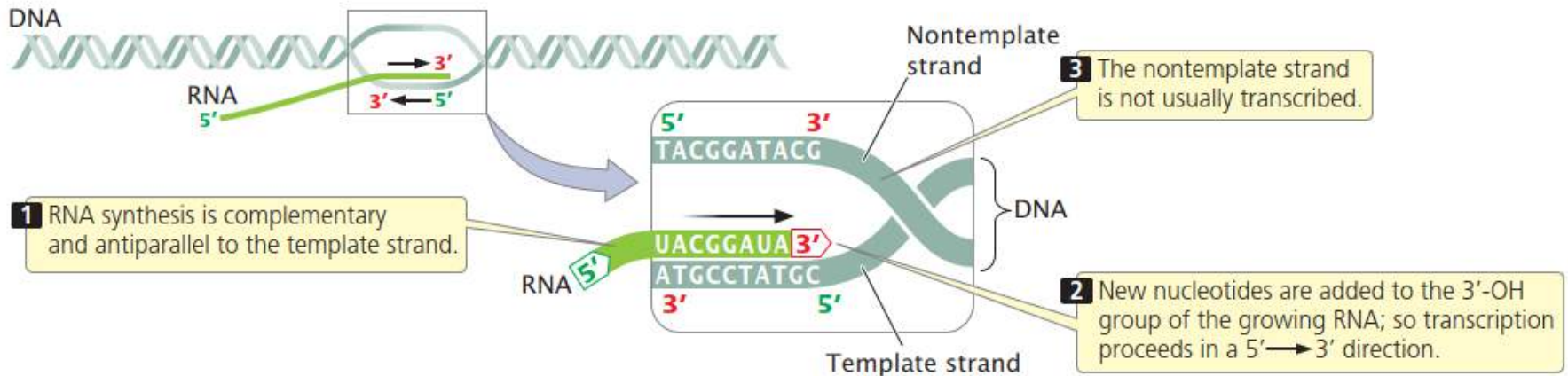
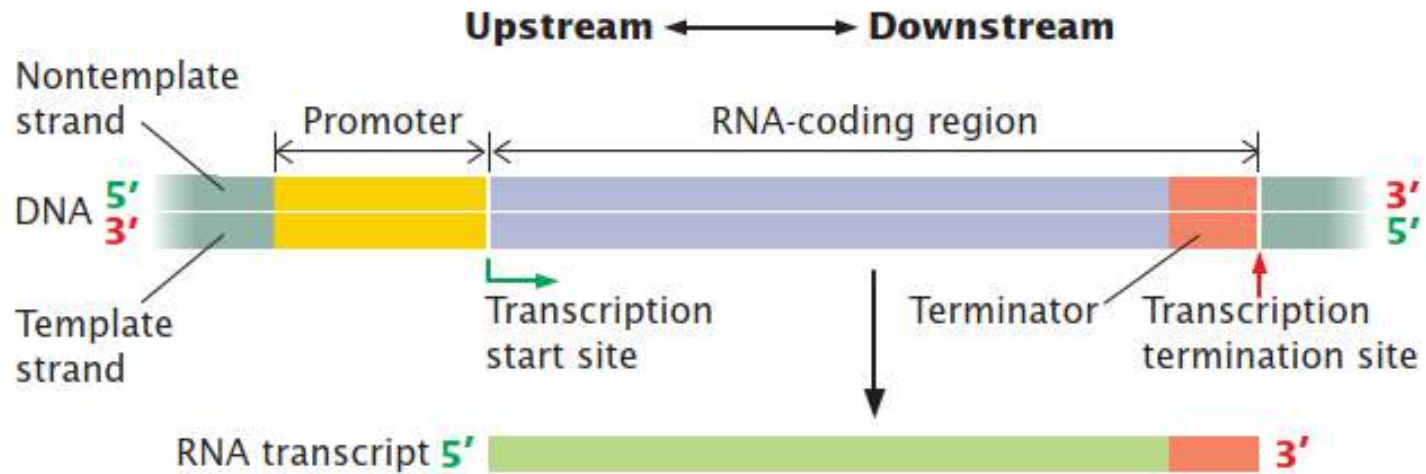
**Transcription:** is the synthesis of RNA molecules using DNA strands as the templates so that the genetic information can be transferred from DNA to RNA.

- This RNA copy is called **messenger RNA (mRNA)**
- Occurs in the nucleus of eukaryotic organisms
- Is catalyzed by RNA polymerase, which pries the DNA strands apart and hooks together the RNA nucleotides
- It follows the same base-pairing rules as DNA, except that in RNA, uracil substitutes for thymine

**It requires:**

- RNA polymerase, four NTP,  $Mg^{2+}$  and DNA template
- **Promoter sites:** specific sequences of nucleotides at the beginning of a gene where RNA polymerase binds and initiate the process







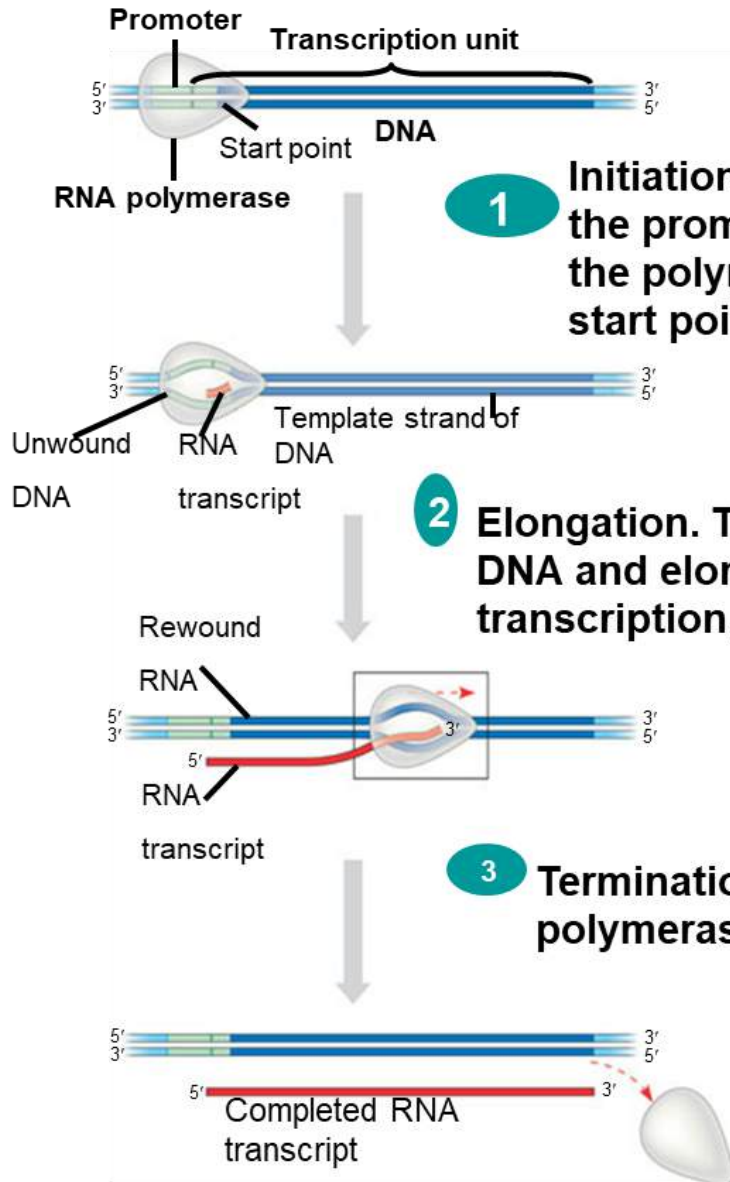
Three main types of RNA are predominantly made during transcription:

1. Messenger RNA (mRNA): a transcript copy of a gene used to encode a polypeptide
2. Transfer RNA (tRNA): a clover leaf-shaped sequence that carries an amino acid
3. Ribosomal RNA (rRNA): A primary component of ribosomes



# Mechanism of Transcription

1. RNA polymerase attaches to a promoter site, part of the DNA strand that codes for a gene, and separates that section of the DNA strand
  2. RNA polymerase uses one DNA strand as a **template** to assemble nucleotides into a **complementary strand of mRNA**. This follows the base pairing rules for DNA replication except that in RNA, uracil rather than thymine, pairs with adenine
  3. Once polymerase reaches the terminator site (end of the gene), mRNA separates from the DNA strand & leaves the nucleus then DNA strands rejoin or zip closed
- The strand of mRNA that is made during transcription has two regions
- ✓ **Exon – code for protein**
  - ✓ **Intron – non coding section**
- In order for the mRNA to be used in translation the non-coding introns need to be removed by spliceosome
  - Then the formed mRNA (with exon only) leaves the nucleus for translation in ribosome



**1** **Initiation.** After RNA polymerase binds to the promoter, the DNA strands unwind, and the polymerase initiates RNA synthesis at the start point on the template strand.

**2** **Elongation.** The polymerase moves downstream, unwinding the DNA and elongating the RNA transcript 5' → 3'. In the wake of transcription, the DNA strands re-form a double helix.

**3** **Termination.** Eventually, the RNA transcript is released, and the polymerase detaches from the DNA.



# Protein synthesis: Translation

- The code for protein synthesis is specified by DNA and has to be sent to the ribosome.

## Events during protein synthesis

- i. mRNA travels from the nucleus to ribosome
- ii. Free amino acids are transported from cytoplasm to ribosome by transfer RNA (tRNA) molecules
- iii. Ribosome read mRNA code and assembles amino acids presented by tRNA into a protein by a process called translation



## Process of Translation

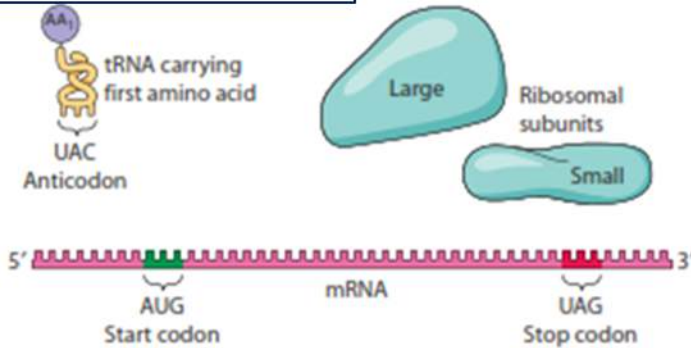
1. Translation begins with the binding of small ribosomal sub unit to mRNA strand **upstream** of the start **codon**
  - Each amino acid is brought to ribosome by specific tRNA
  - The type of amino acid is determined by the anticodon sequence of the tRNA
2. Complementary base pairing occur between the codon of mRNA and anticodon of tRNA (initiator tRNA)
3. After the initiator tRNA binds to the start codon, the large ribosomal sub unit binds to form translation complex and initiation is completed
  - In the large ribosomal sub unit there are sections called E, P and A site
4. During elongation individual amino acids are brought to mRNA strand by tRNA molecule through complementary base pairing of codon and anticodon
  - Each anticodons of tRNA molecule corresponds to a specific amino acid





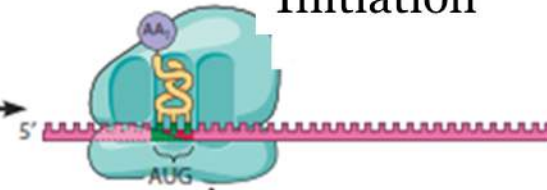
# Steps of protein synthesis

## Ingredients

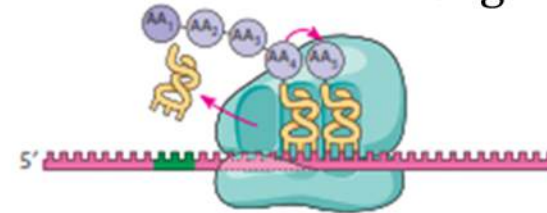


1 During initiation, the components of the translational apparatus come together with an mRNA, and a tRNA carrying the first amino acid (AA) binds to the start codon (AUG).

## Initiation

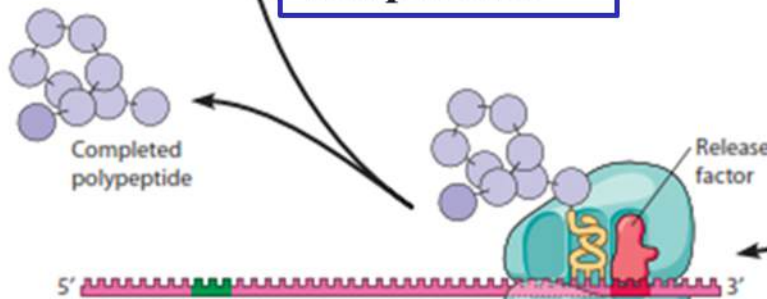


## Elongation



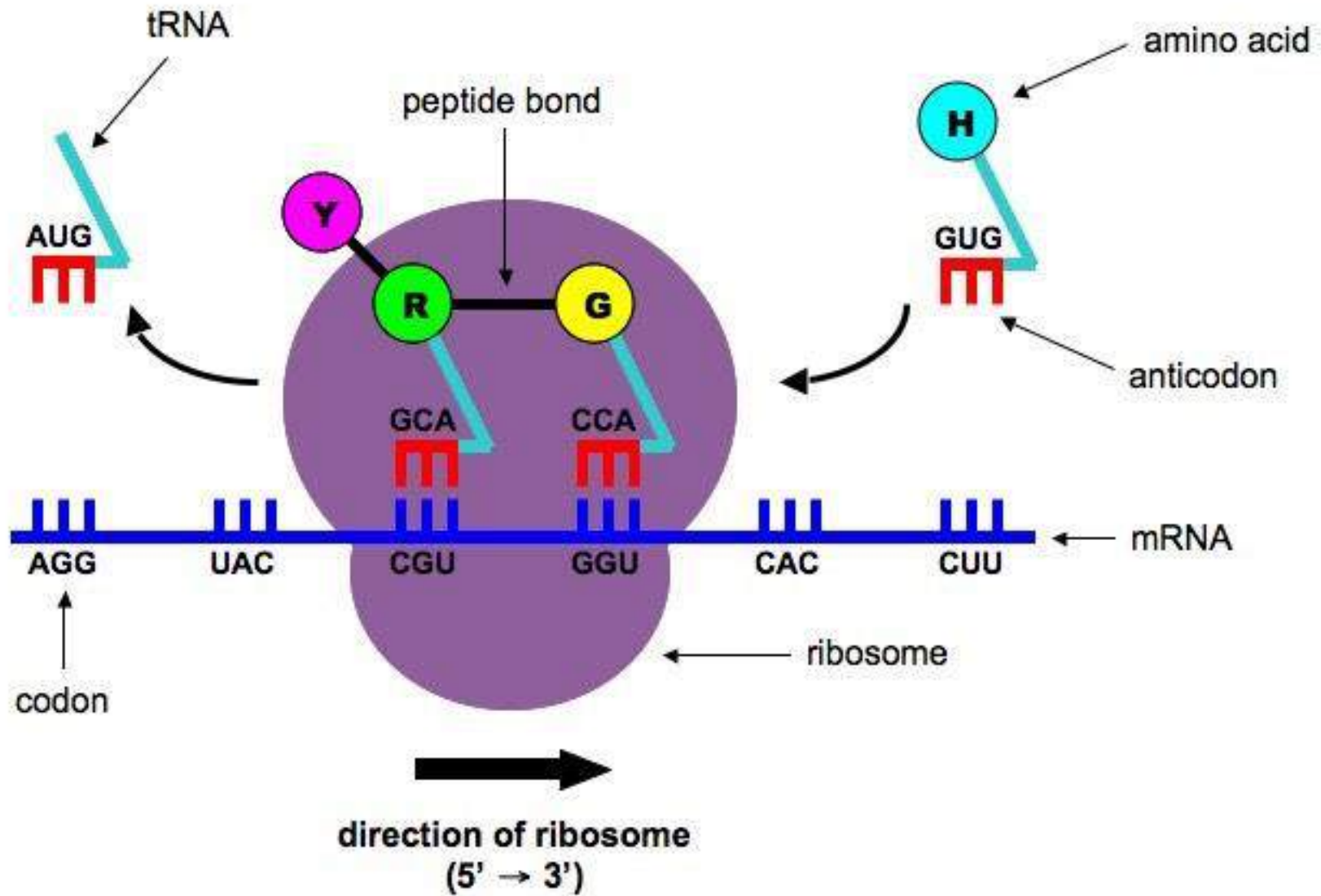
2. During elongation, amino acids are brought to the mRNA by tRNAs and are added, one by one, to a growing polypeptide chain

## Recycling of translational components



## Termination

3 During termination, a stop codon in the mRNA is recognized by a protein release factor, and the translational apparatus comes apart, releasing a completed polypeptide.





# The Genetic Code

Genetic code (codon) is triplets in the DNA molecule that carries a code for a protein-protein synthesis

1. Codons are triplet : Each group of 3 nucleotides on the mRNA is a codon. Since there are 4 bases, there are  $4^3 = 64$  possible codons, which must code for 20 different amino acids :
2. It is nonover-lapping and “commaless.” The codons are aligned without overlap and without empty spaces between them. Each base belongs to one and only one codon.
3. It contains 61 amino acid coding codons and the three stop codons UAA, UAG, and UGA. One of the amino acid coding codons, AUG, is also used as a start codon.
4. It is unambiguous. Each codon specifies one & only one amino acid
5. It is degenerate. More than one codon can code for an amino acid.
6. It is almost universal. With the minor exception



# Codons

**The Genetic Code<sup>a</sup>**

|                        | Second letter                                              |                                          |                                                                          |                                                                |                  |
|------------------------|------------------------------------------------------------|------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------|------------------|
|                        | U                                                          | C                                        | A                                                                        | G                                                              |                  |
| First (5') letter<br>U | UUU } Phe (F)<br>UUC }<br>UUA } Leu (L)<br>UUG }           | UCU }<br>UCC } Ser (S)<br>UCA }<br>UCG } | UAU } Tyr (Y)<br>UAC }<br>UAA Stop (terminator)<br>UAG Stop (terminator) | UGU } Cys (C)<br>UGC }<br>UGA Stop (terminator)<br>UGG Trp (W) | U<br>C<br>A<br>G |
|                        | CUU }<br>CUC } Leu (L)<br>CUA }<br>CUG }                   | CCU }<br>CCC } Pro (P)<br>CCA }<br>CCG } | CAU } His (H)<br>CAC }<br>CAA } Gln (Q)<br>CAG }                         | CGU }<br>CGC } Arg (R)<br>CGA }<br>CGG }                       | U<br>C<br>A<br>G |
|                        | AUU }<br>AUC } Ile (I)<br>AUA }<br>AUG Met (M) (initiator) | ACU }<br>ACC } Thr (T)<br>ACA }<br>ACG } | AAU } Asn (N)<br>AAC }<br>AAA } Lys (K)<br>AAG }                         | AGU } Ser (S)<br>AGC }<br>AGA } Arg (R)<br>AGG }               | U<br>C<br>A<br>G |
|                        | GUU }<br>GUC } Val (V)<br>GUA }<br>GUG }                   | GCU }<br>GCC } Ala (A)<br>GCA }<br>GCG } | GAU } Asp (D)<br>GAC }<br>GAA } Glu (E)<br>GAG }                         | GGU }<br>GGC } Gly (G)<br>GGA }<br>GGG }                       | U<br>C<br>A<br>G |

Third (3') letter

= Polypeptide chain initiation codon  
 = Polypeptide chain termination codon

<sup>a</sup>Each triplet nucleotide sequence or codon refers to the nucleotide sequence in **mRNA** (not DNA) that specifies the incorporation of the indicated amino acid or polypeptide chain termination. The one-letter symbols for the amino acids are given in parentheses after the standard three-letter abbreviations.



# Mutations

Mutation is the alteration of the nucleotide sequence of the genome of an organism, virus, or extrachromosomal DNA.

- ✓ May occur in somatic cells (aren't passed to offspring)
- ✓ May occur in gametes (eggs & sperm) and be passed to offspring

Causes of mutation

- ✓ Replication errors
- ✓ Mutagens, such as UV rays and chemicals

## Types of mutations

1. **Point mutations**- changes that involve a single base.

There are several types of point mutations:

- ✓ Substitution- one base is replaced by another base
- ✓ Deletions-a base is missed out during DNA replication.
- ✓ Addition – an extra base is added.





## 2. Chromosomal Mutations: part of a chromosome is missed/added

➤ can happen in either of the following:

1. **Deletion:** loss of an inner chromosomal fragment/end of a chromosome breaks off.
2. **Insertion:** presence of a chromosome segment more than once in the same chromosome
3. **Inversion:** the change of direction of a chromosomal segment when it breaks out and is re-inserted backwards
4. **Translocation:** movement of chromosomal segments from one chromosome to another, **non-homologous chromosome**

**NB. Euploidy/polyploidy**/-is a condition where a cell has one complete set of extra chromosomes

**Aneuploidy** – is a condition where a cell lose or add one or more chromosomes from/to the normal set of chromosome





# Consequence of mutations

## 1. Useful mutation: is the fuel for evolution

- Can create new species

## 2. Harmless mutation

- ✓ Silent mutation: different codon but still encodes for same amino acid
- ✓ Neutral mutation: Missense mutation that changes amino acid sequence, but does not alter the function of protein

## 3. Harmful

- Cause disease
- Damage the cell
- kill the cell



# Manipulation of DNA : Genetic engineering

- Genetic engineering is a process by which the genome of an organism is altered, usually by having extra gene from different organisms
- The organism is termed **genetically modified organism (GMO)**, transgenic organism or genetically engineered organism
- The immediate importance of genetic engineering may depend on the type of organism that **gets engineered**

## A. Bacteria

- ✓ to produce useful products
  - Insulin
  - Antibiotics
  - Enzyme for food processing industry
  - Human growth hormone
  - Vaccines



- ✓ Plants are also genetically modified
  - To absorb more CO<sub>2</sub>
  - Disease (drought) resistant
  - Improved yield
  - To produce specific product
- ✓ Animals are genetically modified for the following purposes
  - high growth rate
  - high yield of protein
  - Detection of water pollution
  - Disease resistant



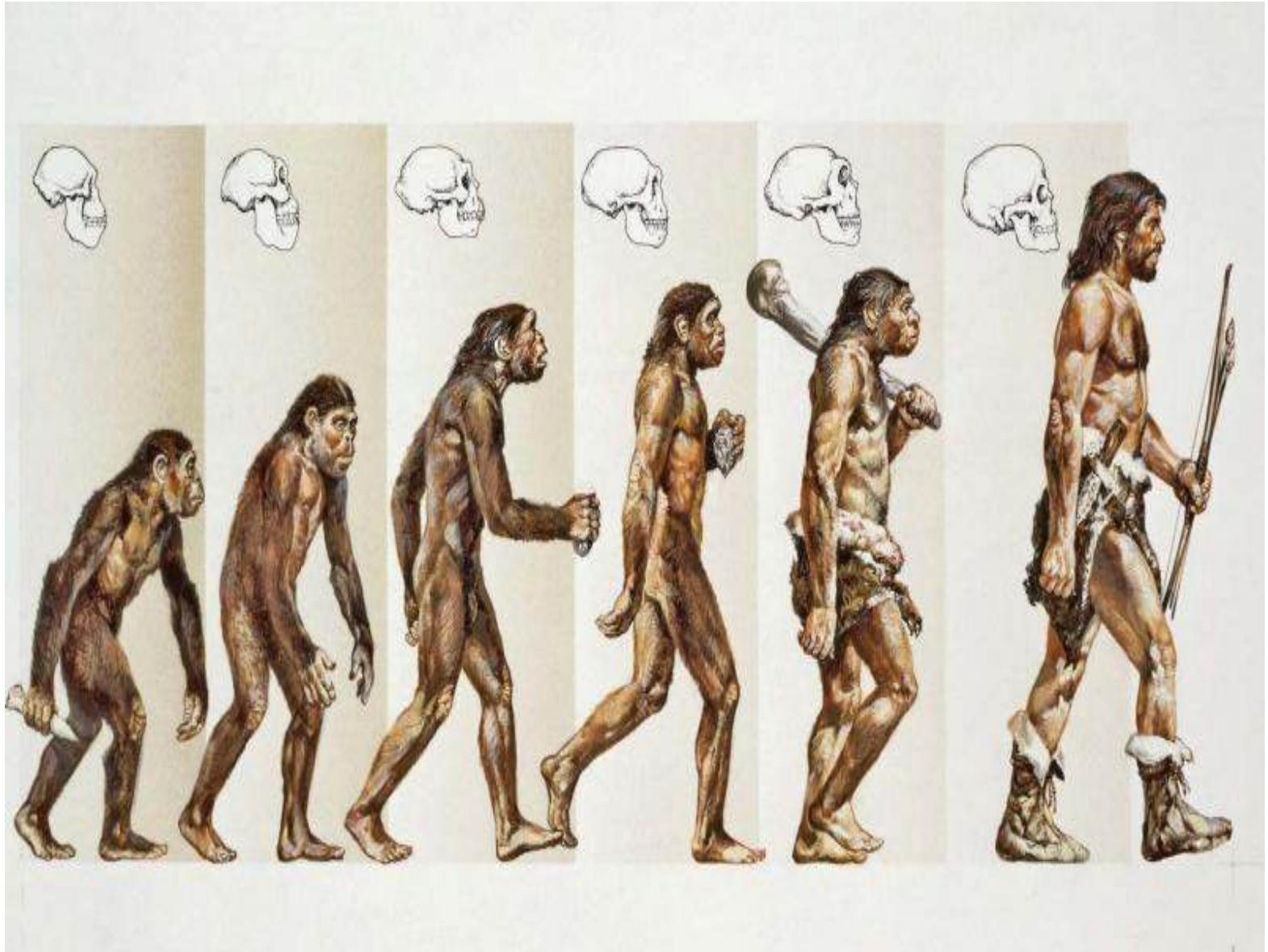
# Introduction to Evolution

## Evolution

- Gradual change of organisms on the earth over long periods, with new forms replacing old ones biodiversity of the planet increasing, larger and more complex organisms replaced the smaller and some species has get extinct

## Theories of Evolution

- Theories of evolution explains
  1. How does evolution happen?
  2. what drives the population to become a new species?





# A. Lamarck theory of evolution

- Lamarck published a paper entitled 'Philosophie Zoologique' (1809)
  - ✓ described two-way mechanism by which change is gradually introduced into the species and passed down through the generations
- The two parts of Lamarck theory are
  1. Use and disuse
    - ✓ structure or process in organism that can be used continuously will become enlarged or more developed but any structure that is not, will be reduced
      - Neck of giraffe
      - wings of penguins
  2. Inheritance of acquired traits
    - ✓ Lamarck believed that traits acquired during an organism's lifetime could be passed to its offspring

Example: Giraffe that had acquired long necks would have offspring with long neck





- NB. Nowadays, Lamarck's theories are not accepted
  - Because the environmental changes that were believed by Lamarck have brought about the changes in the phenotypes (Physical appearance) of the organisms have no effect on their gametes and hence their heredity

## **B. Charles Darwin and Natural Selection**

- Charles Darwin and Alfred Wallace jointly published a scientific paper
  - ✓ species were modified by natural selection
- Darwin studied finches found on five of the Galapagos Islands, made drawings, and collected species
- He noted there were many similarities between them but have some obvious differences



- Darwin concluded that an ancestral finch had colonized the Islands from mainland and been able to adapt to the different conditions on the islands and evolve into different species
  - ✓ E.g. He suggested that some finches had evolved into insect eaters (pointed peak), other in to seedeaters (crushing peak)

### Darwin summarized his observations in two main ideas

- All species tend to produce more offspring than can possibly survive (Fecundity)
- There is a variation among the offspring

From these observation Darwin concluded that:

- There will be a struggle for existence between members of a species because they are over reproduced and resources are limited
- Some members of a species will be better adapted than others to the environment because there is a variation in the offspring



## Darwin proposed the theory of natural selection

- Those members of a species, which are best adapted to their environment, will survive and reproduce in greater number than other less adapted (died out)
- ***Survival of the Fittest***: Individuals with certain heritable adaptive characteristics survive and reproduce at a higher rate than other individuals.
- ***Natural selection*** increases the adaptation of organisms to their environment over time.
- ***Speciation***: If an environment changes over time, natural selection may result in adaptation to these new conditions and may give rise to new species.



## C. Neo – Darwinism Theory

- Neo-Darwinism is a modification of Darwin's original theory that takes into account genetics and ethology (behavioral change)
- Charles Darwin knew very little about genetics and did not propose how a variation in the population was passed to the next generation
- Now day's genes and gene action are the driving force of evolution in the theory of Natural selection
- A Gene pool is all the alleles in the population. A population might evolve into a new species



- ❖ Suppose an allele determines a feature that gives an organism an advantage in its environment. Then, the followings will happen:
  - ✓ Those individuals with the advantageous allele of a gene will survive to reproduce in greater number than other types
  - ✓ Advantageous alleles pass to their offspring in greater numbers than other genes (alleles)
  - ✓ The frequency of the advantageous allele will be higher in the next generation of a population
- Mutations are important in introducing variation into population

# **Chapter 6**

## **Infectious diseases and Immunity**



# Outline of the chapter

- **Infectious diseases**
- **Infection and disease**
- **Disease development and stages**
- **Pathogenicity (primary and opportunistic pathogen)**
- **Types of infectious diseases**
- **Modes of transmission**
- **Host defenses against infectious diseases**

# What are infectious diseases?

- **Infectious diseases** are illnesses caused by harmful agents (*pathogens*) that get into the body. The most common causes are **bacteria, viruses, fungi, protozoa, and helminthes**.
  - Usually spread from person to person, through contaminated food, water and through bug bites
  - It occurs as a **result of interactions between pathogenic microorganisms and the host**
- ❑ **In order to cause infectious disease, a pathogen must accomplish the following steps**
- 1. It must enter the host**
  - 2. It must metabolize and multiply on or in the host tissue**
  - 3. It must resist host defenses**
  - 4. It must damage the host**

Infectious disease cause host damage by entering cell and **directly causing cell death, releasing toxin that kill cell at a distance, releasing enzyme that degrade tissue components, damage blood vessels & etc.**

# Infection and disease

The terms "**infection**" and "**disease**" are **not synonymous**

**An infection** is the successful colonization of the host by a microorganisms.

- ✓ It results when a pathogen enters and begins growing within the host
- ✓ It can lead to **disease**, which causes **signs** and **symptoms** resulting in a deviation from the normal structure or functioning of the host.

**Disease** is any condition in which the normal structure or functions of the host body cells are damaged or impaired.

- ✓ as a result of the infection and signs and symptoms of an illness appear

# Signs and Symptoms of Disease

## ❖ Sign

The **signs** of disease are *objective* and *measurable*, and can be directly *observed* by a clinician as a result of disease.

- ✓ *Vital signs which are used to measure the body's basic functions.* E.g. body temperature, heart rate, blood pressure, breathing rate and etc.
- ✓ *Changes in any of the body's vital signs may be indicative of disease.* E.g. having a fever
- ✓ *Observable conditions.* E.g. the presence of antibodies in a patient's serum

## ❖ Symptoms

Symptoms are *subjective*. A change in body function that is *felt or experienced by the patient* as a result of disease.

- ✓ Examples of symptoms include nausea, loss of appetite, and pain.

## ❖ Syndrome

- ✓ A specific group of signs and symptoms that occur together and characterize a particular abnormality or condition.

# Classification of infectious diseases

❑ *Number of criteria can be used to classify infectious diseases*

## 1. Based on transmission potential

### ■ *Communicable disease*

- ✓ a disease that can spread from one host to another. Some communicable disease is **contagious** meaning easily spread; it spreads through direct, bodily contact with an infected person, their discharges, or an object or surface they've contaminated.
- ✓ the degree to which a disease is contagious usually depends on how the pathogen is transmitted

E.g. Measles, Influenza, Salmonella and etc.

### ■ *Non-communicable disease*

- ✓ **Aren't caused by microorganisms**, but by genetics, anatomical differences, getting older and the environment you live in.
- ✓ a disease that is not transmitted from one host to another

E.g. Cancer, diabetes, congestive heart failure and Alzheimer's disease

### 2. Based on occurrence/based on a disease's rate of spread

A variety of terms are used to describe the occurrence of an infectious disease within a specific geographic area or population.

#### ■ Sporadic disease

- ✓ disease that occurs occasionally and unpredictably in a population

#### ■ Endemic disease

- ✓ when disease is consistently present but limited to a particular region
- ✓ *E.g. Malaria considered endemic in certain countries and regions.*

#### ■ Epidemic disease

- ✓ an unexpected increase in the number of disease cases in a specific geographical area.
- ✓ disease acquired by many hosts in a given area in a short time

#### ■ Pandemic disease

- ✓ when a disease's growth is exponential, affecting several countries and populations; worldwide epidemic



# Measures of disease frequency: prevalence and incidence

## ❖ Prevalence

- fraction of a population *having a specific disease* at a given time

✓ number of existing cases

$$\text{Prevalence} = \frac{\text{number of people in sample with characteristics}}{\text{total number of people in the sample}} \times 100$$

E.g. In a population of 10,000 people, 500 persons are reported to be affected by a certain disease. So what is the prevalence of this disease in this population?

## ❖ Incidence

- fraction of a population that contracts a disease during a specific time

✓ number of new cases of a disease or disorder

$$\text{Incidence} = \frac{\text{new cases}}{\text{total population}}$$

## Cont'd

### 3. Severity or duration of infectious disease

*The duration of the period of illness can vary greatly depending on the pathogen, effectiveness of the immune response in the host, and any medical treatment received.*

#### ■ Acute

- ✓ *Pathologic changes occur over a relatively short time (e.g., hours, days, or a few weeks) and involve a rapid onset of disease conditions.*

*E.g. Influenza (the incubation period is approximately 1–2 days)*

#### ■ Chronic

- ✓ *Pathologic changes can occur over longer time spans (e.g., months, years, or a lifetime)*

*E.g. chronic gastritis caused by *Helicobacter pylori*.*

## Cont'd

### ■ Subacute

- ✓ symptoms appear between acute and chronic

### ■ Latent

- ✓ The causal *pathogen goes dormant for extended periods of time with no active replication*
- ✓ Period of ***no symptoms*** when the causative agent is inactive

**E.g. *herpes simplex viruses*** evade the host immune system by residing in a latent form within cells of the nervous system for long periods of time,  
but they can ***reactivate to become active infections during times of stress and immunosuppression.***

## Cont'd

### 4. Extent of infection in the host

#### 1. Local infection

- ✓ Pathogens are limited to a small area of the body

#### 2. Systemic infection

- ✓ An infection throughout the body by blood and/or lymph
  - **Bacteremia**: presence of bacteria in the blood
  - **Septicemia**: presence and growth of bacteria in the blood
  - **Toxemia**: toxins in the blood
  - **Viremia**: viruses in the blood

#### 3. Focal infection

- ✓ Systemic infection that began as a local infection and moves via blood or lymph to set up a new infection at another site
  - tooth infection

## Cont'd

### 5. Pathogenicity

**Pathogenicity** is the ability of a microbial agent to cause disease. Pathogens can be classified as either *primary pathogens* or *opportunistic pathogens*.

- **Primary pathogen**

- ✓ A primary pathogen can ***cause disease in a host regardless of the host's resident microbiota*** or **immune system**.
- ✓ Microbe or virus that causes disease in healthy individual

- **Opportunistic pathogen (opportunistic)**

- ✓ Can ***only cause disease in situations that compromise the host's defenses***, such as the body's protective barriers, immune system, or **normal microbiota**.

# Laboratory Diagnosing Infectious Disease

**Diagnosing** is identifying the causative agent of disease. The following are methods of diagnosing infectious disease.

- *Microscopy and in vitro culture*
- *Molecular diagnosis*- molecular techniques such as nucleic acid sequencing and protein analysis by mass spectrometry are rapidly replacing traditional methods
- *Serologic diagnosis*- immunologic techniques are used to detect, identify, and quantitate antigen in clinical samples.
  - ✓ The specificity of the antibody-antigen interaction and the sensitivity of many of the immunologic techniques make them powerful laboratory tools



# Stages of Pathogenesis

To cause disease, a pathogen must successfully achieve four steps or stages of pathogenesis: **exposure (contact), adhesion (colonization), invasion, and infection.**

## ■ **Exposure (contact)**

- ✓ *encountering with a potential pathogen. The food we eat and the objects we handle are all ways that we can come into contact with potential pathogens.*

## ■ **Adhesion (colonization)**

- ✓ *following the initial exposure, the pathogen adheres at the portal of entry using **adhesion factors**.*
- ✓ *E.g. Molecules (either proteins or carbohydrates) called **adhesins** are found on the surface of certain pathogens and bind to specific receptors (glycoproteins) on host cells.*
- ✓ *Adhesins are present on the fimbriae and flagella of bacteria, the cilia of protozoa, and the capsids or membranes of viruses.*

## Cont'd

### ■ *Invasion*

- ✓ *Once adhesion is successful, invasion can proceed.*
- ✓ *Invasion involves the dissemination of a pathogen throughout local tissues or the body.*
- ✓ *Pathogens may produce exoenzymes or toxins, which serve as virulence factors that allow them to colonize and damage host tissues as they spread deeper into the body.*

### ■ *Infection*

- ✓ *successful multiplication of the pathogen leads to infection.*
- ✓ *Infections can be described as local, focal, or systemic, depending on the extent of the infection.*

# Disease development and stages

- If the immune system is able to fight off the pathogen before it can establish an infection, no signs of disease will occur.
- The infectious disease process is categorized into different stages of infection.
- **When someone contracts an acute infection, they will experience five stages of infection:**

## 1. Incubation period

- ✓ time interval between initial infection and first appearance of signs and symptoms. A person does not show any signs or symptoms of sickness.

## 2. Prodromal period

- ✓ characterized by **appearance of first mild signs and symptoms** that signal onset of some diseases.
- ✓ But the symptoms are mild and not very specific or diagnostic.

## Cont'd

### 3. Period of illness

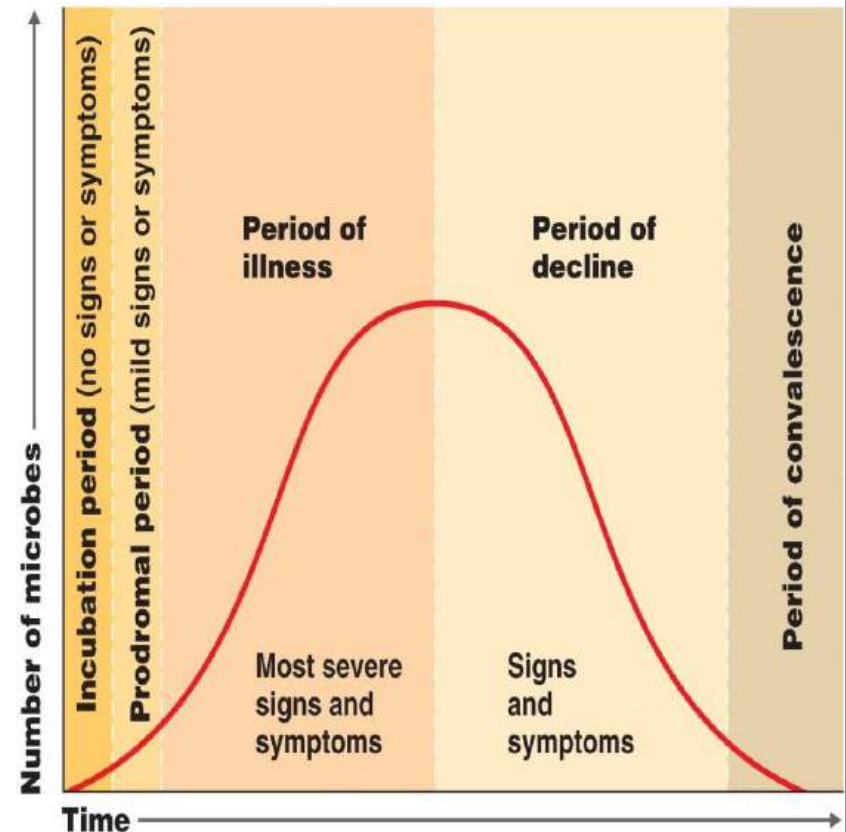
- ✓ **disease at its height**, when it is highly contagious and person experiencing fulminant symptoms.

### 4. Period of decline

- ✓ **signs and symptoms subside**, invading pathogens decrease in the body as the immune system combats the infection.

### 5. Period of convalescence

- ✓ **body returns to pre diseased state**, health is restored, microbial replication has halted by immune system.



## Causative agents of infectious disease

- **Pathogen** is an infectious agent responsible for development of disease.
- The common pathogens known as causative agents for different types of infectious diseases are:
  - ✓ **bacteria, viruses, fungi, protozoa and helminths**

# Bacteria

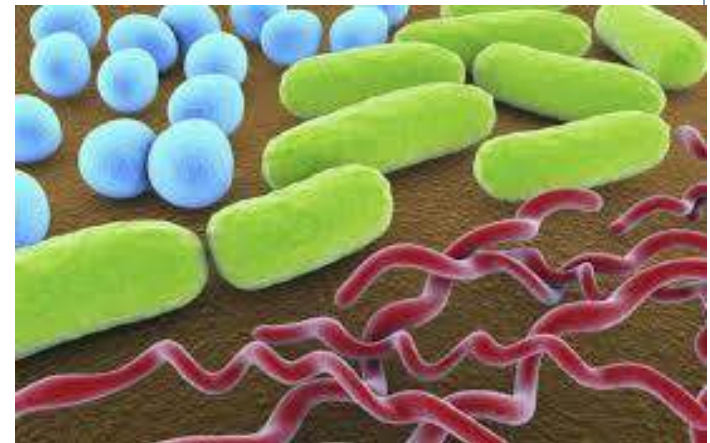
- Are ***unicellular prokaryotic organisms*** reproduce by **binary fission**
- Their cell walls are made from peptidoglycan
- **The three most common shapes of bacteria are:**
- **Bacillus** (rod-shaped), **coccus** (spherical), or **spirillum** (helical rods) Based on their **cell wall structures**, which influences their Gram stain reaction, bacteria can be:

## 1. Gram-negative

- ✓ Appear **pink** after the staining procedure
  - *Salmonella typhi* : causes typhoid fever
  - *Yersinia pestis*: causes plague

## 2. Gram-positive bacteria

- ✓ Appear **purple** after Gram staining procedure
  - *Staphylococcus aureus*
    - causes skin, respiratory and wound infections
  - *Clostridium tetani*
    - produces a toxin that can be lethal for humans





## Bacterial association with the host

- Some of them live inside the human body **without causing harm (normal flora)**. E.g in mouth, intestine
  - ✓ have symbiotic relationship with the host
  - ✓ the **host provides** a place to colonize, nutrients,
  - ✓ while the microbes stimulate innate immunity and prevent colonization with unwanted pathogens
  - ✓ *Not all body organs harbor normal flora*; some are free of any microbes. Eg. Kidney, heart, lung & etc
- Some bacterial cause diseases like:
  - ✓ E.g. cholera, pneumonia, tuberculosis, typhoid, and etc.
  - ✓ Bacterial infections can be treated with **antibiotics**

# Viruses

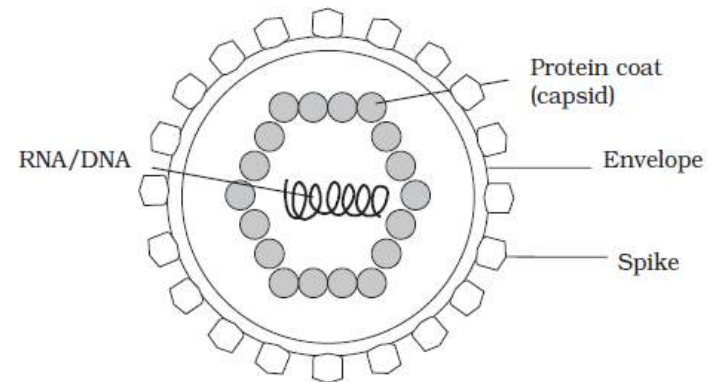
- ✓ **Are not living organisms** because, unless occupying an appropriate living host cell **cannot reproduce**
- ✓ **They have no metabolic capability**
- ✓ All viruses **are obligate intracellular parasites**
- ✓ They contain **either DNA or RNA**, but not both
- ✓ They have no internal cellular structure

## What viruses are composed of?

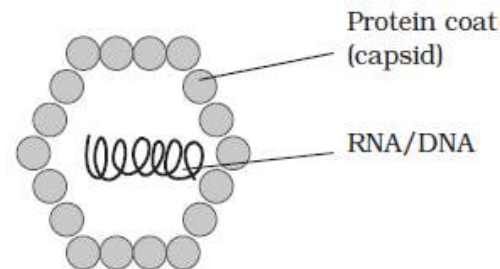
An intact viral particle, or **virion**, has in essence just **two components**: a core of nucleic acid, surrounded and protected by a protein coat or capsid.

- **Enveloped viruses**- composed of nucleic acid DNA or RNA, capsid (protein) and an envelop)

E.g. HIV a retrovirus, Influenza virus,  
SARS-CoV-2 virus

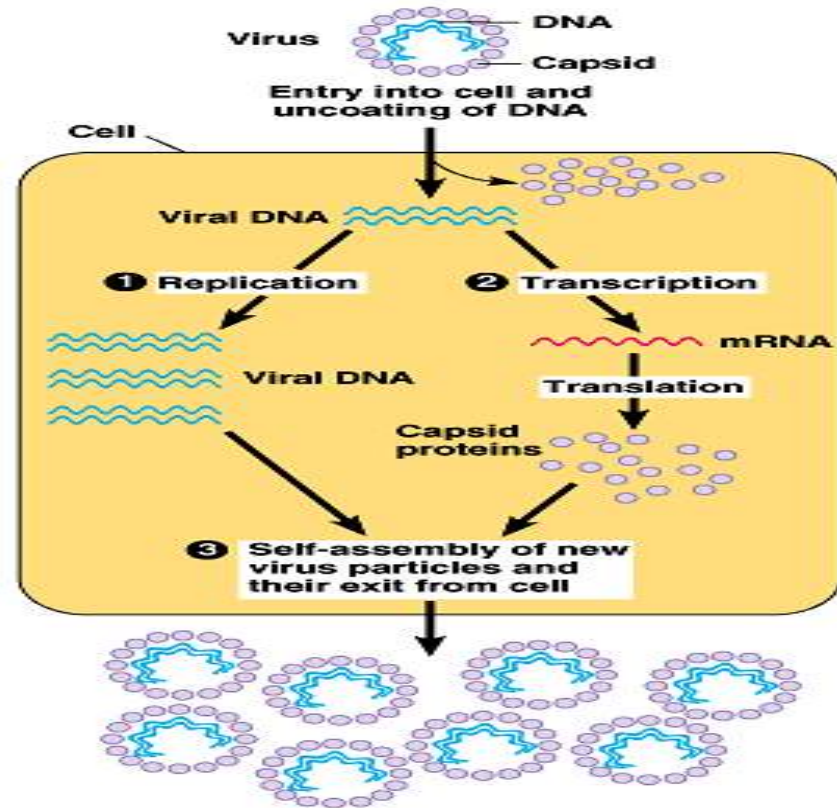


- **Non- enveloped viruses**- composed of nucleic acid DNA or RNA and capsid.



# The general viral process of replication and infection

1. Viral particle first attaches to a specific host cell
2. The viral genome is then inserted into the host cell,
3. Replication, then
4. The new viruses are released from the host cell



❖ **The new viruses can be released through either of the two situations**

1. In some cases, virus-derived enzymes destroy the host cell membranes, **killing the cell and releasing the new virus particles**
2. In other cases, new virus particles exit the cell by a **budding process**, weakening but not destroying the cell

## Cont'd

**Table** Major groupings of viruses based on the Baltimore system

|           |                                                     |
|-----------|-----------------------------------------------------|
| Group I   | dsDNA viruses                                       |
| Group II  | ssDNA viruses                                       |
| Group III | dsRNA viruses                                       |
| Group IV  | (+) sense ssRNA viruses                             |
| Group V   | (-) sense ssRNA viruses                             |
| Group VI  | Single-stranded (+) sense RNA with DNA intermediate |
| Group VII | Double-stranded DNA with RNA intermediate           |

- **Single-stranded RNA viral genomes can be divided into two types: (+) sense and (-) sense RNA.**
  - ✓ (+) sense is able to act as mRNA, attach to ribosomes and become translated into the relevant proteins within the host cell. As such, it is infectious in its own right.
  - ✓ (-) sense RNA is only infectious in the presence of a capsid protein possessing RNA polymerase activity. This is needed to convert the (-) RNA into its complementary (+) strand, which then acts as a template for protein production, as described above.

## RNA viruses

- Genetic material can be used directly as messenger RNA to produce viral proteins
- RNA viruses that cause human disease include
  - ✓ rhinoviruses: cause most common colds
  - ✓ myxoviruses and paramyxoviruses that cause influenza, measles, and mumps
  - ✓ rotaviruses: cause gastroenteritis
- **Retroviruses**, use a unique enzyme called *reverse transcriptase* to copy the RNA genome into DNA
  - ✓ This DNA then integrates itself into the host cell genomeE.g. the human immunodeficiency virus (HIV)



## Cont'd

- **Viruses cause disease by disrupting normal cell function.** They do this in a variety of ways
  1. **Some viruses make repressor proteins** that stop the synthesis of the host cell's proteins, RNA and DNA
  2. **Some viruses change the function of the host cell**
    - ✓ E.g. human papillomavirus can lead to cancer by forcing cells to replicate in an uncontrolled way
  3. Viral activity may **weakens cell membranes** and **lysosomal membranes**, leading to **cell autolysis**
- Antiviral medications help in some cases
  - ✓ They can **either prevent the virus from reproducing or boost the host's immune system**

# Fungi

- Fungi are **eukaryotic**, **heterotrophic organisms** that have cell walls made from **chitin**
- Reproduce primarily by forming spores
- Most fungi are **multicellular** (such as moulds have long, branched, threadlike filaments called **hyphae**, which aggregate together to form a **tangled mycelium**), although some, such as **yeasts**, are **non-flamentous unicellular**
- Most fungi are **saprobic**- they obtain their nutrients from decaying matter by secreting enzymes extracellularly to break down complex molecules

## Fungi cont'd

- **They are of great importance economically and may have beneficial effects** (e.g. the production of bread and alcohol, cheese making process, antibiotic production & etc) or
- ***Detrimental effects*** (e.g. destroy materials, some fungi may cause disease; huge damage is caused to crops and other commercially valuable plants, **while a number of human diseases, particularly of the skin and scalp are also caused by fungi.** ).
- Many fungi are **dimorphic**, they exist in **two distinct forms**. Some fungi that cause human infections can change from the yeast form in the human to a mycelial form in the environment in response to changes in nutrients, and environmental factors such as CO<sub>2</sub> concentration and temperature.

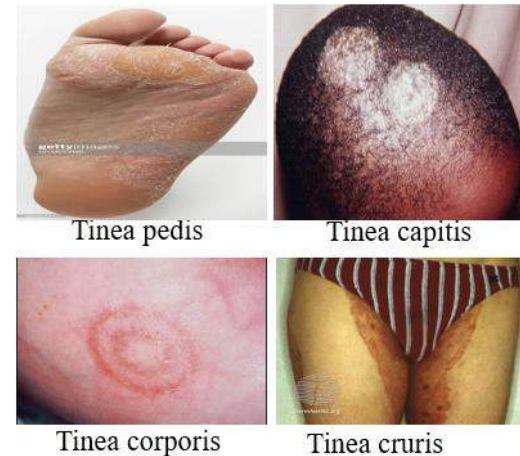
## Fungi cont'd

- A limited number of fungi are pathogenic to humans. **Mycoses** is diseases caused by fungi to humans and animals. Mycoses in humans may be **cutaneous**, or **systemic**.

- **Cutaneous mycoses-** caused by fungi known as *dermatophytes*

- ✓ Utilize the keratin of skin, hair or nails by secreting the enzyme *keratinase*.

E.g. ringworm and athletes' foot



- **Systemic mycoses-** infections deep within the body

- ✓ spores generally enter the body by inhalation, but subsequently spread to other organ systems via the blood, causing serious, even fatal disease.
- ✓ much more serious, and include conditions such as **histoplasmosis** and **blastomycosis**.

# Protozoa

- ✓ **Unicellular, heterotrophic eukaryotes** (typically ingesting articulate food such as bacteria, and digesting them in phagocytic vacuoles) that include amoeba and paramecium
- ✓ Morphologically there are diverse group, lack cell walls
- ✓ Majority of them are free-living, the group also includes commensal forms and some parasites of animals and humans.
- ✓ Have **contractile vacuole**- *to pump out excess amounts of water that enter the cell by osmosis*
- ✓ *The structural features used to achieve locomotion (e.g. cilia, flagella) are among the characteristics used to classify the protozoans.*
- ✓ Can be acquired through
  1. Contaminated food or water
  2. Bite of an infected arthropod E.g. Malaria

## Group of protozoans are

- **Zooflagellates**- characterized by the long flagellum. May be free-living, symbiotic or parasitic.
  - ✓ **Parasitic zooflagellates (E.g. *Trypanosoma brucei*- the causative agent of African sleeping sickness in humans; *Giardia lamblia*, the causative agent of the intestinal disease giardiasis).**
- **The ciliates (Ciliophora)**- Most are free-living types such as paramecium
  - ✓ The only ciliate known to cause disease in humans is ***Balantidium coli***, which causes a form of dysentery.
- **The amoebas (Sarcodina)**- characterized by the possession of pseudopodia
  - ✓ **Most amoebas are free-living in aquatic environments** however, are some important parasites, including ***Entamoeba histolytica***, which causes amoebic dysentery in human.
- **The sporozoans (Apicomplexa)**- characterized by a spore-like stage called a **sporozoite**
  - ✓ Are all parasitic, infecting a range of vertebrates and invertebrates.
  - ✓ E.g. ***Plasmodium***, the causative agent of malaria



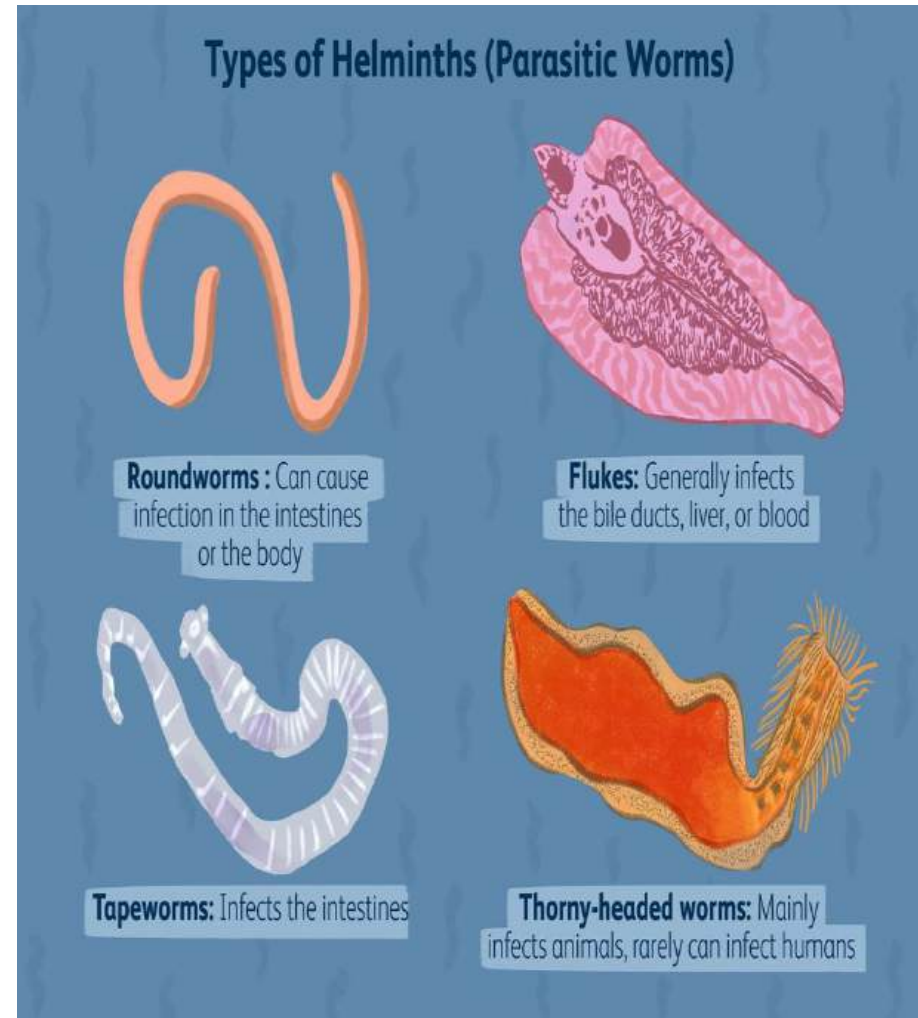
# Helminths

- Helminths are worm-like parasites
- Simple, *invertebrate characterized by elongated, flat or round body*. Some of them are **infectious parasites**
- They are multicellular and have differentiated tissues
- Human helminthic infections can be acquired through consumption of food products from infected animals and plants, drinking contaminated water and etc,
- Their physiology is similar in some ways to human
  - ✓ This makes parasitic helminth infections difficult to treat because drugs that kill helminths are frequently very toxic to human cells
- Many helminths have complex reproductive cycles that include multiple stages, many or all of which require a host
- The common helminthes are ascaries, tape worm, hook worm, schistosoma & etc.

## Helminths cont'd

Classification is based on the external and internal morphology of egg, larval and adult stages.

- **Flukes (Trematodes)**- adult flukes are leaf-shaped flatworms. Are hermaphroditic except for blood flukes, which are bisexual. The life-cycle includes a snail intermediate host.
- **Tapeworms (Cestodes)**-adult tapeworms are elongated, segmented, hermaphroditic flatworms that inhabit the intestinal lumen.
- **Roundworms (Nematodes)**- adult and larval roundworms are bisexual, cylindrical worms. They inhabit intestinal and extra-intestinal sites.



# Prions

- **Prion** is protein that contains no genetic material
  - ✓ do not replicate or feed on the host but trigger abnormal behavior in the body's cells and proteins
- Cause disease in humans and animals by triggering normally healthy proteins in the brain to fold abnormally.
- Once a misfolded prion enters a healthy person potentially by eating infected food, it converts correctly folded proteins into the disease associated form.

## Cont'd

- The known prion diseases include
  - ✓ **Creutzfeldt-Jakob disease** (in humans)- fatal neurodegenerative disease of the brain
  - ✓ **scrapie** (in sheep) and
  - ✓ **bovine spongiform encephalopathy** ("mad cow disease" in cattle)-neurodegenerative disease
- Mode of transmission can be:
  1. inherited
  2. infection from
    - ✓ eating infected tissue or
    - ✓ medical procedures such as tissue transplants

# Modes of transmission

- Communicable diseases are transmitted from source of infection to susceptible individual in many different ways depending on infectious agent, portals of entry, and local ecological condition.
- Mode of transmission may be classified as:

| Direct transmission          | Indirect transmission                     |
|------------------------------|-------------------------------------------|
| ■ Direct contact             | ■ Vehicle borne                           |
| ■ Droplet infection          | ■ Vector borne ( mechanical & biological) |
| ■ Contact with soil          | ■ Air borne                               |
| ■ Inoculation into skin      | ■ Fomite borne                            |
| ■ Trans-placental (vertical) | ■ Unclean hands and fingers               |

## Direct transmission

- **Direct contact-** transmitted direct contact from skin to skin and etc.  
**E.g. AIDS, leprosy, skin infection**
- **Droplet infection-** direct projection of a spray of droplets of saliva and nasopharyngeal secretion during coughing, sneezing...  
**E.g. common cold, TB, whooping cough**
- **Contact with soil-** direct exposure of susceptible tissue to the disease agent in the soil.  
**E.g. hookworm larvae, tetanus**
- **Inoculation into the skin or mucosa-** disease agent may be inoculated directly into the skin. **E.g. Rabies**
- **Transplacental or vertical**



Transmission of disease from parent to child during reproduction and fetal development

**E.g. herpes encephalitis (which occurs when an infant contracts the herpes simplex type II virus during birth)**



## Indirect transmission

Occurs when a pathogen can withstand the environment outside its host for a long period of time before infecting another individual.

- **Vehicle borne-** transmission through water, food, blood, plasma...  
E.g. water & food borne- typhoid fever, cholera, hepatitis A
- **Vector borne-** (mechanical vector and biological vector)
- **Air borne-** float in the air and disseminated by air currents from the point of their origin.
- **Fomite borne-** inanimate substances such as clothes, towels, books other than water or food.
- **Unclean hands and fingers-** common medium by which pathogens are transferred to food from d/t sources

## What are the parts of immune system?

- ❑ **Our immune system is made of up a *complex collection of cells and organs.***
  - **White blood cells**
  - **Lymph nodes:-** contain immune cells that analyze the foreign invaders brought into body,
  - **Spleen:-**store white blood cells,
  - **Tonsils and adenoids:-** can trap foreign invaders as soon as they enter the body,
  - **Thymus:-** helps mature a certain type of white blood cell,
  - **Bone marrow:-** makes new blood cells and releases them into blood stream,
  - **Skin, mucous membranes and other first-line defenses**
  - **Stomach and bowel:-** stomach acid kills many bacteria soon after they enter your body.

# Host defenses against infectious diseases

## What is the Immune System?

- Immune System is a *large network of organs, white blood cells, proteins (antibodies) and chemicals* which works together to protect us from **foreign invaders**.
- Set of *cellular* and *humoral* components
  - **Function:** to keep germs out of our body, destroy them or limit the extent of their harm if they get in.
  - Immunity is the ability to ward off damage or disease
- Immune system has two lines of defense against pathogens
  - ✓ **Innate** and
  - ✓ **Adaptive**, both divided into **cell mediated defense** and **humoral defense factors**
- ❖ These two systems work together in order to destroy invaders or to trigger defense processes

## Cont'd

- ❑ Since we live in microbial world, our bodies are constantly being exposed microbes and must restrict the normal flora from entering into sterile tissue sites.
- **The initial defense mechanisms** are barriers such as the **skin, acid** and bile of the gastrointestinal tract, and mucus that inactivate and prevent entry of the foreign agents
- If these barriers are **compromised** or **the agent gains entry in another way**, then the innate responses must quickly rally to the challenge and prevent expansion of the invasion.
- The innate responses then activate a major campaign ***specifically directed against the invader*** by **antigen specific immune responses** (B cells, antibody, and T cells).

## Innate Immune system

- Comprises **cells** and **mechanisms** that defend the host from infection by other organisms in a **non-specific manner**
  - ✓ designed to **prevent microbes from entering the body** and **to help eliminate those that do gain access**
- Have the following characteristics
  - ✓ **Faster**
  - ✓ **Non-specific**
  - ✓ **No memory component developed**/retains no memory of previous infection
  - ✓ Duration of the immune **response is over a short period**

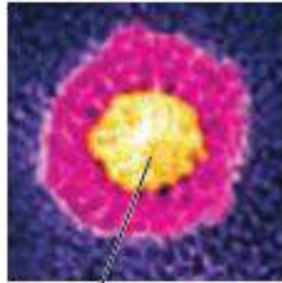
cont'

## How do immune cells recognize pathogens?

**Innate recognition** Each of a small set of receptors recognizes a molecule absent from animals, but common to a type of pathogen, as shown by these examples.

*Pathogen*

Virus



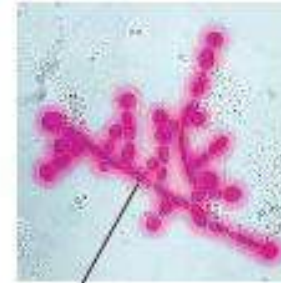
dsRNA (nucleic acid in genome)

Bacterium



Flagellin (protein in flagella)

Fungus



Mannan (oligosaccharide in cell wall)

*Example of  
recognized  
molecule*

➤ Innate immune system commonly divided into four compartments

- 1. Mechanical barriers**
- 2. Humoral parameters**
- 3. Cellular components**
- 4. Fever and inflammation**



# 1. Mechanical Barriers

## ■ Skin and mucous membranes

- ✓ provide both physical and chemical barriers **that discourage pathogens from penetrating the body** and **causing disease**
- ❖ Skin- epidermis of skin provides a formidable physical barrier to the entrance of microbes
- ❖ Mucous membranes- the epithelial layer of mucous membranes secretes a fluid called **mucus** which traps many microbes and foreign substances
  - ✓ the mucous membrane of nose has mucus-coated hairs
  - ✓ upper respiratory tract contains **cilia** to **trap** and **filter microbes, dust, and pollutants** from inhaled air
  - ✓ coughing and sneezing accelerate movement of mucus and its entrapped pathogens out of the body

## 2. Humoral component (chemical barriers)

Contains a variety of **proteins capable of destroying or inhibiting growth of infectious microorganisms**, which include:

- **Lysozyme**
  - ✓ Acts on the peptidoglycan of pathogens cell wall
- **Complement system proteins**
  - ✓ Comprised of soluble and membrane proteins that destroy pathogens through membrane injuries
- **Interferons**- interfere with viral replication
- **Anti-proteases**- against microorganism proteolytic proteins
- **C-reactive protein**
  - ✓ promoting pathogen opsonization, complement activation and phagocytosis

### 3. Cellular components

- **Phagocytes**- specialized cells that perform phagocytosis
  - ✓ The two major types of **phagocytic cells** in mammalian body
  - I. Neutrophils** (circulate in the blood, are attracted by signals from infected tissues and then ***engulf and destroy the infecting pathogens***) and
  - II. Macrophages** (“*big eaters*”)-phagocytosing foreign antigens such as bacteria.
- In addition to **these cells**, a number of more specialized cell types also contribute to **innate immune defenses**:
  - **Dendritic cells** mainly populate tissues, such as skin, that contact the environment.
    - ✓ They stimulate adaptive immunity against pathogens that they engulf.

## Cellular components cont'd

- **Eosinophils**- often found beneath an epithelium, are important in defending against multicellular invaders, such as parasitic worms.
  - ✓ Upon encountering such parasites, **eosinophils** discharge destructive enzymes.
- **Natural killer cells**- circulate through the body, detect the abnormal surface proteins found on some virus-infected and cancerous cells.
  - ✓ **Do not engulf** stricken cells. Release chemicals that lead to cell death, inhibiting spread of the virus or cancer.
  - ✓ Immune cells present in blood, spleen, lymph nodes, and bone marrow
- **Mast cells**- are found in connective tissue and make key contributions to the inflammatory response as well as to allergies.

## 4. Fever and Inflammation

- **Fever** is an inflammatory response results in overall increase in body temperature.
  - ✓ Enhances innate immune defenses by stimulates leukocytes and rise in body temperature may inhibit growth of many pathogens.
  - ✓ It also reported that fever may stimulate the release of iron sequestering compound from the liver thereby starving out microbes that rely on iron for growth.

## Cont'd

- **Inflammation-** a local inflammatory response begins when activated macrophages discharge **cytokines**, **signaling molecules that recruit neutrophils to the site of injury or infection**.
- ✓ In addition, **mast cells** release the signaling molecule **histamine** at sites of damage.
- ✓ Histamine triggers **blood vessels to dilate and become more permeable. The resulting increase in local blood supply produces the redness** and increased skin temperature typical of the inflammatory response.
- ✓ During an inflammation dilation of blood vessels occurs to:
  - increase blood volume in affected areas
  - allow invasion of white blood cells into the affected area
- ✓ Characterized by redness, swelling, fever, and pain



# Adaptive immunity

- When **non-specific** mechanisms fail, the body initiates a second, **specific line of defense**.
- The **adaptive immune response**, also known as the **acquired immune response**, is activated after the innate immune response and **develops more slowly**
- **Have the following characteristics**
  - ✓ **Specific**
    - for particular foreign molecules (antigens)
    - distinguish self from non-self molecules
  - ✓ **Memory component developed**
  - ✓ **Duration of the immune response is over a long period (some times lifetime)**

**Adaptive recognition** Each of a vast number of receptors is specific for a particular part of a protein in one pathogen, such as a surface protein of the influenza (flu) virus. In this example, a different adaptive immune receptor would recognize each of the flu strains based on the proteins' amino acid sequences, which differ at the positions colored orange.



## Cont'd

- Adaptive immunity relies on **T cells** and **B cells**, which are types of **white blood cells** called **lymphocytes**

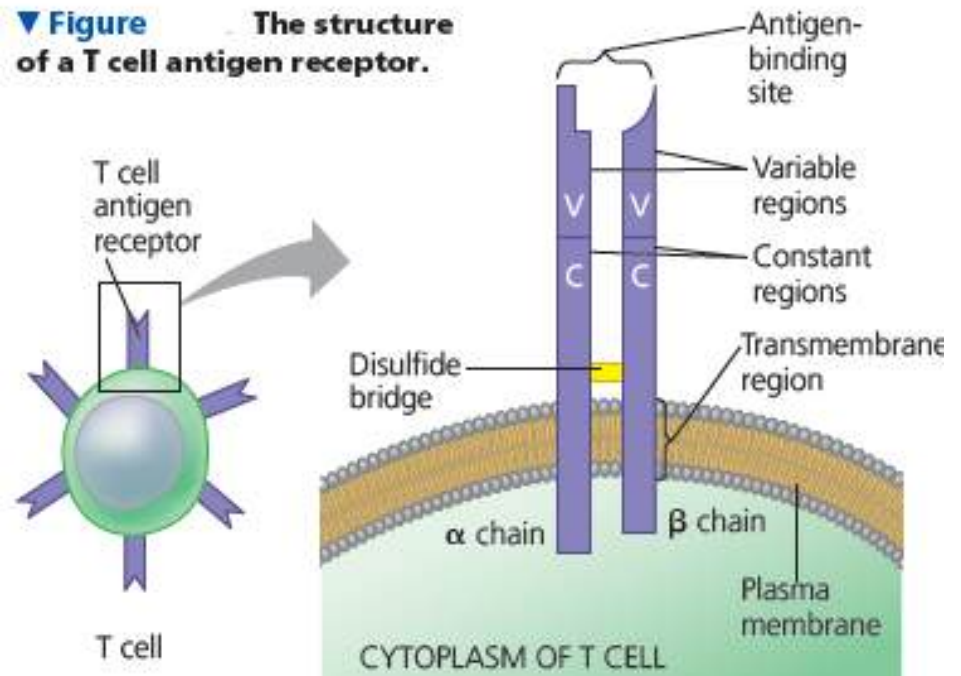
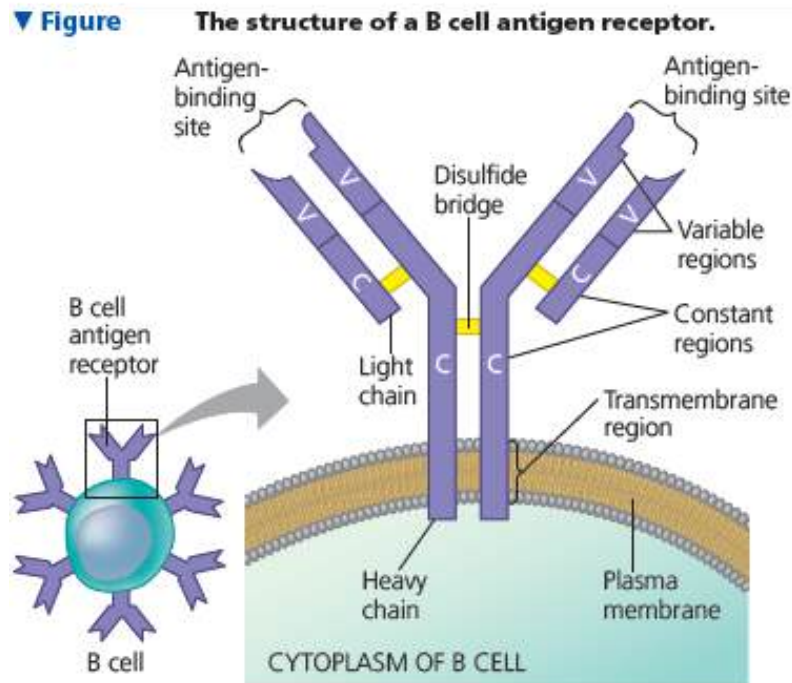
Comparison of B and T Cells

| Property                       | T Cells                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | B Cells                                                                                                                                                                                                                                                                                    |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Origin                         | Bone marrow                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Bone marrow                                                                                                                                                                                                                                                                                |
| Maturation                     | Thymus                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Bone marrow, Peyer patches                                                                                                                                                                                                                                                                 |
| Functions                      | Varied (see section Subsets)                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Antibody production</b><br>Antigen presentation to T cells                                                                                                                                                                                                                              |
| Protective response            | Enhance and control innate and immune responses; resolution of intracellular and fungal infections                                                                                                                                                                                                                                                                                                                                                                                   | Antibody protects against rechallenge; blocks spread of agent in blood, opsonize, etc.                                                                                                                                                                                                     |
| Products <sup>a</sup>          | Cytokines, growth factors, cytolytic substances (perforin, granzymes)                                                                                                                                                                                                                                                                                                                                                                                                                | IgM, IgD, IgG, IgA, or IgE                                                                                                                                                                                                                                                                 |
| Distinguishing surface markers | CD2 (sheep red blood cell receptor), TCR, CD3                                                                                                                                                                                                                                                                                                                                                                                                                                        | Surface antibody, complement receptors, class II MHC antigens                                                                                                                                                                                                                              |
| Subsets                        | CD4 TH0: helper precursor<br>CD4 TH1: activates B-, T-, and NK-cell growth; activates macrophages, CTLs, DTH responses, and IgG production<br>CD4 TH2: activates B- and T-cell growth; promotes IgG, IgE, and IgA production<br>CD4 TH17: antibacterial, inflammation<br>CD4 Treg, Tr1: suppression and regulation<br>CD8: cytotoxic T cells (CTL), cytokine production<br>NKT, MAIT, $\gamma\delta$ T; rapid response to infection<br>Memory cells: long-lived, anamnestic response | B cells (IgM, IgD): antibody, antigen presentation<br>B cells (IgG or IgE or IgA): antibody, antigen presentation<br>B-1, Marginal zone B cells: natural antibody production<br>Plasma cell: terminally differentiated antibody factories<br>Memory cells: long-lived, anamnestic response |

<sup>a</sup>Depending on subset.

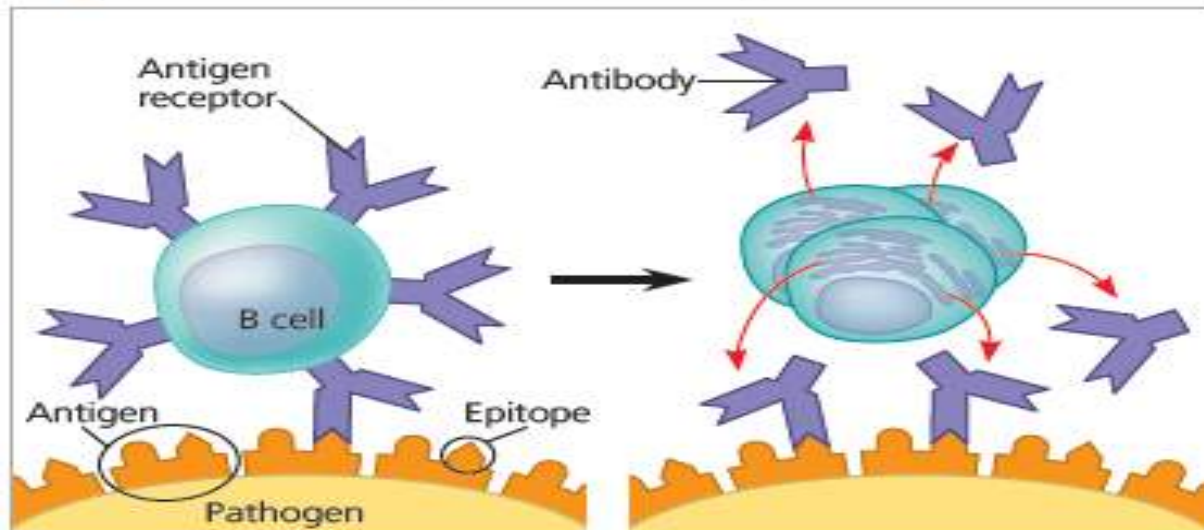
CD, Cluster of differentiation; CTL, cytotoxic lymphocyte; DTH, delayed-type hypersensitivity; Ig, immunoglobulin; MAIT, Mucosal associated invariant T cells; MHC, major histocompatibility complex; NKT, natural killer T (cell); TCR, T-cell receptor; TH, T helper (cell); Treg, regulatory T cell; Tr1, Type 1 regulatory cell.

- Any substance that elicits a B or T cell response is an antigen.
- Recognition occurs when a **B cell or T cell binds to an antigen**, such as a bacterial or viral protein, via a protein called an antigen receptor.
- Each antigen receptor binds to just one part of one molecule from a particular pathogen, such as a species of bacteria or strain of virus.

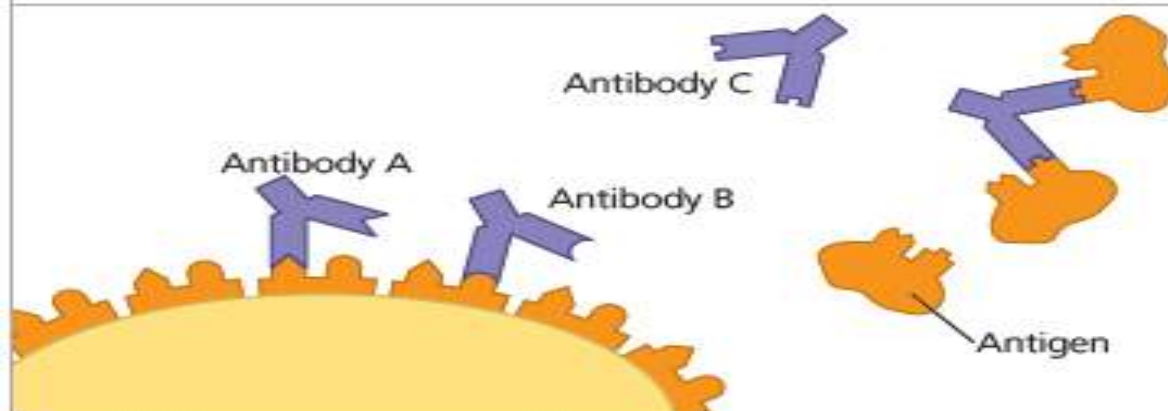


▼ **Figure**

**Antigen recognition by B cells and antibodies.**



**(a) B cell antigen receptors and antibodies.** An antigen receptor of a B cell binds to an epitope, a particular part of an antigen. Following binding, the B cell gives rise to cells that secrete an antibody: a soluble form of the receptor that is specific for the same epitope as the original antigen receptor.



**(b) Antigen receptor specificity.** Different antibodies can recognize distinct epitopes on the same antigen. Furthermore, antibodies can recognize free antigens as well as antigens on a pathogen's surface.

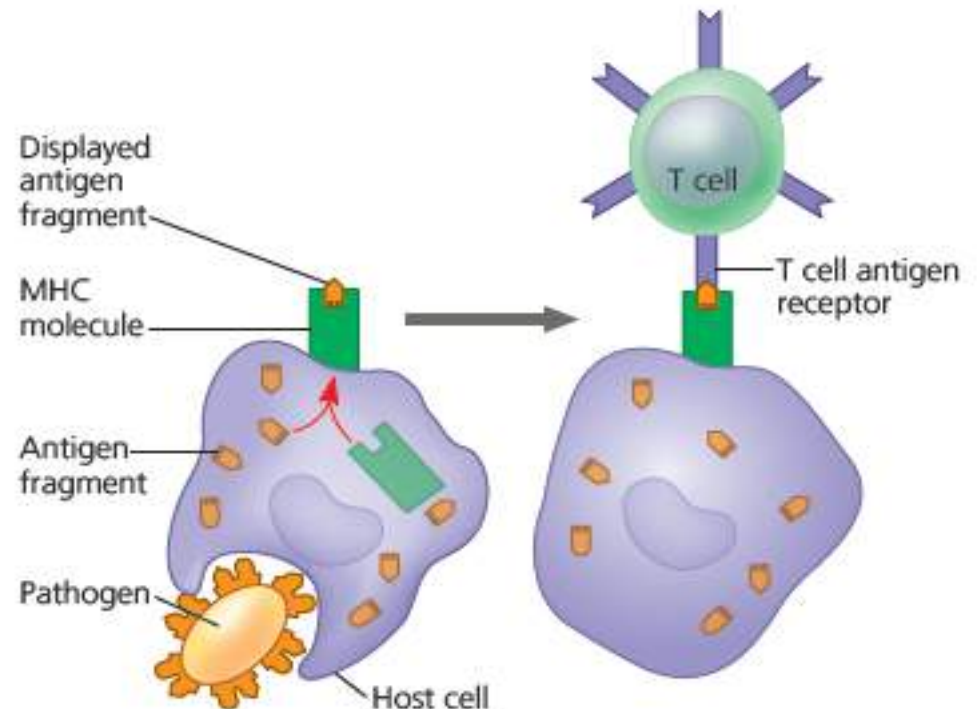


- The host protein that displays the antigen fragment on the cell surface is called a **major histocompatibility complex (MHC) molecule**. By displaying antigen fragments, MHC molecules are essential for antigen recognition by **T cells**.

▼ **Figure**      **Antigen recognition by T cells.** Inside the host cell, an antigen fragment from a pathogen binds to an MHC molecule and is brought up to the cell surface, where it is displayed. The combination of MHC molecule and antigen fragment is recognized by a T cell.

➤ There are **two major types of mature T-cells** that exit the thymus

- ✓ **Helper T cells (CD4)**- help the activation of cytotoxic T cells, B cells and other immune cells
- ✓ **Cytotoxic T cells (CD8)**- responsible for removing pathogens and induce cell death (kill cell harboring pathogens)



# Components of adaptive immunity

- The defenses provided by **B and T lymphocytes** can be divided into **humoral and cell-mediated immune responses**.

## 1. Humoral immune response

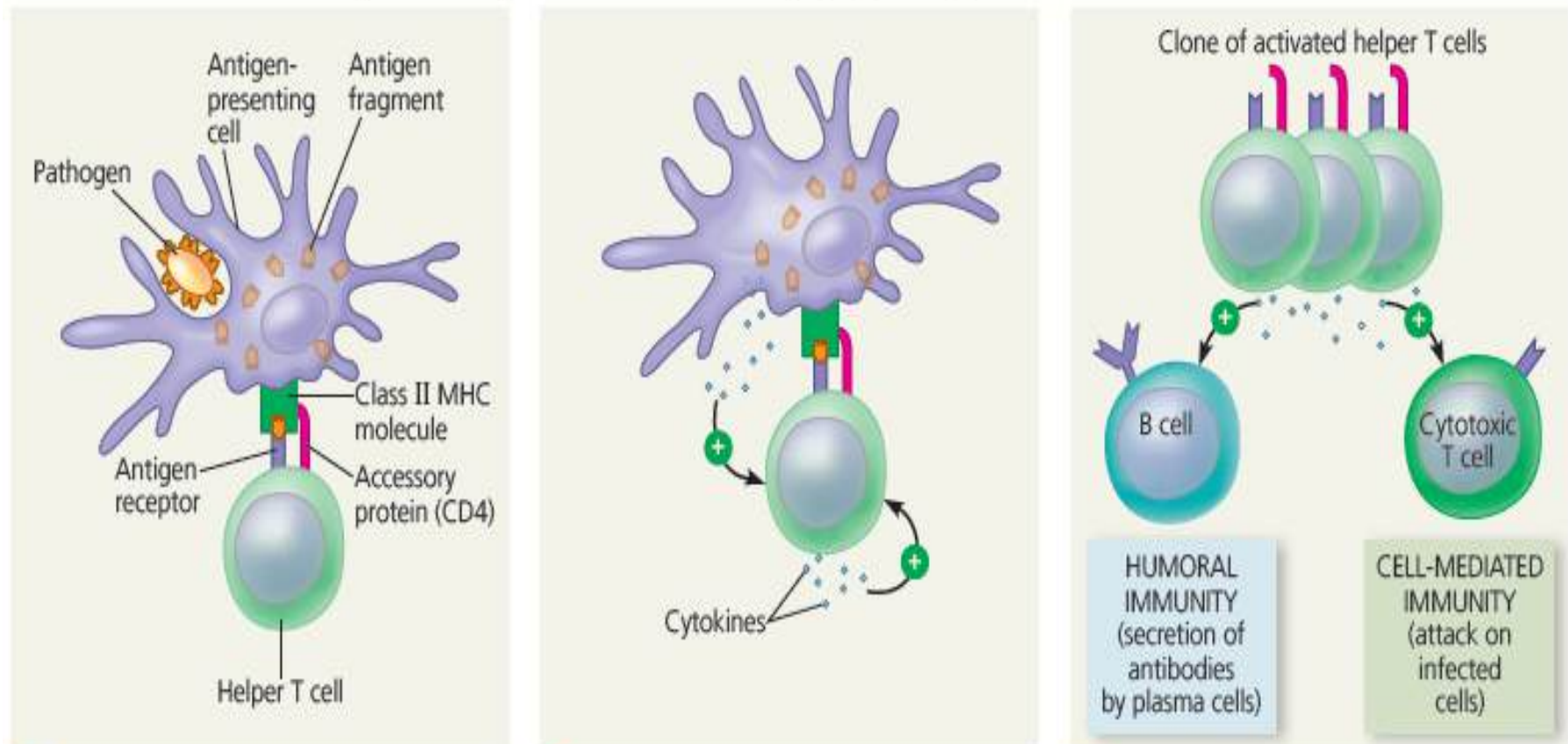
- ✓ protects the blood and lymph (once called body humors, or fluids).
- ✓ Primarily driven by **B cells**
- ✓ In this response, **antibodies help neutralize or eliminate toxins and pathogens in body fluids**.

## 2. Cell-mediated immune response

- ✓ Specialized **T cells destroy infected** host cells.
- ✓ Unlike humoral immunity, **does not depend on antibody**.
- ✓ Both humoral and cellular immunity can include a primary and a secondary immune response, with memory cells enabling the secondary response



▼ **Figure**      **The central role of helper T cells in humoral and cell-mediated immune responses.** Here, a helper T cell responds to a dendritic cell displaying an antigen.



1 An antigen-presenting cell engulfs a pathogen, degrades it, and displays antigen fragments complexed with class II MHC molecules on the cell surface. A specific helper T cell binds to this complex via its antigen receptor and an accessory protein called CD4.

2 Binding of the helper T cell promotes secretion of cytokines by the antigen-presenting dendritic cell. These cytokines, along with cytokines from the helper T cell itself, activate the helper T cell and stimulate its proliferation.

3 Cell proliferation produces a clone of activated helper T cells. All cells in the clone have receptors for the same antigen. These cells secrete other cytokines, which help activate B cells and cytotoxic T cells with the same antigen specificity.

## Cont'd

- Both **the cell-mediated** and **humoral mediated responses** are initiated after **macrophage engulfs** a pathogen and then **display antigens from the pathogen on their surface**.
  - ✓ This display helps the macrophages stimulate specific **helper T-cells** to release signal molecules called **lymphokines**
  - ❖ **Lymphokines** are cytokines produced by T cells
- The **lymphokines**, in turn, stimulate the **cell-mediated** and **antibody-mediated responses**

## Cont'd

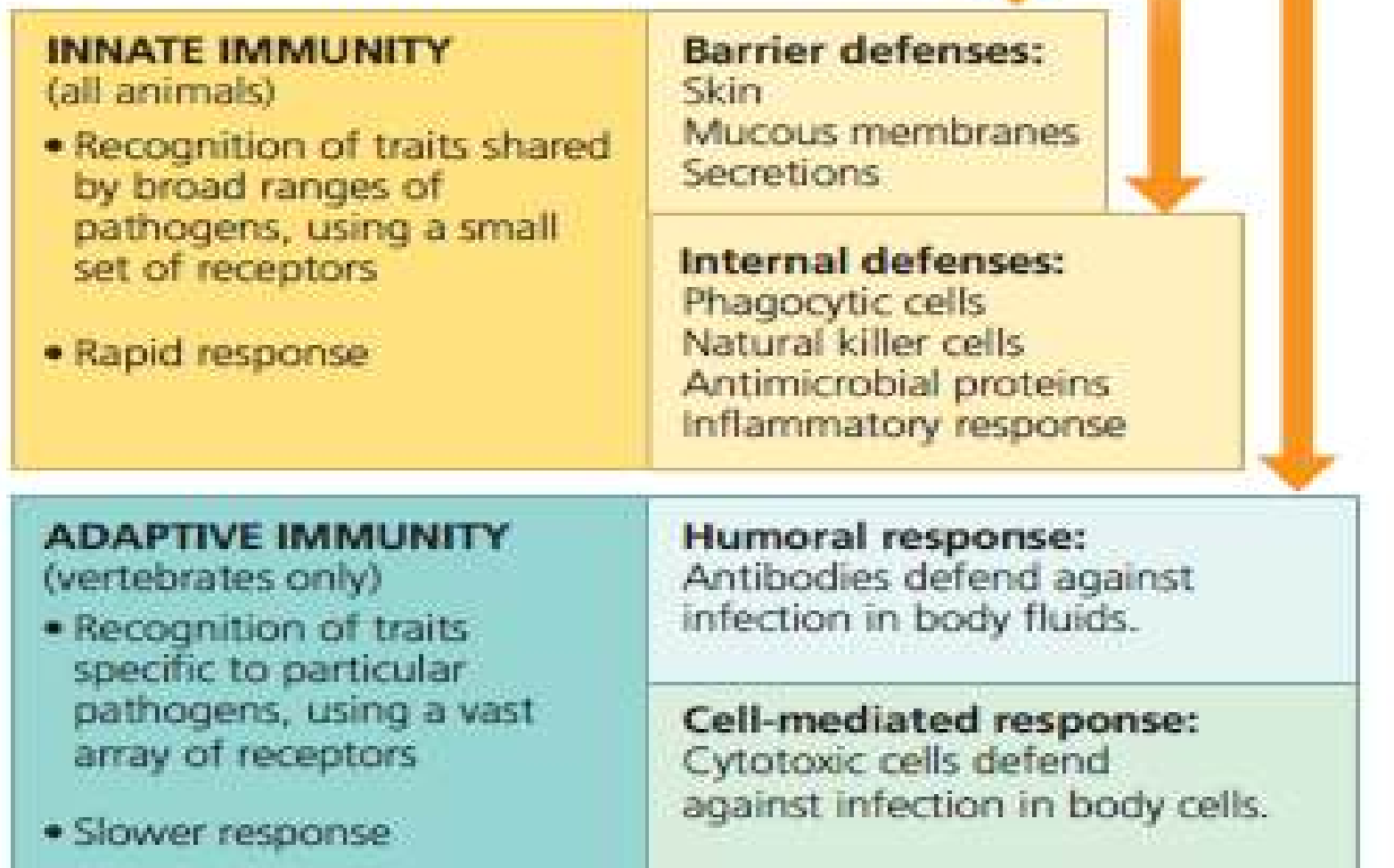
- The **humoral/antibody mediated** response occurs **when the lymphokines** activate **specific B-cells** to produce antibodies
  - ✓ These antibodies attach to antigens on the surface of the pathogens and signal attack by phagocytic cells and complement system.
  - ✓ Other **B-cells** go on to become memory B-cells, which respond quickly by producing more antibodies upon subsequent infection.

- **Both innate immunity and adaptive immunity work together**
  - ✓ through direct contact with each other, and through interactions with chemical mediators, the **cytokines**.
  - ✓ **cytokines** *are hormone-like proteins that act on cells to activate and regulate the innate immune response*

### **Examples**

- **Macrophages** are responsible to **phagocytosis the infectious microbes** and at the **same time produce important cytokines** that help to induce the adaptive immune response.
- **Complement components** of the innate immune system can be activated directly by microbes, but can also be activated by antibodies, molecules of the adaptive system.

**Overview of animal immunity.** Innate immunity offers a primary defense in all animals and sets the stage for adaptive immunity in vertebrates.



# Immunological disorders

- The aim of the immune responses is **to protect the body from damage** that may be caused by any environmental agents including pathogens.
- However, sometimes adverse immunological reactions can occur and cause adverse effects to the host due to:
  - ✓ exaggerated immune response and/or producing inappropriate immune response against self-components
  - ✓ failure of appropriate recognition mechanism
- In general, there are three types of immunological disorders
  1. Hypersensitivity
  2. Autoimmune diseases
  3. Immunodeficiency



# Hypersensitivity reactions

- **Hypersensitivity** is an immunological state in which the **immune system “over-reacts” to foreign antigen** such that the immune response itself is more harmful than the antigen.
  - ✓ causes inappropriate and/or excessive tissue damage
- Hypersensitivity reactions are responsible for many of the symptoms associated with microbial infections.
- **All types of hypersensitivity involves:**
  - ✓ the adaptive immune response (highly specific reactions via T or B cells)
- Prior exposure to the antigen, the initial exposure sensitizes the individual but does not cause a hypersensitive reaction
  - ✓ hypersensitivity is only seen on secondary exposure

- **Hypersensitivity reactions** occur to people with an established immunity to the antigen. The mediator and the time course primarily distinguish the four types of hypersensitivity responses

### Type I:

- ✓ is caused by IgE and is associated with allergic, atopic, and anaphylactic reactions)
- ✓ rapid-onset reactions (2-30 minutes)

### **Type II: cytotoxic reactions**

- ✓ cell damage due to complement activation via IgM or IgG

### **Type III: immune complex reactions**

- ✓ responses result from activation of complement by immune complexes
- ✓ cell damage due to excess antibody/antigen complexes

### **Type IV: delayed cell-mediated reactions**

- ✓ T cells & macrophages mediated inflammatory response
- ❖ Types I-III are all antibody-mediated, Type IV is cell-mediated responses.

## Hypersensitivity Reactions

| Reaction Type | Onset Time                          | Key Features                                                                                                 | Beneficial Effects                                                                     | Pathologic Effects                                                                                                                                                |
|---------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type I        | <30 min                             | Soluble antigen-triggered, IgE-dependent release of vasoactive mediators followed by late-phase reaction     | Antiparasitic responses and toxin neutralization                                       | Localized allergies (e.g., hay fever, asthma)<br>Systemic anaphylaxis                                                                                             |
| Type II       | <8 hr                               | Cell-bound antibody promoting C'-mediated cytotoxicity; antibody binding and modulation of receptor function | Direct lysis and phagocytosis of extracellular bacteria and other susceptible microbes | Destruction of red blood cells (e.g., transfusion reactions, Rh disease)<br>Organ-specific tissue damage in some autoimmune diseases (e.g., Goodpasture syndrome) |
| Type III      | <8 hr                               | Soluble antigen-antibody complexes activate C'                                                               | Acute inflammatory reaction at site of extracellular microbes and their clearance      | Arthus reaction (localized)<br>Serum sickness and drug reactions (generalized)<br>Systemic autoimmune diseases                                                    |
| Type IV       | 24-72 hr (acute); >1 week (chronic) | Phagocytized protein antigen presented to CD4 T cells activates macrophages and inflammation                 | Protection against infection by fungi, intracellular bacteria, and viruses             | Acute: contact dermatitis, tuberculosis skin test<br>Chronic: granuloma formation                                                                                 |

# Autoimmune disorders

- *A condition that occurs when the immune system mistakenly attacks and destroys healthy body tissue.*
  - ✓ immune system fails to recognize a protein as “self” and launches an attack
- Autoimmune disorders are **chronic** and usually **irreversible** but drugs can help slow the progress of these diseases
- **Examples autoimmune disorders**
  - **Multiple sclerosis-** a disorder in which the body's immune system attacks the protective covering of the nerve cells in the brain, optic nerve...
  - **Lupus** – inflammatory disease affect the joints, skin, kidneys, blood cells, brain, heart and lungs.
  - **Rheumatoid arthritis-** mainly affect joints

# Immunodeficiency

- **Loss or inadequate function of various components of the immune system**
- It may result from **genetic deficiencies, starvation, drug-induced immunosuppression** (e.g., steroid treatment, cancer chemotherapy, chemotherapeutic suppression of tissue graft rejection), cancer (especially of immune cells), or disease (e.g., AIDS) and naturally occurs in neonates and pregnant women.
- Deficiencies in specific protective responses put a patient at high risk for serious disease caused by infectious agent.
- **Can occur in any part of the immune system**
  - ✓ physical barrier, phagocytes, B lymphocytes, T lymphocytes, complement, natural killer cells

# Classified as primary or secondary immunodeficiency

## 1. Primary (congenital) immunodeficiency

- ✓ Genetic abnormality caused by mutations affecting any of a large number of genes that control the expression and activities of immune responses
- ✓ can affect either the lymphoid or the myeloid lineages
- **Defects in lymphoid lineage**
  - ✓ involve B-cell, T-cell or both
    - B-Cell immunodeficiency, T-cell deficiencies
- **Defects in myeloid lineage**
  - ✓ affect innate immunity, especially phagocytic activity.
    - Chronic granulomatous disease (CGD)- genetic disorder in w/c phagocytes are unable to kill certain type of bacteria and fungi
    - Complement deficiencies

## 2. Secondary (Acquired) immunodeficiency

- ✓ results from infections, nutritional deficiencies or treatments
  - E.g. AIDS, chronic leukemia



# Reading Assignment

## Tumor Immunology

- **Tumor immunology** refers to the relationship between immune function and tumor cells
- Immune surveillance is a concept that envisages prevention of the development of the most tumors through early destruction of abnormal cells by the host's immune system.



# **General Biology (Biol1012)**

## **Lecture Slides**

### **Chapter 7. Taxonomy of Organisms**



# Unit 7. Taxonomy of organisms

- ❖ Taxonomy is derived from two Greek words:
  - ✓ "taxis", which means arrangement and
  - ✓ "nomos" which means law
- ❖ Thus, taxonomy means **the law governing the classification or arrangement of organisms into taxa (taxonomic groups)**
- it is mainly concerned with description, classification, nomenclature and identification of organisms.
- ✓ **Description:** assignment of features to an organism
- ✓ **Classification:** ordering of organisms into hierarchy of ranks based on similarities/phylogenetic relationships
- ✓ **Nomenclature:** naming of taxonomic groups/taxa according to some standardized system (i.e. allocation of name of the taxa produced)
- ✓ **Identification:** assignment of an organism or specimen to an existing classification/by reference to an already existent classification



# Early Attempts to Classify Organisms

- ❖ The development of modern taxonomy has not a linear path.
- Naming, describing & classifying living organisms are natural & integral activity of humans. Without such knowledge, it would be difficult to communicate.
- ❖ Phases of developmental stages of taxonomy:

## A. Pre-Linnaean

### i) Ancient taxonomy

- Ancient people have grouped/ categorized living things into categories & gave common names = Folk Taxonomy
- Characterized by **existent & no-existent** of common names
- Important for communication of the people



# Early Attempts to Classify Organisms ...

## ii) Greek and Roman Philosophers=Essentialists

- Tried to classify organisms based on selected/more essential characteristics

### a) Aristotle (384–322 BC)

- ✓ Aristotle viewed life was hierarchical and he classified organisms

- the first to classify living things into plants & animals

❖ **Animals:** into “animals with blood” & “animals without blood”

❖ animals with blood: live-bearing & egg-bearing,

❖ Animals without blood: insects, crustacea and testacea (molluscs)



# Early Attempts to Classify Organisms ...

## b) Theophrastus (370-285 BC)

- ❖ He was the first taxonomist to write down a classification in a logical form
- ❖ Classified of all known plants at that time, in his book “De Historia Plantarum”,
- ❖ He classified 480 kinds of plants based on their habit/growth form
- ✓ Plants were classified as trees, shrubs/herbs
- ✓ However, his work incorporates the principle of the rules of Dichotomy or Excluded Middle
  - According to Theophrastus classification system, *there are no any intermediate organisms*
  - ✓ Any given organisms is either A or not A
  - He was typologists, belief that every natural group of organisms has an invariant, generalized or idealized pattern shared by all members of the group





# Early Attempts to Classify Organisms ...

## c) Pliny the Elder (23-79 A.D.)

- Naturalist who wrote books termed *Historia naturalis*
  - ✓ Categorized plants in terms horticultural practices medicinal uses, and timber use practices
  - ✓ described several plants & gave them Latin names.
  - ✓ Many of these names are stil in use (e.g. *Populus alba* & *Populus nigra*)
  - ✓ Some call him the Father of Botanical Latin.

## d) Dioscorides (40–90 AD)- Greek physician

- ❖ Wrote a book entitled “*De Materia Medica*” ≈ 600 plant species with their useful application (mostly medicinal uses).
- Used in medicine until the 16<sup>th</sup> C
- ✓ classification based on medicinal properties

## e) Caesalpinus (1519-1603)

- In his book “*De Plantis*” he illustrated about 1500 plant species
  - ✓ He classified plants based on definite morphological criteria mainly on the basis of habit, fruits, seeds, and ovaries and ovary positions



# Early Attempts to Classify Organisms ...

## f) Bauhin J. (1554-1631)

- ❖ Recognize genera and species as major taxonomic levels
- ✓ Used a binary nomenclature composed of the genus name followed by a single specific

## B. Linnaean Era (1707-1778)= Sexual system of classification

- ❖ Considered as the father of modern taxonomy
- Briefly described the genera and the species in his books “Genera Plantarum (1737) and Species Plantarum (1753)”
- **Binomial nomenclature:** system of nomenclature in which two terms are used to denote a species name (genus name and specific name /specific epithet.
- ❖ His classification system was artificial-which depends only on few characteristics like stamens and pistil numbers & position.
- Linnaeus developed a hierarchical system of classification



## **C. Post-Linnaean Taxonomy= Natural system of classification**

- ❖ The foundation of modern classification comes mainly after the the works of other taxonomists M. Adanson (1727-1806), De Jussieu (1748-1836), and J. de Lamarck (1744-153 1829), A.P. de Candolle and C. Darwin (1859), who never followed artificial system of Linnaeus classification.
- ❖ **M. Adanson (1727-1806)**
  - Championing the idea that in classification one should use a great range of characteristics without placing greater emphasis on some than on others=**Natural system of classification**
  - Empirical approach. Adanson was a severe critic of Linnaeus'



➤ **Classification system of diversified organisms depends up on the combined evidences:**

- ✓ **Morphology**
- ✓ **Ultrastructure and**
- ✓ **Molecular phylogeny**

## **1. Artificial classification system**

- ✓ **In this system, only one or at most a few characters are selected for use in making comparison among organisms**
- ✓ **Ranking is done subjectively**
- ✓ **Artificial system is monothetic**
- **Possession of unique set of characters/features is both sufficient and necessary for membership in the group defined**
- ❖ **Classifications followed by Theophrastus and Linnaeus are good example of artificial system of classification**



## 2. Natural classification system

- This classification system is based up on several to many characters selected for their value
- ❖ This system is therefore polythetic
- ✓ it places together organisms that have the greatest number of shared features
- NO single feature is either essential to group membership or is sufficient to make an organism a member of the group
- ❖ Classification system by Adanson and De Jussieu are good examples of natural classification system



### 3. Phyletic or Evolutionary classification system

- It is a classification that aims at reconstructing a sequence starting with the most primitive character and ending with the most advanced or derived character
- It ensures that each taxon recognized as a monophyletic i.e. has arisen by the diversification of a single ancestor
- According to phyletic classification system,





## 4. Phenetics classification system

- According to Harrison (1960), the word "Phenetic" is to mean a relationship by overall similarity of all available morphological characters
  - ✓ Thus, similarity/resemblance is determined based on a set of phenotypic characteristics of organisms (numerous characters with equal weight)
- The following methods/approaches are used for determining the similarity of the organisms under study
  - ✓ Selection of taxa-called Operational Taxonomic Units (OTUs) for study. These may be species, genera or families.
  - ✓ Selection of character states (minimum of 60 and 80 -100 are desirable for comparisons)
  - ✓ Description or measurement of character states
  - ✓ Ranking of all OTUs into the categories of the taxonomic hierarchy



## 5. Phylogenetic classification system

- is the concepts and methods of determining similarity by constructing branching patterns of evolution
- The method works as follows:
  - ✓ Select monophyletic taxa (descended from common evolutionary ancestor)
  - ✓ Select characters of evolutionary interest
  - ✓ Describe and/or measure character states
  - ✓ Determine homologies of characters and character states
  - ✓ Generate trees or cladograms
- Once the cladogram or tree of life is generated, it is possible to construct classification based upon the cladogram



## Hierarchical System of Classification

- An arrangement of taxa into an ascending series of ever-increasing inclusiveness forms
- In a hierarchical system, we start at the bottom with individuals (species) and end up at the top with one all-embracing taxon (Domain)
- The major levels in the hierarchy are the following

**Domain**

**Kingdom**

**Division/Phylum (pl phyla)**

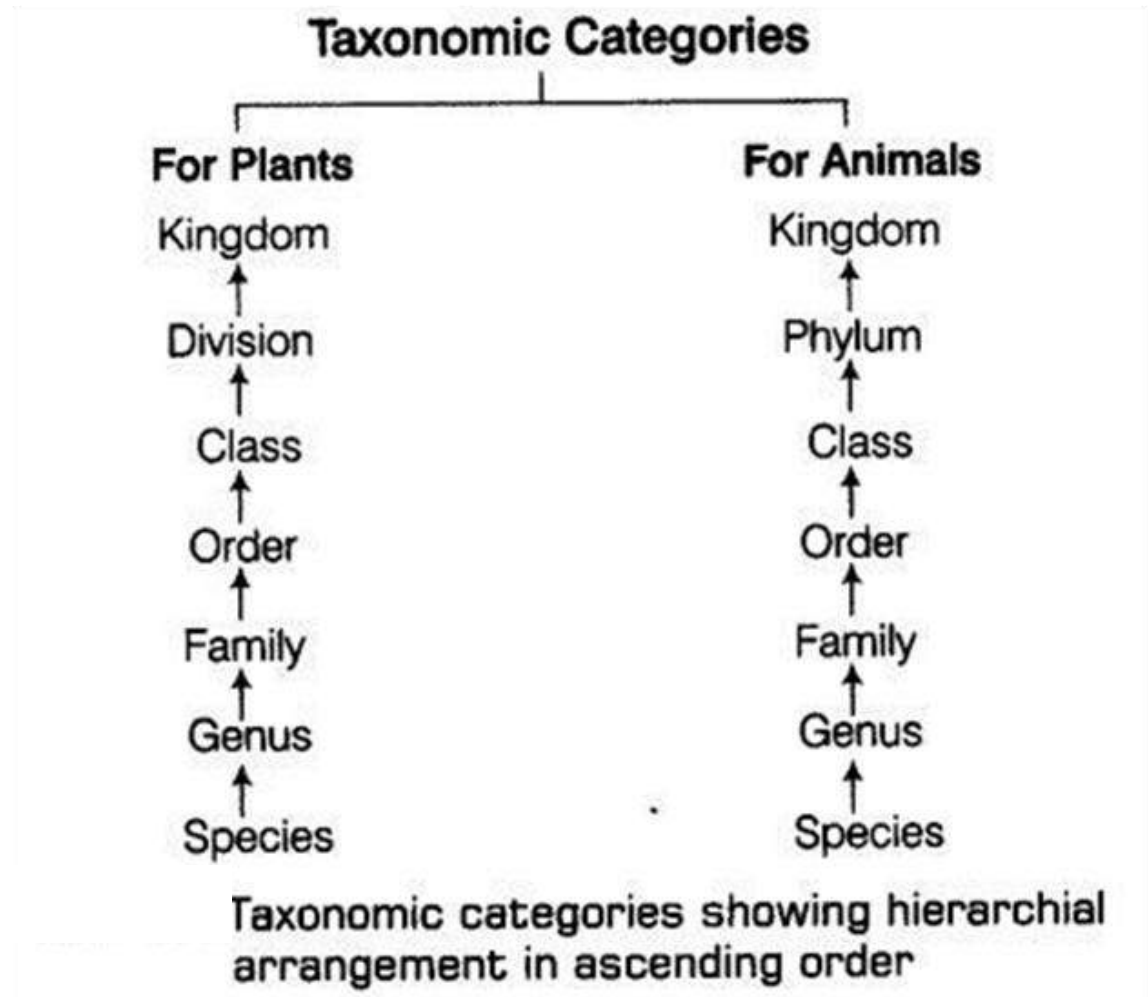
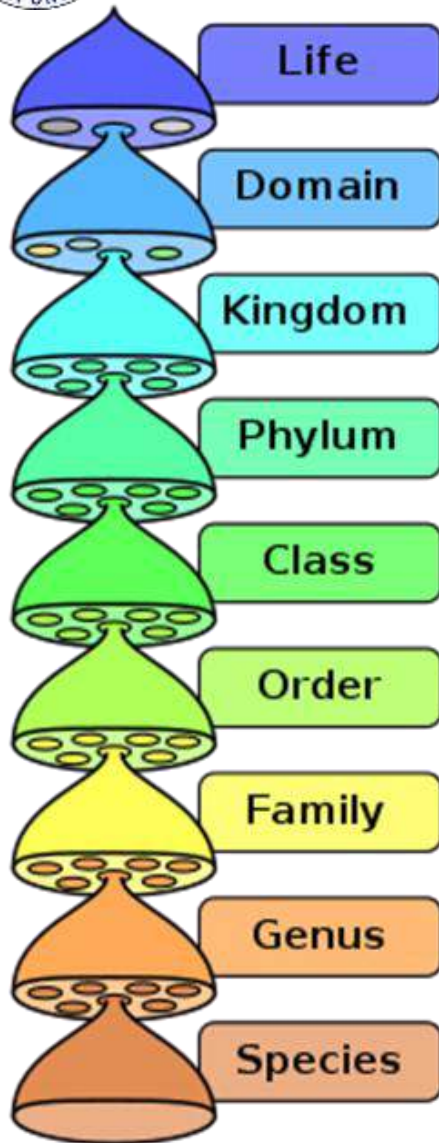
**Class**

**Order**

**Family**

**Genus (plural, genera)**

**Species**



# Concepts of the Kingdom

- Different kingdom systems have been introduced during different developmental stages of taxonomy
- Number of of kingdom has increased from 2 to 5 or 6/7.

## i) Two kingdom approach:

- **kingdom animalia** -living thing that moved, ate, and grew to a certain size
- **Kingdom plantae** - living thing that did not move or eat and that continued to grow
- **Mode of nutrition and presence or absence of cell wall.**

## ii) Three Kingdoms

- A third kingdom, protista (unicellular) was proposed by Ernst Haeckel in 1866.
- Whether organisms are unicellular (**Protista**) or multicellular (**Animals and Plants**).
- **Three Kingdom System**: **Kingdom Protista**, **Kingdom Plantae** and **Kingdom Animalia**.



### III. Four kingdoms

- The result of the discovery of the electron microscope
- A **four-kingdom** was proposed by **Copeland** in 1938
- Prokaryotes (**bacteria and blue green algae** = **kingdom Monnera**) and eukaryotics (**plantae, Animalia & Protista**) were distinguished
- Initiated the rank above kingdom, a **super-kingdom/ empire/ domain** = **Prokaryot vs Eukaryta**

## IV. Five kingdoms

- **Proposed by Whittaker in 1969**
- **Formation of the 5<sup>th</sup> kingdom was based mainly on differences in nutrition; multicellular saprotrophs (saprophytes) formed the kingdom **Fungi****
- **Fungi are prokaryotic, no chlorophyll & they contain chitin in their cell wall.**
- ***Thus, Kingdom Monera (prokaryotic) + Eukaryotics (Kingdoms Protista, Plantae, Fungi and Animalia )***

## VI. Six Kingdoms

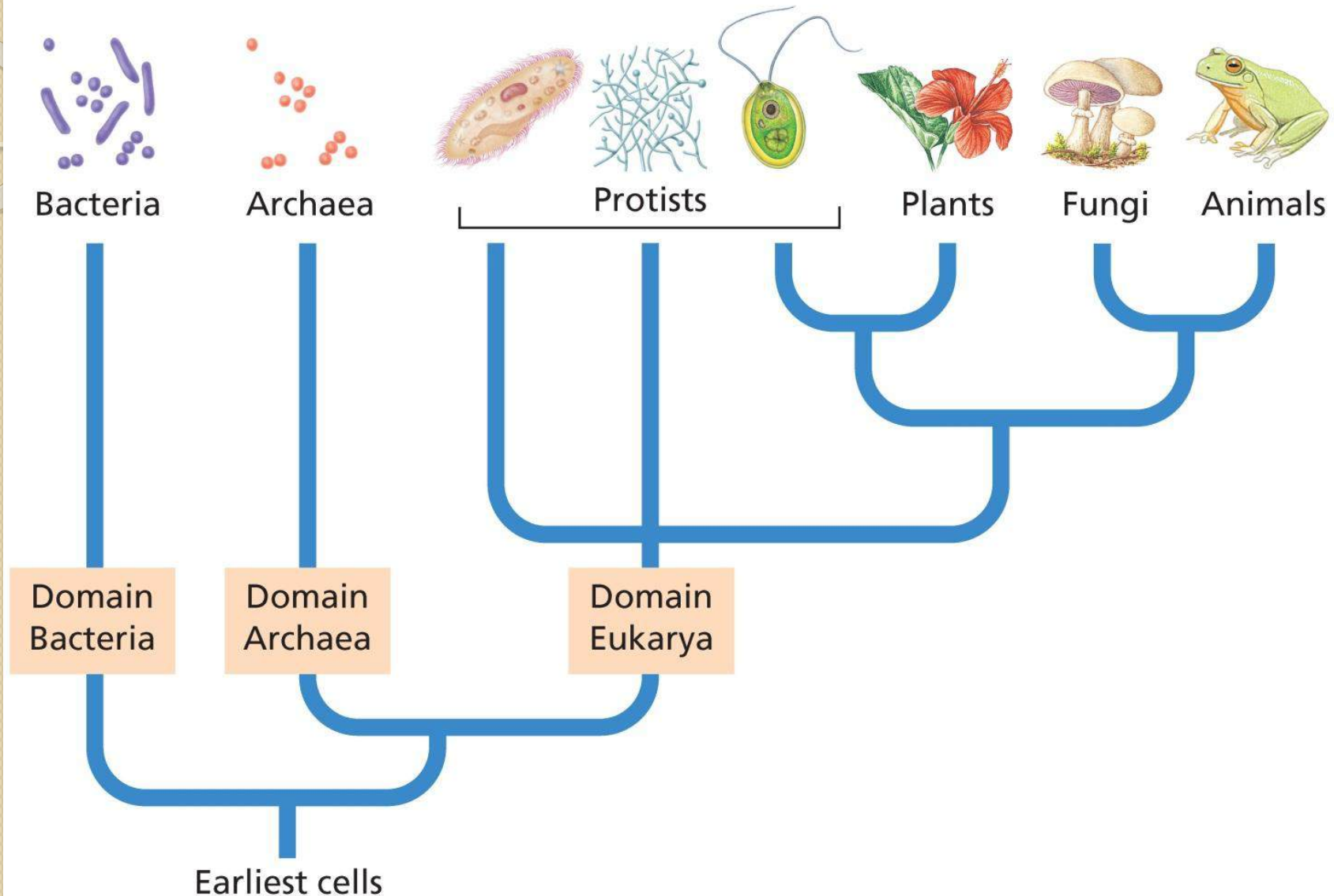
- Based on such RNA studies, Carl Woese (1977) divided prokaryotes (Kingdom Monera) into *Eubacteria* & *Archae bacteria*.
- Kingdom Bacteria, Archae, Protista, Plantae, Fungi & Animalia

|      |                 |                  |
|------|-----------------|------------------|
| Life | Domain Bacteria | Kingdom Bacteria |
|      | Domain Archaea  | Kingdom Archaea  |
|      | Domain Eukarya  | Kingdom Protista |
|      |                 | Kingdom Plantae  |
|      |                 | Kingdom Fungi    |
|      |                 | Kingdom Animalia |
|      |                 |                  |

# The Three-Domain System

- As biologists observed differences between two kinds of prokaryotes, they saw similarities among eukaryotes.
- A new system divided all organisms into three domains: Bacteria, Archaea, and Eukarya.
- Today, most biologists tentatively recognize three domains and six kingdoms.

# Phylogenetic diagram of major groups of organisms



# Kingdom and Domain Characteristics

| Domain   | Kingdom         | Characteristics |                             |                               |                               |                               |
|----------|-----------------|-----------------|-----------------------------|-------------------------------|-------------------------------|-------------------------------|
|          |                 | Cell type       | Cell structure              | Body type                     | Nutrition                     | Example                       |
| Bacteria | Eubacteria      | Prokaryotic     | Cell wall, peptidoglycan    | Unicellular                   | Autotrophic and heterotrophic | Enterobacteria<br>Spirochetes |
| Archaea  | Archaeobacteria | Prokaryotic     | Cell wall, no peptidoglycan | Unicellular                   | Autotrophic and heterotrophic | Methanogens                   |
| Eukarya  | Protista        | Eukaryotic      | Mixed                       | Unicellular and multicellular | Autotrophic and heterotrophic | Amoebas<br>Euglenas<br>Kelps  |
| Eukarya  | Fungi           | Eukaryotic      | Cell wall, chitin           | Unicellular and multicellular | Heterotrophic                 | Yeasts<br>Mushrooms           |
| Eukarya  | Plantae         | Eukaryotic      | Cell wall, cellulose        | Multicellular                 | Autotrophic                   | Ferns<br>Pine trees           |
| Eukarya  | Animalia        | Eukaryotic      | No cell wall                | Multicellular                 | Heterotrophic                 | Birds<br>Earthworms           |





# Binomial Nomenclature

- **Nomenclature is the naming of organisms and the rules governing the application of these names**
- **In nomenclatural system, each organism is designated by two names**
  1. **the first is the name of the genus**
  2. **the second is the specific epithet (in Botany)/specific names (Zoology)**
- **No two species can have the same scientific name.**
- **The names are always Latin (or Latinized) and the genus is capitalized while species name is not**



## The Purpose of giving names to organisms

- Name is a conventional symbol or code that serves as a means of reference

### Purpose

- ✓ Easy communication
- ✓ Avoids the use of an inconvenient descriptive phrase
- ✓ To universalize the organism
- **For this reason sets of rules have been drawn up**
  - ✓ International Code Botanical Nomenclature (ICBN)
  - ✓ International Code of Zoological Nomenclature (ICZN)
  - ✓ International Code of Nomenclature of Bacteria
  - ✓ International Code of Nomenclature for Cultivated Plants
- ❖ International Organizations & Unions for the Stabilization of Changes



# Operative Principles of Nomenclature

## ❖ Operative Principles of Nomenclature

➤ The nomenclature code has set certain provisions, called Operative Principles of Nomenclature. These provisions are

- ✓ Publication
- ✓ Typification and
- ✓ Priority

## ❖ Publication

- ✓ Publication is the means by which scientific names enter to the biological nomenclature system
- ✓ Two basic conditions must be fulfilled before a properly formulated scientific name
  1. The name must be published in works that are printed in permanent and made available to the interested public
  2. A name must be accompanied by a written matter or a reference to such description



## ❖ **Typification**

➤ A type is an element on which the description associated with the original publication of a name was based

➤ In nomenclature, there are different kinds of "Types". Some of these are:

### ❖ **Holotype**

✓ It is the sole element used or designated by the author of a name as a type

### ❖ **Isotype**

✓ The duplicates of a holotype (collected specimen at the same time by the same person from the same population).

### ❖ **Syntype**

✓ Any one of two or more elements used by the author of a name who did not designate a holotype (when more than one specimen are cited by the author without mentioning about the holotype)

### ❖ **Lectotype**

✓ Is an element selected subsequently from amongst syntypes to serve as a type

### ❖ **Neotype**

✓ An element selected to serve as a nomenclatural type when through loss or destruction of the above "Types"



## ❖ **Priority of publication**

- The principle of priority requires when two or more names apply to the same taxon
- In this case, the name is given generally based up on by the oldest one, which is validly published name
- In nomenclature, when two or more names are applied the same taxon, it is known as synonyms (**i.e. the synonyms are rejected name as only one scientific name is accept as correct name**)
- There are two kinds of synonyms
  1. Nomenclatural
- ✓ based on the same ‘ type‘ of the organism and called obligate or homotypic synonyms
- 2. Taxonomic synonyms
- ✓ based on different ‘ type‘ of an organism, and called heterotypic synonyms



# Authorities and Their Citations

- ❖ The scientific names of organisms are often written followed by one or more personal names / author name, who first published the name in the way that satisfies the criteria of valid publication or available names.
- Sometimes the names of the authors are abbreviated  
e.g. *Rumex abyssinicus* Jacq., *Plantago lanceolata* L.
- Sometimes, double citations are required due to change in taxonomic position or rank.  
e.g. *Carissa edulis* Vahl. was later changed to *Carissa spinarum* (Vahl.) L. by Linnaeus .